Heinemann Interactive Science 3

Christine Watson

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return to main menu

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eGlossary

return to main menu

Contents

Introduction vi

About the Author vii

Stage 5 Correlation Grid viii

Chapter 1 Working Like a Scientist

Chapter Outcomes 1
Introduction 1
1.1 Designing experiments 2
1.2 Better designs 4
1.3 Scientists solve problems 6
1.4 Science in Focus. The nature and practice of science: Medical breakthroughs 8
1.5 Developing your graphing skills 10
1.6 Trends, patterns and predictions 12
1.7 Science in Focus. Implications for society and the environment: Human experimentation and the computer body 14
1.8 Student research projects 16
1.9 Science in Focus. Current issues, research and development: Is your future in science? 20
1.10 Chapter Review 22

Chapter 2 Atoms, Molecules and Ions

Chapter Outcomes 23
Introduction 23
2.1 Science in Focus. The nature and practice of science: Changing models of atoms 24
2.2 Molecules and ions 28
2.3 The periodic table 30
2.4 Bonding 32
2.5 Writing chemical formulae 34
2.6 Ions everywhere 36
2.7 Testing for acids and bases 38
2.8 Science in Focus. Applications and uses of science: Maintaining a chlorinated pool 40
2.9 Electroplating 42
2.10 Chapter Review 44

Chapter 3 Chemical Reactions

Chapter Outcomes 45
Introduction 45
3.1 Chemical reactions 46
3.2 Science in Focus. History of science: Down with the phlogiston theory 48
3.3 A balancing act 50
3.4 Types of chemical reactions 52
3.5 Reactions of acids and bases 54
3.6 Reactivity of metals 56
3.7 Science in Focus. Applications and uses of science: Extracting and refining by using chemistry 58
3.8 Energy and changes 60
3.9 Reactions—speeding up and slowing down 62
3.10 Chapter Review 64

eGlossary

return to main menu

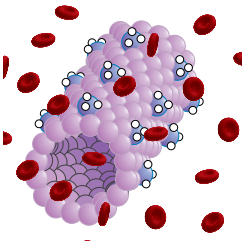
Chapter 4 Electricity

Chapter Outcomes 65
Introduction 65
4.1 Current electricity 66
4.2 Series and parallel 68
4.3 Catching on to the flow 70
4.4 What's the difference? 72
4.5 Ohm's law 74
4.6 Science in Focus. History of science: A history of electricity and magnetism 76
4.7 Electricity and magnetism 78
4.8 Using electromagnetic principles 80
4.9 Science in Focus. Implications for society and the enviroment: How much do we depend on electricity 82
4.10 Chapter Review 84

Chapter 5 Behaviour and Properties of Light

Chapter Outcomes 85
Introduction 85
5.1 Reflection in curved surfaces 86
5.2 Bending light 88
5.3 Using lenses—convex lenses 90
5.4 Using lenses—concave lenses 92
5.5 Eyes that need help 94
5.6 Science in Focus. History of Science: The nature of light and colour 96
5.7 Our coloured world 98
5.8 What colour do you see? 100
5.9 Science in Focus. Applications and uses of science: Photography . . 102
5.10 Chapter Review 104

Chapter 6 Simple and Complex Machines



Chapter Outcomes 105
Introduction 105
6.1 Making life easier 106
6.2 Which lever for the job? 108
6.3 Inclined planes 110
6.4 Wheels, axles and gears 112
6.5 Types of machines 114
6.6 Machines need energy 116
6.7 Science in Focus. History of science: Machines through the ages . . 118
6.8 Science in Focus. Implications for society and the environment: Machines and progress? 120
6.9 Chapter Review 122

Chapter 7 Reproduction

Chapter Outcomes 123
Introduction 123
7.1 Reproduction 124
7.2 Sexual reproduction in flowering plants 126
7.3 Sexual reproduction in animals 128
7.4 Science in Focus. Implications for society and the environment: ARKS 'n' SPARKS 130
7.5 Human reproduction. 1 132
7.6 Human reproduction. 2 134
7.7 Pregnancy and birth 136
7.8 Science in Focus. Current issues, research and development: Reproductive technology 138
7.9 Cloning around 140
7.10 Chapter Review 142

eGlossary

return to main menu

Chapter 8 Genetics

Chapter Outcomes 143
Introduction 143
8.1 The double helix 144
8.2 Genes and chromosomes 146
8.3 Science in Focus. Current issues, research and development: Genetic engineering 148
8.4 Multiplying and dividing 152
8.5 Characteristics and inheritance 154
8.6 Intermediate inheritance 156
8.7 Sex determination 158
8.8 Chromosomal abnormalities 160
8.9 Science in Focus. Applications and uses of science: Genetic counselling 162
8.10 Chapter Review 164

Chapter 9 Fighting Disease

Chapter Outcomes 165
Introduction 165
9.1 In sickness and in health 166
9.2 Infectious diseases 168
9.3 More pathogens! 170
9.4 Defence systems 172
9.5 Science in Focus. The nature and practice of science: Howard Florey and the penicillin story 174
9.6 Food alert 176
9.7 Science in Focus. Implications for society and the environment: Vaccination 178
9.8 Controlling disease 180
9.9 Chapter Review 182

Chapter 10 Energy and Matter Through Ecosystems

Chapter Outcomes 183
Introduction 183
10.1 The biosphere—where life exists 184
10.2 Producers versus consumers 186
10.3 Caught in a web 188
10.4 Science in Focus. Applications and uses of science: Pesticides—poisoning the web? 190
10.5 Decomposition 192
10.6 Cycling matter 194
10.7 Science in Focus. Current issues, research and development: Vermi—what? 196
10.8 The water cycle 198
10.9 Conserving the environment 200
10.10 Chapter Review 202

Chapter 11 Dynamic Earth

Chapter Outcomes 203
Introduction 203
11.1 Windows in the Earth 204
11.2 Earthquakes and tsunamis 206
11.3 Volcanoes 208
11.4 Rocks made of crystals 210
11.5 The tortured crust 212
11.6 Science in Focus. The nature and practice of science: Pieces of a puzzle 214
11.7 Science in Focus. The nature and practice of science: Completing the puzzle 216
11.8 Coping with a restless Earth 218
11.9 Chapter Review 220

Glossary 221
Acknowledgments 230

eGlossary

Introduction

The Heinemann Interactive Science series has been specifically designed to assist schools in the implementation of the New South Wales Science Syllabus Stages 4 and 5. This resource will provide students with a wide range of learning experiences to achieve the knowledge and understanding, skills and values and attitudes outcomes of the syllabus.

Features of Heinemann Interactive Science

Chapters begin with a short **Introduction** which places the content in a **context** which will engage students and stimulate their interest. The chapter context gives coherence to the knowledge and understandings, skills and values and attitudes being developed. The chapter **outcomes** are descriptions of the intended student outcomes which could reasonably be expected to be demonstrated by students who successfully complete the chapter. They address both core and options outcomes from the syllabus.

Each chapter contains two **Science in Focus** double-page spreads which specifically address the mandatory **Prescribed Focus Areas** of the syllabus. These spreads are:

- History of science: 3.2, 4.6, 5.6, 6.7
- The nature and practice of science: 2.1, 9.5, 11.6, 11.7
- Implications for society and the environment: 1.7, 4.9, 6.8, 9.7
- Applications and uses of science: 2.8, 3.7, 5.9, 7.4, 10.4
- Current issues, research and development: 1.9, 7.8, 8.3, 10.7

The double-page spread format of Heinemann Interactive Science provides all the content, experiments, discussion topics, activities and questions for a lesson. Each spread gives teachers the flexibility to treat a topic in whatever way is most appropriate for their students.

Experiments are detailed guided investigations relevant to each topic. Suggested extensions encourage students to investigate further, often using a more open-ended approach.

This icon is used to identify experiments for which safety glasses should be worn.

This icon is used to identify experiments that are regarded as having significant and identifiable risks associated with them. The Teacher's Resource Disk includes extra advice on minimising these risks.

The **Using Science** activities include a wide variety of learning experiences such as model making, the manipulation and interpretation of given data and diagrams, group discussions and debates.

Talk about involves students sharing and developing their personal understandings. These may be used as a lesson introduction, to place the topic in context and establish the students' prior knowledge.

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Each double-page spread contains a range of question types to reinforce students' understanding of the concepts. **Think about** questions ask students to go further with the concept they are learning about, to consider the application of the concept to different situations. **Find out** questions encourage students to research from a range of sources information related to the topic. They are aimed at extending students to obtain a greater depth of knowledge and understanding. Additional **Activities** are provided on many spreads. These additional learning experiences could be used as homework or assignment tasks. Many of the Find out and Activities involve students in writing tasks such as descriptive reporting, recounting, experimental reporting and narration.

Fact files are snippets of information designed to heighten students' interest in the concepts covered in each spread.

Each chapter concludes with a **Chapter Review**. These include exercises to reinforce the development of vocabulary, questions to check knowledge, and higher order questions to check application of knowledge. Some 'challenge' material is also included as additional or extension work.

Christine Watson

Teacher support

The **Teacher's Resource and Assessment Pack** consists of a teacher resource disk written by **Monika Khun** and a student's book. The disk contains assessment and course advice. The safety guidelines are of particular interest to less experienced teachers. A range of worksheets for each chapter is also provided. One worksheet per chapter addresses a functional approach to literacy. An end-of-chapter test with answers provided and answers to textbook questions will also save teacher preparation time.

Heinemann Science Online, an email service which brings relevant current events in science into the classroom, provides student worksheets with questions and activities and teacher notes which provide answers and further suggestions.

About the Author

Christine Watson is currently head teacher science at Colo High School. She has 20 years of teaching experience in both rural and metropolitan high schools. She has been involved in the writing and editing of numerous resources to support science teaching.

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Stage 5 Correlation Grid

The following grid references the **Stage 5** ***knowledge and understanding*** **outcomes** to the relevant chapters of **Book 3** and **Book 4**. The Teacher's Resource and Assessment Disk contains detailed information on mapping core content and assessment. A sample work program is available from hi.com.au/interactivesci.

Outcomes	Book Three Chapter	Book Four Chapter
	bold = principal source *italics* = secondary source	
5.1 explains how social factors influence the development and acceptance of scientific ideas	**3 4 5 6**	**5 8 10**
5.2 describes the processes that are applied to test and validate models, theories and laws	**2 9 11**	**1 3**
5.3 evaluates the impact of applications of science on society and the environment	**2 3 5 6 7 10**	**2 4 5 9**
5.4 discusses evidence supporting different viewpoints	**1 4 6 9**	**1 2 6 7 9**
5.5 analyses how current research might affect people's lives	**1 7 8 10**	**3 6 7 8 10**
5.6 applies basic physical models, theories and laws to situations involving energy, force and motion	**4 5**	*1* **3 4 5** *9*
5.7 relates properties of elements, compounds and mixtures to scientific models, theories and laws	**2 3**	**1** *2*
5.8 relates the structure and function of living things to models, theories and laws	**7 8 9**	**6 7 8**
5.9 relates the development of the universe and the dynamic structure of Earth to models, theories and laws and the influence of time	**11**	**10**
5.10 assesses human impacts on the interaction of biotic and abiotic features of the environment	**10**	
5.11 analyses the impact of human resource use on the biosphere to evaluate methods of conserving, protecting and maintaining Earth's resources		*2* **9**
5.12 describes scientific principles underlying some common technologies	*4* **6** *8 11*	*1* **4** *5*

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CHAPTER 1

WORKING LIKE A SCIENTIST

Some people claim they can find underground water by using a divining tool. The divining tool may be made of metal rods or a forked stick. Water diviners say they can feel the tool tremble or move when it is held over a spot where water is located underground.

The Australian Skeptics Society has offered $100 000 to anyone who can demonstrate beyond chance that they can divine water; the water diviner must demonstrate his or her ability in a controlled experiment. The experiment involves the diviner walking over a wooden box 2 m long, 1 m wide and 15 cm high. The box is divided into four equally sized compartments. A length of sealed pipe completely filled with water is placed in one of the compartments, initially with the knowledge of the water diviner. The diviner then checks that he or she can detect the hidden water with their divining tool.

The testing then begins by placing the water pipe in a compartment unknown to the water diviner. The experiment is repeated 20 times, the position of the water pipe being randomly selected. Chance alone would predict that a water diviner would correctly guess the position of the water around five out of 20 attempts.

So far no one has demonstrated any ability to detect water underground with a divining tool, so the $100 000 prize money is still on offer. Controlled experiments like this one can be used to seek evidence and test claims from the everyday to the slightly weird. In this chapter you will learn more about designing experiments and develop your skills in problem solving.

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Design and perform controlled experiments.
- Use appropriate strategies to solve problems.
- Interpret and analyse information and make conclusions.
- Think critically to evaluate investigations.
- Display data in an appropriate form by using column, sector or line graphs.
- Identify trends and patterns in data and make predictions.
- Discuss issues involving science which have implications for society and the environment.
- Identify possible career paths in science.

Web links

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1.1

Designing experiments

FIGURE 1.1
Experiments must be well planned if they are to be successful.

If you want to find out about something, you usually ask questions. Science involves asking questions and then finding answers to the questions.

Talk about

How many different methods can you use to find answers to questions. Brainstorm in a group and record all ideas on a sheet of paper. Compare your list with those of other groups. Do you have a preferred method of finding answers to questions?

Experimenting to find answers

One of the best ways to find answers is to **experiment**.

Experiments should be planned so that you can:

- observe how something behaves;
- investigate an observation;
- test an idea, prediction, inference or hypothesis; or
- find a solution to a problem.

If an experiment is to find answers that are reliable and true, it is essential that it be a fair test. In a simple experiment you change something and observe what happens. The thing you change on purpose is called the **independent variable**. The change that follows is called the **dependent variable**. The dependent variable is the factor which is measured as the results of the experiment. For example, an experiment is designed to test the hypothesis that cola-flavoured soft drinks contain more sugar than lemonade or orange-flavoured soft drinks. The independent variable in this case is the flavour of the soft drink and the dependent variable is the mass of sugar extracted from each.

The flavour of the soft drink is not the only factor or variable that can influence the dependent variable. Other **variables** such as the volume of soft drink used and the methods of collecting and measuring the sugar content also influence the results. These other variables must be kept the same or constant each time the test is repeated. This process is called controlling the variables.

Controlled experiments

Many experiments are **controlled experiments**. In a controlled experiment, observations or measurements are made on two groups: the experimental group and the control group. The **control group** is a standard by which any change can be measured. In the **experimental group** all conditions are kept the same as in the control group, except one.

In a controlled experiment, only one variable at a time is investigated. That way, if any differences are observed between a control group and an experimental group, then the difference must be due to the variable being investigated.

Controlling the variables in an experiment makes the results far more reliable. But in many experiments the results cannot be taken as proof of something. For an experimenter to say that the results collected are due to one particular 'cause', the experimenter must find out whether the same results are found when the cause is removed. This involves carrying out a controlled experiment. For example, to test the hypothesis that taking vitamin C tablets daily reduces the number of colds people catch would require a controlled experi-

FACT FILE

In science, it is important to record and report the results of your experiments in a clear way so that other people can understand what you have found out. A good scientific report usually has a title, an aim, a hypothesis, a list of materials used, a method, results, a discussion and a conclusion.

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USING SCIENCE 1 LINDA'S CONTROLLED EXPERIMENT

When Linda visited her aunt one day, she saw her empty her teapot into the garden. 'It makes my beans grow much taller,' explained Aunt Gertrude. Linda was not sure about her aunt's hypothesis, so she set up a controlled experiment.

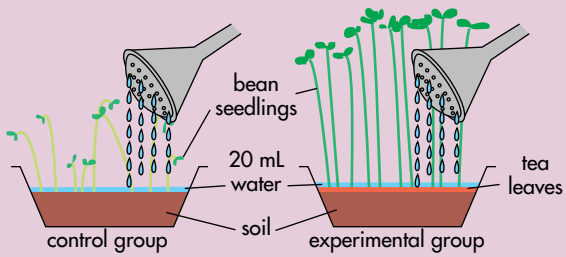

FIGURE 1.2

In Linda's experiment:

- 10 bean seedlings were planted in each pot
- each pot held the same amount and type of soil
- each pot received the same amount of sunlight
- each pot was watered with 20 mL of water every day
- the experimental group of bean seedlings had a teaspoon of used tea leaves added to the soil every day.

After 2 weeks, the bean seedlings which had tea leaves added seemed much healthier and taller than those which did not have tea leaves added.

DISCUSSION

1 What was the hypothesis Linda was testing?
2 What was the independent variable in Linda's experiment?
3 What was the dependent variable in Linda's experiment?
4 Which variables were controlled?
5 Did the results of the experiment support the hypothesis?
6 What could Linda conclude from this experiment?

ment. A control group of people who did not take vitamin C could be observed and compared with an experimental group who did take vitamin C tablets daily.

Questions

1 Explain why it is necessary to control variables in experiments.
2 What is the difference between the independent variable and the dependent variable?
3 Explain what is meant by a 'hypothesis'.
4 For each hypothesis listed below write down an independent variable, a dependent variable and at least two other variables to be controlled for a fair experiment.
a Black hair is stronger than blonde hair.
b A cup of coffee contains more caffeine than a cup of tea.
c A long pendulum swings faster than a short pendulum.
5 An experiment is set up to find out which of three different anti-rust treatments protects new cars the longest. The amount of rust which appears is measured and recorded over 5 years. Describe how a control could be set up for this experiment.

Think about

Alan plays tennis, but he is fussy about the tennis balls he uses. Consider the desirable characteristics of tennis balls, then design and perform a series of tests to find out, for each characteristic:
a which tennis balls Alan should buy for a grass court
b which tennis balls perform best on different surfaces.
Write a report for your experiments.

Foundation worksheet 1

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1.2

Better designs

FIGURE 1.3
Repeating experiments can highlight errors in measurement or faulty design.

Designing simple or controlled experiments is just the beginning of learning about experimenting. There are several other things to consider when designing experiments, for example, errors in measurements and repetition.

Talk about
Is it possible to say that an object is exactly 1.2 m tall?

Errors and accuracy

Mistakes or errors can occur because an instrument is not working properly or because a person is not using it correctly. Errors also occur when a mistake is made when reading a scale or writing down measurements. In each of these examples, the results obtained will not be **accurate**. By using measuring instruments correctly and being careful we can cut down on the number of errors made. However, you cannot get a better answer than your measuring instrument allows you. Scales used on instruments are marked into divisions. The smallest division used limits the **precision** of the instrument. Careful choice of measuring equipment is essential to good experiment design. For example, in one experiment involving the measurement of mass, a spring balance accurate to 1 g may be suitable but in another experiment an electronic balance accurate to 0.01 g may be needed. The electronic balance gives a more precise reading.

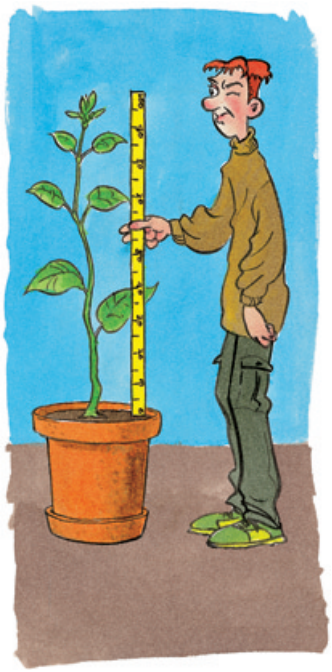

FIGURE 1.4
Make sure you line up the zero point on the ruler with the edge of the object and keep your eyes level with the scale when reading it.

Repetition

Once an experiment is completed, the first important step is to repeat it. The idea is to see if it is possible to obtain the same results. If the first set of results cannot be repeated then it is possible that there is an error in the measurements or a flaw in the design of the experiment.

Advanced experimental techniques

It is not always possible to control all the variables in an experiment. This is particularly true for experiments involving people or other living things. It is almost impossible to control the behaviour of living things or people's attitudes. Scientists can use design techniques which overcome these problems such as **counterbalancing** (sometimes called cross-over design) and blind experiments.

Using counterbalanced design involves doing a controlled experiment twice with the same subjects. In the second experiment the control group becomes the experimental group and the experimental group becomes the control group. This technique is used to reduce the effect of any

FIGURE 1.5
'Oh no, not another blind experiment!'

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EXPERIMENT 1 NOISE AND CONCENTRATION

This is a whole class experiment which uses a counterbalanced design to test a hypothesis.

AIM

To find out if loud music reduces the ability of students to concentrate on a task and recall information.

MATERIALS

- cassette player
- heavy-metal music tape
- two lists of ten simple words, sufficient copies for each student

METHOD

1 Form two equal groups called A and B.
2 For the first trial, group A is the experimental group and group B is the control.
3 Issue word lists to group A and allow 5 minutes to learn words while loud music is playing.
4 Collect word lists then ask each group member to write down words from memory.
5 Repeat for group B without loud music.
6 For the second trial, group B becomes the experimental group and group A the control.

RESULTS

Record the number of words correctly recalled for each group in trials 1 and 2.

DISCUSSION

1 In the first trial which group A or B correctly recalled the most words?
2 On the basis of the first trial only, what can you conclude about the hypothesis?
3 What other factors besides loud music could affect someone's ability to concentrate and recall information?
4 In the second trial which group A or B recalled the most words?
5 Do the results of the second trial change your conclusion about the hypothesis?
6 Explain why scientists use counterbalanced design for experiments involving people.

uncontrolled differences that may exist in the two groups.

One of the biggest problems when experimenting with people is that their ideas and attitudes can influence the results. This can be overcome if the subjects do not know if they are in the control group or the experimental group. This is called a **blind experiment**. For example, a drug company wishes to test a new product which they claim can increase muscle development. An experimental group and a control group are selected from volunteer body builders. All the volunteers are given tablets to take—real ones to the experimental group and fake ones to the control group. However, none of the body builders knows which he or she has been given. The fake tablet is called a **placebo**: it appears to be exactly the same as the real thing. After many months, changes in muscle development is compared between the two groups.

Questions

1 Write down three errors which could occur when using a triple-beam balance to measure mass?
2 Why is it so important to repeat experiments?
3 When are blind experiments necessary?

Find out

1 What is the placebo effect?
2 How can we tell how accurate a measurement is? (Hint: find out about uncertainty.)

Activity

Test the hypothesis 'Eggs get heavier when boiled'. Select the most suitable measuring instrument for this task.

FACT FILE

In some experiments, researchers use a double-blind design where not only the subjects but also the researchers do not know who is in the experimental group and who is in the control group. How might a double-blind design improve the results of an experiment?

Foundation worksheet 2

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1.3

Scientists solve problems

FIGURE 1.6
Finding new ways to keep our air clean is one of many problems that scientists are working on.

Talk about

What do you think are the big problems of modern society? Could science help to solve some of these problems? Are there some problems that science cannot help solve?

Being curious and seeking answers to questions is part of being human. Since ancient times people have sought to answer questions about themselves and their surroundings. The challenges of surviving, finding food and shelter, and keeping predators away were the first problems to be solved. Today people are faced with problems and questions that are far more complex. Is the Earth's atmosphere warming up? How do we cure AIDS? Is there life on other planets?

Scientists work to solve problems in many different ways. History shows that some problems were solved when scientists had a hunch or a flash of inspiration, while other scientists used **trial and error**, doggedly working on a problem year after year. Thomas Edison (1847–1931) tested thousands of different materials before he successfully invented the light bulb. Some problems have even been solved by accident.

The problem-solving process

Some solutions to problems are discovered through experimenting. Other problems are solved by designing and constructing a new device or invention. The two methods of solving problems involve a similar process.

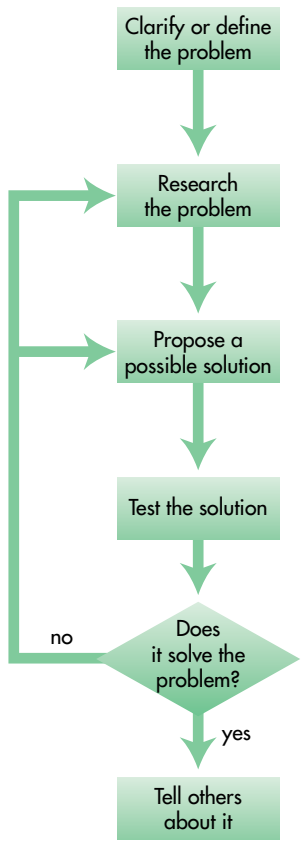

FIGURE 1.7
A flow diagram showing possible steps in solving a problem.

FACT FILE

Science is about making discoveries or finding new knowledge. Technology often uses scientific knowledge to invent new devices or improve existing ones. The two go hand in hand. With improved technology, scientists can make new discoveries!

Questions

1 Describe three ways problems can be solved.

2 What are the similarities and differences between science and technology?

3 Generate a list of problems by completing the following phrases. Be as creative as possible.
What would happen if ____?
Is there another way to ____?
Will it still work if ____?

Find out

Select one of the following scientists and find out what accidental discovery he made that led to a problem being solved: Edouard Benedictus, Charles Goodyear, Alexander Fleming, Louis Jacques Mande Daguerre.

Extra worksheet

Internet assignment

Web page

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USING SCIENCE 2 PROBLEM SOLVING THROUGH EXPERIMENT

Work in a group and select one of the following problems.
Are expensive brands of aluminium foil really better than cheap brands?
Which drink could do the most damage to children's teeth, soft drink or fruit juice?
Some brands of products sold as solutions are labelled as 'super strength' or 'highly concentrated'. Are they really different from standard brands?

STEP 1 CLARIFY THE PROBLEM

The problem which you select should be made more specific or narrowed down to a single variable to investigate. For example, what do we mean by 'better aluminium foil'? Stronger, thicker, shinier or more able to insulate food? Which ingredients in drinks could damage teeth? Sugar, food acid, artificial colouring? What is added to solutions to change their concentration or strength? Choose one specific variable to investigate and rewrite the problem.

STEP 2 RESEARCH THE PROBLEM

It may help to talk to other people about the selected problem or gather more information from sources such as books, CD-ROMs or the Internet.

STEP 3 PROPOSE A POSSIBLE SOLUTION

Make a hypothesis, a specific statement that answers your question. Then design an experiment to test the hypothesis. Decide on the independent variable (what you will change). Work out how you will measure the dependent variable. Consider how you will make your experiment fair by controlling the other variables.

STEP 4 TEST THE SOLUTION

Carry out your experiment, taking care with all measurements. Record your measurements in a data table. Repeat your experiment as many times as possible.

STEP 5 DOES IT SOLVE THE PROBLEM?

Look at your results; do they support your hypothesis? If the results are unclear, think about the design of your experiment. How could it be improved? Your results may not support your hypothesis. If possible propose an alternative hypothesis.

STEP 6 TELL OTHERS ABOUT IT

Report your findings or solution to others by writing a report and/or giving an oral presentation.

USING SCIENCE 3 PROBLEM SOLVING THROUGH DESIGN

Work in a group and select one of the following problems.
Emma wants to find out how well people can judge short time intervals. She needs a device which can operate for 90 s.
Last year's winning entry in a balloon-powered vehicle competition travelled 6.5 m in a straight line. Design and construct a balloon-powered vehicle that can travel further.
Mrs Klopper suffers from arthritis in her knee joints and has great difficulty bending to pick up small objects off the floor. Design and construct a device which she can use to avoid unnecessary bending.

STEP 1 CLARIFY THE PROBLEM

Discuss your selected problem and rewrite the task as a single statement (design brief).

STEP 2 RESEARCH THE PROBLEM

Find out more about the topic. Make sketches of construction ideas and investigate materials and techniques.

STEP 3 PROPOSE A POSSIBLE SOLUTION

Choose your best idea and develop your design on paper.

STEP 4 TEST THE SOLUTION

Build your device and test it.

STEP 5 DOES IT SOLVE THE PROBLEM?

Evaluate your solution against your design brief. How well does it work? Compare your solution with those of other groups.

STEP 6 TELL OTHERS ABOUT IT

Give an oral presentation and demonstration of your solution to your class.

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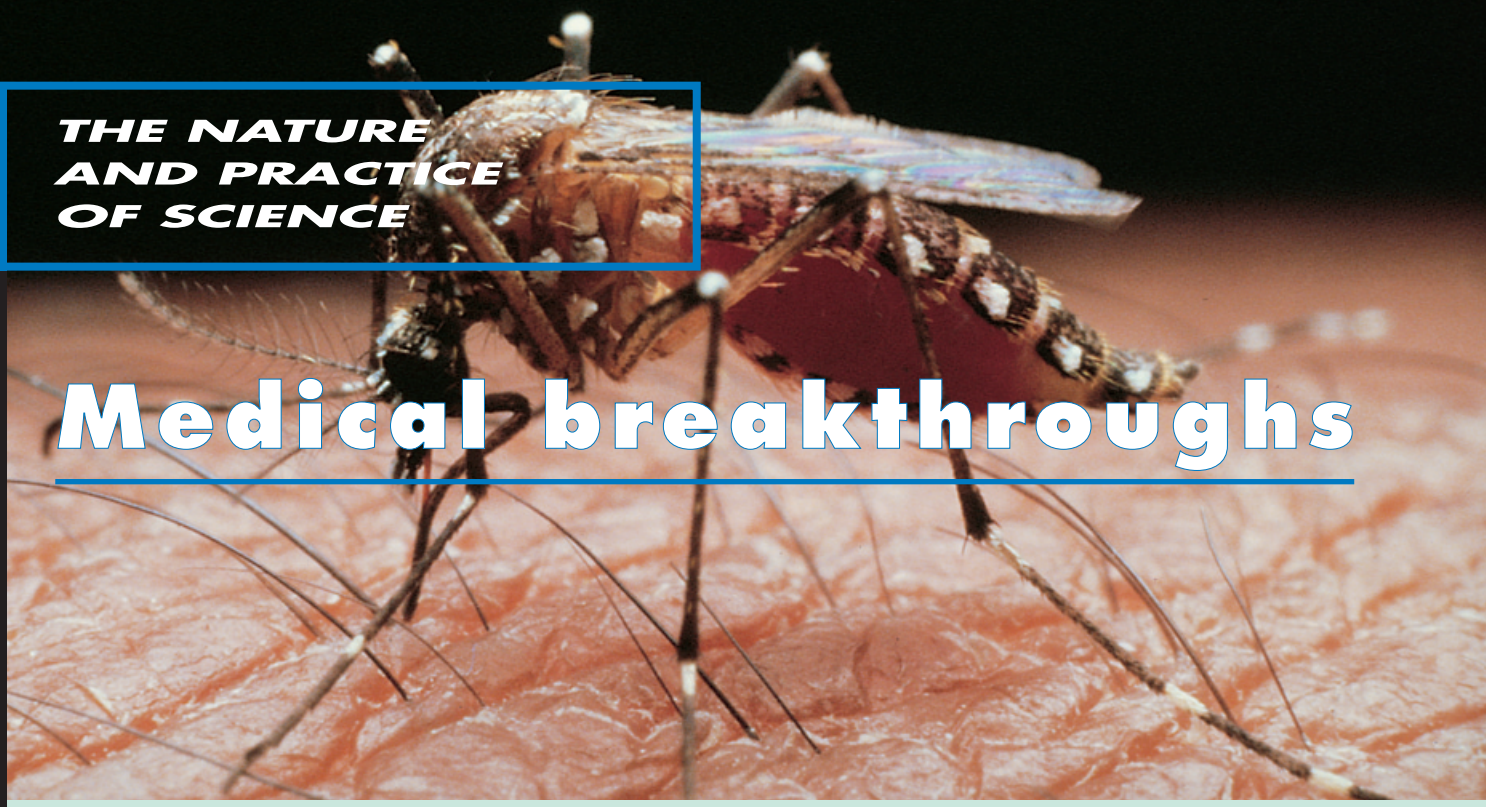

Medical breakthroughs

FIGURE 1.8
The mosquito *Aedes aegypti* can carry yellow fever.

Yellow fever is a disease that has killed thousands of people living in tropical regions. The disease produces fever, bleeding, vomiting and jaundice (a yellowing of the skin). Death may occur a few days after the onset of symptoms. It has now been controlled in most places in the world.

Experimenting to find the cause of yellow fever

The Panama Canal was built between 1882 and 1914 in the Central American country of Panama. It is a canal which connects the Atlantic and Pacific Oceans.

During the building of the Panama Canal, yellow fever killed so many people that the project had to be postponed for many years. It was important to find the cause of this deadly disease. A team of **doctors** worked together to investigate the problem.

Before 1900, it was thought that the disease was spread through contact with the person or clothes of a person who had the disease. In 1900, one of the doctors in the investigation team believed that a certain kind of mosquito, *Aedes aegypti*, spread yellow fever by biting people.

The doctors, led by Dr Walter Reed, investigating the cause of yellow fever found that they had three problems to solve.

Problem 1: Is yellow fever carried by the clothing of people who have the disease?

To investigate this idea, the doctors put on the clothes of a man who recently died of the disease and slept on his bed. They screened the building completely.

Observation: Nothing happened. The doctors did not contract the disease. The same experiment was tried several times with many other well people. None of these people contracted yellow fever.

Conclusion: Yellow fever is not carried by clothing. People do not 'catch' yellow fever by coming into contact with the victims or their clothing.

Problem 2: Do *Aedes aegypti* mosquitoes cause yellow fever?

To answer this question, the doctors hatched some of these uninfected mosquitoes and allowed them to bite healthy people who were kept away from other people.

FACT FILE

Some cases of yellow fever have occurred where *Aedes aegypti* mosquitoes are not present. Scientists are now searching for another carrier of a special form of yellow fever called 'jungle yellow fever'.

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Observation: Nothing happened. No symptoms of the disease ever appeared in the healthy people.

Conclusion: The experiment showed that the *Aedes aegypti* mosquito will not transmit the disease to a well person if the mosquito has not been infected with the disease.

Having shown that neither clothing of an infected person nor uninfected mosquitoes could cause the disease, the doctors now turned to the third problem.

Problem 3: What happens when an infected mosquito is allowed to bite a well person?

An American army doctor, Jesse Lazear, suspected that yellow fever was transmitted when an infected *Aedes aegypti* mosquito bit a human. He tested this idea by allowing himself to be bitten.

Observations: The doctor came down with yellow fever a few days after being bitten. He died 3 weeks later. A soldier who volunteered survived after a serious bout of yellow fever.

Conclusion: Infected *Aedes aegypti* mosquitoes carry yellow fever.

The practical application of these experiments was at once clearly seen. If the *Aedes aegypti* mosquito could be destroyed, at least one cause of yellow fever epidemics could be removed. The actual cause of yellow fever has since been found to be a virus.

The Panama Canal was built only after draining all the swamps in that region where these mosquitoes bred. Now, yellow fever has been completely eliminated from this region.

Australian scientists

Australia has produced a remarkable number of scientists who have found practical solutions to many life-threatening medical problems. The work of Australian scientists has improved the lives of millions of people around the world; however, very few Australians have even heard their names.

One of these unsung heroes was Dr Norman McAlister Gregg (1892–1966). He solved the problem of why some babies were born blinded by cataracts. Dr Gregg was an ophthalmologist (a scientist investigating the structure, function and diseases of the eye). He was a very busy doctor treating many patients. He observed a large number of young children, all around the same age, all suffering from cataracts. Some of the children also had heart disorders and deafness. Dr Gregg suspected a common cause. He set about meticulously searching for something that the children had in common by checking and cross-checking medical records from children across the country.

Finally, in 1941, Dr Gregg found a '**pattern**', one common factor in all the children's lives. Their mothers had caught German measles (also called rubella) early in their pregnancies. As a result of Dr Gregg's work, women of child-bearing age are now protected against rubella by a vaccine. Dr Gregg received a knighthood for his work in 1953.

Questions

1 When the scientists investigated whether wearing the clothes of an infected person caused yellow fever, why did they screen the building completely?

2 Identify any dangerous actions undertaken by the team of doctors, and suggest a safer way to solve the problem.

3 For each problem investigated by the team of doctors, identify:

a the aim of the investigation
b the materials used
c the method followed
d the observations
e the conclusion.

Find out

What medical problem did each of the following Australian scientists help solve?

a Donald Metcalf
b Susan Beal
c Graeme Clark
d Stran Sutherland

FACT FILE

Many deaf people have had their hearing partly restored because of the work of a team of Australian scientists led by Professor Graeme Clark. This team, working at the University of Melbourne, developed the first bionic ear (cochlear implant). This device consists of a speech processor which converts sound waves into radio waves for transmission to a stimulator. The stimulator is surgically implanted behind the patient's ear.

The stimulator converts the radio waves into electrical impulses which travel to the cochlear nerve fibres. A sensation of hearing is then produced in the brain.

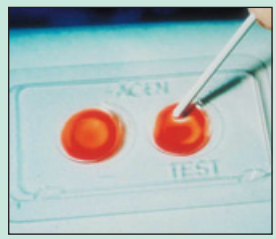

FIGURE 1.9
Australian medical invention SimpliRED, a hepatitis B antigen test.

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1.5

Developing your graphing skills

FIGURE 1.10
Scientists keep careful records of their observations and experimental results. The data collected can then be organised in a number of ways, including tables, graphs and diagrams.

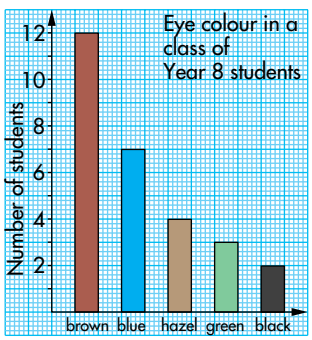

FIGURE 1.11
Eye colour as a column graph.

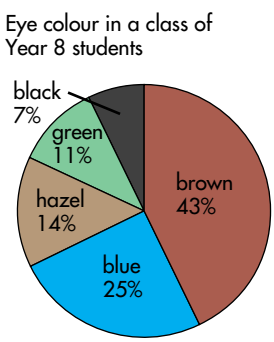

FIGURE 1.12
Eye colour as a sector graph.

Talk about

Which is the busiest street in your suburb? Work in groups of at least eight. Each person should choose a different street to survey. At an agreed time, count and record the number and type of car that passes you for half an hour. Present your results in the form of a table, graph or diagram.

Managing data

When you have performed an experiment, you will have collected information, or **data**. Often, data is easier to handle and analyse if it is presented in a table or a graph. A graph shows numerical data as a diagram.

There are three types of graphs: **column graphs**, **sector graphs** and **line graphs**. Each type of graph shows data in a different way. Graphs should be drawn by pencil if a computer is not used.

Table to column and sector graphs

Data is often collected and entered directly into a table.

In a class of 28 Year 8 students, for example, a survey was taken of eye colour. The results were presented in Table 1.1. These results can be presented in a column graph as shown in Figure 1.11.

Column graphs allow you to compare measurements—such as length, height or weight—or the sizes of objects. Column graphs also allow you to compare amounts and changes. All column graphs have a title, a scale and equal divisions.

TABLE 1.1

Eye colour	Number of students	Percentage of students
Blue	7	25
Black	2	7
Brown	12	43
Green	3	11
Hazel	4	14

The data for class eye colour could also be presented as a sector graph (Figure 1.12). A sector graph can be thought of as a circle graph which is divided. The full circle represents the entire sample (100%) you have surveyed. When the circle is divided, you can see what fraction, percentage or share each part represents.

Line graphs

Line graphs show changes in data and help you see trends or patterns in data. Unlike column and sector graphs, several different sets of data can be plotted on one line graph so that comparisons can be made.

When drawing line graphs, points representing the data are plotted on the graph. The points are then connected by a line.

Questions

1 Look back at the graph you constructed for your traffic survey.
a How could your graph be improved?
b Use a different type of graph to represent your results.

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USING SCIENCE 4 LOOKING AT LINE GRAPHS

Binh heated 100 mL of water with a Bunsen burner for over 1 minute. The following is a **line graph** which shows his results:

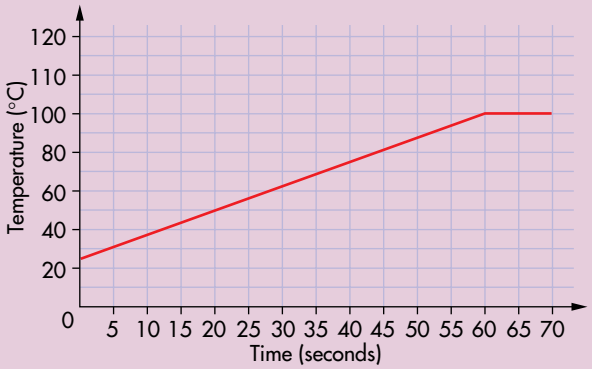

FIGURE 1.13
Line graph: the temperature of heated water.

DISCUSSION

1 Complete the following table of Binh's results:

TABLE 1.2

Time (seconds)	Temperature (°C)
10	
20	
30	
40	
50	
60	
70	

2 Why didn't the graph start at 0°C?
3 What happened after 60 seconds?
4 Why is this graph drawn as a line graph rather than a column graph or a sector graph?

EXTENSION

Look at the line graph in Figure 1.13. What rules do you think should be followed when drawing line graphs? Check your answers with your teacher.

2 Which type of graph would you use to represent:
a the speeds of the ten fastest animals in the world
b the change in Australia's population over the last decade
c the average number of children in Australian families
d a Year 8 class's favourite subjects.

3 The average temperatures of five Australian capital cities were announced as being 25°C, 17°C, 35°C, 29°C and 13°C.
a Which cities could these temperatures represent?
b Draw a graph to show this information.

4 Emily bought four pet mice. She weighed them when she bought them, then weighed them again 2 months later. This is what Emily wrote in her diary:
***10 March:* I'm so excited! I've just bought four pet mice! Zig weighs 25.6 grams, Zag weighs 27.5 grams, Zoe weighs 22.4 grams and Zac weighs 28.8 grams.**
***10 May:* My mice have learned to do lots of tricks! Zag is the most active and Zoe seems to learn the fastest! When I weighed them, I found that all of them had gained weight. Zig now weighs 27.8 grams, Zag weighs 29.1 grams, Zoe weighs 25.9 grams and Zac weighs 31.2 grams.**
a Write Emily's data in a table.
b Show Emily's data in a graph.
c Which mouse had gained the most weight?
d Which mouse had gained the least weight?

Think about

Look at the column graph in Figure 1.11 and the sector graph in Figure 1.12. What rules do you think should be followed when drawing column graphs and sector graphs? Check your answers with your teacher.

Extension worksheet

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1.6

Trends, patterns and predictions

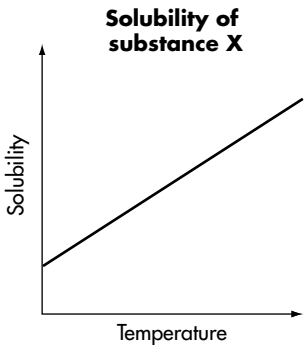

FIGURE 1.14
The trends is, as the temperature increases, the solubility of substance X increases.

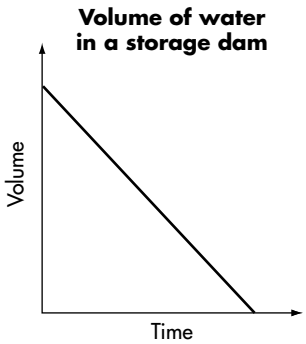

FIGURE 1.15
The trend is, as the time increases, the volume of water in a storage dam decreases.

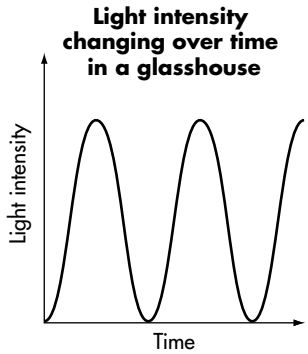

FIGURE 1.16
The pattern is, as the time increases, the light intensity in the glasshouse increases then decreases in a regular cycle.

Line graphs are probably the most useful type of graph used in science. They are often used when analysing the results of an experiment. Correctly drawn line graphs can be used to describe **trends** or **patterns** in data and make **predictions**.

Describing trends or patterns

When scientists analyse data they look for trends or patterns. A trend is both the direction and rate of change in data. The shape of a graph can show a trend. When a trend is repeated over time we can describe a pattern. Patterns can be easier to find when data is graphed.

Rules for drawing line graphs

Selecting the correct axes

The **independent variable** (the one changed by the experimenter) is placed on the horizontal axis. You can usually tell which is the independent variable because the measurements change by a regular amount.

Putting the scales on the axes

The scale you put on an axis must be a sequence of numbers with equal intervals between them. The size of the scale intervals should allow you to plot all the measurements, producing the largest graph possible in the space available. Always begin each scale with zero.

Labelling

Each axis should be labelled with the quantity and the units being plotted. The graph should also be given a title.

Plotting the points and drawing a line

If there is a connection between the two variables (a change in one variable causes the other to change) then a line of best fit is used. This is a smooth curve or straight line which passes as close as possible to all points (it may pass through some points but not necessarily all). Try to have roughly equal numbers of points on both sides of the line of best fit.

If there is no direct connection between the two variables (a change in one variable does not cause a change in the other) then a line may be drawn which simply joins the plotted points. Never connect the line to a zero point unless zero is one of the measurements to be plotted.

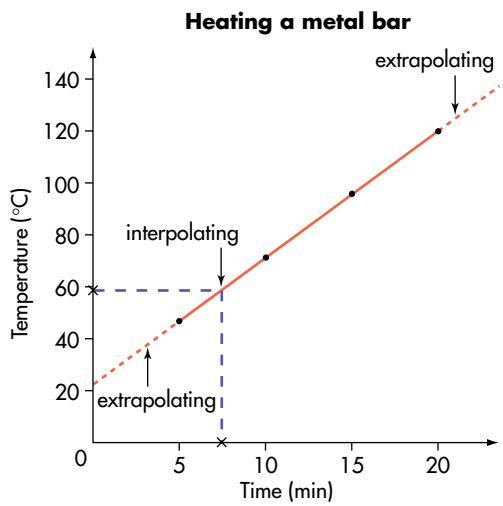

FIGURE 1.17
Interpolating is making a prediction within a range of measurements. Extrapolating is making a prediction beyond a range of measurements.

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USING SCIENCE 5 WHAT'S WRONG WITH THIS?

Ben investigated the effect of changing the length of a pendulum on the time taken for it to swing forwards and back.

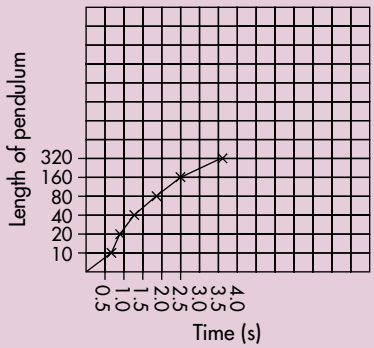

FIGURE 1.18
Ben's line graph of results.

TABLE 1.3

Length of pendulum (cm)	10	20	40	80	160	320
Time for complete swing (s)	0.6	0.9	1.3	1.8	2.5	3.6

Ben has not followed the rules of drawing line graphs. He has made many mistakes.

1 Make a list of all the mistakes made in this graph.

2 Re-draw the graph, correcting all the mistakes.

3 Write a statement which describes the trend between the length of a pendulum and time.

4 Compare your re-drawn graph with another student's and discuss any differences.

Questions

1 For each set of data in Tables 1.4 and 1.5 say which variable should be placed on the horizontal axis and whether the plotted points should be joined or a line of best fit used.

TABLE 1.4

Mass added to spring (g)	50	100	150	200	250
Distance spring stretches (cm)	2.5	5.1	7.5	9.9	12.5

TABLE 1.5

Time	9am	10am	11am	12pm	1pm	2pm
Distance from home (km)	0	10.3	19.2	28.7	28.7	41.3

2 Describe the trend or pattern shown in each graph in Figures 1.19 and 1.20.

3 Draw a graph of the data in Table 1.4. Use the graph to predict the distance the spring would stretch if the mass added was

a 125 g

b 275 g.

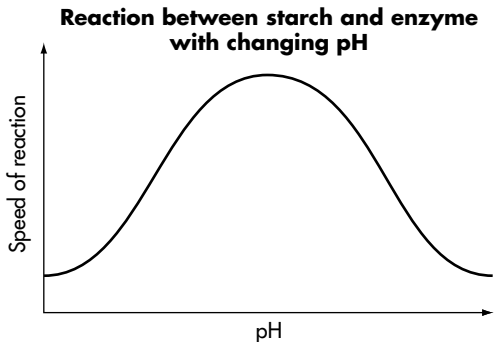

FIGURE 1.19

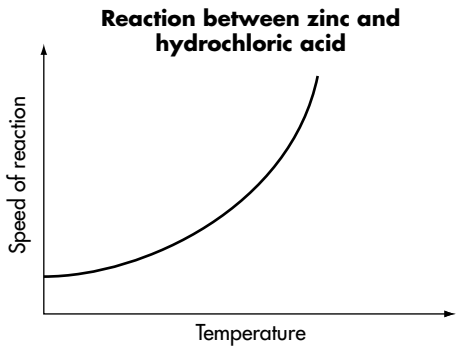

FIGURE 1.20

Find out

What are scatter graphs and when are they used in science?

FACT FILE

Graphs are drawn by using actual measurements. It is possible to use graphs to make predictions about a value not measured. If the prediction is about a value between two measurements, the process is called interpolating. If the prediction is about values beyond the last measurement the process is called extrapolating.

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IMPLICATIONS FOR SOCIETY AND THE ENVIRONMENT

Human experimentation and the computer body

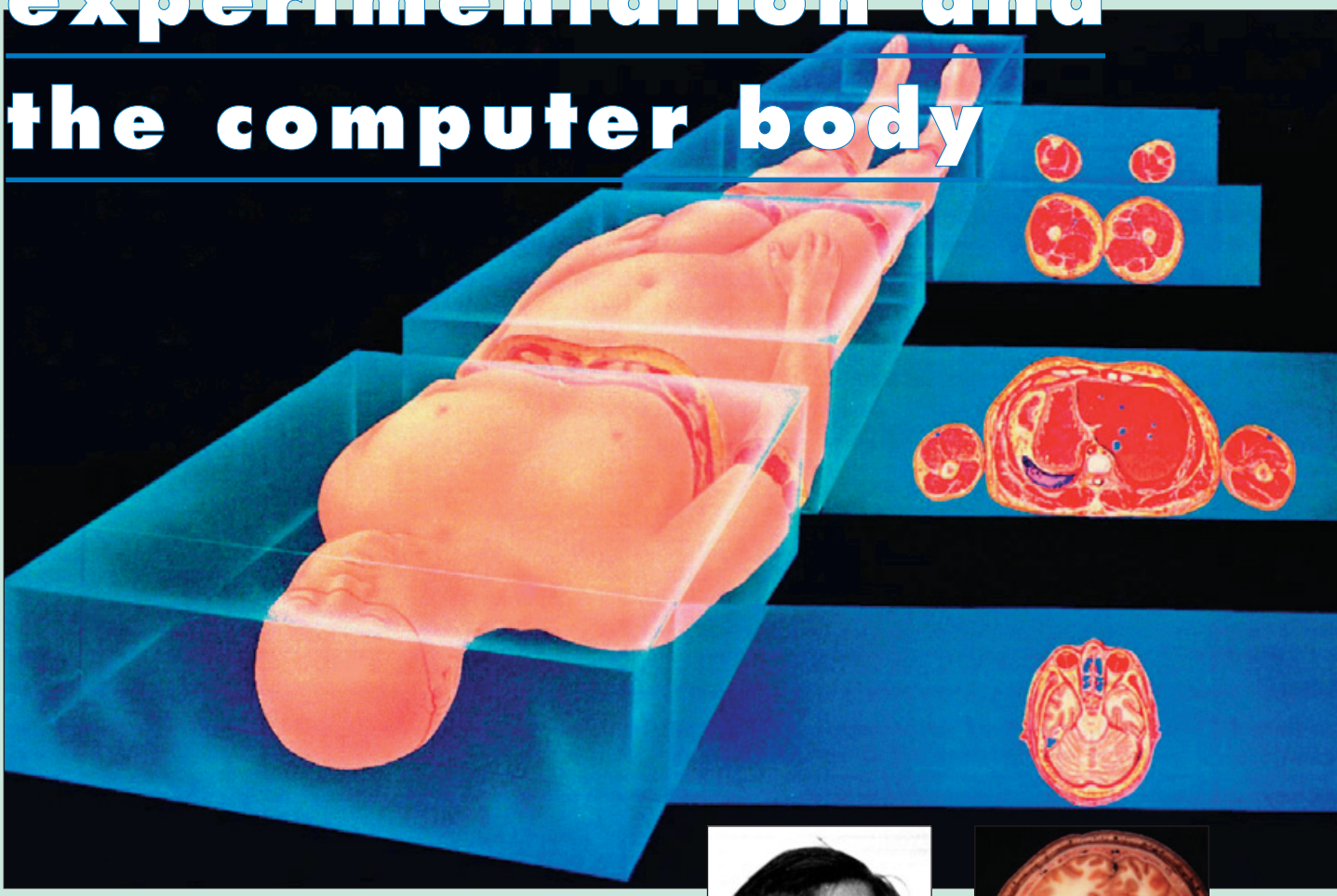

FIGURE 1.21
What is the world's greatest medical teaching tool? A book? A patient? No … it's the 'Visible Man' — a man you can take apart and put together again on a computer screen!

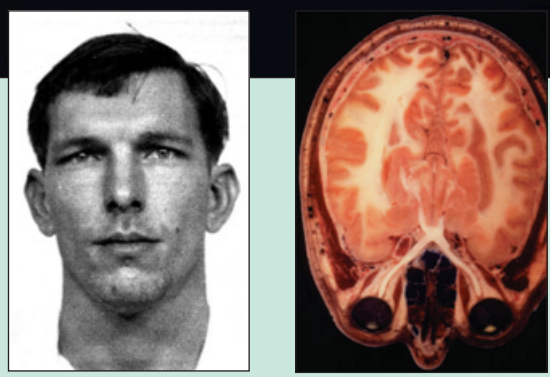

FIGURE 1.22

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Talk about

Discuss the advantages and disadvantages of a dissection of a body image on a computer screen rather than an actual body on a dissection table.

What happens when a criminal who has a healthy body is executed and donates his body to science? What was once a human body has now become a detailed computerised atlas of the human body known as the 'Visible Man'.

The body was sliced up by machine into 1870 1-mm strips, then pieced together on computer screen by using thousands of X-rays and magnetic photo and photo images from head to toe.

Now the Visible Man lives forever on computer screens around the world as the first three-dimensional computerised dead body. The information from the body is so detailed that it uses 15 gigabytes of computer storage space.

An extremely healthy specimen of man was needed since doctors will use the Visible Man's organs to compare diseased organs from living people. It took scientists over 2 years to find a suitable man who was happy to donate his body to science.

Every part of the Visible Man's body has been linked in words and pictures so researchers can just type in a name and the required picture will appear on screen. Medical students study the human body by dismembering the electronic image of a human. Surgeons can use the Visible Man to practise surgical procedures. Doctors will be able to use a program to simulate cancer in the body of Visible Man and learn how best to fight it. Scientists will use a program to simulate ageing to see exactly how and why it occurs.

A scaled-down version of the Visible Man on CD-ROM is used by schoolchildren to learn about the human body.

Human and animal experimentation

Humans and, more often, animals are used for many scientific experiments, including testing for drug safety and effectiveness, cosmetics testing and environmental monitoring. Both human and animal experiments are controlled by law.

Think about

Use your library to find an example of animal experimentation. In groups of four, discuss these issues:

- **What are the advantages and disadvantages of using humans or animals in scientific research?**
- **Can you think of a scientific experiment which involves the use of humans or animals? If so, suggest a different way the experiment could be done without using humans or animals.**
- **Who should decide whether humans and animals can be used in scientific research?**

In your group, develop a policy for the humane use of humans or animals in scientific research.

FIGURE 1.23
Instead of operating on donated bodies, medical students study the human body by pulling apart the electronic image of a human on a computer screen.

FACT FILE

In addition to the Visible Man, a computerised atlas of a 59-year-old woman has been completed. The subject died of a heart attack and willed her body to science. Her body was sliced into segments 0.3 mm thick, even finer than the Visible Man. This project was under taken by the United States National Library of Medicine. For more information on the Visible Human Project search the Internet for http://www.nlm.nih.gov

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1.8

Student research projects

A student research project is an excellent opportunity to apply the knowledge and skills you have gained during your science course. By this stage you should be able to undertake a more difficult research project. Your project could use more advanced experimental design, measuring and graphing techniques such as those discussed in this chapter.

Getting started

Choose a topic

Propose a topic of your own or select one from the list provided. The topic you select should be one that you find interesting. This will stimulate your curiosity and creativity. You must discuss your topic with your teacher to check that your topic is workable. A workable topic is one which takes into account your knowledge, skills, the time available and the resources required. Your topic must not be explosive, involve cruelty to animals or damage the environment.

Working in a group

Scientists often work in groups and it may be possible that you can work with others to undertake your research project. If group work is to be successful it will be essential that the role and responsibilities of each group member is clearly determined before you begin. Group work can be fun and provide advantages as more ideas are pooled and discussed. More research can be completed as the work load is shared. However, problems can occur if all members of the group do not communicate clearly and contribute equally.

Starting a log book

At this stage you will need to obtain a small exercise book to use as a log book or diary. Your log book will be used to record all your preliminary planning, research notes, results and a rough draft of your report.

FIGURE 1.24
How could you prevent this problem?

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Library research

It is essential that you do some background reading on your project topic before you clarify the problem and design an experiment. There are a range of books, journals and multimedia sources of information to scan for useful ideas. Before you begin to look for information make a list of key words related to your topic. Use these key words to search indexes, catalogues and the Internet. When you find useful information make brief notes and record the sources of the information in your log book. It may be necessary to contact experts directly for assistance. Write a letter or use e-mail to request information.

Planning your project

Time lines

A student research project will test your time-management skills. You will be given a set amount of time to complete your research and submit a report. A good way of meeting work dead lines is to plan a time line and stick to it. To do this you will need to make a list of tasks to be completed with an estimate of the amount of time needed. These tasks can then be written onto a calendar.

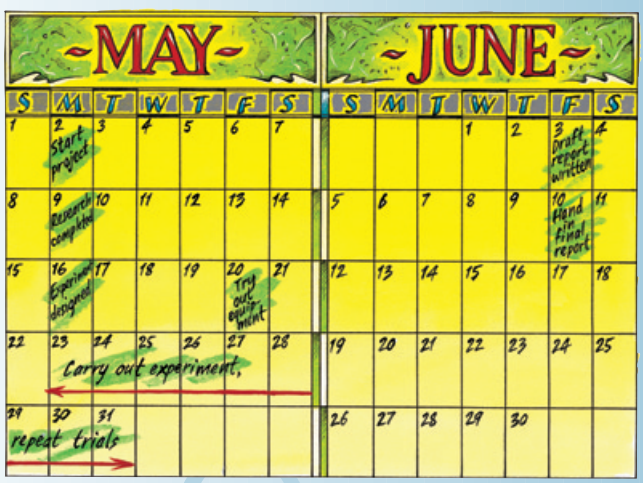

FIGURE 1.25

Designing and trialling your experiment

At this stage you will need to clarify your research project by asking a specific question. The question you ask will influence the equipment you will need and the measurements you take. If specialised equipment is not available you will need to modify your question. It is a good idea to set up a trial run of your experiment. It is better to discover any problems early and modify your design before you proceed further. Remember to record all your problems and modifications in your log book so you can show how your project develops over time. When you are happy with the design of your experiment, have your teacher check it again or use the experimental design check list.

Experimental design check list

Run through these quick questions before you begin your experiment. If you cannot answer any of the questions seek help.

- ☐ Do you have a simple question to investigate?
- ☐ What is your hypothesis?
- ☐ What is your independent variable?
- ☐ What is your dependent variable?
- ☐ What variables will you control to make a fair test?
- ☐ Is your measuring equipment precise enough for your experiment?
- ☐ Have you carried out a trial run to check your equipment?
- ☐ Has your teacher given you the go ahead?

FIGURE 1.26

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Student research projects (continued)

Carrying out your experiment

1 Repeat, repeat, repeat

Once the design of your experiment has been checked you will begin to carry out your experiment and collect the results. Repeat your experiment as many times as possible, in the time allocated, recording all results in your log book.

Analysing the results

Before you can make any conclusions you must organise and analyse the results you have collected. Organise the results in a table so that you can look for trends or patterns in the data. Compare the results from each trial. Are they consistent, with only small variations, or are there big differences? You may wish to calculate **average** results from repeated trials. Drawing graphs may also help you identify and describe any trends or patterns in your results. Finally, decide if your results support your hypothesis and make a conclusion.

Presenting your research project

The final stage of your student research project is the writing of the report. This should not be a difficult task if you record all your work in your log book as you proceed. Your log book will contain your question, hypothesis, method, results etc. These can be used to write a draft report which you can edit and improve before you produce a final copy for assessment. In your final report you should include diagrams showing equipment set-up, results tables and graphs.

Use the discussion section of your report to describe any problems you experienced with your project. Explain how you overcame these problems or how you would overcome them in the future. Also suggest further research which could be undertaken on your topic.

Include a bibliography in your report, listing in alphabetical order all the sources of information which may have helped you in your project. For books used write the name of the author or editor, year of publication, title, publisher and place of publication. For example,
Watson, C. (1999). *Heinemann Interactive Science 3*. Heinemann. Port Melbourne, Victoria.

Encyclopaedias, periodicals, newspaper and multimedia sources of information are recorded in slightly different ways. Talk to your librarian for more information.

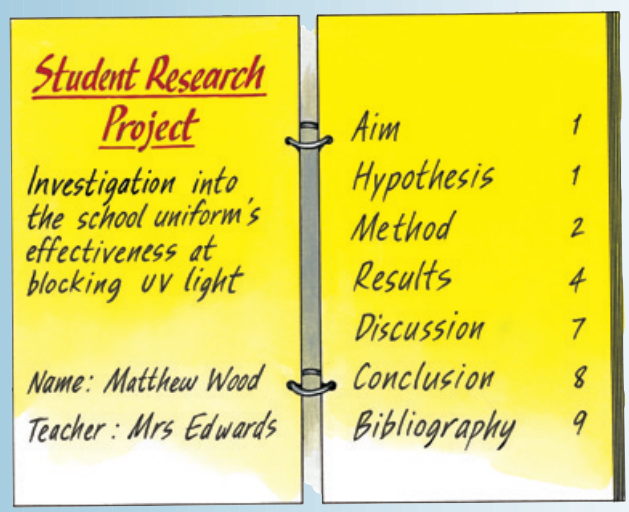

FIGURE 1.27

Multimedia

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Project ideas and starters

- Compare brightness of home lighting devices.
- Investigate methods of noise control.
- Study variations in air quality.
- Investigate the effect of air resistance on velocity.
- Investigate communication between animals.
- Study the range of algae found in water courses.
- Investigate the effects of storage temperatures on batteries.
- Study plant-cloning techniques.
- Investigate medicinal uses of plants.
- Compare the strengths of building materials.
- Study fire-retarding building materials.
- Investigate the effects of ultraviolet radiation on cells.
- Investigate the effects of colour on plant growth.
- Compare methods of preventing corrosion.
- Investigate the factors affecting the growth of crystals.
- Study the magnetic properties of crystals.
- Study the effects of age on peripheral vision.
- Compare methods of preventing mould.
- Investigate how an oil spill is cleaned up.
- Compare the quality of water from different sources.
- Compare different windmill designs.
- Investigate commercial uses of yeast.
- Test sports equipment.
- Study the antibacterial effects of garlic.
- Study the biodegradability of plastics.
- Compare the effectiveness of hair sprays.
- Investigate the food colours used in powdered drink mixes.
- Investigate techniques for removing graffiti.
- Study muscular strength and fatigue.
- Study the salt tolerance of plants and seeds.
- Study the ripening of fruit.
- Investigate flower colour and insect attraction.
- Compare reaction times.
- Compare the effectiveness of blankets, doonas and sleeping bags.
- Investigate the effectiveness of swimming pool blankets.
- Compare the volume of carbon dioxide produced by alka-seltzer tablets.
- Investigate the absorption of water by crystals.
- Investigate the best catalyst for the reaction between zinc and hydrochloric acid.
- Compare various window-cleaning techniques.
- Investigate the colourfastness of various natural dyes.
- Investigate the spacing between a line of dominoes and the time taken for them to topple.
- Compare plant growth in various potting mixes.
- Investigate the effectiveness of stain-removal products.
- Investigate colour preference in caged birds.
- Compare strength of different coloured hair.
- Investigate the effect of microwaves on seeds.
- Investigate various methods of wrapping school lunches.

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CURRENT ISSUES, RESEARCH AND DEVELOPMENT

Is your future in science?

FIGURE 1.28
Peta Thiel, chemist, analyses a water sample to ensure it is safe to drink.

FIGURE 1.29
Nicole Mays, marine biologist, studies introduced marine pests so they can be controlled or possibly eliminated.

The field of science is broad. It provides many opportunities. It has room for every kind of ability and every kind of personality. What is in it for you? Read the following sections. You may find your future in science.

Is chemistry for you?
Chemists can work outdoors, monitoring pollution and helping to explore mountains, valleys and seabeds for precious metals, oil, gas or coal. Chemists also work in laboratories, producing medicines to cure illnesses and testing the purity of molten metals. Chemists can be found working in paper mills, museums, archaeological sites, zoos, and in factories whose products range from chocolate to automobiles.

Chemists are involved in processes that change one substance into another. Careers in chemistry include pharmacy and materials manufacture and design. Is chemistry for you?

Is biology for you?
In the Amazon rainforest biologists monitor the numbers of animals and plants flourishing in the dense jungle. In the Gibson Desert and on the Great Barrier Reef, biologists search for animals and plants of the past and present. Some are fossil hunters who may bring back the remains of a dinosaur or marine 'monster'. In hospitals biologists study the causes of diseases in the hope that they may eventually find a cure. Biologists may study cells, genetics, the oxygen uptake of athletes or the body functions of insects.

Biologists study living things and their activities. Careers in biology include medicine, nursing, optometry, dentistry, zoology, genetics and agriculture. Is **biology** for you?

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Is physics for you?
The work of physicists ranges from working out the flight path of a rocket to developing better colour television screens. In automobile plants, physicists use powerful machines to test the strength of the steel used to make the cars. Physicists may be involved in splitting the atom, testing for radiation leakage, seeding clouds to produce rain or determining the energy efficiency of motors.

Physicists study matter and energy. Careers in physics include engineering and aeronautics. Is **physics** the field for you?

FIGURE 1.30
Jennifer Sai, electrical engineer, works to ensure that people receive a constant and reliable supply of electricity.

Is geology for you?
Most expeditions, whether they be to the South Pole, the Himalayas or the Nullarbor Plain, include a **geologist**. The geologist might collect the rocks of the region, explore caves, hike up and down rivers, climb mountains or take samples from different depths in the soil. The geologist may go deep into a mine to study coal, copper, silver or oil deposits or may go scuba-diving to examine the ocean floor.

A geologist studies our earth, its rocks, its mountains, its rivers, its seas and its oceans. Do you think you might want to be a geologist?

FIGURE 1.31
Andrew McDonald, geologist, locates deposits of useful minerals and evaluates their suitability for mining.

Is astronomy for you?
While geologists study the Earth, astronomers study other planets and their moons, comets, meteors and stars. They are interested in all the bodies in the sky and may investigate reported sightings of UFOs.

Does astronomy interest you?

FIGURE 1.32
Bryan Gaenster, astronomer working for NASA on groundbreaking research into exploding stars. He was the 1999 Young Australian of the Year.

Find out

Investigate a career in science that interests you. Collect as much information as you can about your chosen field of science.

You might start by examining pamphlets from your school's careers department or discussing the career with your parents, teachers and the careers counselling officer. Try to talk to someone who does the kind of work you want to do. (You may even be able to do work experience in the field of your choice.)

Find out what a typical day's work involves in your chosen field and present a poster or 'photo essay' that summarises your findings.

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Using scientific language

Why does the sea always look so clean even though rivers and waves are always adding mud to it? Consider the following hypotheses.

i The salt in ocean water makes mud settle out quickly.
ii One of the several minerals in sea water makes the mud drop out.
iii The oceans are just so big that the muddy water doesn't make much difference.

a What other hypotheses can you propose?
b Design and perform an experiment to test one of these hypotheses.
c Write a report of your findings.

Check your knowledge

1 Chas performed an experiment and drew the following graph of his results:

Spreadsheet

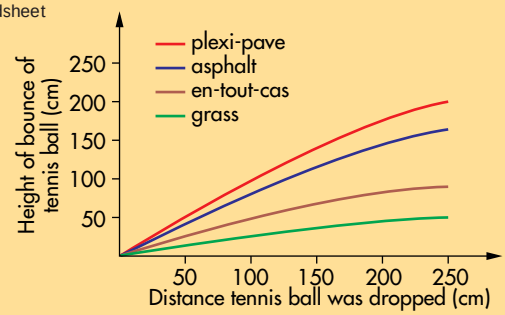

FIGURE 1.33
Graph of Chas's experimental results.

a Suggest a hypothesis that this experiment was testing.
b What was the aim of this experiment?
c Suggest a title for the graph.
d Before drawing his graph, Chas recorded his measurements in a table. Draw the table that you think Chas could have drawn.
e Write a method and conclusion for this experiment.

2 Glenise, a keen gardener, found that there were five different types of insects on her rose plants. One day she counted the insects on one plant and found that there were 15 aphids, 22 red spiders, 29 thrips, 12 beetles and 17 other insects which she couldn't identify.

a Draw a column graph to show this information.
b Draw a sector graph to show this information.
c Explain how Glenise could find out what the unknown insects were.

Apply your skills

1 Three students were discussing the results of an experiment.

'The plants in the first container grew much better than those in the other two containers,' claimed Mandy.
'That's because they were closer to the sun than the plants in the other two containers,' stated Quoach.
Misha seemed puzzled. 'Well, I think that the sun evaporated more water from the first container so that the soil was drier. Maybe the plants in the first container don't like too much water?'

a What do you think the students were trying to find out in the experiment?
b Identify the observations that were made by the three students.
c Identify the inferences that were made by the three students.
d Explain why this experiment is not an example of a controlled experiment.
e Outline, using words or diagrams, how the experiment could have been set up so that it was a controlled experiment.

2 Decide what kind of graph you would use to show the proportion of time you spend each day sleeping, eating, playing sport, watching television, going to school.

a Draw the graph.
b Explain why you chose the form of graph you drew.
c Identify other types of graphs that you could have drawn.
d Identify the types of graphs that could not be used to present your information, giving reasons why they would not be suitable.

3 For each of the items below:

a estimate the quantity in brackets, then measure and compare
b choose a different quantity to measure, take the measurements and record your results in a table.

i your pen (mass)
ii a table (height)
iii a cut banana (time taken to go brown)

Challenge yourself

Choose an Olympic distance event. Draw a graph which shows how the winning times for the event have changed over the last ten Olympic Games.

eTester

CHAPTER 2

ATOMS, MOLECULES AND IONS

People stand and stare, fascinated by the glowing squiggly lines which dance inside the glass ball. Hesitantly, someone raises her hand and touches the glass. A bright plume of charge jumps across the sphere below the hand, making the glass ball glow brighter still. Plasma balls can entertain and amaze for hours. However, they are more than just novelty items. Watching a plasma ball is like watching the basic particles of matter being torn apart and put back together again.

A plasma ball consists of a large glass sphere filled with a mixture of gases such as nitrogen, neon, krypton and xenon. In the middle of the glass sphere is a small metal ball which can be charged with a very high voltage. Sparks jump from the metal ball across to the glass sphere. This causes electrons to be stripped from the hot gas atoms inside the sphere. The atoms become charged ions. When the ions re-unite with their electrons they emit light. Touching the plasma ball earths the glass, causing the spark to discharge at that point.

Plasma is a mixture of neutral atoms, molecules, ions and electrons. This so-called fourth state of matter can be artificially created by using high temperature or high voltage. Artificial plasma can be found in operating neon lights and arc welding units. However, plasma is the most common state of matter in the universe. Every star including the Sun is composed of plasma. In this chapter you will learn about the difference between atoms, ions and molecules and their chemical behaviour.

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Describe how scientists use models of atoms to explain observations.
- Describe the structure of the atom and distinguish between atoms, molecules and ions.
- Explain what the periodic table is and how it is organised.
- Explain the difference between metallic, covalent and ionic bonding.
- Use symbols and valencies to write formulae for a range of ionic compounds.
- Explain the behaviour of ionic compounds in solution in terms of charged particles that are free to move.
- Identify the properties of acids, bases and salts.
- Describe the role of indicators and use the pH scale to compare solutions.

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Web links

Changing models of atoms

FACT FILE

Over a period of one and a half thousand years the Christian church rejected the idea of indivisible atoms but did accept Aristotle's four-element model of matter. During this time individual scholars kept the idea of atoms alive until the 16th and 17th centuries when people began to investigate the properties of substances by experimenting.

Scientific models are ideas used to help us explain our world. Models are often built up over time from the observations and experiments of many different scientists. Each scientist contributes a small piece of information to help construct a mental picture. The story of the model of the **atom** is a very good example of how scientific understanding changes over time.

Ancient models

Models of the atom began with the ancient Greek philosophers. Democritus argued that atoms were the building blocks of **matter**. Other philosophers said matter was made of four elements: earth, air, fire and water. Neither model was based on experimental evidence. Since the four-element model had more important supporters it lasted for over 2000 years and Democritus's ideas were largely ignored in the western world. However, other ancient cultures put forward the idea that matter was made of atoms. These ideas were linked with religious beliefs. In India and in the Arab world various groups developed explanations of matter which included descriptions of miniscule, indestructible particles.

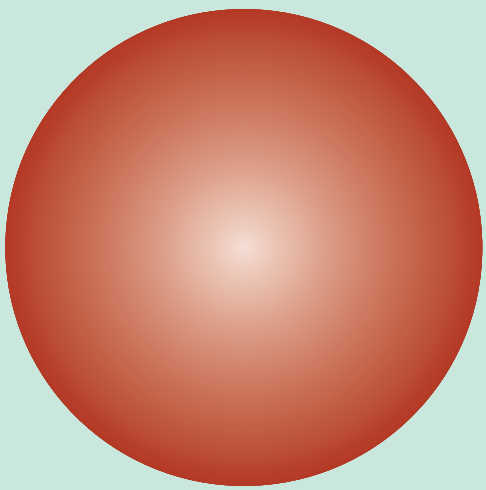

FIGURE 2.1
Dalton said that each element had its own characteristic atoms and that when compounds were formed different atoms joined together.

Dalton's model

It was in 1803 that John Dalton developed a model based on extensive observations of pure substances. He said that matter was made of tiny indivisible particles which he called atoms. His atoms could join together to form **molecules** in a regular predictable way. Dalton's model was gradually accepted by other scientists of the time as it helped them explain their own discoveries about elements and compounds.

Modern ideas

By the end of the 19th century great strides were being made into the study of electricity. New observations of the electrical behaviour of matter meant that Dalton's model of the atom was no longer adequate.

In 1897 J.J. Thomson discovered tiny negatively charged particles known as **electrons**. Thomson found the particles when he subjected gases in glass tubes to

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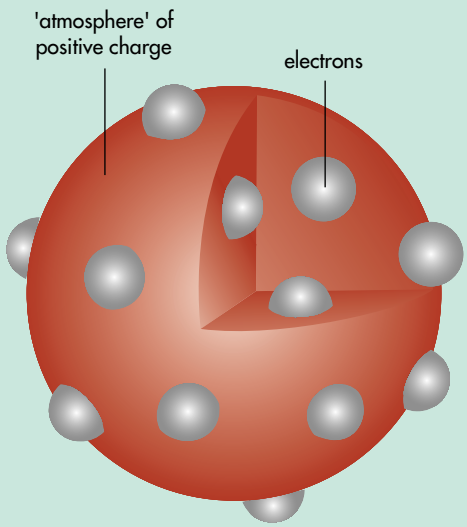


FIGURE 2.2
Thomson thought of atoms as small solid particles with negative charges embedded in them. His ideas are often called the plum pudding model. Can you explain why?

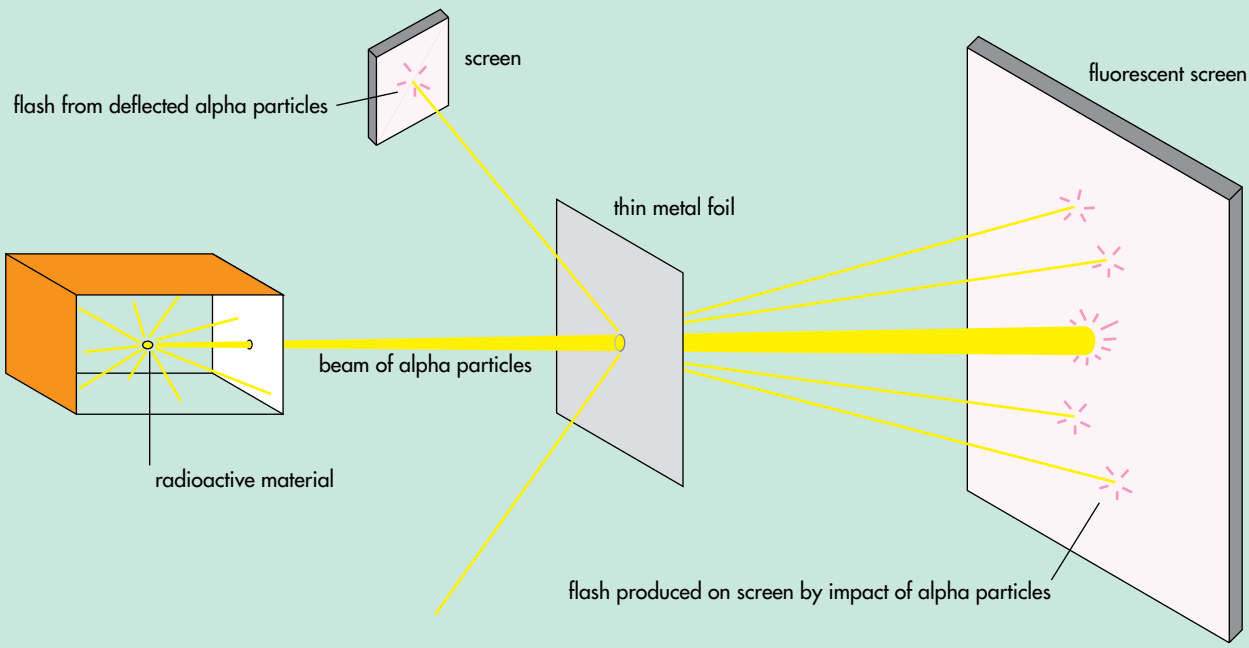
high voltages. The presence of negatively charged particles meant that there must be positively charged particles in atoms too.

Another scientist, Ernest Rutherford, discovered positively charged particles, which he named **protons**. Developments in radioactivity allowed Rutherford to conduct experiments which led him to put forward a completely new model of the atom. In his most famous experiment Rutherford fired **radioactive alpha particles** at a thin sheet of gold. To his surprise most of the particles passed straight through. A few alpha particles were deflected back at large angles.

This observation led Rutherford to conclude that atoms were mostly empty space with a very small positive centre which he called the **nucleus**. So Rutherford suggested a model of the atom which consists of a small positive nucleus surrounded by electrons. He thought the electrons would travel around the nucleus in the same ways as planets travel around the Sun. However, this model did not fit the existing laws of physics which suggested that the electrons would spiral into the nucleus.

It was Niels Bohr in 1913 whose work on light led him to suggest that the electrons didn't spiral into the nucleus because they were arranged in layers or shells. The model of the atom could now be used to explain all the chemical properties and

FIGURE 2.3
Rutherford found that most of the alpha particles passed straight through the gold foil as though nothing was in their path.



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Changing models of atoms (cont.)

behaviours of substances. It is this model we will use when learning about **ions** and **molecules**.

A third particle called the **neutron** was discovered in 1932 by James Chadwick. The model of the atom has continued to change as technology provides the means to explore its inner structure further. Further changes to the model of the atom will be discussed in *Interactive Science 4*.

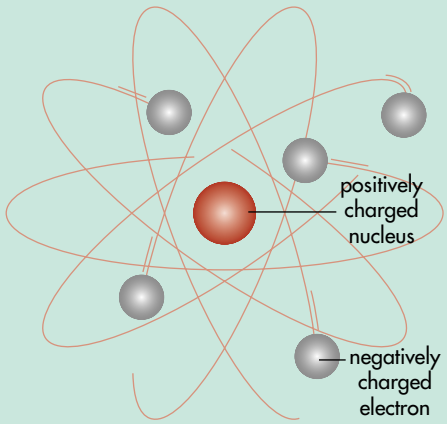

FIGURE 2.4
Rutherford's model is often called the solar system model because he thought the electrons orbited a positive nucleus.

USING SCIENCE 1 EVIDENCE FOR THE ELECTRICAL ATOM

PART A

You can demonstrate the presence of electrical charges in matter by investigating static electricity.

WHAT YOU NEED

- woollen cloth
- ebonite rod
- polystyrene balls
- aluminium foil 5 cm by 15 cm
- retort stand

WHAT TO DO

1 Charge the ebonite rod by rubbing it with the woollen cloth then lower the rod close to the polystyrene balls.
2 Hang the aluminium foil strip from the retort stand and bring the charged rod close to it.
3 Turn a tap on to give a small steady stream of water. Bring the charged rod close to the stream (do not allow the rod to get wet).
4 Record your observations.

PART B

Your teacher may set up a demonstration to show you what J.J. Thomson saw when he subjected gases to a high voltage.

Draw a diagram of the effects produced in the cathode ray tube.

DISCUSSION

1 When ebonite is rubbed with a woollen cloth a negative charge is produced on the rod and a positive charge on the cloth. Explain this in terms of electron transfer and use your explanations to say why each effect observed in Part A occurred.
2 The greenish rays seen in the glass tube by J.J. Thomson were called cathode rays because they were produced by the cathode or negative terminal. The cathode rays were in fact effects caused by a flow of electrons in the gas atoms. The other terminal in the glass tube is called the anode. What charge would be on the anode?

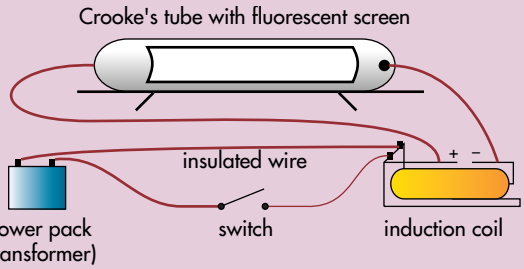

FIGURE 2.5

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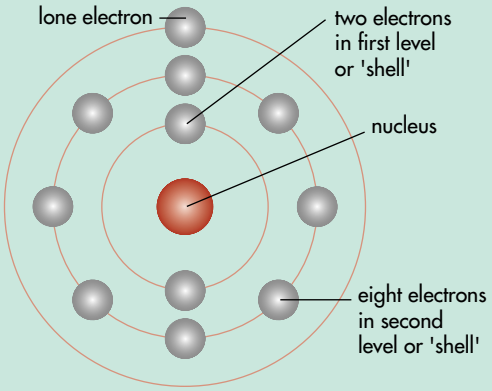

FIGURE 2.6
In Bohr's model of the atom, the electrons move in energy levels or shells. The electrons can move between energy levels by absorbing or emitting energy.

TABLE 2.1 Atomic particles.

Particle	Location	Mass	Charge
Proton	nucleus	1 atomic mass unit	+1
Neutron	nucleus	1 atomic mass unit	0
Electron	outside the nucleus	1/1840 of an atomic mass unit	–1

Questions

1 Explain the term 'scientific model'.
2 How are protons similar to neutrons? How are they different?
3 How are electrons different from protons?
4 What development or improvement in technology allowed J.J. Thomson to change Dalton's model of the atom?
5 What scientific discovery allowed Rutherford to change Thomson's model of the atom?

Think about

Would it be fair to say that Dalton's model of the atom was wrong in 1803?

Find out

1 Select one of the following scientists and find out what contribution they made to our understanding of the atom: William Crookes, Marie Curie, Antoine Henri Becquerel, Willhelm Roentgen, Frederick Soddy, Henry Moseley, Robert Millikan.
2 Several ancient Greek philosphers discussed ideas about matter. Some supported the idea that matter was made of atoms, others totally rejected this suggestion. Research the ideas of the following philosophers and conduct a debate on the question 'Do atoms exist?'.
For: Leucippus, Democritus and Epicurus.
Against: Empedocles, Aristotle and Thales.

Activity

Create an illustrated timeline for the developing ideas of the atom starting with Democritus in 350 BC. Include a drawing of the model, the person involved and the date.

FACT FILE

Ernest Rutherford was born in 1871 on the South Island of New Zealand. He showed great promise at school and gained a scholarship to attend New Zealand University where he studied physics. In 1895 while digging potatoes on his father's farm he received news that he had won a scholarship to Cambridge University. At Cambridge he worked for J.J. Thomson who appreciated his experimental skills. Rutherford worked in the new field of radioactivity. He made many important discoveries which changed the model of the atom. He was awarded the Nobel Prize for chemistry in 1908, knighted in 1914, elected President of the Royal Society in 1925 and made a baron in 1931.

Extra worksheet

Animation

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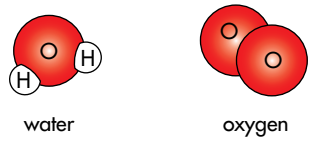
2.2

Molecules and ions



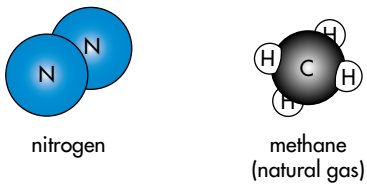


FIGURE 2.7
Some simple molecules and what they are thought to look like. Which of these molecules are elements? Which are compounds?

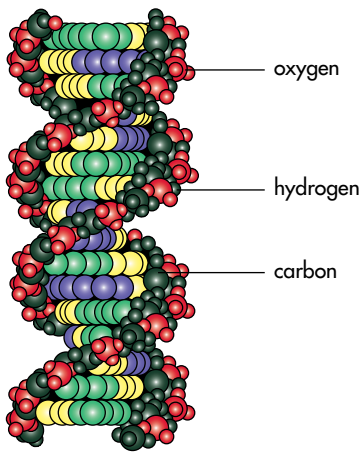


FIGURE 2.8
A very complex molecule. DNA makes up the chromosomes in our cells. A single DNA molecule can consist of millions of atoms.

In nature there are only 92 different kinds of atoms to be found. Twenty more can be made in laboratories.

Substances made up of atoms of the same kind are called **elements**. Most of the atoms on Earth are found in combination with other atoms and usually these atoms are different types. Substances composed of a combination of different types of atoms are called **compounds**.

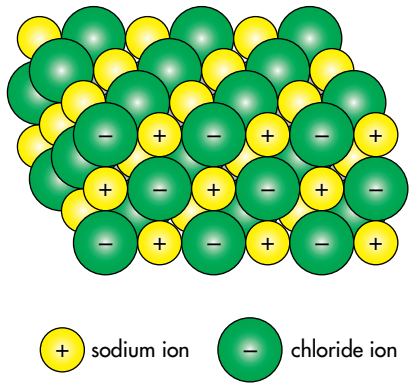


FIGURE 2.9
Sodium chloride is a typical ionic substance. It is composed of positively charged sodium ions and negatively charged chloride ions in a one-to-one ratio.

The three ways that atoms can be combined or bonded will be covered in section 2.4, Bonding.

One type of combination of atoms is the **molecule**. There is a set number of atoms in a molecule. For example, a hydrogen molecule is composed of two hydrogen atoms joined together. The atoms making up a molecule may be the same or different, but they are always non-metals.

We are surrounded by molecules. They are in the air, sea and even our food. Our body is mostly molecules, and uses special molecules to break food into smaller molecules to provide us with energy.

Some substances made of simple molecules are water, oxygen, carbon dioxide and methane. Examples of more complicated molecules are those in plastics, proteins and even our own DNA.

EXPERIMENT 1 FITTING THEM TOGETHER

AIM
To investigate how the shapes and sizes of molecules can cause unusual effects large enough to be seen.

MATERIALS
- water
- ethanol
- 2 × 100 mL measuring cylinders
- 2 beakers
- whiteboard marker
- ruler
- handful of large plastic beads and handful of small plastic beads

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EXPERIMENT 1 CONTINUED

PART A: METHOD

1 Pour exactly 50 mL of water into one measuring cylinder and 50 mL of ethanol into the other.

2 Now pour the ethanol into the water.

RESULTS

1 How much liquid do you now have? Is the cylinder warmer or colder than before? Are there any bubbles rising in it? Note any other observations.

2 Draw a measuring cylinder, showing clearly how much liquid you actually have.

DISCUSSION

What has happened here? How is it that 50 mL + 50 mL doesn't equal 100 mL?

PART B: METHOD

1 Put the large beads into one of the beakers so it is less than half full. Mark the level and measure the depth with a ruler. Record this.

2 Put the same depth of small beads into the other beaker.

3 Pour the small beads into the first beaker and mix the two sizes of beads together.

RESULTS

Measure the depth of the mixture of beads. Draw a picture and explain what has happened.

DISCUSSION

Can you add to your answer to the discussion in Part A?

Experiment 1 models with plastic beads the kind of thing thought to be happening with the ethanol and water molecules. **Ethanol** and water molecules don't fit together very well with themselves, but each molecule is just the right shape to fit together and fill the gaps left by the other type of molecule. The way substances behave has helped us to infer the shape of their molecules.

Ions

The number of protons in an atom determines what kind of element it is. Usually atoms have the same number of electrons as they have protons, so that their overall electrical charge is zero. Often, though, atoms can lose electrons to other atoms or gain extra electrons from other atoms. This makes them become positively or negatively charged overall because they now have either more or fewer electrons than protons. When an atom has an overall electrical charge it is called an **ion**. An ion is still the same element, but being an ion makes it behave in different ways. In particular, positive and negative ions attract each other, and as a result can stick together very strongly to make what is called a **lattice**. This is known as **ionic bonding** and we will be looking more at this later.

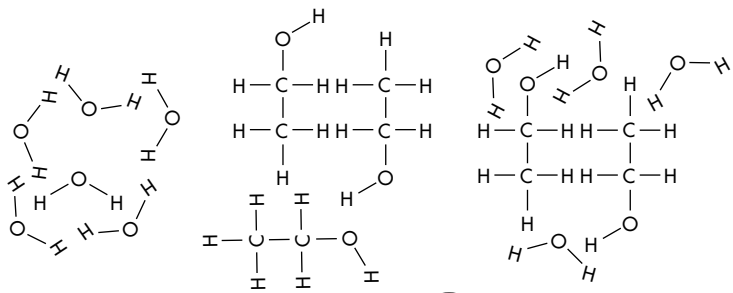

FIGURE 2.10
Ethanol and water, snug as bugs in rugs.

Questions

1 In what ways do molecules differ from atoms?

2 A molecule of carbon dioxide is composed of one carbon atom and two oxygen atoms. Draw this molecule.

3 What is an ion? What is an ionic compound?

Foundation worksheet 1

Activity

If you have access to a molecular building kit, try to model some of the substances mentioned in this section.

Find out

What are the main types of elements found in humans? Choose one molecule found in humans and prepare a short report on its major features and function(s).

Multimedia eGlossary

FACT FILE

One atom of element 112 was made on 9 February 1996. It lasted just one-third of a millisecond before it decayed.

2.3

The periodic table

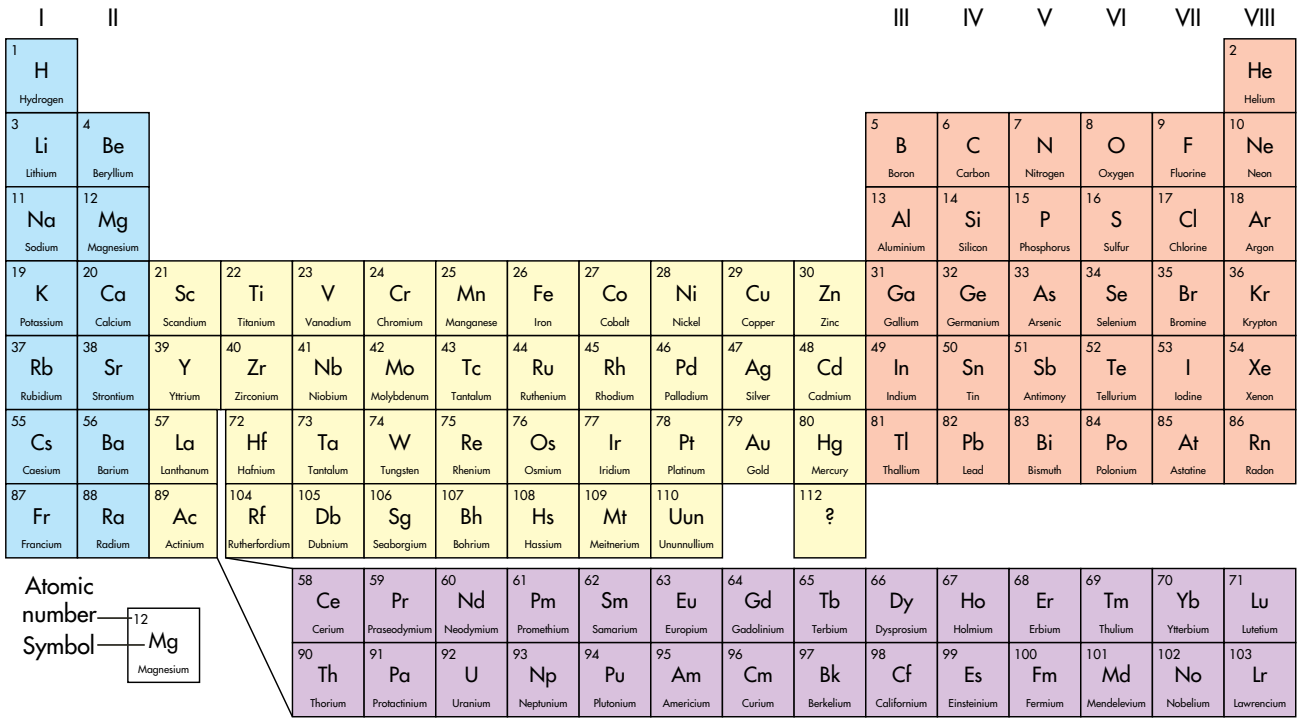

FIGURE 2.11
The modern periodic table is still growing!

Many of the different elements have much in common. For instance, the metals have many similar properties, such as **electrical conductivity** and **malleability**. Certain gases, like neon, argon and krypton, are very stable and relatively unreactive.

In 1869, a Russian scientist called Dmitri Mendeleev proposed a way of arranging the list of elements which showed how they were related to each other. You will look in more detail at the history and development of the **periodic table** in *Interactive Science 4*.

The periodic table organises the elements according to three basic factors.

1 How many protons the element has in its nucleus. This is called the **atomic number** and determines how far down the list of elements any particular atom appears. For example oxygen has 8 protons so is further down the list than carbon, which has 6.

2 The number of **electron shells** the element's atoms have. The number of electron shells can vary from 1 to 7 and determines which row the element appears in. Each of the seven kinds of shells has a maximum number of electrons it can contain. The first shell can contain up to 2 electrons, while the second shell can contain up to 8 electrons. Eight electrons go into the third shell, and then the fourth shell starts to fill. Carbon, which has 6 electrons, has 2 in the first shell, and 4 in the second shell. Chlorine has 17

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electrons: 2 in its first shell, 8 in its second shell, and 7 in its **outer** or **valence shell**. The electrons in the outermost shell are called the **valence electrons**.

3 The number of electrons in the **outer shell** determines how the element will react with other atoms. Elements which have the same number of electrons in their outer shells behave in similar ways. The number of electrons in the outer shell determines which column the element appears in. Lithium, sodium and potassium are all violently reactive metals. This is because they each have just 1 valence electron. Lithium has 3 electrons, sodium has 11, and potassium has 19 electrons (find them on the periodic table). Each has 2 electrons in the first shell, leaving lithium with 1 valence electron; sodium has 8 more in its second shell, leaving 1 valence electron in its third shell; potassium has 2 in its first and 8 in both its second and third shells, leaving only 1 in its fourth shell.

The elements with full outer shells, called the '**noble gases**', are in the last column and are very stable substances. This makes them extremely useful in lighting, since if a filament in a light bulb was exposed to ordinary air it would react with oxygen and burn to a frazzle in a matter of moments. Instead light bulbs are filled with argon. Very intense lights, such as spotlights and photographic flashes, are filled with xenon.

Questions

1 Use the periodic table to draw diagrams similar to those in Figure 2.12 for nitrogen, magnesium and chlorine.

2 Name all the elements which have 2 valence electrons. Which group are these in?

3 The *Hindenburg* was an airship filled with hydrogen that blew up in 1937. What can you tell about hydrogen from its place in the periodic table that can help explain this disaster?

4 Helium is now used to inflate airships. Why?

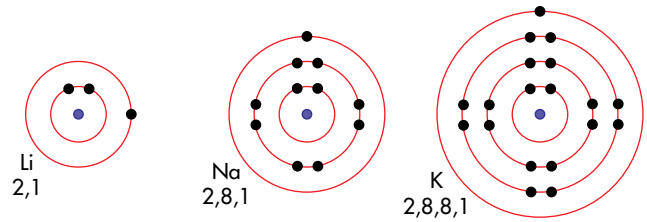

FIGURE 2.12
Li, Na and K all have similar properties, and all have just one valence electron. Because all three have just one valence electron they all appear in the first column. Lithium has two shells, so it is in the second row, sodium has three shells so it is in the third row, and potassium has four shells so it is in the fourth row.

FIGURE 2.13
Without the noble gases these lights wouldn't have been possible.

Think about

1 Why do you think the noble gases are called 'noble'?

2 Make up a mnemonic or rhyme to help you remember the first 20 elements.

Foundation worksheet 2

Activity

Design a game of Periodic Table Bingo using the first 20 elements.

FIGURE 2.14
The German-made *Hindenburg* was filled with hydrogen. A similar airship calamity had stopped any further construction of airships in Britain 7 years earlier.

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2.4

Bonding

FIGURE 2.15
Salt is formed from large crystals of NaCl.

Talk about
Read the Fact File. If atoms are mostly empty space, why don't we just fall through things?

What holds molecules together and stops other atoms from passing through them? This is where electrons become very important. Electrons are all negatively charged and so repel each other very strongly. We don't fall through other atoms because their electrons repel ours. But electrons also cause certain atoms to stick together to form molecules and lattices.

FACT FILE

Atoms are mostly empty space. If an atom was blown up to the size of a football field the nucleus would be about the size of a pea.

Metallic bonding

A typical metal can be bent and shaped, will conduct heat and electricity, and tends to be shiny.

Metals are also elements. How are the atoms in a metal held together? Can this explain their properties?

As mentioned earlier, the outer or valence electrons determine how an atom will behave with other atoms.

Metals have valence electrons that are easy to remove, thus creating a stable outer shell. In a piece of metal the atoms lose their valence electrons, and this results in the formation of positive ions which are fixed in position, surrounded by the lost electrons which remain free to move. The attraction of the positive metal ions to all these electrons is strong enough to hold the **metallic lattice** together.

This kind of bonding is called **metallic bonding**.

This helps explain why metals can be hammered into different shapes or stretched out into thin wires. Their atoms can be forced to move without breaking up the lattice because the valence electrons which hold the structure together are free to move around.

Covalent bonding

Non-metals hold on tightly to their valence electrons and even try to catch extra electrons to fill their outer shells. Atoms with full outer shells are more stable.

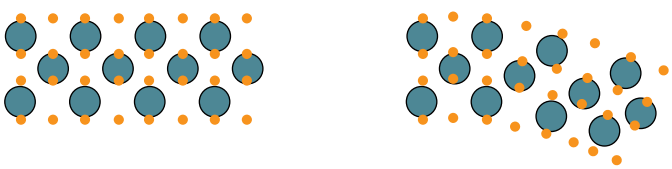

FIGURE 2.16
When enough force is applied the valence electrons in metals allow the ions to be moved around without the structure breaking apart.

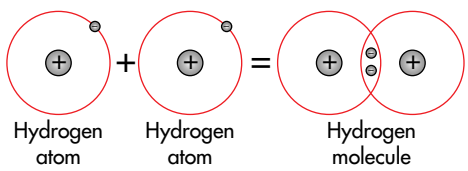

FIGURE 2.17
The two atoms which make up a hydrogen molecule are held together by covalent bonds. This type of bond is the result of sharing valence electrons to complete each atom's outer shell.

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EXPERIMENT 2 CHECKING OUT THE BONDS

AIM

To investigate some metallic, covalent and ionic bonds.

MATERIALS

- 2 petri dishes
- 3 × 100 mL beakers
- Bunsen burner, tripod and gauze mat
- tongs
- 2 stirrers
- 3 crucibles
- water
- some salt, sugar and soft metal solder
- hand lens or stereomicroscope

METHOD

1 Pour 50 mL of water into each of the beakers.
2 Add salt, sugar and solder to separate beakers.
3 Add more of each substance to its beaker and stir until no more dissolves. Record your observations.
4 For each of the substances that dissolved, pour just enough solution into a petri dish to cover the bottom. Place somewhere warm to dry out.
5 Set up your Bunsen burner and tripod.
6 Place a sample of each substance into separate crucibles.
7 Heat each substance and record what you observe before, during and after heating.
8 Try to crush some of each substance on your bench top. Record your results.
9 Once the liquid in the petri dishes has evaporated, scrape the bottom of each dish and record what you see. You may need a hand lens or stereomicroscope.

DISCUSSION

After reading about metallic, ionic and covalent bonds, which substances do you think have these types of bonds? Give reasons for your answer.

Non-metals need to capture only a few electrons to fill their valence shell and become more stable, so when two non-metal atoms get close enough they can share their outer electrons to complete their outer shells. This makes the atoms stick together very strongly, and is called **covalent bonding**. The bonds that hold the atoms together in molecules are covalent bonds. Covalent bonds are also found in very hard substances like diamonds.

Ionic bonding

Another important kind of bonding is **ionic bonding**. This takes place between metals and non-metals. In this case the metal becomes more stable by giving away one or more of its outer electrons. This leaves it positively charged. The non-metal becomes more stable by capturing the metal's discarded electron or electrons, making it negatively charged overall because of the extra electrons.

Remember, an atom which has a charge is called an ion.

Now, the two stable ions are strongly attracted to each other because of their positive and negative charges. Large numbers of positive and negative ions will form an ionic lattice held together by this strong electrical attraction (Figure 2.9).

Ionic substances tend to form hard crystals which are very rigid and brittle. Salt is one of these, with strong attraction between the metal sodium (Na^+) ions and non-metal chlorine (Cl^-) or chloride ions, which is why it is also called sodium chloride (NaCl).

Interactive

Internet assignment

Questions

1 Explain the difference between metallic, covalent and ionic bonding. Examples and diagrams should be used where possible.
2 Why do metals lose electrons easily and non-metals gain electrons easily?
3 Why can't an ionic compound be bent and shaped?
4 What are the major differences between molecules and lattices?

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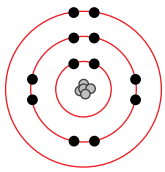
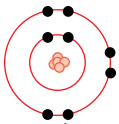
2.5

Writing chemical formulae



magnesium has an atomic number of 12, so it has 12 electrons



oxygen has an atomic number of 8, so it has 8 electrons

FIGURE 2.19
Magnesium and oxygen atoms.

TABLE 2.2 Valencies of the first 20 elements.

Element	Valency
H	+1
He	0
Li	+1
Be	+2
B	+3
C	+4
N	–3
O	–2
F	–1
Ne	0
Na	+1
Mg	+2
Al	+3
Si	+4
P	–3
S	–2
Cl	–1
Ar	0
K	+1
Ca	+2

Ionic compounds

In order to understand chemical substances better, we need to examine how chemicals react with each other. In order to do this, we need to express a reaction as an equation.

The burning of magnesium ribbon can be described by the equation:

$$2Mg + O_2 \rightarrow 2MgO$$

Magnesium oxide is an **ionic compound**. This means that both the magnesium and the oxygen in the compound are **ions**. An ion is a charged particle. Magnesium and oxygen became ions during the reaction because an exchange of electrons occurred. This exchange of electrons occurred because oxygen required two electrons to complete its outer shell of electrons, and magnesium needed to lose two electrons to have a full outer shell of electrons.

If an atom does not have a complete, or full, outer shell, it tends to lose or gain electrons in order to have a full outer shell of electrons and become more stable.

Look at the structure of magnesium and oxygen (Figure 2.19). Magnesium has two electrons in its outer shell. If it could get rid of these, then it would have a complete outer shell of electrons. Oxygen needs two electrons in order to fill its outer shell of electrons. Magnesium can give its two electrons to the oxygen, so that both now have a complete outer shell of electrons. The oppositely charged ions will attract each other and become a compound—magnesium oxide.

By using a **valency chart**, we can determine how many electrons are needed or how many are in excess, and therefore what **valency**, or **charge**, each ion in a reaction has.

From the valency chart we can see that a magnesium ion, when formed, has a charge of +2, as it is in the +2 column. This means that it has two electrons to give away. Oxygen is in the –2 column, so when it becomes an ion, it has a charge of –2. It can also be seen that oxygen has six electrons in its outer shell. Therefore oxygen needs another two electrons in order to fill its outer shell. This is why it has a charge of –2 when it becomes an ion.

FIGURE 2.18
Reaction between magnesium and oxygen.

In order to write the formula for the compound we need to follow three steps:

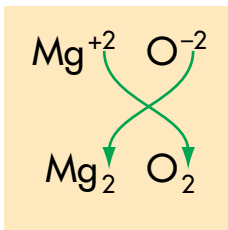


FIGURE 2.20
1 Write the ions with their charges.
2 Swap each of the charges and make them subscripts.
3 If the subscripts are both the same then they can be left out (ionic compounds only).
MgO is the formula for the compound. Check: (+2) + (–2) = 0

Note the **cation** (positive ion) is named and written first, and the **anion** (negative ion) is written and named second.

For example, in the reaction between sodium and chloride to form sodium chloride, a sodium ion has a charge of +1 and a chlorine ion has a charge of –1, so:

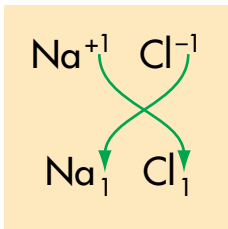


FIGURE 2.21
NaCl
Check:
(+1) + (–1) = 0.

Sometimes the ionic valencies are not equal. Consider magnesium chloride. A magnesium ion has a charge of +2 and a chloride ion has a charge of –1. This means that two chloride ions are needed for each magnesium ion.

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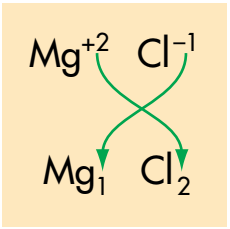

FIGURE 2.22
The formula is $MgCl_2$. Check: $(+2) + (2 \times -1) = 0$.

Totally radical

Ions can be single atoms with more electrons than protons (anions) or fewer electrons than protons (cations). Ions can also be groups of atoms with a charge. These **polyatomic ions** are also called **radicals**. Radicals can form compounds with ionic bonds in the same way as other ions.

When writing the **chemical formulae** for compounds containing radicals the same process is followed. However, if multiples of a radical are involved brackets are placed around it in the formula and the number placed outside.

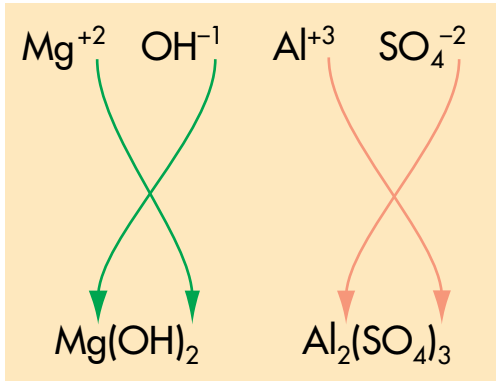

FIGURE 2.23
The formulae for magnesium hydroxide and aluminium sulfate.

TABLE 2.3 Formulae and valencies of radicals.

Name	Formula	Valency or charge on ion
Ammonium	NH_4	+1
Carbonate	CO_3	–2
Hydroxide	OH	–1
Nitrate	NO_3	–1
Phosphate	PO_4	–3
Sulfate	SO_4	–2

Questions

1 Write formulae for the following ionic compounds:

a lithium chloride
b hydrogen chloride
c lithium oxide
d calcium chloride
e hydrogen fluoride
f magnesium chloride
g calcium hydroxide
h sodium carbonate
i aluminium nitrate
j ammonium chloride

2 What charge do the following atoms have when they become ions?

a hydrogen
b sodium
c beryllium
d aluminium
e nitrogen
f lithium

3 How many electrons do each of the following atoms have in their outer shell?

a oxygen
b hydrogen
c neon
d carbon
e helium

4 Atoms and their outer shell electrons can be represented as shown in Figure 2.24. Draw the atoms below in this way:

a chlorine
b magnesium
c neon
d nitrogen
e sodium

FIGURE 2.24

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2.6

Ions everywhere

FIGURE 2.25
A range of products containing electrolytes and non-electrolyte mixtures.

FIGURE 2.26
Some common household salts.

Many compounds dissolve in water to form **solutions**. These compounds are described as **soluble**. When compounds dissolve they may form ions. It is not possible to tell if a solution contains ions just by looking at it, but there is a way of testing the solution. A solution containing ions will allow an electric current to flow through it. Solutions that conduct an electric current are called **electrolytes**. Those that do not are called **non-electrolytes**.

Electrolytes

Many compounds which are electrolytes belong to one of three groups: **acids**, **bases** and **salts**. Salts usually contain metal atoms and non-metal atoms. In the solid form they exist as a crystal lattice of positive and negative ions held together by ionic bonding. When a salt is dissolved, the ions separate and are free to move in the solution. For example:

$$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$$
$$MgSO_4(s) \rightarrow Mg^{2+}(aq) + SO_4^{2-}(aq)$$

Acids contain hydrogen atoms covalently bonded to other non-metal atoms. When acids dissolve in water the covalent bonds are broken. The hydrogen forms positive hydrogen ions and the non-metal forms negative ions.

Bases often contain positive metal or ammonium ions bonded to negative hydroxide ions. Bases that are soluble are called alkalis.

EXPERIMENT 3 — CARRY THE CHARGE

AIM
To test a range of soluble compounds and classify them as electrolytes or non-electrolytes.

MATERIALS
- stainless steel electrodes with inbuilt lamp
- variable power source
- leads
- 250 mL beaker
- 150 mL of each solution
- sugar, dilute hydrogen chloride solution, dilute sodium hydroxide, sodium chloride and ethanol

METHOD
1 Connect the stainless steel electrodes to the DC terminals of the variable power source. Set to 6 V.
2 Check that the lamp is working by connecting the two electrodes together.
3 Pour 150 mL of a test solution in the beaker.
4 Place the electrodes in the beaker so that they are partially covered with solution.
5 Observe the lamp.
6 Remove the electrodes, rinse in them water and test another solution.

RESULTS
Design a results table to record the name of the solution, whether the lamp glowed and whether the test solutions were electrolytes or non-electrolytes.

DISCUSSION
1 Why is it necessary to rinse the electrodes in between testing solutions?
2 Some compounds are better electrolytes, they produce more ions than others. The more ions the more current flows through the solution. Using only the materials supplied how could you tell which solutions were the better electrolytes?
3 From your knowledge of bonding predict the type of bonding present in each compound tested.

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Carrying on

Electrolytes can conduct electricity because their ions are free to move in the solution. Positive ions will be attracted to the electrode with the negative charge where they will receive electrons. Negative ions will be attracted to the electrode with the positive charge where they will give up electrons. This process is called **electrolysis**. Electrolysis has many everyday uses, including:

- producing chemicals for industry, e.g. sodium, chlorine
- extracting metals from naturally occurring compounds, e.g. aluminium, magnesium
- purifying metal elements, e.g. copper
- electroplating metal surfaces.

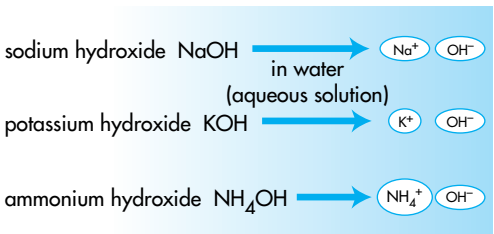

FIGURE 2.27
Alkalis produce hydroxide ions when they dissolve in water.

Questions

1 Describe in detail what happens to a salt crystal when it is placed in water.
2 Distinguish between electrolytes and non-electrolytes.
3 Use chemical formulae to show the formation of ions by each of the following compounds:
a calcium hydroxide
b hydrogen carbonate
c magnesium nitrate
d calcium chloride
e hydrogen fluoride
f magnesium hydroxide.
4 Classify each of the compounds in question 3 as acids, bases or salts.
5 What is electrolysis?

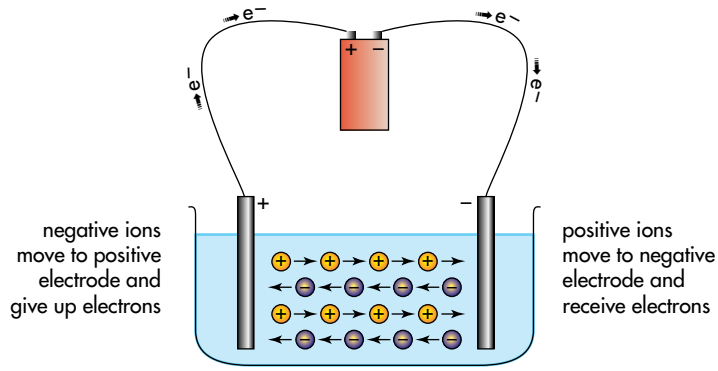

FIGURE 2.28
Chemical reactions take place at each electrode during electrolysis.

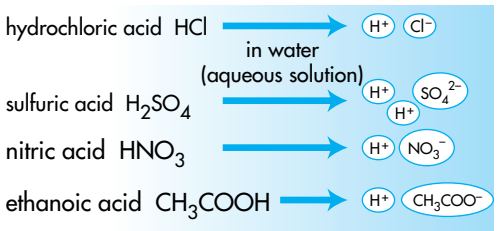

FIGURE 2.29
All acids produce hydrogen ions in solution; it is this that gives them their unique properties.

6 In the electrolysis of water predict where the hydrogen gas and oxygen gas will form.

Think about

1 Use your knowledge of electrolytes to explain why it is dangerous to be in a swimming pool during a storm.
2 If these substances were made during electrolysis would they form at the positive or negative electrode?
a sodium b fluorine
c copper d chlorine

Find out

Some sports drinks advertise that they contain electrolytes. What electrolytes are in sports drinks and why do athletes want to drink them?

Activity

If your school has a voltameter your teacher may demonstrate how water molecules can be broken up by electrolysis.

FACT FILE

Pure water is not a good electrolyte. If a small amount of acid is added to the water, some of the water molecules will ionise (form ions). During electrolysis hydrogen ions and hydroxide ions in the water will react at the electrodes to form hydrogen gas and oxygen gas.

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2.7

Testing for acids and bases

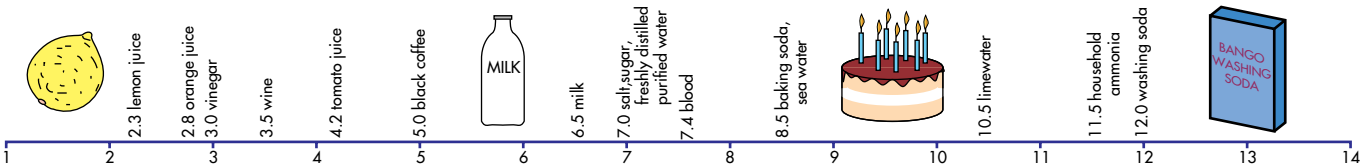

FIGURE 2.30
The pH scale.

FIGURE 2.31
The litmus paper in this jug of vinegar indicates that vinegar is an acid.

FIGURE 2.32
The strength of an acid or a base solution can be distinguished by using a universal indicator colour chart.

Indicators

Acids and bases can be detected by using a group of chemicals called **indicators**. There are many naturally occurring indicators, such as red cabbage, blueberries and cherries. **Litmus** is made from lichen, but many indicators used today are synthetic.

A simple indicator to use is **litmus paper**, which can be red or blue.

- In an acid, red or blue litmus paper is red.
- In a base, red or blue litmus paper is blue.

Universal indicator can be used in paper or liquid form. The colour of the indicator should be compared with a colour chart to determine the pH of a substance.

Strong or weak acids and bases

Acids may be strong (e.g. hydrochloric acid) or weak (e.g. acetic acid). The difference between strong and weak acids is simply the number of hydrogen ions in a solution of the acid. Strong acids ionise to produce many hydrogen ions, whereas weak acids produce few.

Bases may be strong (e.g. sodium hydroxide) or weak (e.g. ammonium hydroxide). A strong base ionises to form many hydroxide ions in solution. Weak bases ionise to produce few hydroxide ions.

What is pH?

Scales are used in measurement. The **pH scale** is used to measure the strength of an acid or a base.

A pH of less than 7 indicates that the substance is an acid.

A pH of more than 7 indicates that the substance is a base.

If the pH is 7, the substance is said to be **neutral**—neither an acid nor a base.

Strong acid: pH is between 0 and 3.
Weak acid: pH is between 3 and 7.
Strong base: pH is between 11 and 14.
Weak base: pH is between 7 and 11.

FIGURE 2.33
Colour changes for indicators.

Indicator	Colour of indicator (pH 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14)
Bromothymol blue	
Litmus	
Methyl orange	
Phenolphthalein	

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EXPERIMENT 4 OF CABBAGES AND INDICATORS

Red cabbage is a natural indicator which can be used to test whether a substance is an acid or a base.

AIM

To investigate the acidity or alkalinity of a range of solutions using three different indicators: red cabbage indicator, litmus paper and universal indicator.

MATERIALS

- red cabbage
- heating apparatus
- universal indicator
- red and blue litmus paper
- 400 mL of hot water
- 3 test-tubes and rack
- various solutions to test, such as orange juice, lemon juice, saliva, washing powder solution, vinegar, freshly distilled water, tap water, wine, bleach, bicarbonate of soda solution, toilet cleaner, shampoo

METHOD

1 Make a cabbage indicator by adding two handfuls of chopped red cabbage to 400 mL of hot water in a beaker. Boil for 10 minutes.
2 Place 1 mL of the first solution to be tested into three test-tubes.
3 Add 1 mL of purple cabbage water indicator to the first test-tube, a piece of litmus paper to the second test-tube and 2 drops of universal indicator to the third test-tube.
4 Observe the colour and record your results in a table.
5 Repeat steps 2–4 for each solution to be tested.

RESULTS

Identify each solution as an acid or a base and determine the pH, if possible.

CONCLUSION

List each substance you tested in order from lowest to highest pH.

DISCUSSION

1 How you could test the pH of solid substances?
2 Compare the advantages and disadvantages of using cabbage indicator, litmus paper and universal indicator in determining whether a substance is acidic, basic or neutral.

EXTENSION

Flowers can also be used as indicators. Choose a flower and make an indicator. (Each class member could try a different flower.)
1 Use your indicator to test different substances.
2 Make a pH colour chart for your flower indicator.

The lower the pH, the more acidic a substance is. The higher the pH, the more basic a substance is.

Questions

1a What is the function of an indicator?
b In what situation would it be preferable to use:
i litmus paper instead of universal indicator?
ii universal indicator instead of litmus paper?
2 Draw diagrams to show the difference between strong and weak solutions of acids and bases.
3 The pH of a creek in an industrial area was monitored by a class of students over a period of 4 weeks. The following data were obtained:

Day	1	4	7	10	13	16	19	22	25	28
pH	7.4	7.3	7.3	7.4	7.2	5.1	3.9	4.0	6.5	7.2

a Plot these data on a graph.
b A local factory had released acid into the creek accidentally.
i During what period was this?
ii Local authorities took action when they learnt of the spill. What do you suggest the authorities did? On what day did this happen?

Spreadsheet

Extension worksheet

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FACT FILE

Strong acids and alkalis can be dangerous. They are highly corrosive. If you spill acid or alkali on your skin or clothes wash the area under running water immediately. Wipe up all spills with a cloth and rinse out in water.

Maintaining a chlorinated pool

FIGURE 2.34
A person stepping into a pool is 'wearing' more than just a bathing suit. They also take into the water persipiration, sun screen, hair spray, face cream and deodorant.

Talk about
What other substances could be taken into a pool?

Swimming pools need regular upkeep in order to keep the pH at safe levels for swimmers. The pH should be between 7.4 and 7.6. In addition, substances including pollen, twigs, leaves, dust, hair, body perspiration and cosmetics are brought into the pool by swimmers, rain and wind. All of these unwanted items in the pool are called **contaminants**.

Chemicals, as well as filtration, must be used to control or eliminate these contaminants and maintain pH at an acceptable level.

The major problem in swimming pools is the growth of **algae** and **bacteria**. Algae pollute the water and can block filters. Bacteria can cause infections. Together, they make the water basic.

Chlorine compounds are usually used to disinfect the water in pools. This is called **pool chlorination**. The 'liquid chlorine' sold for use in home pools is a solution of sodium hypochlorite. 'Dry chlorine' is solid calcium hypochlorite. When these compounds are added to water, they react with the water to form hypochlorous acid.

Chlorine gas also forms hypochlorous acid when added to water. This chlorination method is sometimes used in large public pools.

Since chlorine is broken down quickly by sunlight, cyanuric acid is added to chlorine pool chemicals to act as a UV filter for the pool. These stabilised chlorine compounds, although more expensive, mean that chlorine needs to be added to the pool less often.

For correct pool maintenance, enough hypochlorous acid must be present in the water. It prevents the growth of bacteria

EXPERIMENT 5 TESTING WATER QUALITY

AIM
To test the pH of water samples to determine whether the water has acceptable pH levels for swimming.

MATERIALS
- water samples
- universal indicator

METHOD
1 Collect a number of pool water samples as well as river and tap water samples.
2 Use universal indicator to test the pH of each of the samples.
3 Record your results in a table.

CONCLUSION
Is the pH level for each water sample acceptable for swimming?

EXPERIMENT 6 NEUTRALISING POOL WATER WITH A HIGH pH

Imagine that your swimming pool is as alkaline as ammonia solution. By adding hydrochloric acid, the pH can be restored to between 7.4 and 7.6 (ideal pH conditions for swimming pools).

AIM

To investigate the changes in pH as 'pool water' that has a high pH is neutralised with an acid.

MATERIALS

- 100 mL conical flask
- liquid universal indicator
- 2 × 10 mL measuring cylinders
- dilute, non-detergent ammonia solution ('pool water' sample)
- dilute hydrochloric acid solution

METHOD

1 Measure 10 mL of ammonia solution in a measuring cylinder, and pour into a conical flask.
2 Add 2 drops of universal indicator to the flask and swirl.
3 Record the pH of the solution in a table.
4 Measure 1 mL of hydrochloric acid solution in a measuring cylinder. Add this to the flask.
5 Record the pH of the solution.
6 Repeat steps 4 and 5 until 40 mL of hydrochloric acid has been added to the flask.
7 Graph your results.

Datalogging

QUESTION

What volume of hydrochloric acid was added when the pH of the 'pool water' was ideal for swimming?

and algae. The amount of hypochlorous acid in the water determines the pH.

If the pH is too high (too alkaline), the concentration of hypochlorous acid is low and the pool contains too much base. If the pH is too low (too acidic), the acid content of the water can cause eye and skin irritation. Too much acid can also etch the plaster and corrode the metal piping and filters in the pool.

If the pool pH is too high, it can be lowered by adding an acid. This will neutralise the base. Muriatic acid (a solution of hydrochloric acid) or solid sodium hydrogen sulfate can be used.

hydrochloric acid + base → water + salt

sodium hydrogen sulfate + base → water + salt

If the pool pH is too low, it can be raised by neutralising some of the acid with a base. Sodium carbonate (also called washing soda or soda ash) or sodium bicarbonate (baking soda) can be used.

sodium carbonate + acid → salt + water + carbon dioxide

sodium bicarbonate + acid → salt + water + carbon dioxide

If people have been on holidays, or neglected to maintain correct chlorine or pH levels, the pool water will become unsafe to swim in. Chemical products are available for a quick solution.

- If pool water goes cloudy, products which superchlorinate the pool can be added. These chemicals are simply large doses of calcium hypochlorite.
- If green slime or black spots appear on the wall of a pool, **algicide inhibitors** may be used to kill microscopic algae that have grown.

Questions

1 What may happen if the pH of a pool is not balanced?

2 Prepare an advertising brochure for a new range of pool products which are easy to use. Include instructions for your products.

Find out

Investigate the advantages and disadvantages of alternatives to chlorine for pool maintenance.

FIGURE 2.35
Commercial kits are available to check the chlorine levels and pH of swimming pools.

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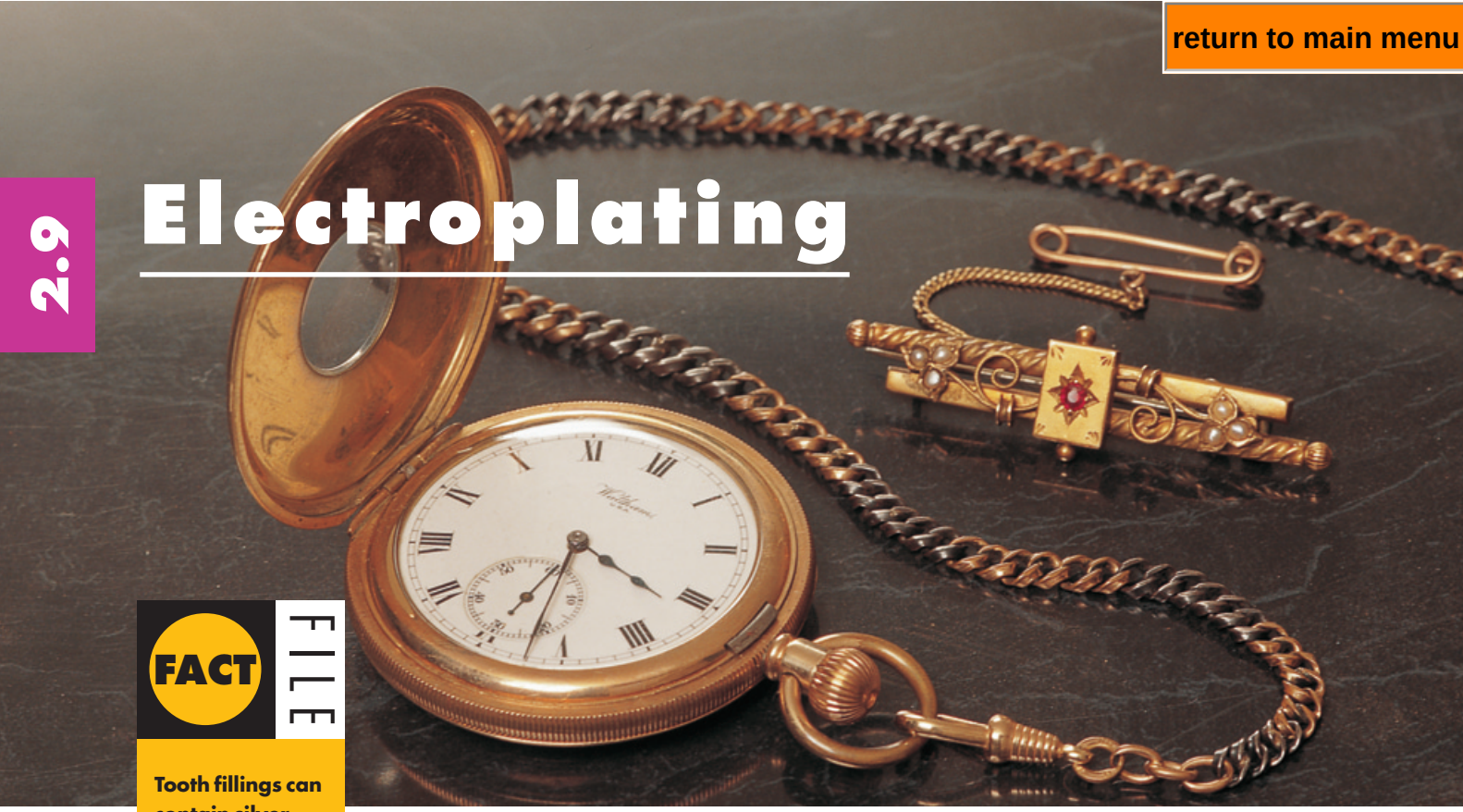
2.9

Electroplating



FACT FILE

Tooth fillings can contain silver, which forms silver oxide on the surface. Aluminium atoms lose electrons more readily than silver atoms. This means that if you allow a piece of aluminium foil to come into contact with a tooth filling, the silver ions in the oxide will gain electrons from the aluminium. Since teeth are filled in order to cover nerves, this flow of electricity is like mildly electrocuting the tooth nerve. Just watch the expression on the face of a person who accidentally chews on a piece of aluminium foil!

FIGURE 2.36
Which is pure gold and which is gold plated?

Gold is one of the most sought after and expensive metals. It is very popular for making jewellery, and a chunky gold bracelet can cost thousands of dollars. But there are cheaper alternatives. It is possible to make the bracelet out of some inexpensive metal, and then coat it in gold by using electricity. This is called **electroplating**, and the bracelet would be described as **gold plated**.

If you did the Activity in section 2.6 you would know that electricity can be used to break water into hydrogen and oxygen. This process is known as **electrolysis** and can also be used to coat metals with other metals (see Experiment 7).

Copper, like most metals, will lose its valence electrons and form positive ions. When a voltage is applied in Experiment 7, the positively charged copper ions are attracted to the negative terminal, which in this case is a nail. There they pick up electrons to replace those they have lost, and copper metal attaches itself to the nail in an even coating. Copper atoms in the positive electrode give up their electrons to form ions and these replace the copper ions lost from the solution.

Electroplating can be used to purify metals that have been mined and which are mixed together with other substances. When electroplating is used to refine copper, the negative terminal is a piece of pure copper and the positive terminal is the impure copper. Copper ions form from the impure sample and are transferred to the piece of pure copper.

Electrolysis can also be used to polish intricately shaped pieces of metal. In this case the object is attached to the positive terminal so that its ions will be removed. The high spots are dissolved first, leaving a very flat, unscratched, unstained surface.

Electroplating is used in hundreds of ways, especially for decoration and protection. Iron can be coated with different metals to stop it rusting, for example. Shiny parts of cars have often been chrome plated, printers blocks are prepared by electroplating, 'master' audio disks used for pressing out copies are made by using electrolysis, and 'tin' cans are actually steel which has been tin plated.

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EXPERIMENT 7 ELECTROPLATING

AIM
To use electrolysis to electroplate an iron nail with copper.

MATERIALS
- 200 mL of 2 M copper sulfate solution
- piece of copper
- large iron nail
- accurate scales
- 6 V power supply
- 2 leads with alligator clips
- steel wool

METHOD
1 Clean the piece of copper and the nail thoroughly with the steel wool.
2 Record the masses of the piece of copper and the nail. You could use a table and also record final masses here.
3 Connect up the circuit as shown. Make sure the nail is attached to the negative terminal of the power supply.
4 Turn on the power for a few minutes, being very careful not to let the nail touch the copper.

RESULTS
1 Turn off the power and remove the copper and nail. Let them dry in the air. (Will blowing on them help to dry them? Why?)

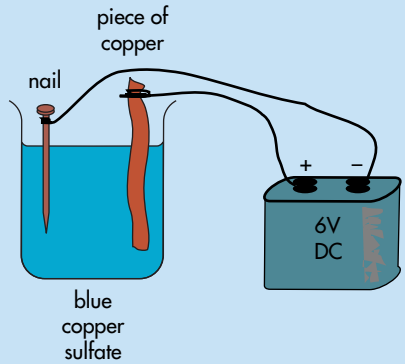

FIGURE 2.37
Copper plating a nail.

2 Note their appearance.
3 Re-weigh the copper and the nail. Record your results.

DISCUSSION
Explain your results. A diagram may help.

EXTENSION
If possible, leave the experiment to run for several hours. How much copper would you expect to be deposited on the nail overnight?

Questions

1 Is there a difference between electroplating and electrolysis? Explain.
2 Why are objects to be plated attached to the negative terminal?
3 Why is gold-plated jewellery cheaper than pure gold jewellery?
4 What are some uses of electroplating?

Think about

1 Why do tin cans rust over time if they have been scratched deeply?
2 How could you tell if something is pure gold and not just gold plated?

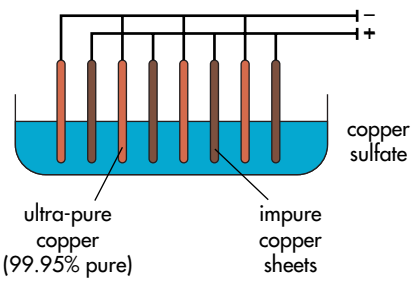

FIGURE 2.38
Copper can be purified by using electricity.

Find out

1 Write to a mining company, asking how they purify aluminium and copper.
2 Find out what audio masters are, how they are used, and how they are made.

FIGURE 2.39
Electroplating is only skin deep!

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Using scientific language

Design and construct a crossword using the following words: atom, proton, molecule, ion, lattice, shell, element, compound, bond, conductor, insulator, electrolysis, electrolyte. Exchange your crossword with another student and complete it.

Check your knowledge

1 a Draw a labelled diagram of an atom that has 9 protons and 9 electrons and 9 neutrons.
  b What type of atom is this?
  c Which other element would you expect to behave most like this element?
2 a Draw and label a molecule of water and a molecule of oxygen.
  b Both are molecules but only one is an element. Explain this statement.
3 An atom of sodium has 11 protons and 11 electrons. How does an ion of sodium differ from this?
4 What is an electron shell? What is the valence shell?
5 What are the major similarities and differences between protons, neutrons and electrons?
6 Why can metals be bent and shaped when ionic substances cannot?
7 Explain in your own words what pH means.
8 The pH of five substances are listed below. For each substance, identify:
  a whether it is weak acid, strong acid, weak base, strong base or neutral
  b what colour the solution would be if three drops of universal indicator were added to it
  c what colour the solution would turn red or blue litmus paper.

Substance	A	B	C	D	E
pH	7	1	5	8	13

Apply your skills

1 Look at Figure 2.40 then answer the questions that follow. (Refer to the periodic table.)
  a What type of atom is it?
  b Is it a metal or non-metal?
  c Which group is it in? Name one other element in this group.
  d Which row of the periodic table is it in? Give the name of the atom that has one more proton than the atom in the diagram.
2 Which element has 4 valence electrons and 3 electron shells?
3 Boric acid is sometimes used as an eye rinse. Why can boric acid be used in this way when other acids, such as sulfuric acid, could cause permanent blindness?
4 Design and conduct an experiment to test various soils for pH levels by using universal indicator.
5 Why does salty water conduct electricity when pure water does not?
6 Explain using diagrams what type of bond you would have between the two atoms of nitrogen that make up a nitrogen molecule.
7 Potassium and chlorine will react to form an ionic lattice. Look first at the periodic table and then draw what you think a small part of this lattice would look like. Do you think it is correct to call this a molecule? Explain.
8 Why do metals tend to form positive ions and non-metals form negative ions?

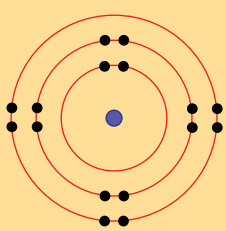

FIGURE 2.40

Challenge yourself

Figure 2.41 shows a circuit through which an electric current is flowing. The current is the movement of charged particles through the circuit. Draw and label diagrams to show what is happening at points A and B of the circuit. You should clearly show which particles are moving and which are not.

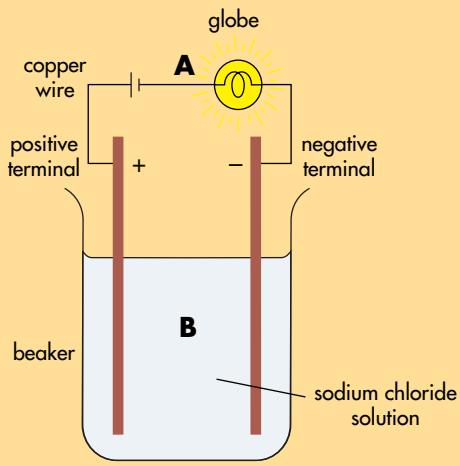

FIGURE 2.41

eTester

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return to main menu

CHAPTER 3

CHEMICAL REACTIONS

The development of the humble match parallels the development of modern chemistry. Before the beginning of modern chemistry people believed that natural processes were controlled by magic and spirits.

The story begins in Germany around 1670 with the discovery of the element phosphorus. Hennig Brand, like most of his peers, was searching for a magical elixir which could turn lead into gold. Instead he found a white, waxy substance that glowed in the dark. The glow was a result of the slow combustion of phosphorus.

Robert Boyle discovered in 1680 that thick paper covered in phosphorus would ignite twigs covered in sulfur, when rubbed together. The first commercial use of his idea 100 years later was the 'phosphorus candle'. This consisted of glass tubing containing thick paper covered in wax, its end soaked in phosphorus. When the glass tube was broken, the phosphorus burst into flames. Many similar inventions followed which were in fact quite dangerous. It was discovered that white phosphorus produced poisonous fumes, and caused many accidental fires.

The first safety matches were invented in 1844 by Gustave Pasch. These matches used the safer red phosphorus which would only ignite if struck against a rough surface. Many changes took place over the years as other chemicals were developed. Modern matches have a head made from antimony sulfide, potassium chlorate and either sulfur or carbon. A mixture of red phosphorus and sand makes up the striking surface on the box. The friction produced by rubbing the match head across the striking surface provides the energy to ignite the match and start the chemical reaction.

In this chapter you will investigate a range of chemical reactions and the energy changes they produce.

Web links

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Distinguish between physical and chemical changes.
- Describe how theories can be modified or rejected as a result of new evidence.
- Write balanced equations for a range of chemical reactions.
- Distinguish between synthesis, decomposition, single and double replacement reactions, giving examples of each.
- Describe reactions involving acids, bases, metals and carbonates.
- Sort metals into an order of activity.
- Describe how chemical reactions can be used to obtain metals.
- Describe energy changes which take place in chemical reactions, and changes of state.
- Give methods of altering the speed of a reaction.

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3.1

Chemical reactions

FIGURE 3.1
Water boiling in a beaker.

Talk about

When water boils in a beaker, bubbles of water vapour burst and escape from the surface. In a voltameter, electrical energy causes water to produce bubbles of hydrogen and oxygen gas. How are these two changes similar? How are they different?

FIGURE 3.2
In a voltameter, electrical energy causes water to produce bubbles of hydrogen and oxygen gas.

Types of changes

Matter is continually changing around us. Living things grow, iron turns into rust, rocks turn into soil, and clouds turn into rain. All changes to matter can be classified as physical or chemical.

A **physical change** involves changes to the physical properties of a substance. The size, shape or state of a substance can alter but no new substance is formed. For example, ice is solid water. When it melts it is still water, it has just changed to a liquid state. When a substance dissolves in water, a mixture is made. No new substance forms, so this too, is a physical change.

Chemical changes are different. Chemical changes produce new substances. The decomposition of water by an electric current is a chemical change because two new substances, hydrogen and oxygen, are formed. Chemical changes are also called **chemical reactions**. You will recall that if a chemical reaction occurs there is usually evidence that a new substance is present. Evidence for a chemical reaction includes:

- a new gas forming
- a solid (or **precipitate**) being produced
- a permanent colour change occurring.

Equations

All chemical changes can be summarised as an **equation**. This is a shorthand way of showing the **reactants** (the chemicals which take part in the reaction) and the **products** (the chemicals the reaction produces). Equations can be written by using the names or the formulae of the substances involved. Symbols are used to give added information (see Table 3.1).

TABLE 3.1

Symbol	Meaning
(s)	solid
(l)	liquid
(g)	gas
(aq)	in solution in water
+	plus
→	gives

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EXPERIMENT 1 DESCRIBING CHANGES

Multimedia

AIM

To observe and classify changes as physical or chemical.

MATERIALS

- 5 test-tubes and test-tube stand
- crucible
- Bunsen burner
- tripod
- pipe-clay triangle
- test-tube holder
- sodium carbonate solution
- copper sulfate solution
- sulfuric acid
- hydrochloric acid
- sodium carbonate
- sodium chloride
- paraffin wax
- granulated zinc
- granulated tin

METHOD

1 Draw up a results table (use a full page).
2 Describe the appearance of the substances before carrying out the tests.
3 Describe any changes which take place.
4 Classify each change as physical or chemical.
a Add 2 mL of sodium carbonate solution to 2 mL of copper sulfate in a test-tube. Leave in stand.
b Put $^1/_4$ teaspoon of copper oxide in a test-tube with 5 mL of sulfuric acid. Warm gently.
c Place 1 teaspoon of paraffin wax in a test-tube and warm gently. Leave to cool in stand.
d Place 1 piece of zinc with 1 piece of tin in a crucible. Heat until metals melt, then allow to cool.
e Add $^1/_4$ teaspoon of sodium carbonate to a test-tube with 5 mL of hydrochloric acid.
f Add $^1/_4$ teaspoon of sodium chloride to 5 mL of hydrochloric acid in a test-tube, shake and place in stand.

RESULTS

Test	Appearance of substances before change	Description of changes	Type of change

DISCUSSION

1 Which chemical changes produced a gas?
2 Which chemical changes produced a new solid (precipitate)?
3 Which chemical changes resulted in a colour change?
4 Which changes involved a change of state?

For example,

$$\text{water} \rightarrow \text{hydrogen} + \text{oxygen}$$
$$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$$

Physical changes can also be shown in an equation. For example,

$$\text{water} \rightarrow \text{water}$$
$$H_2O(l) \rightarrow H_2O(g)$$

Questions

1 Distinguish between physical and chemical changes.
2 Read each equation and say whether it is a physical or chemical change.
a $AgNO_3(aq) + HCl(aq) \rightarrow AgCl(s) + HNO_3(aq)$
b $S(s) + O_2(g) \rightarrow SO_2(g)$
c $NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$
d $CaCO_3(s) \rightarrow CaCO_3(l)$
e $PbO(s) + CO(g) \rightarrow Pb(s) + CO_2(g)$
3 A small sample of solid green copper carbonate is heated in a test-tube until it turns into a black solid called copper oxide. In the process carbon dioxide gas is given off.
a Give two pieces of evidence that this is a chemical change.
b What is the reactant?
c What are the products?
d Write an equation which summarises the same information.

Think about

1 Explain why dissolving sugar is a physical change.
2 Why do chemists write equations?

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HISTORY OF SCIENCE

Down with the phlogiston theory

Talk about

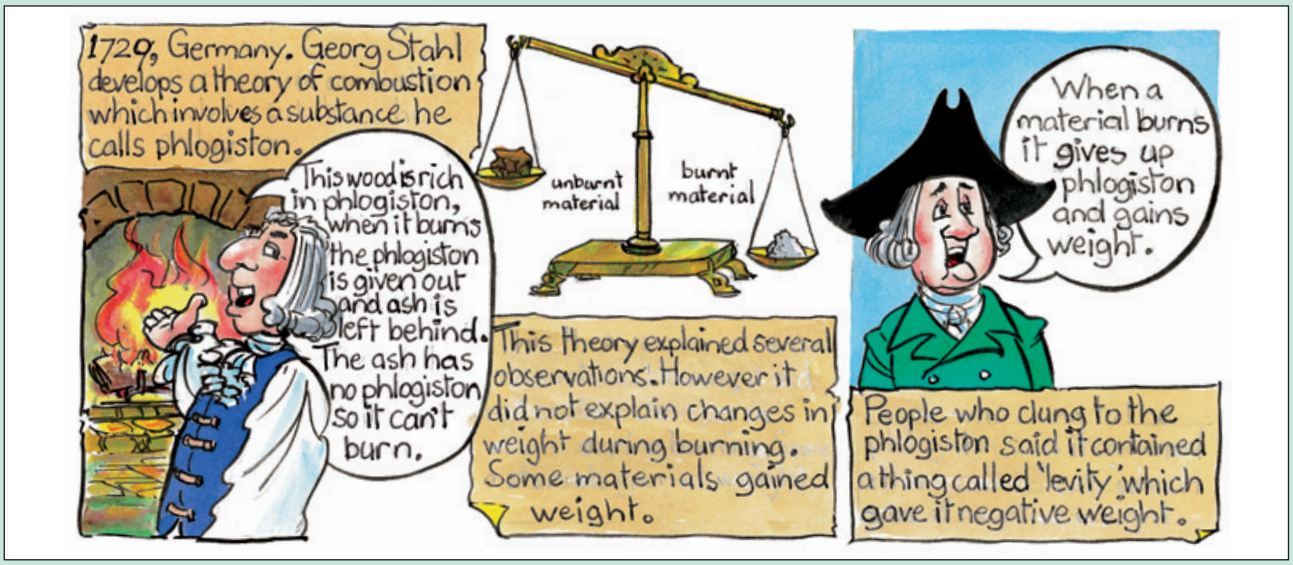
Light a candle and watch it burn. Discuss the changes you observe taking place to the burning candle.

Burning or combustion is a chemical change which has long fascinated humans. For thousands of years fire has been central to human survival and civilisation. Fire has been used to keep warm, cook food and produce metals from ores. No wonder that people wanted to understand what happens when a substance burns.

Ideas or theories about burning have changed over time. Early ideas were based on the ancient Greek teaching of the four basic elements of earth, air, water and fire. During the middle ages alchemists used the four-element model to explain what they observed as they searched for a way to make gold. The alchemists believed that when a material burned it gave up something to the air.

Lavoisier changed the way chemical research was carried out (see Figure 3.3). From an early stage he realised how important accurate measurements were in chemical investigation. Instead of relying solely on what he could see, his measurements allowed him to develop a better understanding of the changes he observed.

In 1798 he published a book containing his new theories. It included his law of the conservation of mass. Lavoisier wrote: 'We must lay it down as an incontestable axiom, that in all operations, nothing is created, an



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equal quantity of matter exits both before and after the experiment'.

Over 200 years later we can rephrase this as 'in a chemical change, matter is not created or destroyed'.

Despite his great scientific contributions to chemistry, Lavoisier was put on trial during the French revolution. Unfortunately he had married the daughter of a tax collector and was accused of many wrongdoings.

Questions

1 What observations of burning would lead people to believe that something was given out?

2 What observations put doubt on the phlogiston theory of combustion?

3 Write a word equation for the heating of mercury in air.

Think about

1 How did Lavoisier's methods of research lead him to develop the law of conservation of mass?

2 What was 'dephlogisticated air'?

3 At Lavoisier's trial, the arresting officer said 'The Republic has no need of scientists'. What do you think of this statement for 1792 France?

Find out

Who were the alchemists? How were they different from chemists?

FIGURE 3.3

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3.3

A balancing act

FIGURE 3.4
Atoms are not created or destroyed in a chemical reaction, but nuclear reactions are very different.

Chemical reactions

We have learnt that in a chemical reaction a new substance is formed. The law of conservation of mass states that matter is not created or destroyed in a chemical reaction. The new substance or product must therefore be formed from the atoms present in the reactants. In a chemical reaction the products are formed by re-arranging the atoms of the reactants.

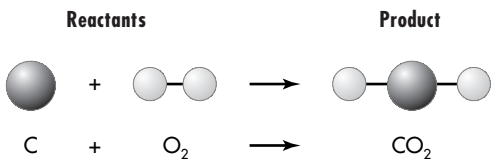

FIGURE 3.5
The same number and type of atoms must be present before and after the reaction.

Balancing equations

When you write **chemical equations** using formulae and symbols you have to show that atoms are not created or destroyed. The number of atoms of each type on the reactants side must be equal to the number of atoms of each type on the products side. If the atoms are equal in type and number then we say the equation is balanced.

Rules for writing equations

1 Write a word equation of the reaction. This may be based on observations you have made or information supplied.

2 Write correct chemical formulae for each substance involved by using valency tables. Set out the formulae according to the word equation. Add symbols to show the state of each substance.

3 Count the number of atoms of each element on the reactant side and the product side. If they are the same, the equation is balanced.

4 When the number of atoms of an element is different on the reactant and product side, place numbers in front of the formulae. This then multiplies all the atoms in the formula by that amount. Continue to place numbers in front of formulae until the equation is balanced.

5 Never try to balance an equation by adjusting a correct chemical formula. For example, when magnesium reacts with hydrochloric acid, hydrogen gas is given off and magnesium chloride is left behind in solution.

(i) Magnesium + hydrochloric acid → magnesium chloride + hydrogen

(ii)The correct formulae are:

$Mg(s) + HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

(iii)

Reactants		Products	
Mg	1	Mg	1
H	1	H	2
Cl	1	C	2

(iv) The equation can be balanced by placing a 2 in front of HCl(aq):

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

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EXPERIMENT 2 CONSERVING MASS

AIM
To measure total mass before and after a chemical reaction.

MATERIALS
- 250 mL conical flask
- rubber stopper to fit flask
- small test-tube to fit inside flask
- string
- sodium carbonate
- hydrochloric acid
- electronic balance

METHOD
1 Put 1 teaspoon of sodium carbonate into the flask.

2 Pour 10 mL of acid into the small test-tube.

3 Tie the string around the neck of the test-tube and use the string to lower it into the flask without spilling any acid.

4 Put the rubber stopper into the neck of the flask, trapping the string so the test-tube hangs upright.

5 Weigh the entire set-up, then tilt the flask on its side so the acid runs out of the test-tube and mixes with the sodium carbonate.

6 Weigh the entire set-up again when no more change takes place.

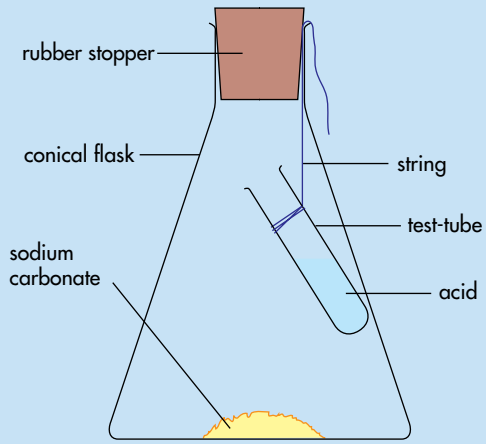

FIGURE 3.6

RESULTS
Record the mass before and after the reaction.

CONCLUSION
Based on your results, was mass conserved?

DISCUSSION
1 Write the name of the reactants.

2 What evidence was there that a new substance was formed?

3 The products were sodium chloride, water and carbon dioxide. Use this information to write a full balanced equation.

Questions

1 Write balanced equations for each of the following reactions.

a Solid copper is produced, with carbon dioxide, when a mixture of carbon and copper oxide is heated.

b Solid iron chloride is formed when iron powder is heated in a stream of hydrogen chloride gas. Hydrogen gas is also produced.

c Solid lead nitrate decomposes to form solid lead oxide and a mixture of two gases. Brown nitrogen dioxide and oxygen are given off.

d Solid ammonium carbonate decomposes, giving off a mixture of three gases, water, ammonia and carbon dioxide

e Magnesium ribbon is added to sulfuric acid, forming magnesium sulfate. Hydrogen gas is given off.

2 Why is it not possible to change a chemical formula so that an equation can be balanced?

Foundation worksheet 1

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3.4

Types of chemical reactions

FIGURE 3.7
Not all reactions occur in the science laboratory. Reactions are happening all around us.

FACT FILE

The space shuttle uses hydrogen and oxygen, held in separate containers, as its fuel. They are mixed when needed to react and provide power for the propulsion of the shuttle. The hydrogen burns in oxygen to produce water as steam, according to the equation:
$2H_2(l) + O_2(l) \rightarrow 2H_2O(g)$
What type of reaction is this?

Chemical reactions can be grouped in many ways. Most chemical reactions can be divided into four main types. These are **synthesis reactions**, **decomposition reactions**, **single replacement reactions** and **double replacement reactions**.

Synthesis reactions

A synthesis or **combination** reaction occurs when two substances react to form a third substance. For example, hydrogen and oxygen can react to form water:

$2H_2 + O_2 \rightarrow 2H_2O$
hydrogen + oxygen → water

Another example is the burning of coal:

$C + O_2 \rightarrow CO_2$
carbon + oxygen → carbon dioxide

Corrosion of metals such as zinc and iron occurs when the metal combines with oxygen from the air:

zinc + oxygen → zinc oxide
$2Zn(s) + O_2(g) \rightarrow 2ZnO(s)$

Many synthesis reactions take place in living things as simple molecules combine to form large complex molecules. For example in photosynthesis, carbon dioxide and water combine in a complex series of steps to produce glucose and oxygen.

Decomposition reactions

A decomposition reaction is one where a single substance breaks down into simpler compounds or elements. Decomposition reactions are the opposite of synthesis reactions. For example, water may be decomposed to hydrogen and oxygen:

$2H_2O \rightarrow 2H_2 + O_2$
water → hydrogen + oxygen

Another example of decomposition is:

copper carbonate → copper oxide + carbon dioxide
$CuCO_3(s) \rightarrow CuO(s) + CO_2(g)$

Single replacement reactions

A single replacement reaction is one where one element or group of elements replaces another element or group of elements in a compound. Single replacement reactions are also called displacement reactions. For example, zinc reacts with copper sulfate according to the equation:

$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
zinc + copper sulfate → zinc sulfate + copper

Is this reaction balanced?

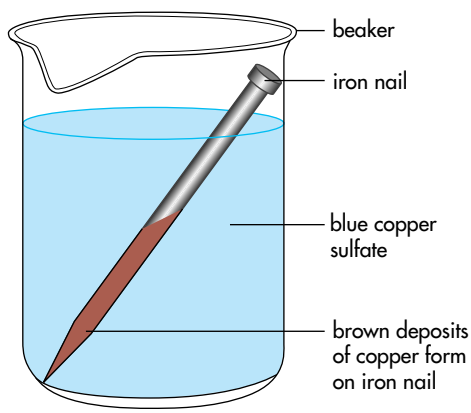

FIGURE 3.8
Copper is being replaced by iron in solution.

Double replacement reactions

When two compounds react to form two different new compounds, this is called a double replacement reaction. Sometimes when two soluble solutions of ionic compounds are mixed, an insoluble salt is produced. This insoluble salt appears as a

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solid which will settle out from the solution. The insoluble salt is called a **precipitate**. For this reason, these chemical changes are sometimes known as precipitation reactions. For example, in the reaction between lead nitrate and potassium iodide:

$PbNO_3$ + KI → PbI_2 + KNO_3

lead nitrate + potassium iodide → lead iodide + potassium nitrate

the balanced equation looks like:

$PbNO_3 + 2KI \rightarrow PbI_2 + 2KNO_3$

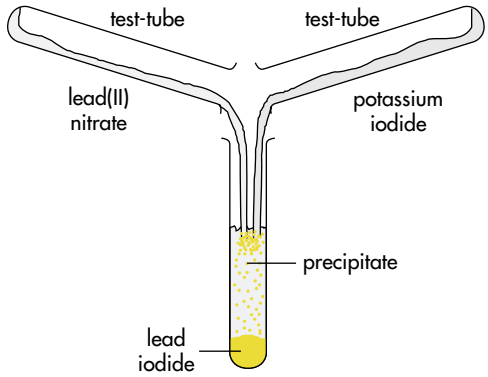

FIGURE 3.9
The precipitate lead iodide is formed in this double replacement reaction.

EXPERIMENT 3 CHALK

Calcium carbonate is a compound which can be found in different forms, such as limestone, sea shells, coral reefs, stalactites and stalagmites in caves, and chalk. This chalk is different from blackboard chalk. It is used as a gentle scouring powder; for example, in toothpaste it is used as an abrasive.

AIM

To make chalk.

MATERIALS

- 2 × 250 mL beakers
- filter paper
- funnel
- retort stand and ring
- distilled water
- 20 mL of 1.0 M calcium chloride
- 2.5 g sodium carbonate

METHOD

1 Place 2.5 g of sodium carbonate into a beaker and add 20 mL of distilled water. Swirl the beaker until the solid has dissolved.

2 Pour this solution into a second beaker containing 20 mL of 1 M calcium chloride solution. Swirl to mix.

3 Set up the apparatus in Figure 3.10.

4 Moisten the filter paper with distilled water.

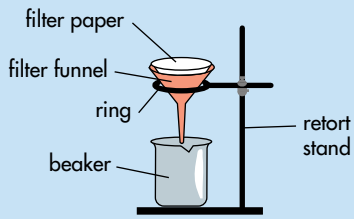

FIGURE 3.10

5 Pour the mixture into the filter. Each time, fill the filter to about half full.

6 After all the mixture is filtered, rinse the solid in the filter paper with 10 mL of distilled water.

7 Place the filter paper in a beaker to dry.

DISCUSSION

1 Is your chalk smooth or gritty?

2 The equation for this reaction is:

calcium chloride + sodium carbonate → calcium carbonate + sodium chloride

a Rewrite this equation using chemical formulae. Remember to balance the reaction.

b What type of reaction is this?

3 Place some of the chalk in a dirty dish. Moisten it with a small amount of water. Use it to scour the dish. Does it work?

Questions

1 What are the four main types of chemical reactions?

2 Decide the type of each of the reactions below:

a $C + O_2 \rightarrow CO_2$

b $2H_2O \rightarrow 2H_2 + O_2$

c $Zn + CuSO_4 \rightarrow Cu + ZnSO_4$

d $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

3 Balance the following equations:

a $CuO + HCl \rightarrow CuCl_2 + H_2O$

b $NaI + Pb(NO_3)_2 \rightarrow NaNO_3 + PbI_2$

c $AgNO_3 + NaCl \rightarrow NaNO_3 + AgCl$

d $Na_2SO_4 + BaCl_2 \rightarrow NaCl + BaSO_4$

e $NaOH + CuSO_4 \rightarrow Na_2SO_4 + Cu(OH)_2$

Internet assignment

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3.5

Reactions of acids and bases

FIGURE 3.10
Strong acids can react violently. Many also give off corrosive fumes. Why might the protective clothing be necessary?

Neutralisation

When an acid and a base are mixed, they react to form salt and water in a process called **neutralisation**:

acid + base → salt + water

The products which form are neither acids nor bases. The acids and bases have neutralised, or cancelled out, each other.

Indigestion and antacid tablets

Have you ever had a burning sensation in your chest? It's probably indigestion caused by stomach acid backing up into your oesophagus. You take an **antacid** tablet. You feel better. What has happened? An antacid contains a base. For example, when sodium hydroxide, which is a base, combines with hydrochloric acid, the following reaction occurs:

sodium hydroxide + hydrochloric acid → sodium chloride + water

The products, sodium chloride and water, are neither acidic nor basic. They form a harmless solution of salt water.

Of the many bases that can neutralise acids, only a few weak bases are safe enough to be used in antacids. The labels on over-the-counter antacids tell you which weak base they contain. Common weak bases in antacids include calcium carbonate and magnesium hydroxide. Each compound differs in the amount of acid it can neutralise and the rate at which it acts.

Acids and carbonates

When acids are mixed with substances containing **carbonates**, carbon dioxide gas is formed, as well as salt and water.

acid + carbonate → salt + water + carbon dioxide

For example,

$2HCl(aq) + Na_2CO_3(s) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$

EXPERIMENT 4 MOTHBALL MOTION

AIM

To investigate the reaction between an acid and a carbonate.

MATERIALS

- beaker of water
- 1 tablespoon of vinegar
- 1 tablespoon of bicarbonate of soda
- 3 mothballs

METHOD

1 Add the vinegar and bicarbonate of soda to a beaker of water.

2 Drop the mothballs into the beaker.

RESULTS

Describe what happened to the mothballs and draw a picture to illustrate your observations.

DISCUSSION

1 Which gas is formed in this reaction?
2 Describe how you could test for the gas.
3 If bicarbonate of soda is a base, what type of substance is vinegar?

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EXPERIMENT 5 ETCHING

Acids can react with metals:
zinc + hydrochloric → zinc + hydrogen gas
acid chloride

AIM
To etch a design onto a zinc plate.

MATERIALS
- zinc plate
- paint remover
- needle
- beaker of water
- 10 M hydrochloric acid (**to be handled by your teacher**)
- (**optional:** paper, paper towelling, roller and printing inks)
- bituminous paint
- paintbrush
- tongs

METHOD
1 Paint each side of a zinc plate with bituminous paint, allowing each side to dry thoroughly.
2 Use a needle to scratch out a design on your plate. Make sure that you can see the metal surface where you have scratched it.
3 Give your plate to your teacher to place in a beaker of hydrochloric acid for 2 days.
4 Your teacher will remove your plate from the acid and place it in a beaker of water to rinse it thoroughly.
5 Remove the paint by using a paint remover.
6 (Optional) Use your plate to produce an etching:
a roll ink onto your plate with the roller
b wipe off surface ink with paper towelling, making sure that ink remains in the crevice where the acid has etched out the metal
c place paper over the plate and press hard to produce an etching on the paper.
Experiment with different inks.

DISCUSSION
1 What evidence did you see to show that etching is a chemical reaction?
2 Etching produces hydrogen gas. How could you test for this gas?

EXTENSION
Different acids can produce different effects. Nitric acid 'bites' an etched line more readily on the sides than on the bottom. This makes the line wider. Hydrochloric acid bites deeper into the plate to give a finer line. Produce an etching with fine and wider lines. You could also try iron or aluminium plates instead of zinc.

Acids and metals

Acids react with metals to produce hydrogen gas and a salt:

metal + acid → salt + hydrogen gas

For example,

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

Questions

1 What is meant by neutralisation?
2 Write a general equation which shows a neutralisation reaction.
3 List three different reactions which involve acids or bases, and give an example of each type of reaction.
4 Bee and ant stings contain an acid whereas wasp stings contain a base.
a How does the treatment for a bee or an ant sting differ from that for a wasp sting? (Note: apply ice first.)
b List some household substances which could be used to treat:
i bee or ant stings
ii wasp stings.
5 Complete the following word equations and then write balanced chemical equations for the following;
a calcium + sulfuric acid
b magnesium hydroxide + nitric acid
c copper carbonate + hydrochloric acid
d nickel + sulfuric acid
e potassium hydroxide + nitric acid
f zinc carbonate + hydrochloric acid

Interactive

TABLE 3.2 Common names of some acids.

Common name	Formula
Hydrochloric	HCl
Sulfuric	H_2SO_4
Nitric	HNO_3
Phosphoric	H_3PO_4

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return to main menu

Reactivity of metals

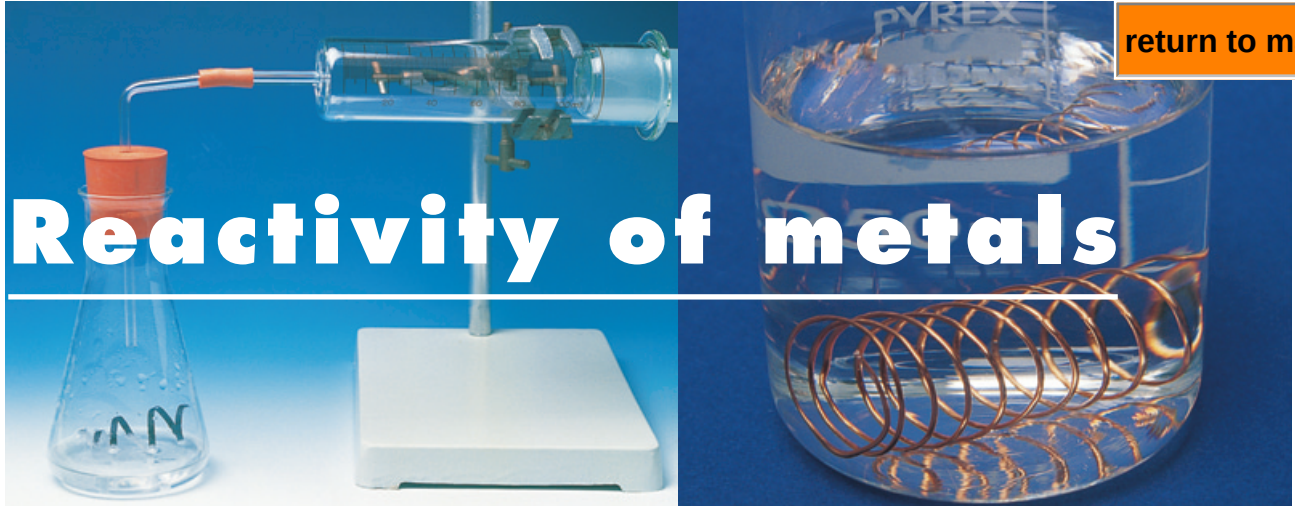

FIGURE 3.11
Different reactivity of metals with dilute hydrochloric acid. Magnesium reacts readily with 2 M HCl, releasing large amounts of hydrogen gas. Copper shows very little reaction.

Some metals are highly reactive, others react hardly at all. Different metals react differently with water and dilute acids. A lump of sodium placed into a trough of water would react according to the following equation:

$$2Na(s) + 2H_2O(aq) \rightarrow 2NaOH(aq) + H_2(g)$$

If the sodium lump is placed into dilute hydrochloric acid, the reaction would become:

$$2Na(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2(g)$$

Very reactive metals like sodium form stable compounds which are difficult to decompose. Moderately active metals such as tin do not react with water but will slowly produce hydrogen gas when mixed with dilute acid. In nature these metals are combined with other elements such as sulfur and oxygen. They can be more easily decomposed. Very unreactive metals like gold will not react with water or dilute acid. These metals rarely combine with other elements in nature and can be found as pure metals.

The activity of metal elements has been compared and the metals have been ranked in order of **reactivity**. This ranking is called the **activity series**. The most reactive metals are found at the top and the least active metals at the bottom of the activity series.

EXPERIMENT 6 RELATIVE REACTIVITY OF METALS

AIM

To determine the relative reactivity of some metals.

MATERIALS

- 5 cm strips of magnesium, copper, lead, iron and zinc
- 250 mL filter flask with side-arm
- one-hole rubber stopper
- 2 retort stands and clamps
- 5 cm rubber tubing
- 100 mL burette
- silicone grease
- distilled water
- forceps
- 100 mL beaker
- graduated 10 mL syringe
- funnel
- bench mat
- 2 M HCl
- watch or clock

METHOD

This is a class activity with different groups testing different metals.

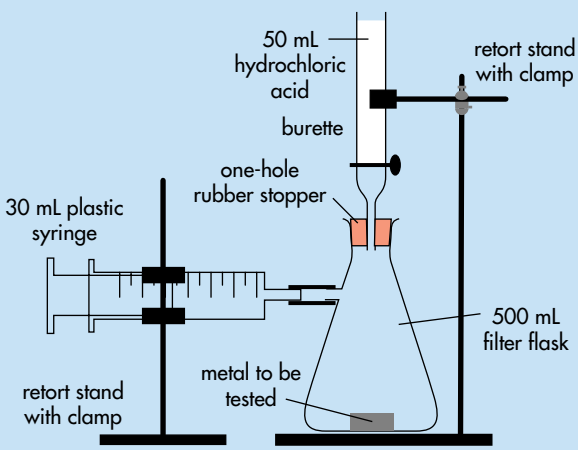

FIGURE 3.12

1 Place the 250 mL filter flask on the bench mat.

2 Using forceps, place the clean 5 cm strip of the metal you are testing into the base of the flask. Do not touch the metal. Seal the top with a one-hole rubber stopper.

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EXPERIMENT 6 CONTINUED

3 Set a retort stand next to the flask and attach a clamp to hold the closed burette in place. Attach the burette to the retort stand. At this stage, do not lower the burette into the hole in the rubber stopper.
4 Lubricate the plunger of a graduated 10 mL syringe with a thin film of silicone grease. Depress the plunger into the syringe and push down to the 0 mL mark.
5 Attach the syringe to the side-arm of the filter flask using a small piece of rubber tubing and support the arrangement as shown in Figure 3.12.
6 Carefully lower the closed burette down into the hole of the rubber stopper to form a tight seal.
7 Using a beaker and funnel, pour 50 mL of hydrochloric acid (HCl) into the burette.
8 Open the burette to release the dilute acid into the flask. Begin timing.
9 Measure the amount of gas produced every minute for 15 min and record the results and your observations in a suitable table in your book.
10 When completed, carefully detach the burette from the flask. Your teacher will tell you how to rinse the flask and burette.

DISCUSSION

1 Transfer your results to a class table containing all the metals tested. Copy the class table into your book.
2 For each metal, graph the class results for time versus volume of gas. Use a different colour for each metal.
3 What is the name of the gas you collected?
4 Write down the equations for the reaction of each metal with acid.
5 What types of reactions have occurred between the metals and dilute acid?
6 Using the collected data, list the metals in order of most gas produced to least gas produced when reacting with HCl.
7 Which of the metals could be classified as:
a most reactive? **b** least reactive?

The activity series can be used to make predictions about reactions. Replacement reactions between metals only occur when a more active metal takes the place of a less active metal.

The reactions the metals have undergone in the experiment are known as **redox reactions** (reduction and oxidation). These reactions occur between the metal and the acid. Materials undergoing **reduction** can gain electrons or hydrogen, or lose oxygen. **Oxidation** occurs when a material loses either electrons or hydrogen, or gains oxygen. For example, copper oxide is reduced to copper in the following reaction:

$$CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(g)$$

In the same reaction the hydrogen is oxidised, as it gains oxygen.

However, not all redox reactions produce oxygen or hydrogen. A more important result of redox reactions is the ability of metals to displace (or replace) each other from their compounds. This result is actively applied in the refining of iron and other metals.

Questions

1 How can we distinguish between metals by using relative levels of reactivity with dilute acid and water?
2 Which of the following metals is the least reactive with water: zinc, magnesium, copper, lead or iron?
3 What metal should you not mix with an acid? Why?
4 What are redox reactions?
5 Water is formed from the oxidation of what chemical?

Think about

Metals such as sodium are known to be highly explosive in water. What precautions should be taken when storing sodium metal?

Find out

Where and how is sodium metal produced? What is the main use of sodium metal?

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APPLICATIONS AND USES OF SCIENCE

Extracting and refining by using chemistry

FIGURE 3.13
Molten iron is poured into a transfer ladle directly from a blast furnace. The iron melts at approximately 1500°C. Very hot!

Metals and metal compounds can react with each other in a redox reaction. These reactions are employed in metal refining.

The extraction of metals from ores and metal refining is a large-scale chemical process. Gold mining is a good example of where a result from chemistry has a direct impact on the wealth of the country and business. How much gold is recovered from a mine depends on the chemical process performed as well as on the quality of the ore.

TABLE 3.3

	Oxidant	Reductant	
strong oxidant (gains electrons readily)	$Au^+ + e^- \rightarrow$	Au(s)	weak reductant
	$Ag^+ + e^- \rightarrow$	Ag(s)	
	$Cu^{2+} + 2e^- \rightarrow$	Cu(s)	
	$2H^+ + 2e^- \rightarrow$	H_2(g)	
	$Pb^{2+} + 2e^- \rightarrow$	Pb(s)	
	$Sn^{2+} + 2e^- \rightarrow$	Sn(s)	
	$Fe^{2+} + 2e^- \rightarrow$	Fe(s)	
	$Zn^{2+} + 2e^- \rightarrow$	Zn(s)	
	$Al^{3+} + 3e^- \rightarrow$	Al(s)	
	$Mg^{2+} + 2e^- \rightarrow$	Mg(s)	
	$Na^+ + e^- \rightarrow$	Na(s)	
	$Ca^{2+} + 2e^- \rightarrow$	Ca(s)	
weak oxidant (does not easily gain electrons)	$K^+ + e^- \rightarrow$	K(s)	strong reductant

Table 3.3 lists the relative redox strengths of some metals.

The reductant is the electron donor and the oxidant is the electron acceptor. A metal ion placed high in Table 3.3 will readily gain electrons from (be reduced by) a metal lower in the table.

For example, if we add copper metal to a solution of the silver salt silver nitrate ($AgNO_3$), the following reaction will occur:

$$Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+} + 2Ag(s)$$

or:

metal + ion of less active metal → ion of more active metal + less active metal

Note that the nitrate ion (NO_3^-) does not take part in the reaction: it is just there to balance the charges.

This type of reaction is called a **replacement reaction**. It is especially useful in the reduction of metal oxides by aluminium powder. Such a process is referred to as the Goldschmidt or Thermit process.

The reactivity of metals—corrosion

Extraction of metals from their minerals involves changing positive **ions (cations)**

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EXPERIMENT 7 COPPER WOOL OR STEEL WOOL?

AIM

To observe the replacement reaction between iron and copper.

MATERIALS

- 1 L beaker
- clean steel wool
- 600 mL of 1 M $CuSO_4$ acidified with 5 mL of 1 M H_2SO_4
- tongs
- thermometer

METHOD

1 Pour 600 mL of 1 M $CuSO_4$ into the 1 L beaker. Place the thermometer into the solution, wait 3 min and record the temperature.

2 Using the tongs, carefully place the steel wool in the solution.

3 Measure the temperature of the solution every minute for 10 min and record your readings in a table.

4 Record your observations of any changes that occur with the steel wool.

5 Do not dispose of the solution down the sink as it is a pollutant. Your teacher will dispose of it appropriately.

DISCUSSION

1 Which chemical was the reductant in this reaction? Explain your answer.

2 What colour change occurred with the solution? Why?

3 What is the pinkish material that may have replaced the steel wool?

4 Draw a line graph showing the change in temperature against time. What was the maximum change in temperature observed?

5 Write the equation for this reaction.

Spreadsheet

Foundation worksheet 2

into neutral atoms. For example:

$$Fe^{2+} + 2e^- \rightarrow Fe$$

Corrosion of metals is the reverse of this process. Iron is a commonly used material which is susceptible to corrosion. When iron corrodes it forms cations:

$$Fe \rightarrow Fe^{2+} + 2e^- \text{ (electrons released)}$$

Another substance is needed to accept the electrons. Oxygen and water supply the right medium for the corrosion to occur.

Questions

1 In the experiment, what other metal could have been used to displace iron instead of copper? Why?

2 Why do we see a temperature change in some reactions?

3 Identify the three ways oxidants differ from reductants in a redox reaction.

4 What metals could be used to recover gold from a solution?

5 After a displacement reaction has occurred between two metals such as silver and copper, what other metal could be used to treat the solution to remove the displaced copper? Refer to Table 3.3.

Think about

Sludge recovered from car engines contains oil that could be recycled. How could the sludge be treated to remove the metallic iron fragments from the oil after it has been recovered from the engine?

Find out

1 What are catalytic converters used for in the exhaust system of the modern car?

2 Magnesium blocks are used as sacrificial anodes to protect iron. Give examples where this is used and explain how the magnesium protects the iron.

FACT FILE

Cadmium is a by-product of the methods used to extract lead and zinc from ore deposits. Some of its uses include pigments for plastics, paints and glazes. Cadmium can affect the movement of our joints and cannot be easily removed from the human body. Cigarette smoke contains traces of cadmium.

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3.8

Energy and changes

FIGURE 3.14
A sports hot pack gets its heat from the energy released in an exothermic reaction.

Chemical reactions involve a change in **energy**. This is because energy may be needed for the reaction to occur, or energy may be produced by the reaction.

When a reaction occurs, bonds are broken and new ones are formed. For example, in the reaction:

$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$$

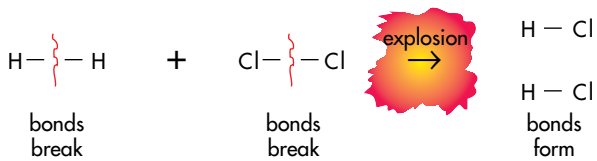

FIGURE 3.16

FIGURE 3.15
Chemical potential energy is converted into heat, light and sound energy in this chemical reaction.

If less energy is stored in the products than in the reactants then the extra 'left over' energy is released as heat.

Every compound contains stored energy. This energy is present in a compound in the bonds within the molecules, holding the atoms together, and in the bonds between the molecules. This stored energy is called its **enthalpy** or heat content. Enthalpy (or heat content) is given the symbol H. The heat content cannot be measured directly so, instead, the change in the heat content, ΔH, is measured. ΔH is measured in kilojoules (kJ). ΔH is a measure of the stored energy of the reaction. It is a measure of the stored energy of the products minus the stored energy of the reactants.

so ΔH = heat of reaction
= H(products) – H(reactants)

When ΔH is positive (i.e. the energy of the products is greater than the energy of the reactants) this means that heat (energy) is absorbed by the reaction.

When ΔH is negative (i.e. the energy of the products is less than the energy of the reactants) then heat (energy) must be released in order for the reaction to occur.

So, reactions may be:

1 **endothermic**—absorb energy, as heat, (these feel cold), or
2 **exothermic**—release energy, as heat (these feel hot).

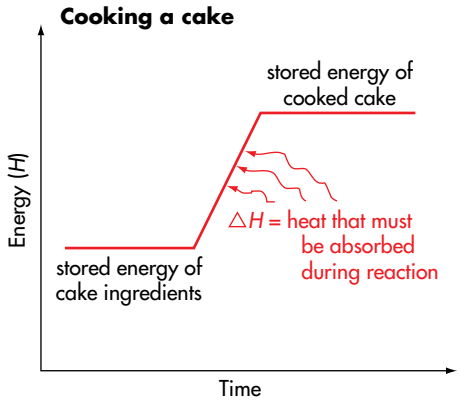

FIGURE 3.17
Energy, or heat, must be added for the reaction to occur. Because heat is added, this is called an endothermic reaction.

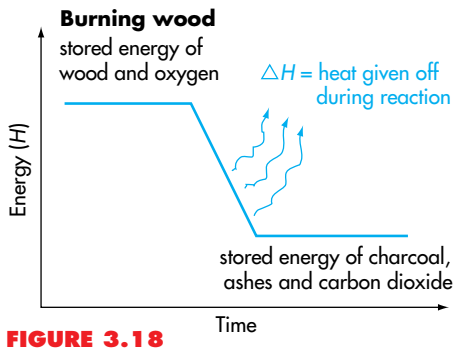

FIGURE 3.18
Energy, as heat, must be released in order for the reaction to occur. This is called an exothermic reaction.

FACT FILE

Physical changes can also involve a change in energy. Heat is absorbed when ice changes to water, but is given out when water changes to ice.

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EXPERIMENT 8 HEAT CONTENT OF COMPOUNDS

AIM

To demonstrate that some changes release energy and some require energy, by adding several substances to water.

MATERIALS

- potassium nitrate
- ammonium nitrate
- ammonium chloride
- sodium nitrate
- sodium thiosulfate
- thermometer (–10 to 110°C)
- measuring cylinder
- spatula
- sodium chloride
- styrofoam cups
- distilled water
- stirring rod

METHOD

1 Design a table to list your results in. Include name of compound, chemical formula, initial temperature, final temperature, exothermic or endothermic.

2 Add 50 mL of distilled water to a styrofoam cup and record the temperature. Add 1 spatula of potassium nitrate.

3 Stir briefly, then record the final temperature reached. From your observations, state whether the reaction is exothermic or endothermic.

4 Add each of the compounds listed in the Materials, in turn, to water, by repeating steps 2 and 3.

DISCUSSION

1 Why were styrofoam cups used instead of beakers?

2 How could this experiment be improved?

CONCLUSION

Write a conclusion for your experiment after rereading the aim.

EXTENSION

Plot each of the substances and their final temperatures on a column graph. Draw a line across your graph to indicate the initial water temperature.

A practical example of an exothermic reaction can be found on the sports field with hot packs. A hot pack contains iron powder, cellulose, salt, charcoal and water. When the pack is squeezed, the contents react. An exothermic reaction occurs, and so the pack feels warm as the heat is released.

You have already learnt that endothermic reactions absorb energy. Sometimes this may happen spontaneously, but often the reactants must be heated before the reaction will begin.

For example, a cold sports pack contains ammonium nitrate and water. When required, these two compounds are mixed by massaging the pack. This causes the ammonium nitrate and the water to react. This is an endothermic reaction, and so the pack feels cold (the heat has been taken from the surroundings by the reaction).

Questions

1 What does the symbol ΔH represent? What is it called?

2 What is the difference between endothermic and exothermic reactions?

3 Which of the following reactions are exothermic? How did you decide?

a $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ $\Delta H = -804$ kJ

b $I_2 + Cl_2 \rightarrow 2ICl$ $\Delta H = +35$ kJ

c $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ $\Delta H = -2043$ kJ

Think about

A liquid fuel used in some rockets is a mixture of hydrazine, N_2H_4, and hydrogen peroxide, H_2O_2, which react according to the equation:

$N_2H_4(l) + 2H_2O_2(l) \rightarrow N_2(g) + 4H_2O(g)$ $\Delta H = -684$ kJ

What two pieces of information in this equation suggest that this reaction would be suitable to use in a rocket?

Datalogging

FACT FILE

When rain falls, water vapour (a gas) is converted to water (a liquid), according to the equation:

$H_2O(g) \rightarrow H_2O(l)$ $\Delta H = -44$ kJ

Can you believe that when it rains, heat is actually released?

Extension worksheet

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3.9

Reactions—speeding up and slowing down

Chemical reactions take place at different speeds. For example, fireworks react quickly once lit, but the rusting of a car will take years. You can speed up or slow down a reaction simply by changing the conditions under which they occur.

These conditions may be:

- the temperature
- the concentration of the reacting substances
- the size of reacting solids
- the presence of a catalyst.

Many reactions increase their **rate** (speed up) when the **temperature** is increased. As the temperature increases, the particles in the substances move faster and come into contact more quickly, so a reaction is likely to occur much faster.

The **concentration** of the substances and solutions involved in a reaction affects the rate of a reaction. The more concentrated a solution, the faster the reaction that will occur.

The concentration of a solution is a measure of how much of a substance is present in a certain amount of water. Imagine making cordial. If you half fill a glass with cordial and top it up with water, your drink would be twice as concentrated as if you only quarter filled your glass with cordial and topped it up with water.

The rate of a reaction depends on the **particle size** of the solid substances; that is, whether the solid is in powdered form or in larger lumps.

The smaller the particles of solid, the quicker the reaction occurs. For example, powdered iron reacts faster with acid than a lump of iron does.

FIGURE 3.19
Food kept in the refrigerator doesn't spoil as quickly as food left at room temperature.

FIGURE 3.20 (far left)
Marble can be slowly dissolved by the chemicals present in rainwater. Large marble statues, however, would take centuries to completely dissolve.

FIGURE 3.21 (left)
Many of the substances we use every day, including synthetic fibres, margarine and ammonia, require the presence of a catalyst for their production.

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EXPERIMENT 9 CONCENTRATION AND REACTION RATE

A 2 M solution of hydrochloric acid contains twice as much acid as an equal volume of a 1 M solution. A 1 M solution of hydrochloric acid contains twice as much acid as an equal volume of a 0.5 M solution.

AIM

To investigate the effect of concentration on the rate of a reaction (magnesium ribbon dissolving in hydrochloric acid).

MATERIALS

- 6 cm magnesium ribbon
- measuring cylinder
- stopwatch
- 0.5, 1 and 2 M solutions of hydrochloric acid
- test-tube rack
- rubber stopper
- 3 test-tubes

Datalogging

METHOD

1 Cut the magnesium ribbon into three equal lengths.
2 Pour 5 mL of each concentration of acid into the test-tubes.
3 Place a piece of magnesium ribbon in each test-tube in turn and time how long it takes for the ribbon to disappear.
4 As the magnesium reacts you may wish to stopper the test-tube and perform a 'pop test' for hydrogen once the reaction is complete.

DISCUSSION

1 Was the reaction time in proportion to the concentration of the acid? For example, was the reaction with the 2 M acid twice as fast as the reaction with the 1 M acid?
2 Draw a graph to illustrate your data.

Questions

1 List four factors that can affect the rate of a reaction.

2 Explain why an animal's chemical processes slow down when it hibernates.

3 Complete the following sentence. The higher the concentration of reactants, the (faster/slower) the rate of reaction.

4 Figure 3.22 shows two graphs. Graph A indicates the rate at which carbon dioxide is produced during a reaction. Graph B represents the same reaction but under different conditions. Suggest two things that could have been altered in order for the reaction shown in graph B.

5 The effect of a catalyst can be dramatic. For example, if hydrogen peroxide is poured into a beaker it may take months to decompose into oxygen gas and water. If a catalyst such as manganese dioxide is added, however, you will soon see bubbles of oxygen gas fizzing out of the solution. Using this example, explain how you could test the idea that catalysts:

a speed up the rate of a reaction

b do not get used up in the reaction.

6 Check your answers to question 5 with your teacher. Perform the experiment and present a report that shows your results and conclusions.

7 Magnesium dissolves in acid solutions. Explain why the rate of the reaction is speeded up when:

a the magnesium is in powdered rather than solid form

b the temperature is increased.

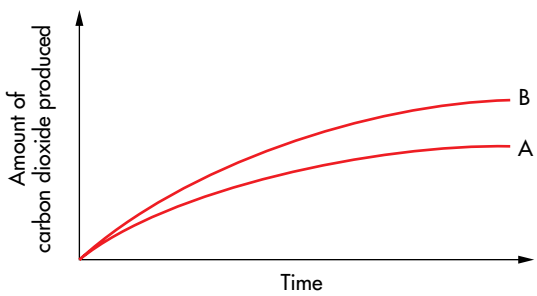

FIGURE 3.22
Graphs A and B show the rate at which carbon dioxide is produced during a reaction under different conditions.

Extra worksheet

FACT FILE

Many chemical reactions can be made to happen much faster by adding a small amount of catalyst.

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Using scientific language

Explain the difference between each pair of terms below.

a physical change and chemical change
b reactant and product
c synthesis reactions and decomposition reactions
d exothermic and endothermic
e oxidation and reduction

Check your knowledge

1 What is the difference between an element and a compound?
2 Why are chemical formulae used?
3 What is the law of the conservation of mass?
4 Why must chemical equations be balanced
5 What is the activity series? How can it be used?
6 Name the four different types of chemical reactions. Give an example of each.
7 What do you look for to decide whether a reaction took place or not?
8 List three methods of changing the speed of a reaction.
9 What is a catalyst? Give two examples.
10 What is neutralisation? Give one practical example of the use of neutralisation.

Apply your skills

1 Consider the chemical reaction:

$CH_4 + O_2 \rightarrow CO_2 + H_2O$

methane + oxygen → carbon dioxide + water

a Name the products and the reactants.
b Balance the reaction.

2 Balance the following equations.

a $C_3H_8(g) + O_2(g) \rightarrow CO_2(g) + H_2O(g)$
b $FeSO_4(aq) + NaOH(aq) \rightarrow Fe(OH)_2(s) + Na_2SO_4(aq)$
c $Na_2CO_3(aq) + Ca(OH)_2(aq) \rightarrow CaCO_3(s) + NaOH(aq)$
d $Fe(s) + H_2O(l) \rightarrow Fe_3O_4(s) + H_2(g)$
e $CaH_2(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)$
f $SO_2(g) + O_2(g) \rightarrow SO_3(g)$
g $NO(g) + O_2(g) \rightarrow NO_2(g)$
h $N_2(g) + H_2(g) \rightarrow NH_3(g)$
i $ZnO(s) + HCl(aq) \rightarrow ZnCl_2(aq) + H_2O(l)$

3 What type of reaction is shown below?

a $NH_4NO_3 \rightarrow N_2O + 2H_2O$
b $CuSO_4 \rightarrow CuO + SO_3$
c $2FeCl_2 + Cl_2 \rightarrow 2FeCl_3$
d $Mg + 2HCl \rightarrow MgCl_2 + H_2$

4 Write balanced equations for the following reactions

a hydrochloric acid + aluminium
b sulfuric acid + iron
c hydrochloric acid + aluminium hydroxide
d nitric acid + magnesium hydroxide
e phosphoric acid + aluminium carbonate
f nitric acid + zinc carbonate

5 Decide if the following reactions are endothermic or exothermic.

a $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O \quad \Delta H = -804$ kJ
b $H_2O(s) \rightarrow H_2O(l) \quad \Delta H = +6$ kJ
c $C + O_2 \rightarrow CO_2 \quad \Delta H = -393$ kJ
d $H_2 + CO_2 \rightarrow CO + H_2O \quad \Delta H = +41$ kJ

6 What changes in technology allowed the phlogiston theory of combustion to be replaced?

Challenge yourself

When copper wire is placed in a solution of dilute nitric acid a vigorous reaction takes place. Large amounts of brown gas are produced. If copper is placed in concentrated nitric acid very little reaction is observed. Why does the reactivity of copper with dilute and concentrated nitric acid differ?

eTester

CHAPTER 4

ELECTRICITY

An analogy is a useful way of developing a mental picture of a difficult idea or concept such as the concept of electric current. Some people compare electric current to the movement of water in pipes. Or it can be compared to people on a water slide ride in a fun park.

Think of a water slide where you sit on a large tyre tube. At the start of the ride a tow chain hooks onto the tyre tube and pulls you to the very top of the ride. The tow chain gives you energy (gravitational potential energy). The water slide winds its way down the slope and has steep sections where you pick up speed and slide easily. (The gravitational potential energy is converted into kinetic energy.) Along the track are deep pools with jets of water and turbulence. These slow the tyre tube down.

When you reach the bottom of the water slide all the energy you gained from the tow chain is gone.

In this analogy the riders on the tyre tubes are like the electrons which make up the electric current. The water slide is like the circuit the electrons flow through. The tow chain is like the battery or generator that give the electrons the energy to move around the circuit. The gravitational field that pulls the tyre tube down the water slide is like the electric field which transfers energy to the electrons in the wire. The deep pools along the water slide are like the components in a circuit that the electrons pass through, losing some of their energy on the way. As in the water slide, the electrons return to the beginning of the circuit and must receive more energy to move around the circuit again.

While analogies can aid our understanding of complex ideas, it is important to realise they may not be perfect in every detail.

In this chapter you will learn more about electric current, how it is measured and how it can be generated.

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- **Use an analogy to explain voltage, current and resistance in a circuit.**
- **Use symbols to draw circuit diagrams.**
- **Construct circuits containing a number of components.**
- **Compare series and parallel circuits.**
- **Describe the relationship between current, resistance and voltage.**
- **Apply the concept of electromagnetism to explain the operation of electric motors and other devices.**
- **Identify some of the scientific ideas that different cultures have contributed to our understanding of electricity and magnetism.**
- **Discuss our dependence on electricity and suggest possible alternative energy sources.**

4.1

Current electricity

FIGURE 4.1
These devices have one thing in common. They depend on current electricity to operate. Each contains conductors and insulators which control the path of moving electrons.

Talk about
What are insulators and conductors? Why do you think they behave in the way they do?

Where would you be without **electricity**? No TV, no CDs, no nightclubs, no air-conditioners, no computers, no microwaves. What a life!

'Current electricity is a flow of free electrons through a conductor'. What does this statement mean?

If a substance will allow some of its electrons to move freely then it can conduct electricity.

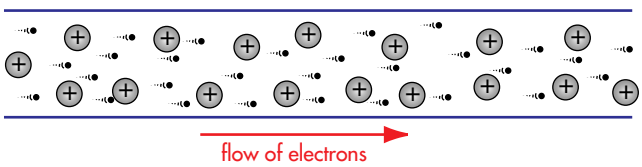
Metals are good **conductors** of electricity. You will recall that metals are made up of a metallic lattice with the positive ions held tightly in place surrounded by electrons which are free to move around.

Electrons are repelled by negative **charges** and attracted to positive charges, so if you imagine a piece of metal (say copper wire) with a big negative charge at one end and a big positive charge at the other, then the free electrons in the wire will move away from the negative charge and towards the positive charge. A battery provides the energy to force the electrons to move. As long as the electrons are flowing you have **electricity** and you can make it do work for you, like turn a motor, or light up a bulb.

Using Science 1 provides a good model of an electric circuit. The ions in the wire (the people) stay in one place and always have their electrons (tennis balls). However, they aren't fussy about which electron they have and are happy to give it away—provided of course they get a replacement from the ion next to them! When the electricity is turned off, the metal still has all of the electrons it needs, but not necessarily the original ones.

Molecules which use all of their electrons to hold the atoms together don't conduct electricity. Imagine what would have happened in Using Science 1 if nobody was willing to let go of their own ball, or to take a ball from anybody else. No electricity would flow. Substances that behave like this are called **insulators**. Wood, plastic, rubber and air are all good insulators. However, if the electrical energy is large enough, insulators can be forced to swap electrons, but only by breaking molecules apart.

FIGURE 4.2
Metals conduct electricity well because the free electrons move through the lattice carrying negative charge.



Circuit diagrams

The requirements for current electricity are a device to push the electrons around (a battery or a generator) and a complete path or **circuit** for the electrons to flow

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USING SCIENCE 1 MODELLING CURRENT ELECTRICITY

AIM
To demonstrate how a metal conducts electricity.

WHAT YOU NEED
About 20 students or more, 30 or 40 tennis balls, and two baskets tied together to hold the balls, one labelled negative and the other one positive.

WHAT YOU DO
1 Everyone starts holding a ball and forms a single file loop with a basket at either end.
2 The rest of the other balls should be in the negative basket.
3 When your teacher yells 'on', the person at the negative end starts to pass the balls along the line one at a time as fast as possible.
4 It is each person's job to always have one tennis ball, but one only, and to pass it on if offered a ball from the next person along.
5 Two people in the middle of the line should call out 'ZAP!!' each time they hand on a ball.
6 The person at the positive end puts the balls into the positive basket.
7 If your teacher calls out 'off' then the person at the negative end stops passing the balls.

through. People who work with electricity draw 'short-hand' plans or maps of their circuits. Symbols are used to represent the **components** in the circuit and the path taken by the electrons. The symbols are connected by using straight lines joined at right angles.

FIGURE 4.4
When the voltage is high enough electricity can break up the molecules of insulators and 'spark' through them.

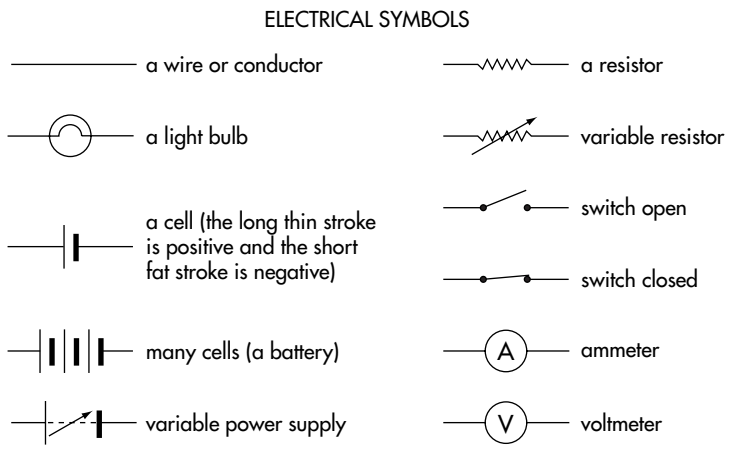

FIGURE 4.3
These are the symbols for the components you will use to construct circuits in this chapter.

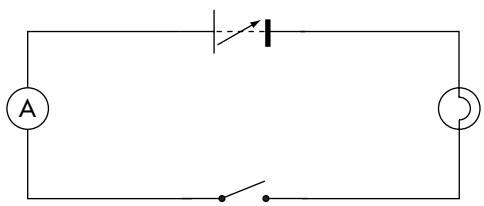

FIGURE 4.5

3 Look at Figure 4.5. Write down instructions for building this circuit.

Questions
1 Why do you think electric wires are usually coated with plastic?
2 Why don't insulators conduct electricity? Draw a diagram to help with your explanation if possible.

Activity
Construct a circuit containing a variable power source, a switch and two lamps. The switch should be placed between the lamps. Draw a circuit diagram for this circuit.

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4.2

Series and parallel

FIGURE 4.6
Diagrams are very important in designing and repairing circuits.

FIGURE 4.7
No matter what the size of the globe, the principles behind the circuits are the same.

EXPERIMENT 1 LAMPS IN SERIES AND PARALLEL

Datalogging

PART A

AIM
To investigate a simple series circuit.

MATERIALS
- variable power source set at 6 V
- 3 lamps
- switch
- 5 connecting leads

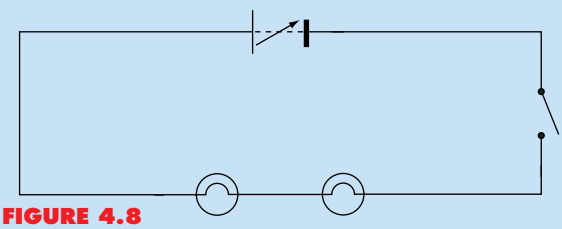

FIGURE 4.8

METHOD
1 Set up the series circuit shown in Figure 4.8.
2 Examine the brightness of the two globes.
3 With the switch off, add one more lamp in series. Turn the switch on and examine the brightness of the three globes.
4 Turn the switch off, remove a globe from one of the lamp holders and turn the switch back on.

PART B

AIM
To investigate a simple parallel circuit.

MATERIALS
As in Part A but with extra connecting leads.

METHOD
1 Set up the circuit shown in Figure 4.9, the lamps are said to be connected in parallel.
2 Examine the brightness of the two globes.
3 With the switch off, add one more lamp in parallel. Turn the switch on and examine the brightness of the three globes.
4 Turn the switch off, remove a globe from one of the lamp holders and turn the switch back on.

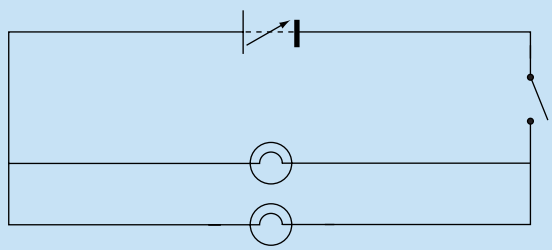

FIGURE 4.9

DISCUSSION
1 In which circuit did adding a third lamp reduce the brightness of the globes?
2 In which circuit did the removal of one globe stop the electric current?
3 In which circuit did the removal of one globe allow the other globes to glow?

CONCLUSION
Summarise the differences between series and parallel circuits.

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In series

Series circuits have only one path for electrons to take. Each component of the circuit is connected along the one path. If one component is turned off or broken, electricity will not flow through the circuit. As the electrons move through each component some of their electrical energy is converted into other forms of energy. Adding extra lamps in series causes the globes to glow less brightly.

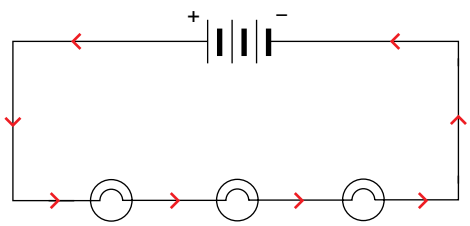

FIGURE 4.10
Flow of 'current' around a series circuit.

Questions

1 What is a series circuit? A parallel circuit? Draw an example of each.

2 Draw a circuit with one cell, one globe and two switches in which both switches must be closed to make a current flow. How are the switches connected in this circuit?

3 Draw a circuit similar to that in question 2 in which only one switch needs to be closed to make the current flow. How are the switches connected in this circuit?

Think about

1 Which arrangement—parallel or series—would you use to connect the lights in a room, for example, your classroom or in your home? Explain your reasons.

2 The car in Figure 4.12 has one broken headlight. How are the bulbs connected, in series or parallel? Give a reason for your answer.

In parallel

Parallel circuits have more than one path for electrons to take. Each component can be connected on a separate path. When one component is turned off or broken the electricity will still flow through other paths of the circuit. Adding extra lamps in parallel does not cause the globes to glow less brightly.

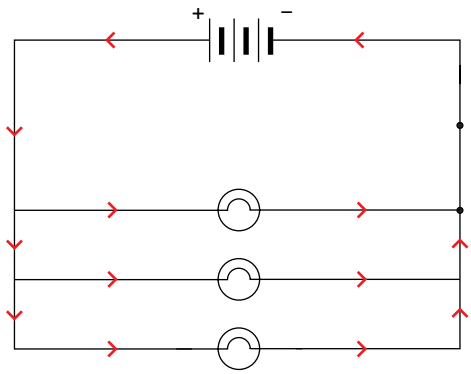

FIGURE 4.11
Flow of 'current' around a parallel circuit.

3 The circuit in Figure 4.13 contains four lamps, A, B, C and D, as well as three switches, X, Y and Z. Predict which lights would glow if:

a X was closed but Y and Z remained open

b Y was closed but X and Z remained open

c X and Z were closed but Y remained open.

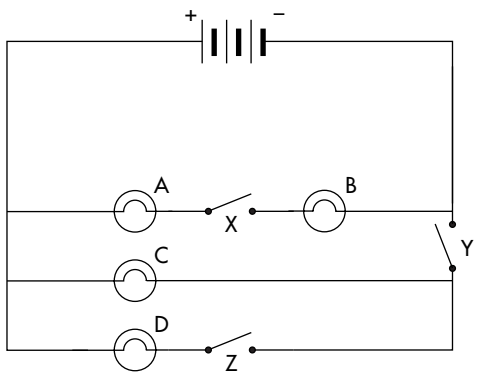

FIGURE 4.13

FACT FILE

'Conventional' current is said to flow from + to – around the circuit. This is opposite in direction to the flow of electrons.

FIGURE 4.12

eGlossary

4.3

Catching on to the flow

FIGURE 4.14
Lightning conductors provide an easier path for lightning than the stone or cement of buildings.

How do we know current is there?

We have used globes to tell us whether a current is flowing, but how reliable is a globe?

If the globe is not lit, does that really mean there is no current? The current might not be large enough to make the globe light up. One globe might be brighter than another even though they have each been connected to the same cell. Globes are often produced very cheaply. They would cost very much more if they all produced the same brightness for the same current.

A more accurate indicator of current uses a pointer. The distance a pointer moves is an indication of the size of the current. In some indicators, the pointer can move in two directions to show the direction the current is flowing. **Current** is measured in **amperes**, often abbreviated to amps, symbol A. The symbol for the current itself is *I*.

If an indicator is used to measure current only, it is called an **ammeter**.

Some ammeters have more than one scale. These ammeters will have one black negative **terminal** and two or three red positive terminals. Each red terminal is labelled for a different scale. Select the largest scale positive terminal of the ammeter first and connect this to the positive terminal of the power supply. Complete the circuit by connecting the black terminal to the next circuit component and turn on the switch. If the needle does not move very much, try a lower scale until you can read the current easily. If the needle of the ammeter goes off the scale switch off the current immediately.

FACT FILE

The unit of electric current, the ampere is named after André Marie Ampère. Ampère was a French mathematician who devoted his life to the study of electricity and magnetism.

FIGURE 4.15
The position of a pointer on a scale indicates the size of the current.

Current

An **electric current** is a flow of very small electric charges, either positive or negative. In metal wire, the current is a flow of negative charges called **electrons**.

Current is the amount of charges which are flowing past a point in a circuit every second.

Charge has the symbol Q and is measured in **coulombs**, symbol C.

$$\text{current} = \frac{\text{amount of charge flow}}{\text{time taken}}$$

or it can be represented by symbols

$$I = \frac{Q}{t}$$

where t is time in seconds (s) taken for the charge (Q) to flow.

Questions

Use circuit diagrams wherever you can in your answers.

1 How does a globe indicate that there is an electric current flowing in a circuit? Can we use a globe to show the size of current? Can a globe show the direction of the current?

2 If a globe is not lit, does this mean that no current is flowing? Explain.

3 How should an ammeter be connected into a circuit?

4 Describe the difference between parallel and series circuits in terms of the currents flowing in each type of circuit (assume that each of the circuits has the same power supply).

5 What charge flows through a globe in 5 min if the current is 0.25 A?

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EXPERIMENT 2 MEASURING CURRENT

Web page

AIM
To use an ammeter to measure current in two circuits.

MATERIALS
- variable DC source (0–12 V) or a number of cells that can be connected in series
- switches
- ammeter
- lamps
- connecting wires

PART A: SERIES CIRCUITS

METHOD
1 Set up the circuit shown in Figure 4.16. Make sure the ammeter is correctly connected so that the positive terminal of the power supply or cell is closest to the positive terminal of the ammeter. Set the voltage at 6 V.

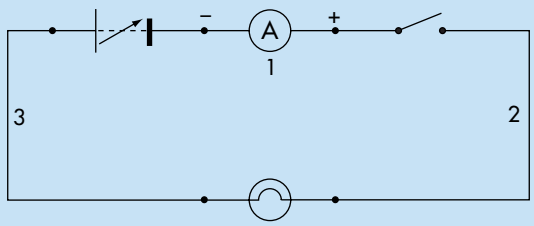

FIGURE 4.16

2 What is the ammeter reading when it is in position 1?

3 Without changing the position of the other components, place the ammeter in each of the numbered positions shown on the circuit diagram in turn. Record the current in each case in a table.

EXTENSION
Find out what happens when the components are rearranged. Does that affect the current measured at different positions in the series circuit?
Vary the voltage or number of cells and find out what effect this has on current.

DISCUSSION
What can you say about the current at different parts of a series circuit?

PART B: PARALLEL CIRCUITS

METHOD
Set up the parallel circuit shown in Figure 4.17.

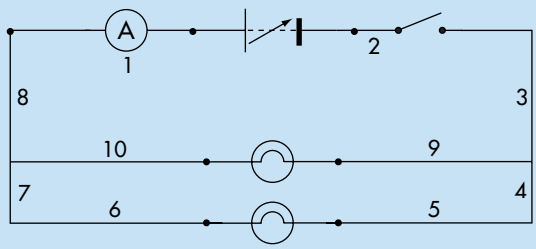

FIGURE 4.17

Measure the current in the different numbered positions. Record your measurements in a table.

DISCUSSION
Compare the currents:
1 in position 3 to the currents in positions 9 and 4.
2 in position 8 to the currents in positions 7 and 10.
What can you say in general about currents in parallel circuits, especially when the current divides in branches?

EXTENSION
Test your ideas by using the parallel circuit in Figure 4.18.

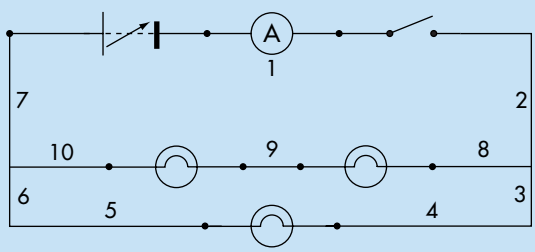

FIGURE 4.18

6 What current flows in a circuit if 240 C pass each point in 2 min?

Think about

Explain this statement: 'A switch can be both a conductor and an insulator'. Use circuit diagrams in your explanation showing the action of a switch.

Find out

Who is the unit 'coulomb' named after?

Extra worksheet

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What's the difference?

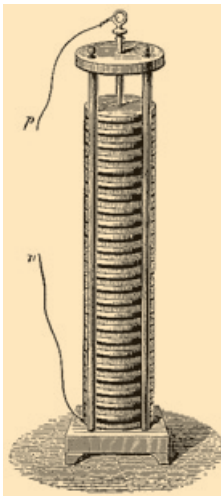

FIGURE 4.19
The first battery, invented in 1800, consisted of a series of zinc and copper discs separated by felt soaked in salt water. The battery provided a continuous supply of electricity for the first time in history.

Potential difference

Energy is transformed in moving electrons around a circuit. The energy needed to move one coulomb of charge around a circuit is called the **potential difference** or **voltage**. It is measured in volts, symbol V.

We use an indicator called a **voltmeter** to measure potential difference. A voltmeter can have different scales just like an ammeter, and, in some, the pointer can point in two directions.

An ammeter is placed in a circuit so that it actually measures the current flowing through it. A voltmeter is connected across the component so that it measures the potential difference across the component.

FIGURE 4.20
Dry cells supplying 1.5 V.

Ammeters and voltmeters are designed so that they draw hardly any current from a circuit

Voltmeters must also be connected so that the positive terminal of the voltmeter is connected to the positive terminal of the power source.

Simple **cells** like those in Figure 4.20 usually supply 1.5 V. By joining cells in series the voltage can be increased. Two 1.5 V cells connected in series will produce 3 V. Three 1.5 V cells connected in series will produce 4.5 V and so on. A variable power supply like those found in school laboratories supply between 2 and 12 V.

EXPERIMENT 3 MEASURING VOLTAGE

AIM
To use a voltmeter to measure potential difference.

MATERIALS
- 2 lamps
- variable power supply
- voltmeter
- connecting leads

METHOD
1 Measure the voltage across the lamps in a series circuit as shown in Figure 4.21.
2 Record your readings.
3 Measure the voltage across the lamps in a parallel circuit as shown in Figure 4.22.
4 Record your readings.

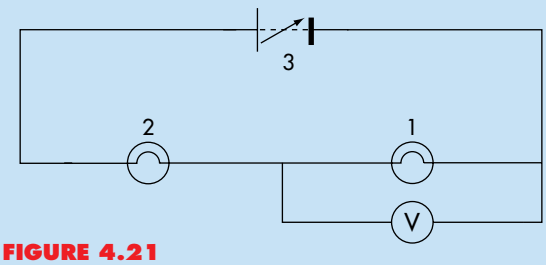

FIGURE 4.21

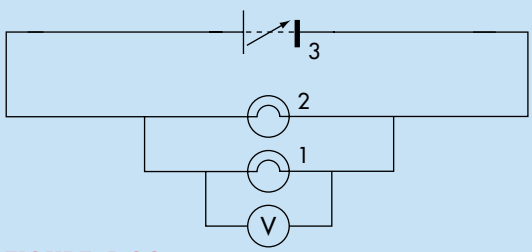

FIGURE 4.22

RESULTS
TABLE 4.1

Position	Voltage around series circuit	Voltage around parallel circuit
1		
2		
3		

DISCUSSION
1 How does voltage change in a series circuit?
2 How does voltage change in a parallel circuit?

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Resistance

Resistance is a measure of how much voltage is required to drive a 1 A current through a component. It measures how difficult it is to push a current through a component. *R* is the symbol for resistance. It is measured in **ohms** which has as its symbol the Greek letter Ω (omega).

As charges flow through a component such as a wire they encounter resistance from the ions of the wire.

Figure 4.23 represents the flow of charges through a component, such as the filament of a globe, connecting wire or a wire. Resistance comes from the collision of the charges with the ions of the lattice. Different materials have different arrangements of atoms so the resistance to the flow of charge through each type of material is different.

The resistance in a circuit can be measured by using a multimeter. A multimeter also measures current and voltage. Resistance can also be calculated if the current and voltage of a circuit are known.

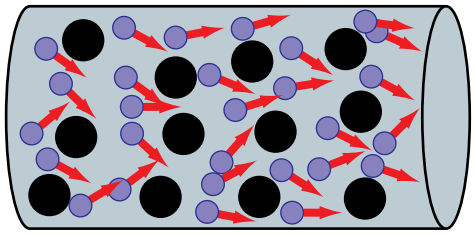

FIGURE 4.23
Resistance explained. Electrons drift past the ions in the metal lattice.

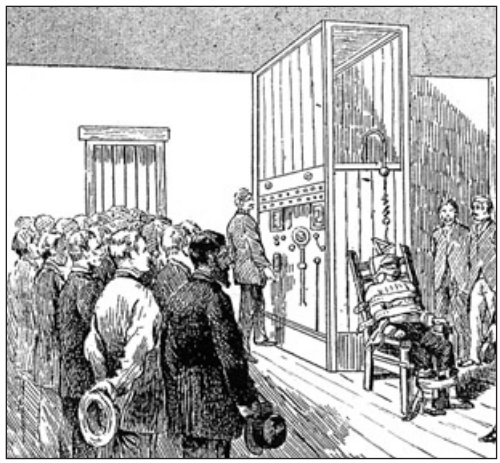

FIGURE 4.24
The first person to be executed by electrocution was murderer William Klemmer in the USA on 6 August 1890. The energy was supplied by an electric generator. Klemmer said he was happy to be contributing to scientific progress!

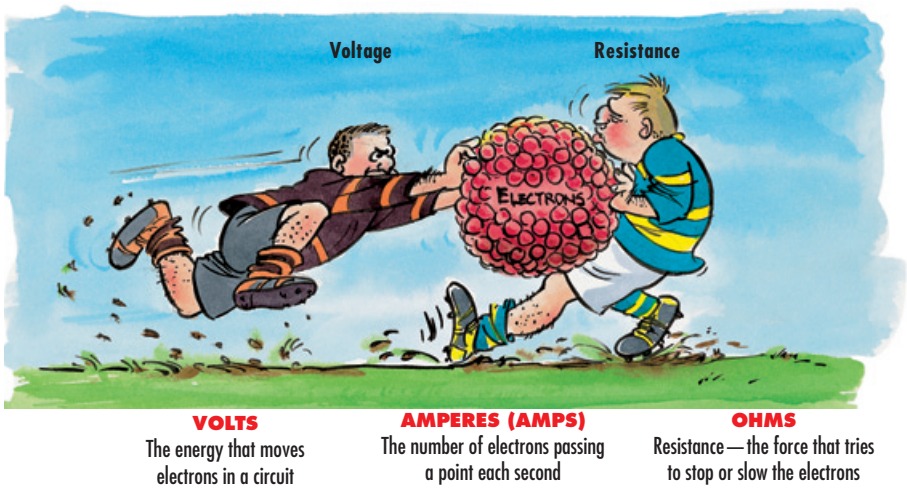

FIGURE 4.25
One way of thinking about current, voltage and resistance.

FACT FILE

The unit of potential difference, the volt, is named after Alessandro Volta. Volta was an Italian physicist. He invented the voltaic pile or battery.

Questions

1 Distinguish between current and voltage.

2 Explain the difference between voltage and resistance.

3 Copy and complete summary Table 4.2.

TABLE 4.2

Quantity	Voltage	Current	Resistance
Name of unit			
Symbol of unit			

4 Draw a circuit consisting of a lamp and a switch with a 6 V battery.

eGlossary

4.5

Ohm's law

EXPERIMENT 4 VOLTAGE, CURRENT AND RESISTANCE

AIM

To find out how current, voltage and resistance in a circuit are related.

MATERIALS

- variable power source
- voltmeter
- ammeter
- 30 cm nichrome wire
- 7 connecting leads, two with alligator clips
- test-tube
- switch
- rheostat

METHOD

1 Wrap the nichrome wire around the test-tube to form a coil.

2 Set up a circuit with the ammeter, switch, rheostat and nichrome wire all connected in series. Connect the voltmeter in parallel with the nichrome wire resistor. (Make sure the positive terminals of the ammeter and voltmeter are correctly connected to the power source.)

3 Turn the power source to the lowest setting to begin with. Switch on the power and adjust the rheostat and the power sources so that the current through the wire reads 0.5 A. Read the voltage and switch the circuit off and allow the nichrome wire to cool.

4 Adjust the rheostat so that the current is 1.0 A. Read the voltmeter as quickly as possible and turn the circuit off.

5 Continue to increase the current in steps of 0.5 A until four readings have been collected. (Remember, it is important to keep the nichrome resistor cool.)

RESULTS

Spreadsheet

TABLE 4.3

Ammeter reading (A)	Voltmeter reading (V)	*V/I*
0.5		
1.0		
1.5		
2.0		

1 Calculate the value of *V/I* for each set of readings.

2 Draw a line graph. Put current on the horizontal axis and voltage on the vertical axis. Draw a line of best fit.

DISCUSSION

1 What happens to the voltage across the nichrome wire resistor as the current increases?

2 If the current is doubled what happens to the voltage?

3 The slope of the line graph and the value of *V/I* is equal to the resistance of the nichrome wire:

resistance = voltage/current

The value of *R* is constant for metallic conductors if the temperature is kept constant. Look at your results for *V/I* and comment on how successful you were at keeping the nichrome wire at the same temperature.

EXTENSION

If the experiment is repeated with a lamp instead of the nichrome wire, the graph will not be a straight line. It will be a curve. This is because as the lamp glows the filament becomes hotter and its resistance increases. Try it out for yourself.

eGlossary

Ohm's law

The relationship between voltage, current and resistance for metallic conductors was discovered by Georg Ohm in 1826. He observed that if the voltage is doubled the current is also doubled. He said that the current flowing is proportional to the potential difference across a metallic conductor. This relationship became known as Ohm's law. It can be written as a formula:

$$R = V/I$$

or $R = V/I$ = a constant

The subject of this equation can be changed so that the same relationship can be shown as $I = V/R$ or $V = IR$.

Using resistance

It is the voltage that pushes the current through a conductor. The resistance of the conductor will determine the size of the current. Resistors are used in many devices to control the current that flows through components. Some resistors are said to fixed; others are variable like the rheostat. Variable resistors are extremely useful in controls such as for the volume in television and radio sets. A variable resistor is also used in a dimmer switch.

Resistors in series

It is harder for current to flow through resistors connected in series. The current has to flow through each one in turn. The total resistance is equal to the sum of the resistances in the circuit.

For three resistors:

$$R_{total} = R_1 + R_2 + R_3$$

If there were four globes, there would be four terms on the right-hand side and so on.

Resistors in parallel

It is easier for current to flow through resistors connected in parallel. Current can flow through either resistor. The smaller resistor will get the most current.

If three resistors are combined in parallel, their combined resistance can be calculated from the formula:

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

If there were four globes, there would be four terms on the right-hand side and so on.

FIGURE 4.26
Insulators are just as important as the materials that carry electric current. Insulators are used to isolate electrical cables from metal pylons.

Questions

1 Table 4.4 lists currents through various globes when the voltage across each of them is 6 V. Copy the table and complete it by calculating the resistance of each globe.

2 If a current of 2 A flows through a coil of wire of resistance 10 Ω, calculate the potential difference across it.

3 An electric iron has a resistance of 75 Ω. What current flows when the potential difference across its heating element is 240 V?

4 Find the resistance of a 12 V lamp that carries 0.5 A.

5 What is the combined resistance of two globes, of resistance 4 and 8 Ω, when connected:

a in series?

b in parallel?

Activity

Construct a circuit from lamps, rheostat and power source to control the brightness of two lamps connected in parallel.

TABLE 4.4

Current (A)	Resistance (Ω)
0.4	
0.1	
0.8	

Extension worksheet

FACT FILE

The unit of resistance, the ohm, is named after Georg Simon Ohm. Ohm was a German mathematician and physicist who was responsible for investigating the relationship between current, voltage and resistance in metallic conductors.

eGlossary

HISTORY OF SCIENCE

A history of electricity and magnetism

People have known about the effects of electricity and magnetism since ancient times. The ancient Greeks could produce a kind of electric charge called static (still) electricity by rubbing amber with fur. The charged amber would attract small objects like feathers. The ancient Greeks also knew about a type of black stone, called lodestone, that attracted some metal objects. The lodestone was a natural magnet. Knowledge of magnetism can be traced back to ancient China. The Chinese used natural magnets to make compasses for navigation at sea. This technology probably spread along an ancient trading route from the Far East to the West.

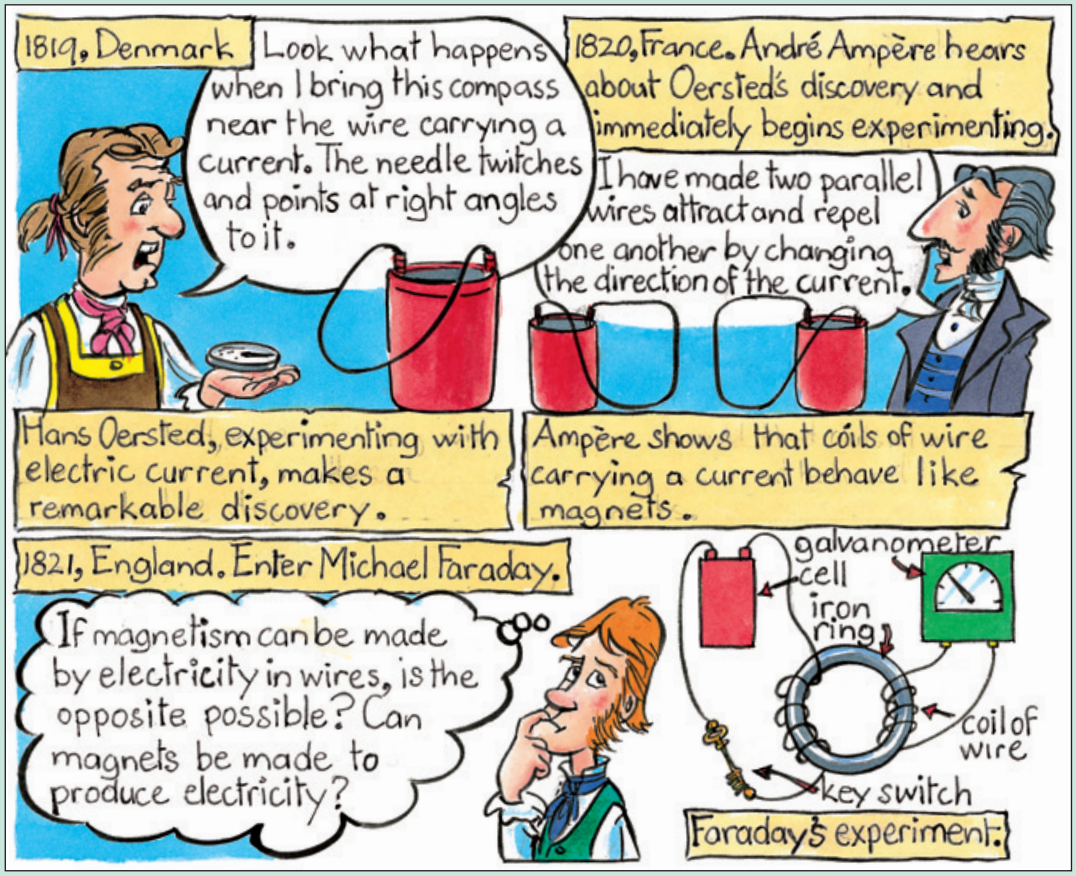

FIGURE 4.27

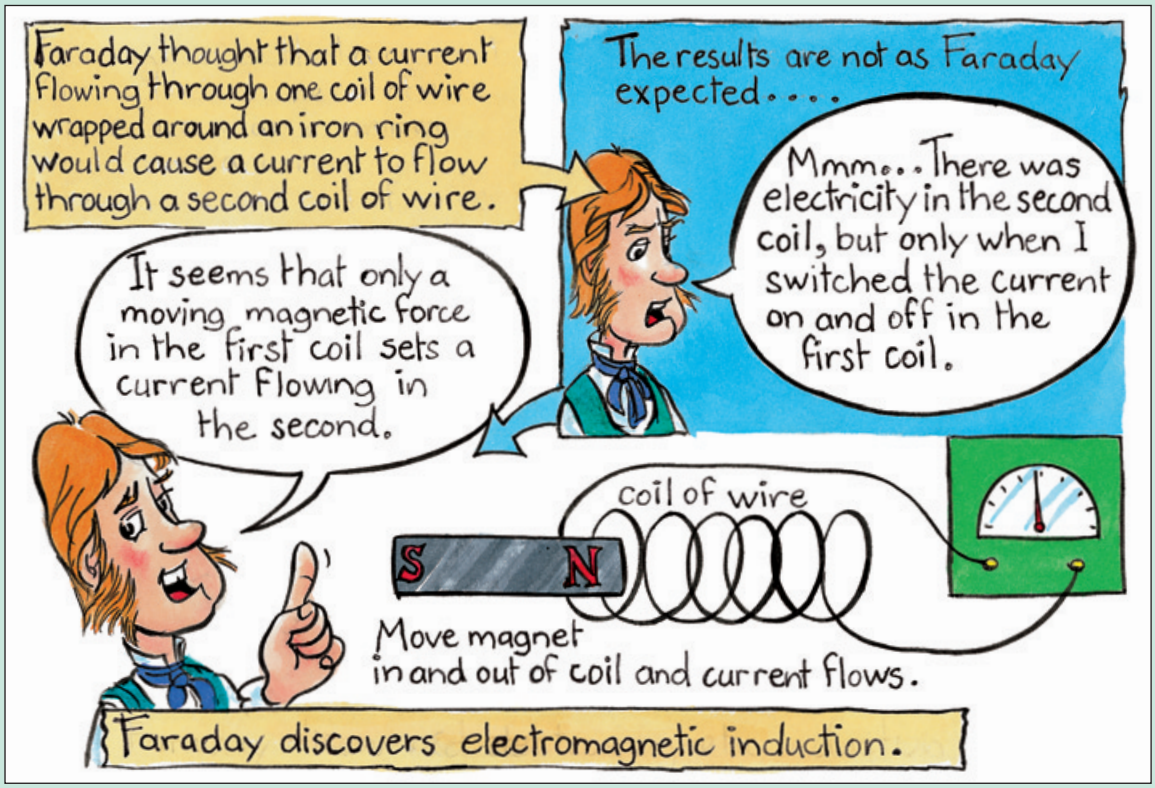

FIGURE 4.28

Throughout the centuries many investigations were carried out into properties of electricity and magnetism. It was suspected that electricity and magnetism were related. It wasn't until the 19th century that any progress was made towards connecting these phenomena.

Faraday built the first transformer and discovered electrical induction. However, the electrical current produced was short lived. He wanted to create a continuous current. So he built a device consisting of a copper wheel with a handle which rotated between the poles of a permanent magnet. Faraday invented the first electric **generator** in 1831. This was probably the greatest achievement in electrical history. The work of Faraday ultimately led to the development of large-scale electricity generation and the **electric motor**.

Questions

1 Different cultures contribute to the development of scientific ideas. Name five different cultures which contributed to the concept of electromagnetism.

2 An electric generator makes an energy conversion. What form of energy did Faraday supply to the copper wheel when he turned the handle? What form of energy was produced?

3 Explain why Michael Faraday's invention of the electric generator is considered to be the greatest achievement in electricity in history.

Find out

1 Michael Faraday spent over 40 years of his life doing scientific research. What other important discoveries or invention is he responsible for?

2 What are transformers, how do they work and what are they used for?

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4.7

Electricity and magnetism

FIGURE 4.29
Many airport security devices are able to detect metal objects because they cause electrical and magnetic changes in electrical circuits.

It has long been known that electric current has a **magnetic effect** and this can be used to produce movement.

The activity below is a good starting point in investigating electricity and magnetism.

USING SCIENCE 2 MAGNETIC PATTERNS AND ELECTRICITY

WHAT YOU NEED

Two bar magnets, plotting compasses, iron filings, fine thread, overhead transparency sheet, variable DC power supply, connecting wires, 3 wide pieces of cardboard.

WHAT YOU DO

1 Hang a magnet by a thread well away from any metal or electric device. In what direction does it point? How do you explain this?

2 What happens if another magnet is brought close? What happens if this magnet is reversed? Try to explain these observations.

3 Sprinkle iron filings onto an overhead transparency placed over a magnet. Tap the sheet gently and record the pattern.

4 Repeat step 3 using two magnets, for example side by side, or pole to pole.

5 Repeat steps 3 and 4 now using a plotting compass instead of iron filings. What happens to the compass needle as it is moved around the magnet(s)?

6 Make a hole in the centre of one of the pieces of cardboard and set up the equipment as shown in Figure 4.30.

7 Sprinkle iron filings on the cardboard and draw the pattern of the magnetic field you see. What happens to the pattern if the current is increased or its direction changed?

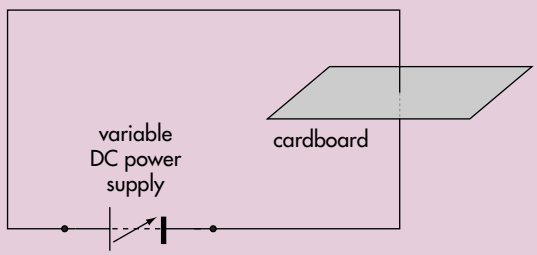

FIGURE 4.30

8 Place the plotting compass at a number of points around the wire, and note the direction it points. What happens if the current is reversed?

9 Make two holes in another piece of cardboard and repeat steps 2 and 3 using the arrangement in Figure 4.31.

10 Make two lines of at least eight holes in the third piece of cardboard and thread the wire through to make a spiral or coil as shown in Figure 4.32. Take care that the loops are as even as possible but do not touch. Repeat steps 2 and 3 with this arrangement.

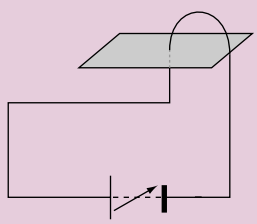

FIGURE 4.31

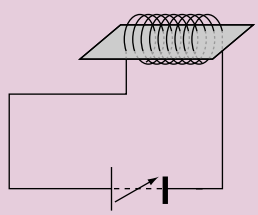

FIGURE 4.32

Datalogging

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USING SCIENCE 3 MAKING ELECTRICITY FROM MOVEMENT

WHAT YOU NEED

DC power supply, a 30 cm length (0.5 cm wide) of aluminium foil, connecting wires, horseshoe magnet, a sensitive voltmeter with a pointer that moves in both directions, insulated wire about 2 m long with the ends bare.

WHAT YOU DO

1 Set up a circuit so that a current flows through the aluminium foil as in Figure 4.33.

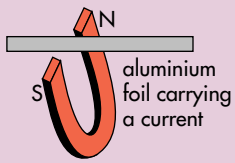

FIGURE 4.33
A current-carrying conductor experiences a force in a magnetic field.

2 What happens when you change the size and/or the direction of the current or switch the poles of the magnet?

3 Connect the insulated wire to the voltmeter.

4 Move a straight length (10 cm) of the wire up and down rapidly between the poles of the horseshoe magnet. What happens to the pointer? What is the effect of reversing the poles?

5 Wind the wire into a coil around a cardboard roll. Make sure that all turns are wound the same way around the coil. Connect the coil to the voltmeter.

6 Move the bar magnet in and out of the coil. What happens to the pointer? Try different speeds, and reversing the poles. What happens if you change the number of turns in the coil?

Magnetic fields

A magnetic field is a way of saying that there are attractive or repulsive forces around a magnet that interact with other magnets or magnetic materials such as iron.

Fields are represented by arrowed lines indicating the direction of the magnetic forces. Closer lines mean a stronger field.

Magnets have two poles, north and south. Like poles repel, unlike poles attract.

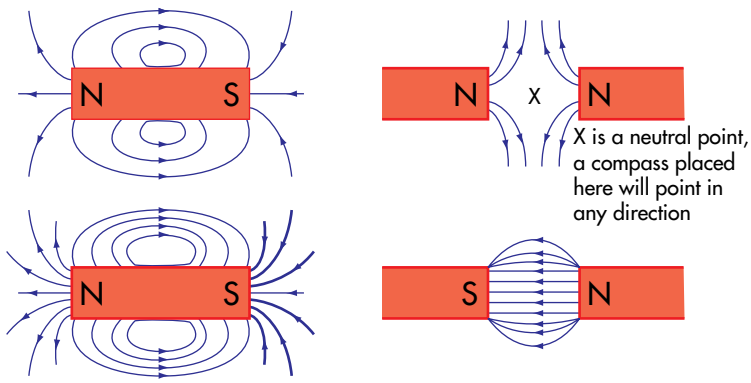

FIGURE 4.34
Magnetic field lines have direction. The closeness of the lines indicates the relative strength of the field.

Questions

1 Draw the field between two bar magnets with south poles facing each other.

2 Draw a diagram similar to Figure 4.30. Show the direction of four compass needles placed on the card around the wire.

3 Copy and complete by selecting one of the two phrases in each of the brackets. The magnetic field due to a current in a straight wire is at (right angles/parallel) to the wire and (changes direction/does not change direction) when the current reverses. The direction of the field at a point near the wire is taken to be the direction shown by the (north/south) pole of a plotting compass at that point.

Think about

1 List the factors that determine the direction and size of the current produced when a magnet is pushed into and out of a coil.

2 The hulls of minesweepers are made of glass fibre. Why might this be so?

Foundation worksheet 1

Internet assignment

eGlossary

4.8

Using electromagnetic principles

Motors

An **electric motor** changes electrical energy into kinetic energy. A **generator** changes kinetic energy into electrical energy. **Loudspeakers** convert electrical energy into sound energy and **microphones** convert sound energy into electrical energy. The principles are the same, it is just the outcomes that are different.

USING SCIENCE 4 LOOKING AT ELECTRIC MOTORS

If you have access to one, build an electric motor from a kit and use it to help you with the following.

DISCUSSION

An electric motor uses the fact that a current in a magnetic field experiences a force.

In Figure 4.35 you can see two carbon brushes. The brushes allow electric current to flow into the coil without the wires getting entangled as the coil turns round.

Copy the diagram and draw arrows on the ends of the coils to show the direction of the current for the present position of the coil. Some arrows have been drawn for you.

What happens to the direction of the current in the part of the coil labelled AB when AB has turned from the left-hand side to the right-hand side?

The brushes also make sure that the current is always flowing in the right direction through the coil.

Explain how the two split rings make sure that the current in the coil is always flowing in the right direction to keep the coil turning.

When the coil is in a horizontal position, a force acts on it. Since the currents on opposite sides are flowing in opposite directions, the forces acting on each of the sides are in opposite directions and they make the coil rotate.

How would you make the coil turn in the opposite direction?

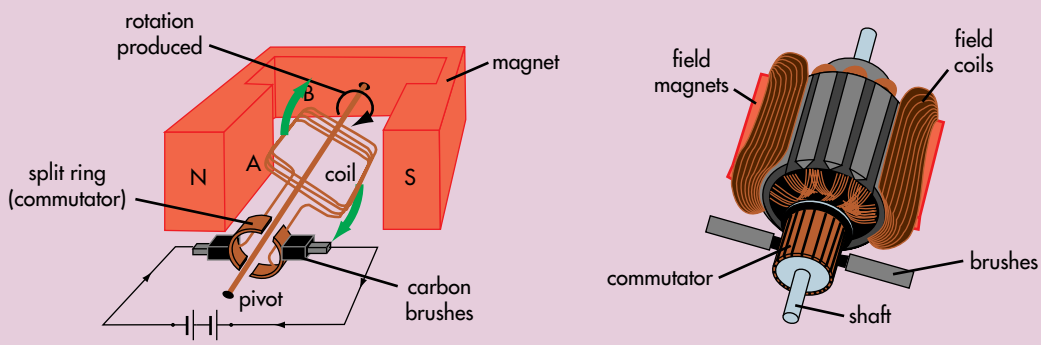

FIGURE 4.35
A simple electric motor and a commercial electric motor. What differences can you see?

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Generators

A **generator** is the opposite of a motor.

Current is produced when a conductor (which is part of a circuit) moves in a magnetic field. The energy comes from whatever is moving the conductor.

A generator makes electricity from other forms of energy. In **power stations**, this energy can come from the burning of coal or from water falling from a dam. In cars, a generator operates the lights and other electrical devices—the energy comes from the combustion of petrol. In a bicycle, the energy comes from the movement of the rider's legs.

Generators can be designed to produce **direct current** (DC) or **alternating current** (AC). DC flows in one direction only, while AC constantly changes its direction. Power stations use AC generators.

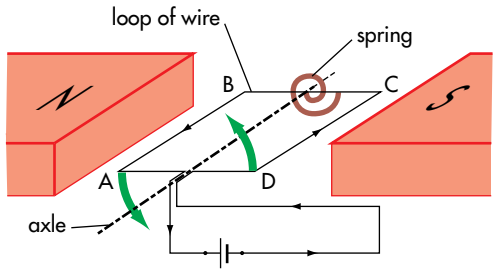

FIGURE 4.36
How a moving coil ammeter works.

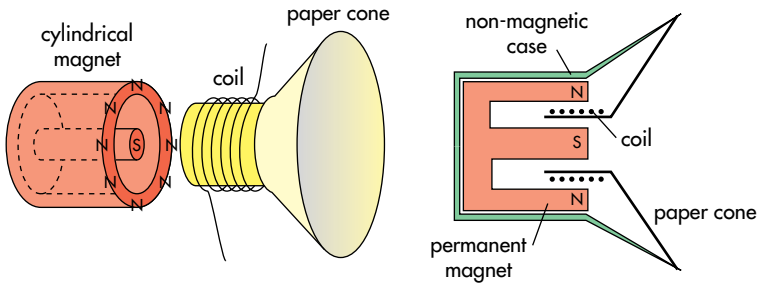

FIGURE 4.37
A moving coil loudspeaker. Loudspeakers in radios and TVs are similar to these.

Current indicators

When an ammeter pointer moves, it is because the current in its coil is experiencing a force. However, as Figure 4.36 shows, the coil cannot rotate because it has to push against a spring. The amount of turn depends on the size of the current and the strength of the spring.

Can you fix a spring to your kit electric motor so that it can become an indicator? If you do not have a spring, is there some other way of making it show the size of the current?

Loudspeakers and microphones

In a **loudspeaker**, the coil moves in and out of the magnet as the current in the coil varies. The coil is fixed to a cone which vibrates to produce the sound waves.

A **microphone** is the opposite to a loudspeaker in that it changes sound waves into currents.

Questions

1 What is the purpose of the brushes in motors and generators? Why do some generators have brushes that press against split rings and others have brushes that press against complete rings?

2 Name three machines that use electric motors.

3 What effect would
a direct current
b alternating current
have on a loudspeaker? Use a diagram to explain your answer.

4 What factors determine how far a pointer moves across a scale on an ammeter that uses a coil and a magnet?

Find out

1 Ask an electrician to tell you what are the most common faults in an electric motor. It may be best to get one person in the class to find this out.

2 How is the movement in an electric bell produced? Is a bell more like a generator or a motor?

Activity

Design and build an electric bell.

Foundation worksheet 2

eGlossary

IMPLICATIONS FOR SOCIETY AND THE ENVIRONMENT

How much do we depend on electricity?

We can take three examples to investigate this question: city rail transport, skyscrapers and computers. The skyscraper is a symbol of our modern cities, the computer is a symbol of our modern society. A skyscraper has lifts, air-conditioning and lighting as well as general office machinery. Without electricity, would we still have skyscrapers? What would be the alternative? How much does our society depend on the many forms of computer in the home, office, factory, school, hospital, bank, transport and the military, for example?

Imagine this future

Suppose the government decided, because of severe shortages, to use **fossil fuel** only for cars, coaches and trucks, and not for producing electricity. What would be the immediate changes we would experience? How long would it take to develop alternative processes that would provide the same amount of electricity as our current system?

These are some alternative sources of energy for electricity.

Solar

The Sun supplies 3000 times more energy to the Earth than we need. A lot of this is absorbed by plants and we obtain this energy by various means.

USING SCIENCE 5 DEBATING THE ISSUES

Choose from the topics below. Write an individual or group report on one of the following responses to the prospect of the crisis raised above. Alternatively you could organise a class debate.

Use the information in the box to help with your report or debate. You should add library research to this once you have decided on a topic.

1 We must use nuclear power as well as fossil fuel. We have to have cities full of people and jobs, and country devoted to large scale-food production with very few people, with big roads to take us where we want to go. Big cities give us great opportunities like lots of jobs, schools, hospitals, public transport, sports, theatre, cinemas, festivals and so on.

2 Let's take advantage of the crisis and scale down so we live in small self-sufficient communities. It is better to live as a member of a community in which all the facilities are close at hand and where we are not dependent on something far away. Why drive when you can walk?

3 We should all find a piece of land and use it to make and grow all that we need. We have done it before and we can do it again.

4 If only we could start now and develop alternatives well before a crisis!

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1 Small solar water heaters on roofs are expensive. Large numbers of people on our planet do not even have a permanent water supply.
2 The Sun's heat can be used to boil water by using mirrors. However, we would need a very large number and these would occupy a considerable area of land which probably would be good farmland.
3 Solar cells produce electricity. They are expensive and not very efficient. Again a vast area of land would be needed to make them worthwhile.
4 Solar cells on satellites could collect the Sun's energy and transmit it by microwave to the ground.

Wind
Windmills need a lot of space in places where there is a lot of wind most of the time. They also make a lot of noise.

Tides and waves
Tides and waves can produce to-and-fro movements to generate electricity. These generators might have environmental effects that could considerably change estuaries and areas near coasts. They could also affect shipping.

Biomass
Plants absorb solar energy and could be used to 'grow' fuels. Organic wastes in tips, as well as decaying plants and sewage, produce methane which can be burnt. However, these sources also use a lot of land.

Geothermal
Heat from inside the Earth can be used to generate electricity. Unfortunately, there are only a few suitable sites around the world.

Hydroelectricity
Falling water turns the turbines that turn the generators of electricity. Large-scale dams have unfortunate environmental effects and a limited life, but they only produce pollution while they are being built.

Sea water
The difference in temperature at different depths can generate electricity.

Nuclear energy
There are plentiful supplies of nuclear fuel. Nuclear power stations do not produce the types of pollution that come from fossil fuels. Nuclear waste, however, lasts for thousands of years. There are still problems with the safety of operating nuclear reactors (both civilian and military) and with disposing of reactors that have outlived their usefulness.

FIGURE 4.38
Can we harness the heat from the inside of our planet?

FIGURE 4.39
Wind can be harnesed to generate electricity.

FIGURE 4.40
Solar cells on satellites could continually harvest the Sun's energy and transmit it back to Earth.

FIGURE 4.41
1980 saw the flight of the first solar-powered aeroplane, the Gossamer Penguin.

FIGURE 4.42
Rubbish tips are a source of methane which can be burned to drive electricity generators.

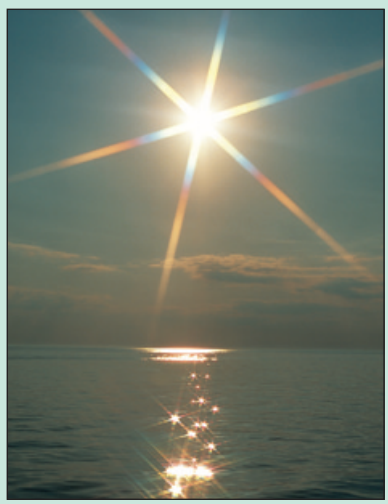

FIGURE 4.43
The oceans absorb nearly 75% of the solar energy that reaches the Earth.

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Using scientific language

Use the following words to complete the statements in questions 1–3: alternating, brushes, coil, current, direct, electricity, motion, movement, opposite, pointer, produce, size, slip, split, spring. Words can be used more than once if necessary.

1 A generator can produce the two types of current: ____ and ____. In one type of generator, the current leaves and enters the coil through ____. These ____ rub against rings which are attached to the ends of the coil. Slip rings are used to produce ____ current and split rings are used to produce ____ current.

2 A motor is the ____ of a generator because electricity is used to ____ motion. A loudspeaker is similar to a motor/generator (choose the correct word) because it converts ____ into ____. A microphone is similar to a motor/generator (choose the correct word) because it converts ____ into ____.

3 An indicator (e.g. an ammeter or a voltmeter) acts like a motor/generator (choose the correct word) because it changes ____ into ____ energy of the coil. The movement of the ____ is resisted by a ____. The amount of ____ depends on the ____ of the ____ and is shown by the movement of a ____ attached to the ____.

Check your knowledge

1 What is the difference between a series and parallel circuit? Using circuit diagrams draw an example of each.

2 Explain using diagrams why a conductor such as copper will conduct an electric current. Suggest why an insulator such as plastic will not.

3 Consider the circuit in Figure 4.44. Which of the following are true? Explain your answers.

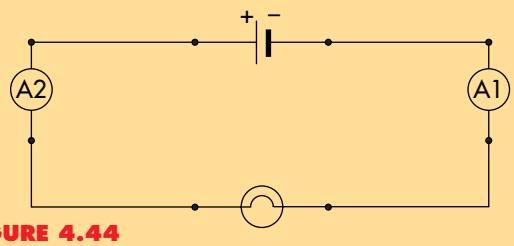

FIGURE 4.44

- a If A1 reads 0.5 A, A2 reads 0.5 A.
- b The current through the cell is zero.
- c In 10 minutes, 5 C of charge flow past any point in the circuit.

4 10 C of charge are stored in a thundercloud. A flash of lightning lasts 0.001 s. What is the current carried by the flash of lightning?

Apply your skills

1 Calculate the resistance of the following combinations:
- a 20 Ω and 40 Ω in series.
- b 20 Ω and 40 Ω in parallel
- c a 10 Ω resistor connected in series to the combination of two 5 Ω resistors connected in parallel.

2 Consider the circuit in Figure 4.45.
- a What are the voltages across the 10 Ω and the 5 Ω globes?
- b Calculate the current flowing through each globe.
- c What is the current flowing through the battery? Explain your answer.
- d What single resistor would draw the same current from the battery as the two globes in parallel?

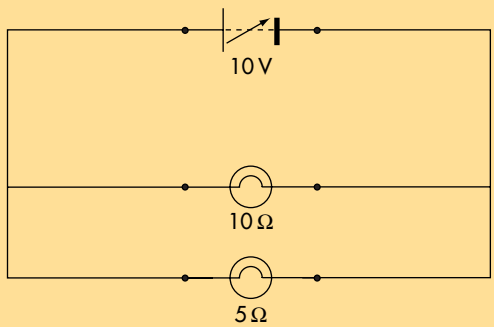

FIGURE 4.45

Challenge yourself

A coil of 10 turns of wire is connected to a voltmeter to form a circuit. The zero of the voltmeter scale is in the centre, so the voltmeter can indicate direction as well as size.

1 What is the reading on the voltmeter when a magnet is held still inside the coil?

2 What motion of the voltmeter pointer do you observe when the magnet is withdrawn? How does this compare with when the magnet is pushed into the coil?

3 How does the speed of the movement of the magnet affect the movement of the pointer?

4 In these experiments, what must be happening in the coil whenever the voltmeter pointer moves?

eTester

eGlossary

return to main menu

CHAPTER 5

BEHAVIOUR AND PROPERTIES OF LIGHT

Our planet, one of the most beautiful in the solar system, is often referred to as the blue planet. It appears blue when viewed from outer space because much of our planet is covered with water. Blue is also the dominant colour we see when we gaze up into the sky.

Why should blue be the most dominant colour seen when the water which makes up our oceans and the air that makes up our atmosphere are both colourless? The answer lies in the behaviour of light.

White light from the Sun, a mixture of different colours, travels towards the Earth at an incredible speed (300 000 km/s). As the light passes through the Earth's atmosphere the gas molecules scatter the light. Scattering is a special kind of process in which the light is reflected in an irregular and diffuse manner. All components of white light are not affected equally by the gas molecules. Blue-coloured light is scattered the most, giving an overall blue effect to our eyes. The blue colour of the oceans is produced in the same way.

In this chapter you will investigate the behaviour of light in many different situations.

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Describe the reflection of light in curved mirrors and give uses of each type.
- Explain refraction of light in various situations.
- Compare the effects of concave and convex lenses on light and explain how they form images.
- Find the focal length of a lens.
- Describe how science can assist people with poor vision.
- Discuss how different cultures developed ideas to explain the nature of light and colour.
- Describe how white light can be split into its constituent colours.
- Explain the interaction of light with coloured objects.
- Explain how scientific ideas about light and colour have been applied to photography.

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Web links

5.1

Reflection in curved surfaces

FIGURE 5.1
Convex and concave mirrors. Different sides of a spoon reflect differently. Mirrors which bulge outwards are known as **convex** (or diverging) mirrors and those that curve inwards are known as **concave** (or converging) mirrors.

In *Interactive Science 2* we saw how light can be reflected. When light rays hit smooth shiny surfaces they bounce off. Using a plane flat mirror, we saw that incident rays strike the mirror at the same angle as the rays are reflected. The **law of reflection** can be stated as follows: the angle of incidence is equal to the angle of reflection. **Reflection** in a **plane mirror** produces an upright undistorted **image** of an object. But what happens to the light if the mirror surface is curved?

Curved mirrors may be spherical, like a netball, or parabolic, like a satellite-receiving dish. The images produced by curved mirrors can be distorted, enlarged, reduced, upright or upside down. These images can be put to good use in everyday situations.

Convex mirrors

When a beam of light hits a **convex mirror**, the rays are spread out after being reflected. The beam of rays **diverges**.

Convex mirrors are used to give a wide area of view. Store detectives use convex mirrors to oversee all parts of a shop.

Concave mirrors

A beam of light striking a **concave mirror** has the opposite effect. The rays are concentrated or **converged** to a point called the **focus** or **focal point**. This can be used to collect and concentrate a beam of light.

If a source of light such as a globe is placed at the focus, a parallel beam of light is formed. This is used in a torch or car headlamp.

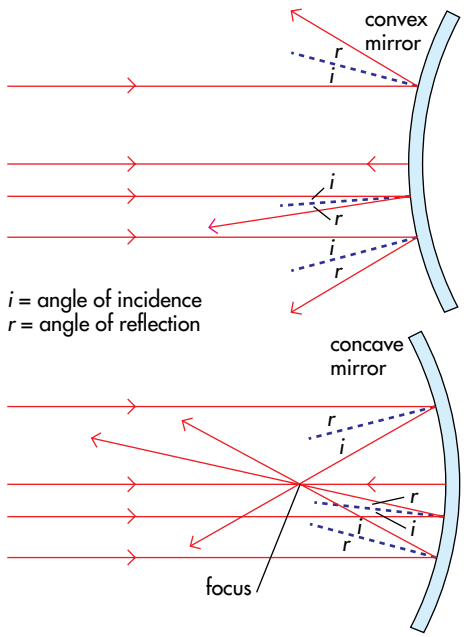

FIGURE 5.2
Rays hitting convex and concave mirrors.

FACT FILE

Space engineers from the USSR experimented with a huge concave mirror in orbit around the Earth. Their aim was to reflect the Sun's light on to areas of the Earth's surface during the hours of darkness. In those areas where winter nights are very long this would save on costly street lighting bills!

USING SCIENCE 1 SPOONS

A metal spoon acts as both a concave and a convex mirror depending on which side you look at. Look very carefully at your reflection produced by both sides of a spoon and describe what your image looks like. Include in your description whether the image is the right way up and its size. Does the distance from your eye matter?

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Spreadsheet

EXPERIMENT 1 REFLECTION AT CURVED SURFACES

AIM

To find out how parallel beams of light are reflected by curved mirrors.

MATERIALS

- concave and convex reflecting surfaces
- ray box—single slit/multiple slits
- power supply
- ruler
- sheet of paper
- protractor and pencil

METHOD

1 Connect the ray box to the power supply. Place a multiple slit in the ray box so that a beam of parallel rays is produced.

2 Place a concave reflecting surface on a clean sheet of paper. Draw along the reflecting surface to locate its exact position.

3 Shine parallel beams of light straight on to its surface. Mark the paths of both the incident and reflected rays by using a pencil.

4 Remove the mirror and use a ruler to connect up the dots you made.

5 Locate and label the focus (where the rays cross after reflection). Do the rays stop here?

6 Replace the concave mirror with a convex mirror. Repeat your ray tracing on a clean sheet of paper. Is there a focus this time?

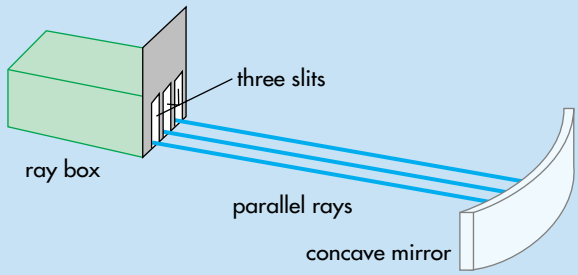

FIGURE 5.3
The experimental set-up.

7 Repeat the previous experiment for either the concave or convex mirror. As carefully as possible, construct normals to the mirror at each point where a ray was reflected. Measure the angles of incidence and reflection and record your results in a table like the one below. Does the law of reflection hold for curved mirrors?

TABLE 5.1

Angle of incidence	Angle of reflection

Questions

1 Distinguish between a plane mirror, a concave mirror and a convex mirror.

Word process

2 Copy Figure 5.4 and extend the rays to show what happens after reflection. Label the angles of incidence and reflection and the position of the focus.

3 If you looked in a mirror and saw a magnified upright image, what type of mirror would you be looking in? What would happen to the image if you moved the mirror further away from you?

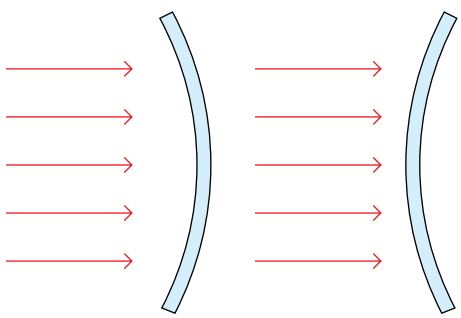

FIGURE 5.4

4 You are the manager of a boutique and you have been worried by losses of stock in the shop, particularly from the areas behind the clothes racks. Explain, with a diagram, how you might locate some convex mirrors to enable you to have all areas visible.

FACT FILE

Concave mirrors are used by dentists to magnify the surface of a tooth. The mirror is fixed to the end of a handle and positioned close to the tooth.

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5.2

Bending light

FIGURE 5.5
Not water on a hot road surface.

FIGURE 5.6
Legs look shorter!

FACT FILE

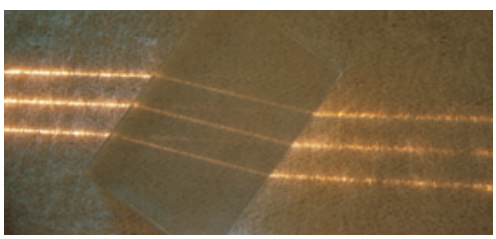
In 1621 Dutch scientist Willebrord Snell found that the ratio of the angle of incidence to the angle of refraction was unique for every transparent substance. He called this characteristic bending power of a substance the refractive index.

Extension worksheet

In section 5.1 we investigated what happens to light when it reflects from shiny objects. Light also behaves in a predictable but different way when it passes through substances.

Have you ever noticed that when you stand near its edge a pool looks shallower than it really is? Or perhaps you have observed a bend in the handle of a paintbrush standing in a jar of water. Even the shimmering of the landscape on a really hot day or the apparent 'pools of water' on a hot road surface that disappear as you get closer are caused by light passing through substances.

When light moves from air into another **transparent** material such as glass or plastic, it slows down. This can make the light change direction abruptly. Similarly, when light leaves the glass or plastic, it speeds back up and the direction of the light changes once more. This changing of direction when light moves from one material to another is called **refraction**.



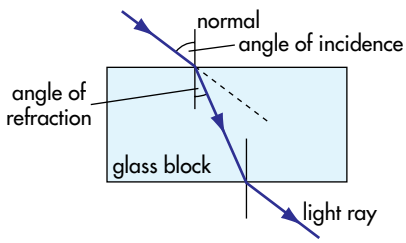


FIGURE 5.7
The light ray refracts as it passes into and out of the glass block.

We can track the direction that light moves by looking at rays in the same way as we investigated reflection at mirrors. When the ray hits the boundary between two materials it is called the **incident ray** and when it leaves the boundary it is called the **refracted ray**. Both the angles of incidence and refraction are measured from the **normal**.

The amount and direction of bending depends on several factors.

- When the light slows down, for example moving from air to glass, it always bends towards the normal.
- When the light speeds up, for example leaving the glass and moving back into the air, it bends away from the normal.
- Some materials slow the light down more than others and cause relatively more bending.
- The closer the ray is to the normal, the less bending takes place—when the ray is exactly perpendicular to the surface no bending takes place.

The bent stick

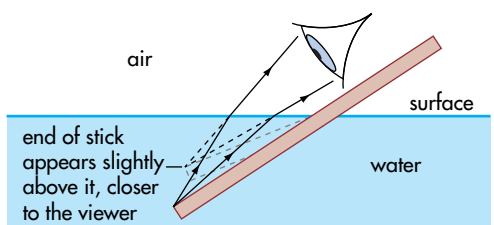


FIGURE 5.8
Drawing the rays helps explain why the stick looks bent. Our eyes can be easily tricked as they expect light to travel in straight lines! The person looking at the stick doesn't realise that the rays have bent when they left the water.

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EXPERIMENT 2 DEMONSTRATING REFRACTION

Spreadsheet

AIM
To demonstrate the refraction of light in a transparent block.

MATERIALS
- ray box (single slit)
- power supply
- transparent glass or plastic rectangular block
- sheet of paper
- ruler, protractor and pencil

METHOD
1 Connect the ray box to a power supply. Place the single slit in the box so that a single beam of light is produced.
2 Place the rectangular block in the centre of the sheet of paper and carefully draw round its edge.
3 Direct the beam of light towards the centre of the longest side. View the path of the ray as it enters, passes through the block and emerges at the far side. Make small pencil marks to trace the ray up to and away from the block.
4 Remove the block and connect up the dots with a ruler to show the passage of the ray. The ray should change direction at both surfaces of the block.
5 Draw a normal at the points where the ray both enters and leaves the block. Measure the angles of incidence and refraction from the normal with a protractor.
6 Repeat for rays entering at a range of other angles.
7 Record your results in a table like the one below.

RESULTS

Ray entering/ leaving block	Angle of incidence	Angle of refraction	Bends away from/ towards normal

CONCLUSION
Can you predict whether a ray of light will refract towards or away from the normal?

The swimming pool appearing shallower

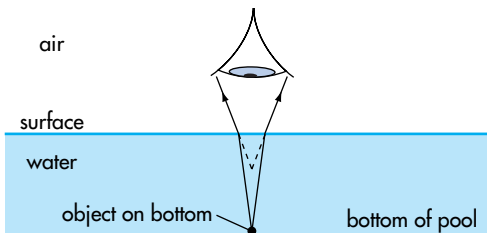
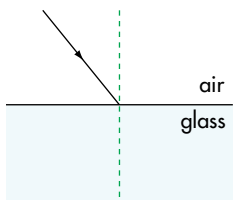


FIGURE 5.9
The pool appears shallower than it really is because the rays seem to come from a point above the true bottom. This is very noticeable in a tiled pool.

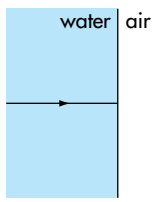
Total internal reflection

Sometimes, when light passes from materials like glass to air, it needs to bend a great deal away from the normal. If light hits the boundary at a sufficiently large angle of incidence, the angle of refraction would be greater than 90°! In this case, the light is reflected inside the substance and does not pass into the air. This is known as **total internal reflection**.



Questions

1 Copy and complete the ray diagrams in Figure 5.10.



2 Name three pairs of materials that will refract light.

3 Explain the refraction of a ray of light in terms of the light speeding up and slowing down.

Find out

Why do cut diamonds sparkle so much? Why is it so easy for an expert to spot a fake diamond?

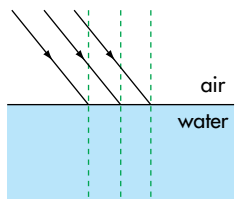


FIGURE 5.10

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5.3

Using lenses—convex lenses

FACT FILE

The word 'lens' comes from the word 'lentil', a small seed used in cooking. Both the lentil and some lenses have a convex shape.

In section 5.1 we saw that it was possible to change the shape of a beam of light by reflecting the light on a curved mirror. We achieve the same sort of thing through refraction by changing the shape of the surface of a transparent object.

A **lens** is a transparent object which has carefully shaped surfaces. There are basically two types of lenses. **Convex lenses** are thicker in the middle than at the edges. The other type, **concave lenses**, are at their thinnest in the centre—but more of this type of lens in section 5.4.

Convex lenses cause parallel beams of light to converge to a **focus (focal point)**. The distance from the centre of the lens to its focus is called the **focal length** and is usually measured in centimetres or metres. In most cases, the thicker the lens at its centre the more powerful it is (the more it bends the light) and the smaller the focal length.

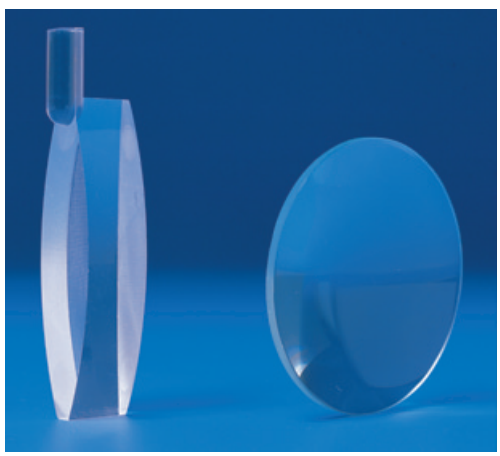

FIGURE 5.11
Convex or converging lenses.

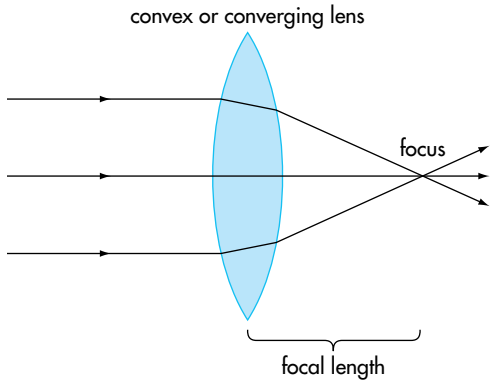

FIGURE 5.12
Parallel rays come to a focus after passing through a convex lens.

One important use of a lens is to produce an **image**. A convex (converging) lens will produce an image on a sheet of paper because it will bring rays of light from objects to a focus.

If you create an image of a distant object (outside the window) on a sheet of paper using a convex lens, the image is easily recognisable, everything is upside down or **inverted**! Try it!

You can also use a convex lens to produce an enlarged or **magnified image** of something. This time, however, the object must be placed close to the lens. The image is not inverted but the right way up or erect.

USING SCIENCE 2 FOCUSING HEAT

As well as bringing light rays to a focus, a convex lens will concentrate the infrared (heat) rays from the Sun at a point. If you focus the light from the Sun on to a piece of dark paper you may be able to get it to smoulder. Be very careful not to burn yourself—and **never look directly toward the Sun itself. This will permanently damage your eyes!**

FIGURE 5.13
A lens can focus the Sun's heat rays and cause a fire to start.

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EXPERIMENT 3 PASSING LIGHT THROUGH A LENS

AIM
To show the effect of different shaped convex lenses on beams of light.

MATERIALS
- ray box with multiple slits
- power supply
- convex lenses of various thicknesses
- sheets of white paper
- ruler and pencil

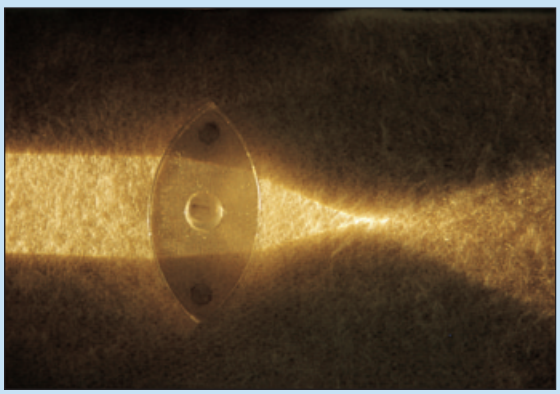

FIGURE 5.14
The paths of the rays passing through the lens can be traced easily.

METHOD
1 Connect the ray box to the power supply. Place the multiple slit in the ray box so that several rays of light are produced.
2 Carefully draw around the outside of one of the lenses. Allow the rays to strike the lens. Observe what happens to the rays as they pass through the lens and emerge at the other side.
3 Mark the path of the rays, remove the lens, and connect up the marks so that the paths of the rays can be seen clearly.
4 Measure the distance of the focus from the centre of each lens—the focal length.
5 Record your observations in a table.
6 Repeat the experiment for different thicknesses of lens.

CONCLUSION
How is the focal length of a lens related to its thickness?

EXTENSION
Find out what happens when a beam of light passes through two lenses of different thickness. Trace the path of the rays through the lenses and draw a conclusion from what you see.

Questions
1 Draw diagrams that show how parallel rays of light appear after passing through convex lenses of different thicknesses.
2 Explain the difference between a convex and a concave lens.
3 How might you make a set of rays coming from a single point change direction to form a parallel beam?
4 Con asks Narim to pass him a convex lens of focal length 25 cm from a box containing many lenses of different focal length. How could Narim find the correct focal length?

Think about
Is it possible to make a magnifying glass strong enough to see atoms?

Find out
1 Lens makers use many different types of materials to construct lenses. Research how a lens is made and the names of materials that are commonly used in commercial lens manufacturing.
2 Using a convex lens you can see the inverted image of a distant object on a piece of paper. If you use the convex lens as a magnifying glass, can you get the magnified image on the paper?

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5.4

Using lenses—concave lenses

FACT FILE

Optometrists often measure the power of lenses in dioptres (symbol D). The higher the number of dioptres, the stronger the lens. Convex lenses have positive powers and concave lenses have negative powers. A fairly strong convex lens would have a power of +5 D. A concave lens of the same focal length would have a power of –5 D.

All the convex lenses dealt with in the previous section had one thing in common. Their centres were thicker than their edges. A **concave lens** is made the opposite way round. The lens is thickest at the edge and gradually becomes thinner towards the centre.

The previous experiments showed us that a convex lens causes rays of light to converge. A concave lens does the opposite, and the rays spread out or **diverge**.

FIGURE 5.15
A concave lens can be recognised easily by its thicker rim.

Because the rays of light are not concentrated at a focus, it might seem strange to think of a focal length for a concave lens. It makes more sense as soon as we allow a parallel beam of light to pass through a concave lens, however. Although the rays are moving apart after leaving the lens, they can be traced back and seem to be coming from a single point. The distance from this point to the centre of the lens is the focal length for this type of lens.

Unfortunately, there is no quick and simple method of measuring the focal length of a concave lens, unlike that of a convex lens.

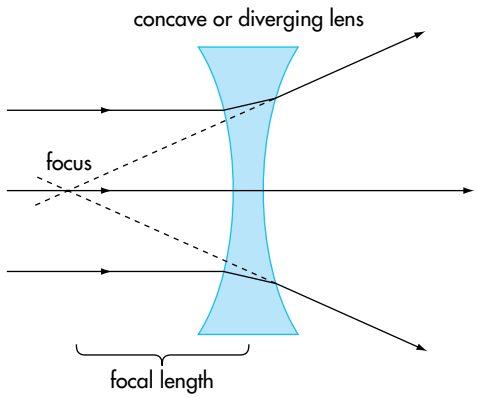

FIGURE 5.16
The diverging rays appear to come from the focus.

USING SCIENCE 3 IMAGES IN A CONCAVE LENS

When you look through a concave lens you will notice two important characteristics of the image. First, the image is always the right way up or erect. You should also notice that the image is always smaller or **diminished**.

Take a sheet of white paper. Can you project an image on to the paper as you did with the convex lens? You will find that it is impossible to get a clear image on the paper. Images that cannot be projected on to something are called **virtual images** and those that can be projected are known as **real**.

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EXPERIMENT 4 PASSING LIGHT THROUGH A CONCAVE LENS

AIM
To show the effect of different shaped concave lenses on beams of light.

MATERIALS
- ray box with multiple slits
- power supply
- concave lenses of various thicknesses
- sheets of white paper
- ruler and pencil

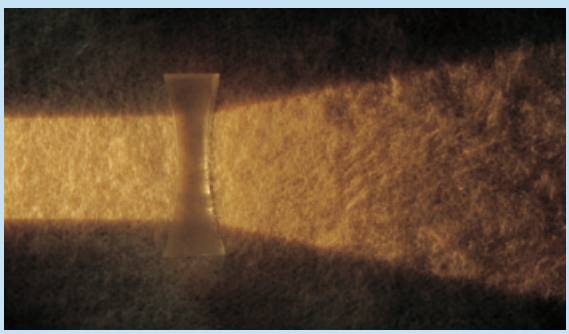

FIGURE 5.17
Rays of light diverge after passing through a concave lens.

METHOD
1 Connect the ray box to the power supply. Place the multiple slit in the ray box so that several rays of light are produced.
2 Carefully draw around the outside of one of the lenses. Allow the rays to strike the lens. Observe what happens to the rays as they pass through the lens and emerge at the other side.
3 Mark the path of the rays, remove the lens, and connect up the marks so that the paths of the rays can be clearly seen.
4 Repeat the experiment for different thicknesses of lens.
5 Trace back to where each set of rays appears to have come from and measure the focal length.

CONCLUSION
Predict how the focal length of a concave lens varies with its thickness.

Questions
1 Explain simply the difference between a virtual and a real image.
2 Draw a diagram to show how a set of rays moving towards each other could be made into a parallel beam of light after passing through a concave lens.

Think about
Sometimes delivery vans have a special type of concave lens attached to the rear window. What would the driver see through the lens and why might this assist safer driving?

Find out
Anton van Leeuwenhoek was born in Holland in 1632 and lived for 90 years, never leaving his native town of Delft. He had very little schooling but was known the world over for 'discovering a new world more amazing than the one discovered by Christopher Columbus'. His hobby was making lenses. Find out the amazing discovery he made!

Activity
Try looking through two or more concave lenses at the same time. Describe the image that you see. Now try using one concave and one convex lens. What do you notice now? Finally try separating the lenses while still looking through them both. Record your observations and share them with other members of your class.

Foundation worksheet 1

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5.5

Eyes that need help

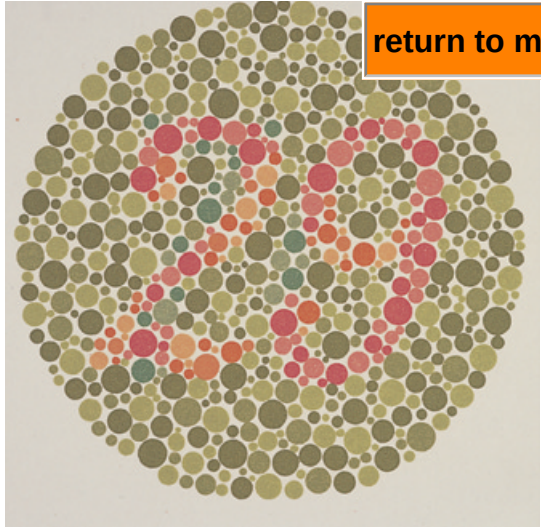


FIGURE 5.18
What number can you see among the dots? (Note that the colours in this book may not be accurate enough to assess your colour vision.)

Our eyes are wonderful optical instruments, but unfortunately, our eyes don't always give us perfect vision.

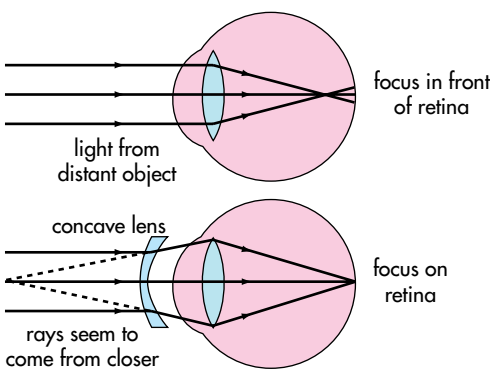
Talk about
What types of problems can people have with their eyes? How are these problems overcome?

Described here are some of the problems humans have with their eyes as well as some of the simpler solutions.

Getting a clear image

- **Short sight (myopia)**—A short-sighted person can see close objects clearly but not distant ones. It may be that the person can read a book easily but writing on the blackboard seems hard to make out, particularly from the back of the room! This happens when the whole eyeball is too long. The rays of light from distant objects are brought to a focus in front of the **retina**. A concave lens in front of the eye diverges the rays of light from a distant object so that they appear to have come from a point close to the eye. The image then falls directly on the retina.

FIGURE 5.19
Myopia and correction.



- **Long sight (hypermetropia)**—A long-sighted person can see distant objects clearly but not close objects. The person would be able to spot birds in the air but be unable to read the print on a novel. This is caused by the eyeball being too short. The rays of light from a close object fail to reach a focus before they hit the retina. A convex lens in front of the eye makes the rays from a close object appear to have come from a point much further away. The eye can then focus the rays on the retina.

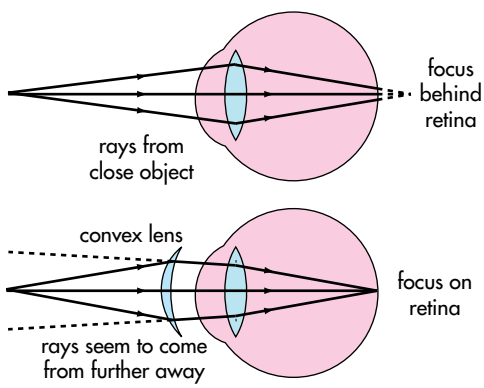


FIGURE 5.20
Hypermetropia and correction.

- **Lack of accommodation (presbyopia)**—As a person gets older, the lens in the eye tends to become stiffer and is not able to change its shape as easily to focus close objects. Have you noticed your parents or teachers holding newspapers at arm's length? This is known as lack of **accommodation**. The remedy is much the same as for long sight—a convex lens in front of the eye. Most people would manage using spectacles just for reading. If the eye's lens becomes very stiff, people may need

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glasses, television-viewing glasses and walking-around glasses! To save the need for three separate pairs, **bi-focal** or **tri-focal** glasses can be prescribed. The lenses are made different strengths according to where light passes through them—the strongest at the lowest edge for reading. Graded focal length lenses gradually change in strength from bottom to top.

FIGURE 5.21
Bi-focal lenses restore the ability to focus objects close and far.

Colour blindness

When the nerve cells in the retina fail to respond normally to colour, we say the person is **colour blind**. There are many different types of colour blindness, a common one being confusion between shades of red and green. An Ishihara test card contains many spots of different colours and intensities (see Figure 5.18). A person with normal colour vision will see a particular number clearly, while someone with colour blindness will see nothing except random coloured dots.

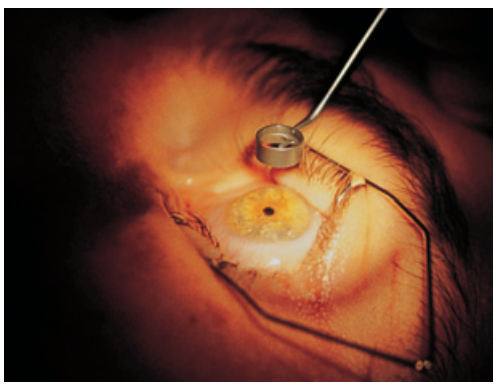

FIGURE 5.22
Laser eye surgery taking place.

Other remedies

Contact lenses are becoming increasingly popular, particularly with sports people. Instead of glasses, a small lens is worn in direct contact with the cornea of the eye. Convex or concave lenses are used according to the particular defect. The lenses are made from either 'hard' plastic or 'soft' water-absorbing materials. Because of the continual contact with the surface of the eye, the lenses must be kept very clean and sterilised regularly. Disposable contact lenses are also available. Contact lens wearers tend to get a wider, clearer field of vision than spectacle wearers, and the lenses are virtually invisible. It is also possible to change your eye colour by having coloured lenses!

Some eye surgeons perform laser surgery on the surface of the **cornea** to correct vision defects. The cornea can be made less strongly curved for people with acute short sight.

Questions

1 A friend can see writing on the board clearly but has difficulty reading from a book. What type of eye defect does your friend have? What type of lens would correct the problem?

2 Draw a diagram to show how the problem in question 1 could be corrected.

3 What is lack of accommodation? How can it be corrected?

4 What is colour blindness? Is it correct to call it a type of blindness?

Find out

Colour blindness is much more common in men than women. Find out why this is the case.

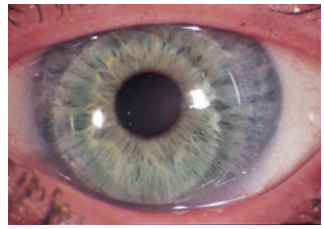

FIGURE 5.23
Many people prefer contact lenses to spectacles.

FACT FILE

The use of spectacles to correct vision problems dates back more than 700 years. The first spectacles were worn by long-sighted people in Europe and China. Around 500 years ago spectacles for short sight were also in use. Then in 1784 Benjamin Franklin invented bifocal glasses.

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return to main menu

HISTORY OF SCIENCE

The nature of light and colour

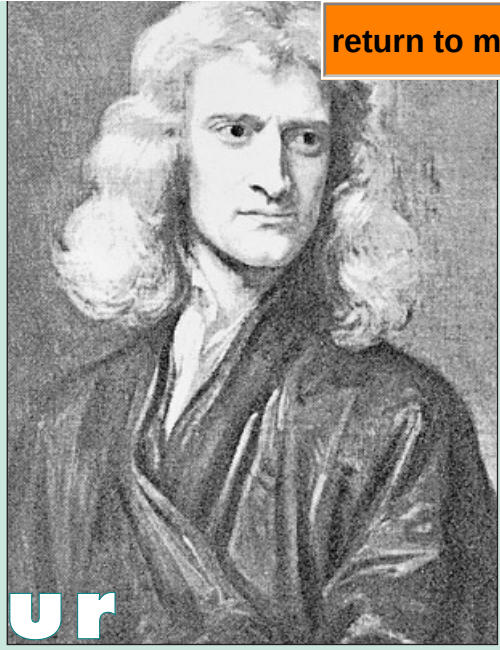

FIGURE 5.24
Sir Isaac Newton.

FACT FILE

When the light given out by stars is passed through a prism the spectrum produced gives us information about the composition of the star. This is how astronomers have come to formulate their theories on how stars are created and what happens to them as they grow old.

Talk about

Imagine that an 8-year-old child asked you these questions:

What is light? How does it travel? How does it allow us to see our surroundings? Discuss how you would answer the child and explain the nature of light.

The history of ideas about light is an amazing story of how scientific understanding develops. Theories about light have been put forward by scientists since around 500 BC. Existing theories were often replaced or changed to make way for better explanations.

In ancient Greece, Socrates and Plato thought that light was made of streamers or filaments which travelled out from the eyes. Pythagoras disagreed and said that light was made of particles which travelled out from the source of the light. Another Greek, Empedocles, taught that light travelled in waves. In the 11th century, an Arabic physicist known as Alhazen said that light travels from objects to our eyes. Leonardo Da Vinci in around 1500 observed the reflection of light and thought it was similar to the reflection of sound. He thought that light must be made of waves just like sound.

Animation

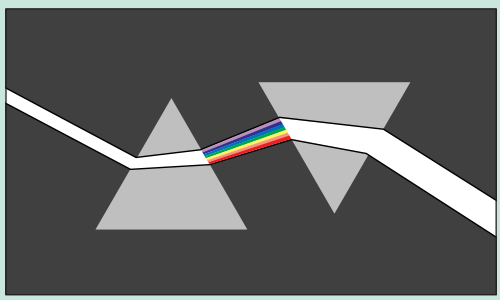

FIGURE 5.25
Newton's demonstration to show that white light is made up of different colours.

The next major step in our understanding of light came in 1666. The English scientist Isaac Newton experimented with light and tried to explain its behaviour. In a famous experiment Newton discovered that a narrow beam of light passing through a glass prism could be split into a band of colours which he called a **spectrum**.

Although Newton saw a complete range of colours, he classified them into seven different groups: red, orange, yellow, green, blue, indigo and violet. Newton chose seven because he thought of them in the same way as the notes in a musical scale. Most people can only really distinguish six! Indigo is very hard to see between blue and violet.

People knew that prisms could produce different colours from sunlight, but they believed that these colours came from the glass, just as coloured light was produced by coloured filters. Newton was able to show that sunlight is made up of the spectrum of colours by using two prisms. The first one split the light into the colours of the spectrum and the second prism joined them back together again to form white light.

The observations Newton made in this and many other experiments led him to believe that light was made of particles not waves. The fact that light travels in straight lines and cannot bend around corners like

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USING SCIENCE 4 MIXING COLOURS

Cut out a disc of fairly stout cardboard about 10 cm in diameter. Divide the disc into six equal sections as shown in Figure 5.26. Colour each section with one of the main colours of the spectrum. Thread a piece of string through the two central holes as shown. Spin the disc by hand until the string coils up tightly. Gently pulling the string will make the disc spin rapidly. Watch the colours merge to produce a rather murky white!

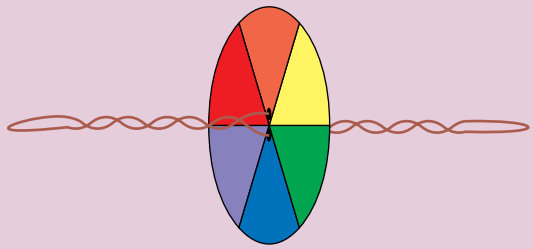

FIGURE 5.26
A colour mixer you can make easily.

other waves seems to fit the theory that light was indeed a stream of particles. Newton called these particles of light corpuscles. He used his corpuscle theory to explain other observations such as reflection and shadows.

Other scientists of Newton's time disagreed, including Christian Huygens, a Dutch scientist who said light was a wave. He supported his theory with evidence that under some circumstances light did bend as it spread out from a narrow slit. This effect is called **diffraction**. Other scientists found more evidence to support the wave theory. The wave theory developed so that it could better explain refraction and the ability of light rays to pass through one another unaffected.

The next major breakthrough came with the realisation that light was a form of **electromagnetic radiation**, along with infrared and ultraviolet waves.

Questions

1 Explain your understanding of the term 'scientific theory'.

2 How did Isaac Newton's experiments on the composition of sunlight change people's ideas about the appearance of coloured light from a glass prism?

3 What observations of the behaviour of light can best be explained by the particle theory?

4 What observations of the behaviour of light can best be explained by the wave theory?

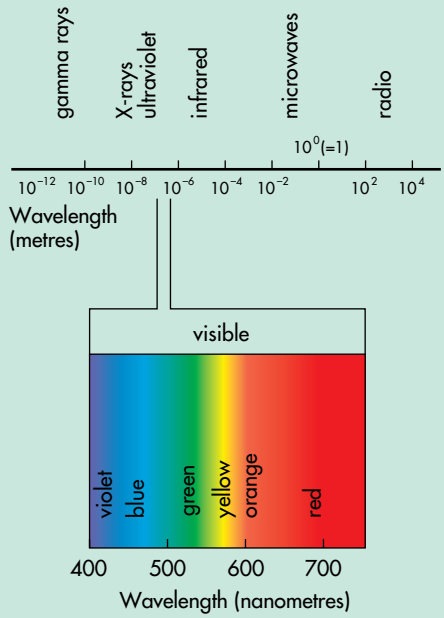

FIGURE 5.27
Visible light makes up a small part of the **electromagnetic spectrum**.

5 Describe a photon. Is it a particle? Is it a wave? Or is it both?

Find out

What contributions did Thomas Young and Augustin Fresnel make to our understanding of the nature of light?

Activity

Hold two pencils parallel to one another about 2 mm apart. Place them just in front of your eye and look through the narrow slit at a light. Gradually bring the pencils closer together. The fine lines which appear can only be explained if we think of light as a wave.

FACT FILE

In 1905 Albert Einstein published a theory bringing together two main ideas about light. He called his theory the photoelectric effect, where light consists of particles, bundles of concentrated electromagnetic energy called photons.

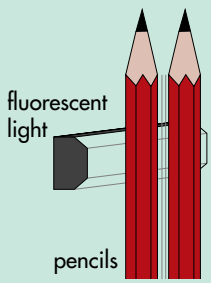

FIGURE 5.28
The effect observed is called diffraction.

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5.7

Our coloured world

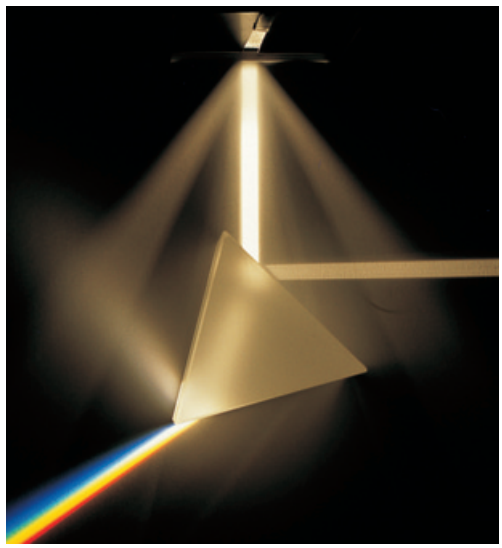
FIGURE 5.29
Scientists believe that rainbows are caused by drops of water in the air breaking up sunlight into the colours of the visible spectrum. Next time you see a rainbow, notice the order of the colours and the position of the Sun in relation to where you are standing. Are they always the same?

White light can split into a **spectrum** when refraction occurs. The process is known as **dispersion**. Scientists now believe that the different colours that make up white light are refracted by varying amounts as they enter a transparent material. Violet light slows down more than red when it passes into a transparent material from air. Consequently, red light is refracted the least while violet light is refracted the most. A triangular prism is good for demonstrating this since the colours are split both going into and coming out of the prism.

Newton's original spectrum was impure since different colours overlapped in places. A **pure spectrum** can be created by passing the light through a convex lens first to bring each colour to a distinct focus. This is shown in Figure 5.30.

The instrument which analyses light by splitting it up is called a **spectrometer**. Forensic scientists can analyse the light from burning a chip of paint found at an accident to find the type of paint it was, and this can sometimes help them trace the make, age and model of the car it came from. Every substance produces a characteristic spectrum when burned—much like a unique 'fingerprint'.

FIGURE 5.31 (right)
White light passing through a triangular glass prism splits into its different colours.



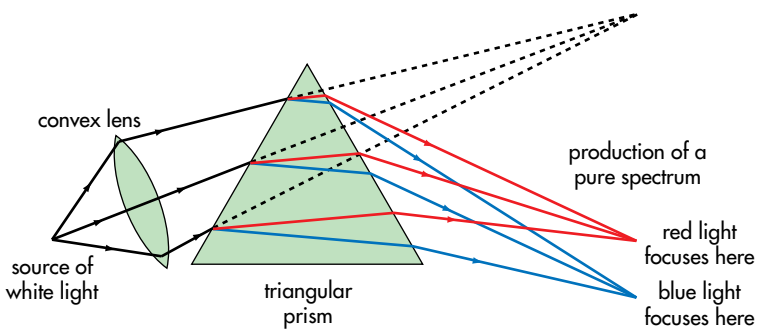


FIGURE 5.30
Different colours are refracted different amounts as they pass through the prism.

Talk about

If sunlight is white, why doesn't everything appear black and white when sunlight shines into a room?

Why are leaves green?

White objects appear white because they *reflect* all the colours of sunlight. Black objects **absorb** all the colours of sunlight, so no colour is reflected back into our eyes.

Questions

1 Why does white light split up into different colours when it is refracted? Which of the colours is refracted the most and which the least?

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EXPERIMENT 5 PRODUCING A SPECTRUM

AIM
To split up white light from an electric globe into its spectrum.

MATERIALS
- ray box (single slit)
- power supply
- triangular glass or acrylic 60° prisms
- sheets of white paper
- ruler and pencil

METHOD
1 Place a 60° prism in the centre of a clean sheet of paper. Carefully draw round its edge.
2 Connect the ray box to a power supply. With a single slit allow a ray of light to fall on the prism. Slowly rotate the paper and prism until you observe a clear spectrum emerging from one side.
3 With a pencil, carefully mark the path of both the white ray and the dispersed rays. Label on the paper which colour is dispersed the greatest amount and which the least amount.
4 Take a second prism and place it near to the first one so that the spectrum entering it will now recombine. When this occurs, draw round the second prism and trace the passage of rays through it.

RESULTS
Complete the sentences:
The colour that was refracted the least by the prism was ____. The colour that was refracted the most was ____.

2 Thin beams of white light fall on to a rectangular block of clear plastic and a 60° prism made of the same material (Figure 5.33).
a What colour is observed at A?
b What colour is observed at B?
c With the block no spectrum is seen. Suggest a reason for this.
3 Many people who live in hot climates wear white clothes. Explain why this would be an advantage.
4 If a piece of paper absorbs indigo, red, blue, green, yellow and violet light, what colour must it be?

Think about
Look carefully at a projected image of a pencil placed on top of an overhead projector. Around the edge you will see slight colouring. Explain why this might be so.

Find out
Find out how a rainbow is produced. Which colour is on the 'top' of the rainbow when viewed from the ground? Research the coloured halo sometimes seen around the Moon. This is another effect of water in the air, but this time as ice crystals.

Multimedia

Interactive

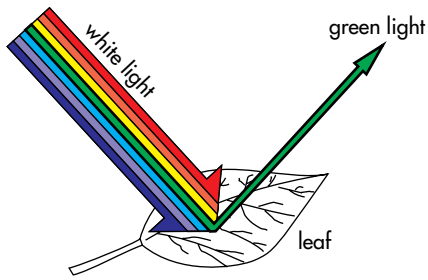
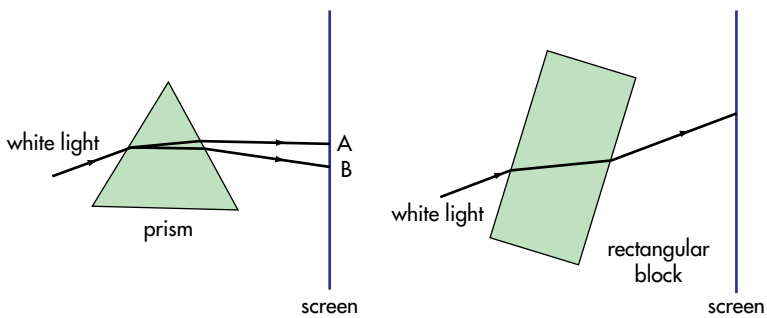


FIGURE 5.32
The colour we see in an object depends on the colours that are reflected off the object back to our eyes. Objects absorb some of the colours in sunlight and reflect others. Green leaves, for example, absorb red, orange, yellow, blue, indigo and violet, but not green. So the green is reflected back to eyes and the leaves look green.

Extra worksheet

FIGURE 5.33



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What colour do you see?

return to main menu

FIGURE 5.34
Yes, red + blue + green = white!

Talk about

Indoor colour photographs taken at night without a flash often look very yellow. When a flash is used, colours look more natural. Why might this be so? What is different about the light from the flash compared to normal lighting?

FIGURE 5.35
Oops . . . the flash didn't work!

FIGURE 5.36
That's more like it!

We have already seen that white light can be split into a spectrum. It is useful to understand how different colours behave when they are mixed or when they fall on different coloured surfaces. Every lamp we use emits particular portions of the spectrum and this can affect the way our clothes, furnishings, make-up and decorations look under artificial or natural light. Perhaps this is the key to why the photographs look so different!

To separate different colours we use **filters**. These are special transparent materials which allow only certain colours to pass through them. Place a green filter in front of a globe and only the green part of the spectrum will be seen. We get a beam of green light.

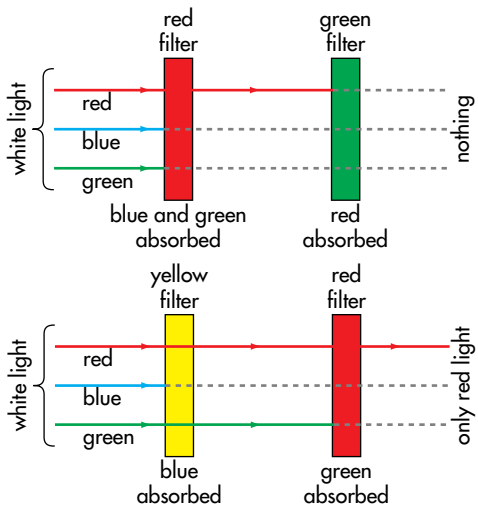

FIGURE 5.37
The red filter absorbs everything but red, and the green filter absorbs the red—nothing gets through!

FIGURE 5.38
The yellow filter passes the red and green but the red filter absorbs just the green, allowing the red through.

A filter works by **absorbing** certain parts of the spectrum. It is the sections of light that it doesn't **absorb** that pass through. Certain filters allow more than one colour to pass through.

On the other hand, we may get a particular colour passing through them all while other colours are absorbed.

And now something you may not believe until you have seen it for yourself! If we shine three beams of coloured light (green, blue and red) on to a white card in a darkened room, we can obtain a pool of white light where they all overlap!

Shine any two of the above colours of light on to the card and we get definite colours too. Red light plus green light gives yellow light; blue light plus red light gives a purplish colour called **magenta**; blue light plus green light gives a bluey green called **cyan** (pronounced sy-ann). Because red, blue and green light mix to form white light, we call them the **primary colours**. Yellow, magenta and cyan are called the **secondary colours** since they are made by mixing two primary colours. This process is called **additive mixing**. Since a secondary colour is a mixture of two primary colours, when a

TABLE 5.2 Primary colour filters.

Type of filter	Red	Blue	Green
Colours passed	red	blue	green
Colours absorbed	blue and green	red and green	blue and red

TABLE 5.3 Secondary colour filters.

Type of filter	Cyan	Yellow	Magenta
Colours passed	green and blue	red and green	red and blue
Colours absorbed	red	blue	green

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EXPERIMENT 6 MIXING AND VIEWING COLOURS

AIM
To mix beams of different coloured light.

MATERIALS
- 3 ray boxes with power supplies
- coloured filters (red, green, blue, yellow, cyan and magenta)
- white paper

METHOD
1 Place the three primary filters (red, blue and green) in the ray boxes so that three different beams of light are produced.
2 In turn, allow two of the beams to overlap on the sheet of white paper. Describe the colour that you see in a table like the ones opposite.
3 If your power supply can be adjusted, dim each of the lamps in turn. Note what happens to the colour of the overlap. This way you are altering the amounts of each primary colour.
4 Now allow all three beams of light to overlap. Again, adjust the brightness of each lamp if you can. Write your findings in your table.
5 Now replace the primary filters with the secondary colours (yellow, cyan and magenta). Repeat steps 2, 3 and 4 and record your results.
6 Describe how the results from the secondaries differ from those from the primaries.

RESULTS

	Red	Blue	Green
Red			
Blue			
Green			

	Yellow	Cyan	Magenta
Yellow			
Cyan			
Magenta			

CONCLUSION
From your observations, complete the following word equations.
Red + Blue + Green = ____
Yellow + Cyan + Magenta = ____

secondary colour is mixed with the third primary, white light results. A primary and a secondary which produce white light are called **complementary colours**. For example, blue and yellow are complementary, as are green and magenta. This is important in the production of colour photographs.

We can now appreciate the effects of filters made from both the primary and secondary colours.

Questions

1 What are the three primary and three secondary colours?
2 What would be the effect of shining green and red spotlights on to a white surface so that the beams partly overlap?
3 A TV logo is coloured green and magenta. If you looked closely at the TV screen, what colours would you see in both the green and magenta areas?
4 Gina and Claire decide to wear their new clothes on the night of the local disco. Gina has a new red top and Claire has a yellow skirt and green shoes. Inside the disco the lighting is nearly all red and blue. What would Gina and Claire's clothes look like in the disco?

Activity

Imagine that you are to produce a play at school. Some of the scenes are set in bright sunshine, others inside a dark, cold, creepy dungeon and another at midnight on a moon-lit night. Write a set of instructions for the lighting technician telling her which coloured lights to use to make the stage look most effective.

Foundation worksheet 2

FACT FILE

Look closely at a picture on your colour TV. You will be able to see the tiny coloured dots or rectangles that provide the colour. They are all either red, blue or green, and all colours produced on the screen are simply different combinations of these three primary colours.

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APPLICATIONS AND USES OF SCIENCE

Photography

Earlier in this chapter we saw it was possible to produce a clear image by using a single convex lens. If this image falls on light-sensitive materials, it is possible to create a **photograph**. It is usual to expose pieces of photographic film to light. Areas exposed to large amounts of light become black whereas those on which little or no light falls remain clear. This produces a 'negative' image since light and dark are reversed. When a negative is printed to produce a photograph, a similar process occurs. What were dark areas on the negative become light on the print etc. Thus what was light in the original scene eventually becomes light again after the two processes, **negative** and **print**.

The camera's job is to produce an image on the film with just the correct amount of light to give a good balance of dark and light areas on the negative. All cameras use the same basic principle. However, more expensive cameras allow greater control over the size and quality of the image and the amount of light entering the camera.

FIGURE 5.39
A disposable camera complete with film. The convex lens is clearly visible.

In a more expensive camera, the **aperture** (f22, f16, f11 etc.) controls the amount of light hitting the film. The **shutter** allows the light to enter the camera for a very short period of time, usually about 1/100 of a second.

Colour photography

These days, most people take colour photographs. Instead of the whole negative being sensitive to all colours of light, **colour film** has three distinct layers, each sensitive to one of the primary colours, blue, green or red.

Figure 5.40 shows a red rose with green leaves against the background of a blue sky. The image of each colour will be

FIGURE 5.40

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USING SCIENCE 5 TAKE THE CAMERA APART

Visit your local photo lab and ask if you can have one of the empty disposable cameras brought in for processing. Carefully take the camera apart and identify all the parts.

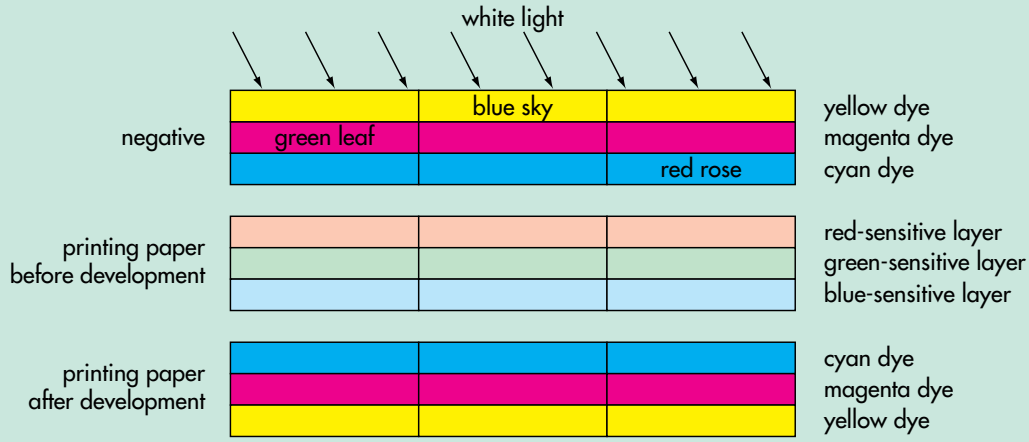
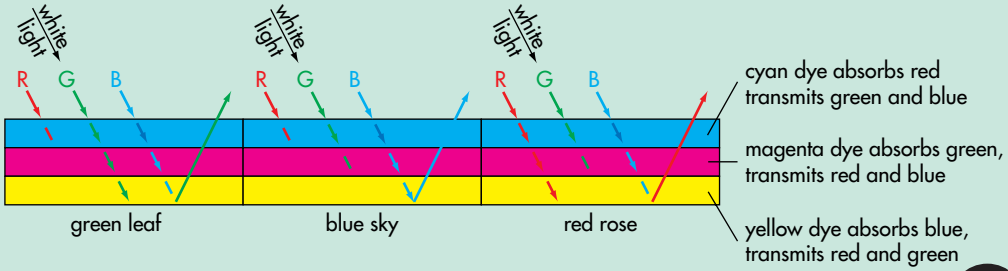


FIGURE 5.41
Follow the light through the different layers of film and paper.

Internet assignment

Viewing the print



formed in a separate layer in the film. When the film is developed, each of the three images is replaced by a coloured dye. The dyes are the secondary colours that would combine with the primary colour to produce white light.

A colour negative is similar to a black and white negative but even the colours are reversed! The image of the blue sky becomes yellow by the release of a yellow dye, the green leaf becomes magenta and the red rose becomes cyan. Compare the colours of the photograph with the colours in its negative (Figure 5.40).

The paper used for the final print also has three layers, each sensitive to one of the primary colours red, green or blue. The process is quite complicated but, put simply, the negative acts as a filter, making sure that light gets to the appropriate area of the paper print. When the paper is developed, cyan and yellow dyes are produced in the red and blue areas.

Questions

1 What colour would the negative be of a yellow flower against a red background?

2 Explain each stage of the process involved in producing a photographic print of a green leaf. You could use a flow diagram or a cartoon strip.

Find out

Form a group of three or four people. Discuss the many different forms of photographic images that we have today. These include photographs, video, cinema films, digital images on CDs, etc. Research one form of photography from either a historical or a technical point of view. Create a poster display of your group findings.

FACT FILE

The *camera obscura*, meaning 'dark room' in Latin, was an enclosure with white walls and a very small hole in one wall. The rays of light entering the room formed an image on the walls. Its use was first described by the Arabian scholar Alhazen in the 10th century. He used it to look at an eclipse of the Sun.

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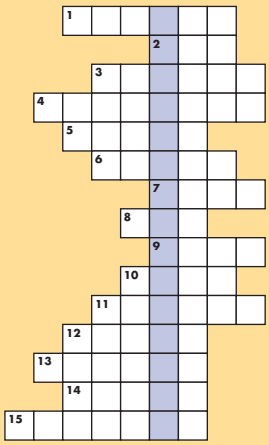
Using scientific language

Copy the grid and then use the clues to fill it in to find out what inspired this chapter!



1 This filter passes red and green but stops blue.
2 Our personal optical instrument!
3 Where the image is formed in the eye.
4 The whole range of colours.
5 This might be long or short!
6 Primary colour—artists might disagree!
7 Light travels in these straight lines.
8 This colour is refracted the least.
9 Another primary colour, perhaps sad!
10 This part of the eye lets you focus close and far objects.
11 Look through this to see only one colour.
12 Something in the eye that looks black—but is really nothing!
13 Hot road surfaces let you see something that isn't really there!
14 This type of energy isn't heavy!
15 This secondary colour is a mixture of red and blue light.

Check your knowledge

1 What can happen when light enters a material that causes it to slow down or speed up?
2 Copy Figure 5.42 and complete the ray diagrams for light passing through the lenses. Mark the focal length in each case.
3 Which type of lens always produces an upright, smaller image?

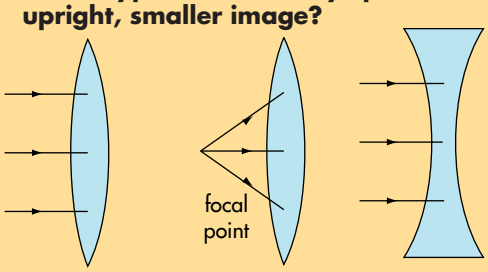


FIGURE 5.42

4 What type of lens would help a student who could see close objects clearly but had difficulty reading the credits at the cinema?
5 Draw a diagram to show how the sight of the student in question 4 could be corrected.
6 What are the colours of the visible spectrum? Why are there seven?

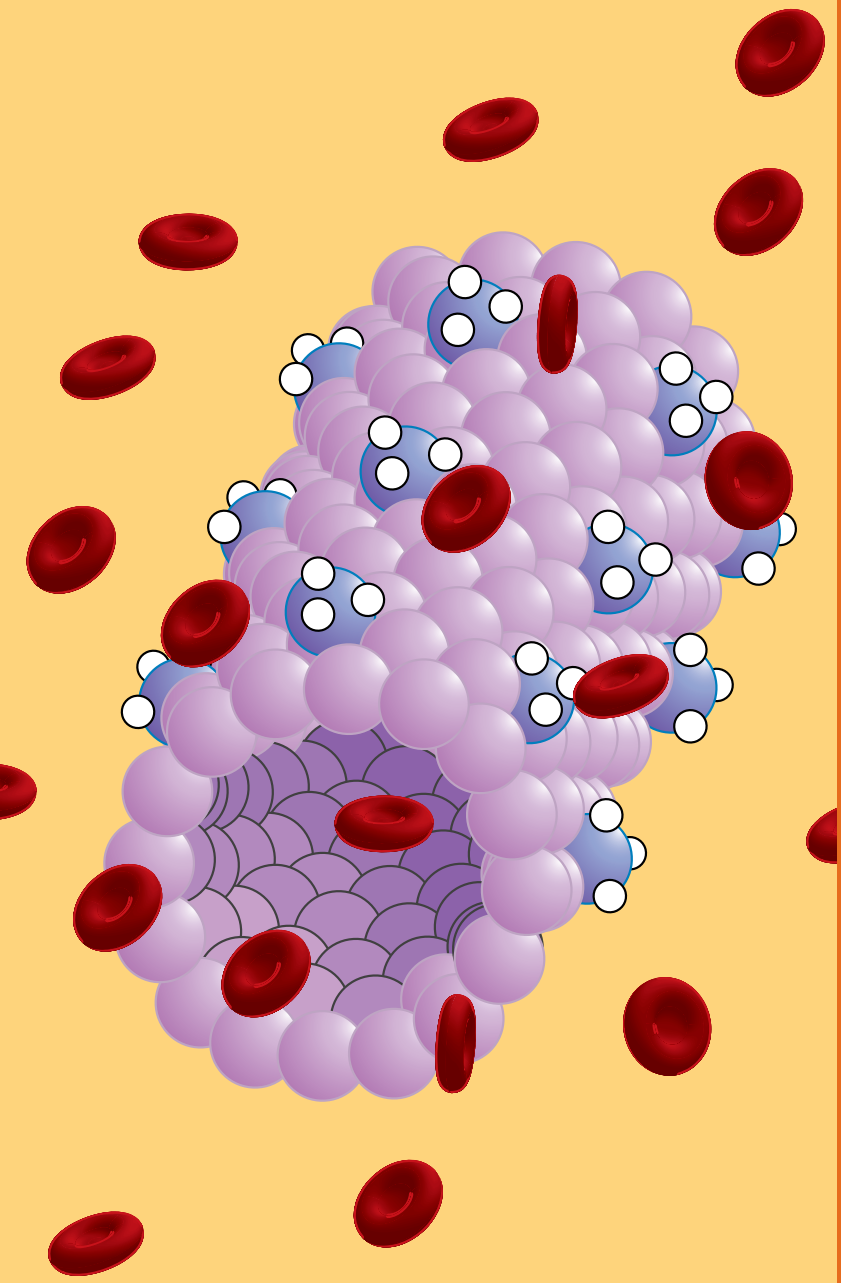
Apply your skills

1 Explain the difference between reflection and refraction.
2 a Draw a large diagram of both a convex and concave lens. On each diagram draw three rays passing through the lens. Carefully show how the rays bend towards the normal when entering the lens and away from the normal when leaving.
   b Explain why a convex lens is also called a converging lens.
   c Explain why a concave lens is also called a diverging lens.
3 When buying a new item of clothing the salesperson suggested I should look at the material in daylight outside the shop. What possible difference could this make to my choice?
4 A parent is complaining that it is becoming difficult to read the newspaper. What might be happening to their eyes and how might the problem be solved?
5 What is a cataract and how can the blindness this causes be cured?
6 Place a coin on the base of an empty ice-cream container which is situated directly below a tap. Move your head downwards until you just cannot see the coin over the edge of the container. Keeping your head as still as possible, get a friend to turn on the tap slowly. As the water level rises the coin will appear! Explain using diagrams why this happens.

Challenge yourself

Tran is red–green colour blind (the colours red and green seem the same to him). Tran is interested in recording music and his tape recorder has coloured lights which tell him the volume level. If the signal is not too loud, green lights come on. When the volume gets too great, red ones light up. Explain how Tran, using coloured filters, could be sure that he always noticed if the volume became too loud. Would it be possible for Tran to check whether the recording level was OK?

eTester



CHAPTER 6

SIMPLE AND COMPLEX MACHINES

Imagine a Boeing 747 aircraft made of a single, huge diamond—as strong as current materials but only $\frac{1}{50}$ the weight. Imagine tiny robots that circulate in our bloodstream, seeking out and destroying disease-causing organisms. Imagine computers that are a billion times more powerful than those we have today.

These are all possibilities brought to us by nanotechnology: combining and arranging materials atom by atom or molecule by molecule. All material is composed of atoms. How these atoms are arranged determines what that the material is. Rearrange the atoms in a lump of coal and you have diamond. Rearranging those in sand produces a computer chip.

To do this, we would need to create very small machines and tools which could use these atoms as building blocks, shifting them around and assembling them in precisely the way we want. These nanomachines or nanobots would be capable of building other nanomachines to help complete tasks quickly and efficiently. A typical nanobot would be about 100 nanometres in length. There are 1 000 000 nanometres in 1 millimetre!

Nanomachines and nanotechnology may sound as if they belong in the far future, but within 30 years nanotechnology is expected to be widely used. Already scientists can move atoms and molecules around to create simple sculptures and they have even created company logos from individual atoms!

In this chapter you will look at machines and consider the impact that these have had and ay have on society.

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Identify that simple machines make tasks easier or more convenient.
- Identify some advantages of levers, pulleys, gears and inclined planes.
- Explain what a machine is and distinguish between simple and complex machines.
- Explain that machines need energy and that different machines have different energy efficiencies.
- Design and construct a complex machine.
- Explore some of the issues associated with the use of machines.
- Identify some of the contributions different cultures have made to the development of machines.
- Give some examples of how different groups in society make decisions about the use of ma

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Web links

6.1

Making life easier

FIGURE 6.1
Machines are all around us. What is a machine?

FACT FILE

Levers have been used for thousands of years. A balance beam for weighing was used in Egypt about 5000 BC, and water was lifted using levers as early as 1500 BC.

The lever

Most of us would find it impossible to lift the lid off a coffee tin by using our fingers. However, a teaspoon placed under the lid will remove it easily. The teaspoon is being used as a lever to apply a large force where normally we could not. Here, the teaspoon is an example of a **simple machine** called a **lever**.

A lever is generally a rigid object, and is usually long such as a length of wood or a metal rod. The lever can be thin like a crowbar or wide like a door—it is still a lever.

In order to use a rod as a lever it is necessary to have something to push against. This is called the **pivot** or **fulcrum**. When a teaspoon is used as a lever, you have to apply a force on one end in order to push against the tightness of the lid on the other. You cannot do this without something to press against. This is the role of the pivot or fulcrum.

The push you apply with your hand is called the **effort**. The force you are working against is called the **load**. An example of a load is the resistance of a lid to coming undone.

A machine can be described as a device which changes the size of a force or the direction of a force or both.

USING SCIENCE 1 SIMPLE AND COMPLEX MACHINES

Word process

Form small groups of three or four to discuss and make lists of machines that you might use, see, hear or hear about in your daily life.

To think about while making the lists:

Machines can be very simple devices that just do one thing, or they can be complicated.

A machine is any device which makes our lives easier and/or more convenient, or does things we cannot do by ourselves.

Most people think of machines as being very complex. They can be, but they do not have to be.

Divide your lists into two parts: **simple** and **complex machines**.

As you build up your list, you might like to consider other ways of dividing them up. For example, machines that use human and animal energy, and those that use fuels like petrol or gas. Another way of dividing them up is according to what they do.

Present your group results on a large sheet of paper. Your group should decide:

a what a machines is

b how to best divide machines into groups.

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EXPERIMENT 1 SEE-SAWS

AIM
To see how a lever is used in balances and see-saws.

MATERIALS
- triangular shaped piece of wood
- metre ruler • eraser • paper clips

METHOD
1 Place the ruler on the triangular shaped piece of wood so that the piece of wood acts as a pivot for your see-saw.
2 Place an eraser on one end of the ruler.
3 How many paper clips have to be placed on the other end to return the ruler to the horizontal?
4 What happens when the pile of paper clips is moved nearer to the centre of the ruler?

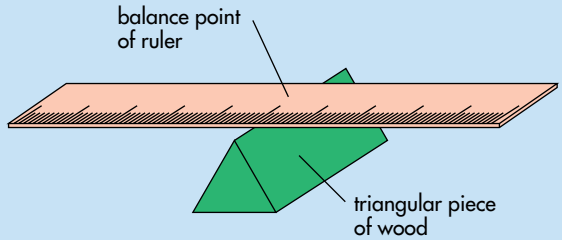

FIGURE 6.2

DISCUSSION
The paperclips and the eraser each cause a turning effect about the pivot. One effect is in the opposite direction to the other. Where did the paper clips have the greatest turning effect, and where did they have the least effect?

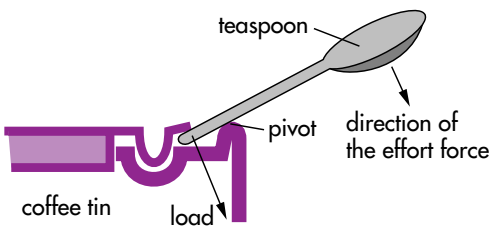

FIGURE 6.3
A lever is a machine which reduces your effort.

The handle is placed as far from the hinges as possible. Attach a rubber sucker to a door as close to the hinges as possible. How easy is it to open the door by using the sucker in this position? How do you explain this?

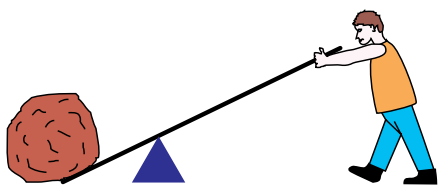

FIGURE 6.4

FIGURE 6.5

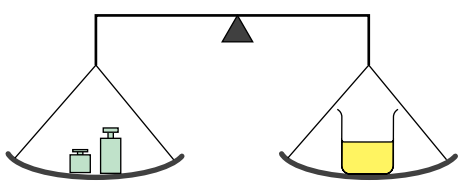

FIGURE 6.6

Questions
1 What is a machine? Give three examples.
2 Copy Figures 6.4–6.6 and show the directions of the effort, the load being working against and where the pivot is in each case.
3 Explain how you can ride on a see-saw with someone who is heavier than you.
4 Give two extra examples of how levers are used to make life easier.

Activity
Doors are also levers with handles to provide us with something to grip.

eGlossary

6.2

Which lever for the job?

FIGURE 6.7
The screwdriver and the door are both examples of levers.

FACT FILE

Doors on hinges save us an enormous amount of effort. Just imagine how long it would take to get into a house if it did not have doors on hinges!

Levers need pivots

When a lever is used, the rod turns about the pivot or fulcrum. In the see-saw, the pivot is in the centre of the rod.

However, the position of the pivot depends on the use to which the lever is put. The pivot is near one end when a teaspoon or screwdriver is being used as a lid opener. In a door, the pivot is at the hinges.

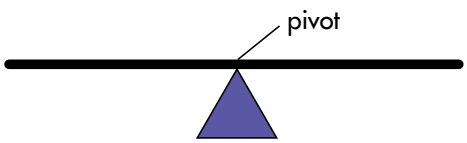

FIGURE 6.8

It is not all gain

Some levers make work easier by reducing the **force** needed to move something. However, in order to use less force, the lever must be moved through a greater distance.

Questions

1 Using diagrams of a door, draw:

a the distance moved by your hand if the handle were placed near the hinges

b the distance you would normally move it when opening a door with the handle in its normal position. How do these distances compare?

2 Using diagrams of a screwdriver opening the lid of a tin, show:

a the distance moved by your hand when opening a lid with a screwdriver

b the distance moved by your fingers if you were to try to open the lid with them.

How do these distances compare?

3 The gain from using a wheelbarrow is that it allows you to carry a heavy load with less effort. What is the loss from using this machine?

4 Some machines are designed to pick up delicate items. Tweezers may be used, for example, to remove splinters from skin. Tweezers are levers in which the force you apply is actually *reduced* to prevent damage

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USING SCIENCE 2 SHIFTING THE LOAD

Figures 6.9–6.13 are not complete. All are missing the pivot, while some of the drawings show the direction of the effort applied and others show the direction of the load.
Copy and complete the drawings by marking the pivot in each case with a *P*, and where necessary, the direction of the load with an arrow marked *L* or the effort with an arrow marked *E*.
In each case think about where you must apply your effort to overcome the load you are working against.

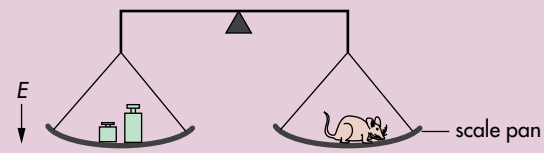

FIGURE 6.9
Lever-arm balance.

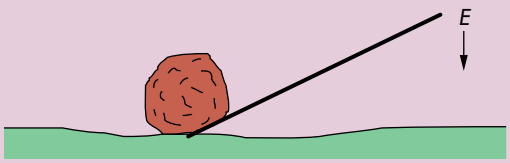

FIGURE 6.10
Crowbar.

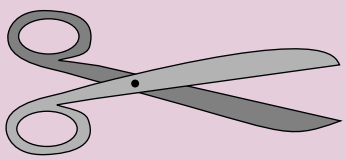

FIGURE 6.11
Scissors.

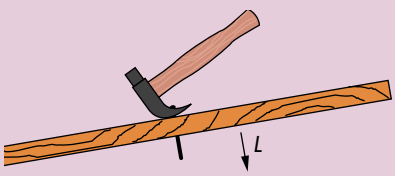

FIGURE 6.12
Claw hammer.

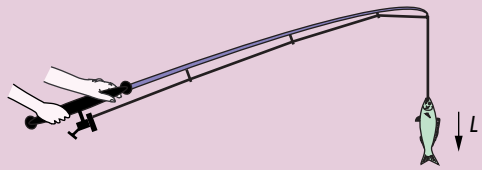

FIGURE 6.13
Fishing rod.

EXTENSION

Using the information below sort the different levers into three groups according to the positions of the pivot and where the load and effort act.

TYPES OF LEVERS

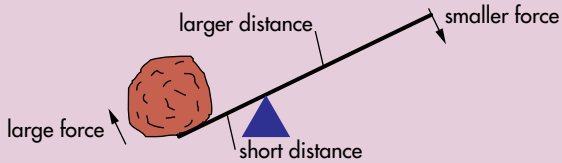

FIGURE 6.14
Class 1 lever: e.g. crowbar. When a lever's pivot is between the load and the effort force, it is called a **Class 1 lever**.

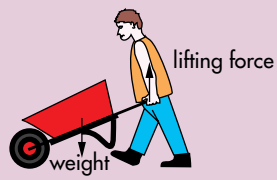

FIGURE 6.15
Class 2 lever: e.g. wheelbarrow. When a lever's load is between the effort force and the pivot, it is called a **Class 2 lever**.

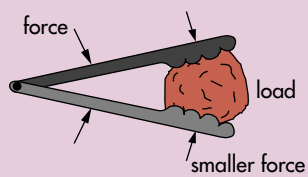

FIGURE 6.16
Class 3 lever: e.g. a pair of tongs. When a lever's effort force is between the load and the pivot, it is called a **Class 3 lever**.

to either the item being picked up or the item from which you are picking something up.
Draw a diagram showing the size and direction of the load and the effort involved in the use of tweezers. Include also the position of the pivot.

Think about

1 What do you think is meant by the turning effect of a lever? Use the screwdriver as an example in your explanation.
2 When do you think a particular lever has the greatest turning effect?

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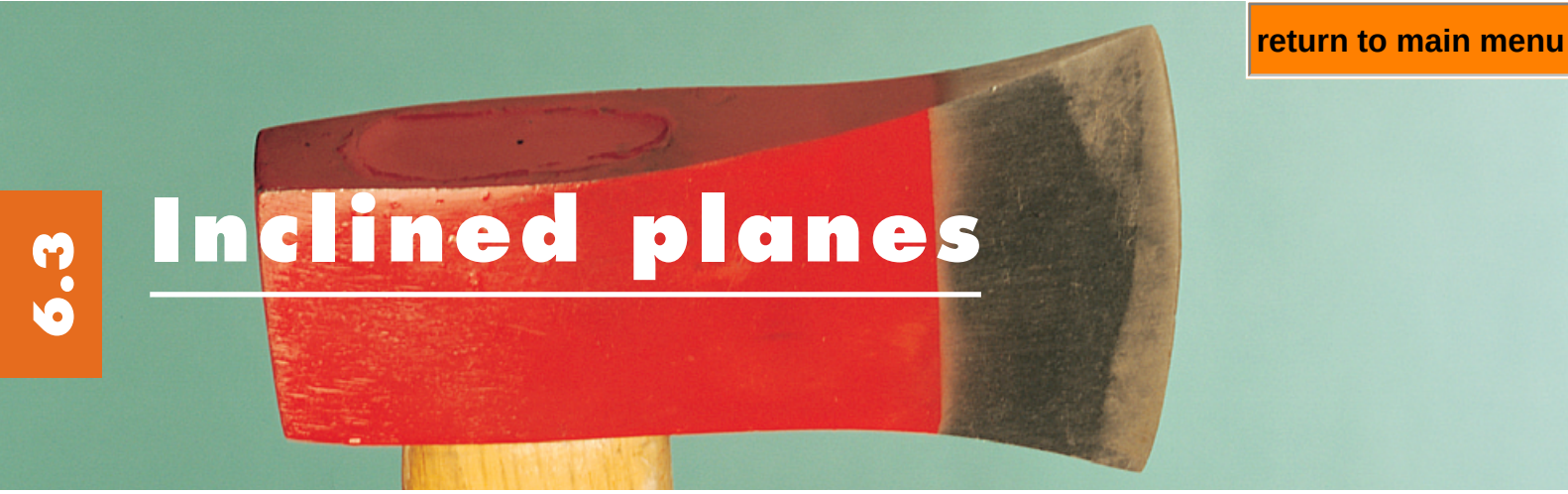

6.3

Inclined planes

FIGURE 6.17
Is this a double ramp?

FIGURE 6.18
Ramps make it easier to roll barrels from one place to another.

FIGURE 6.19
Water skiers gain greater distances by going over a water ramp.

We have examined one machine, the lever, which reduces the effort in doing things like prising open lids or moving heavy objects. There are many other types of machines which can make your life easier!

The ramp—a machine that makes things more convenient

Believe it or not, a **ramp** is a machine. Rolling a small barrel up a ramp is much more convenient than lifting it to place it in the back of a van. Roads that go up mountains are winding because it is much easier for a car to drive up a gentle slope than a straight, steep slope.

The scientific name for a ramp is an **inclined plane**. Inclined planes reduce your effort. However, to do this, you have to move your effort through a greater distance.

FIGURE 6.20
Lifting a car to change a flat tyre would be difficult without a car jack.

FIGURE 6.21
Spiral staircases enable people to move upwards with less effort.

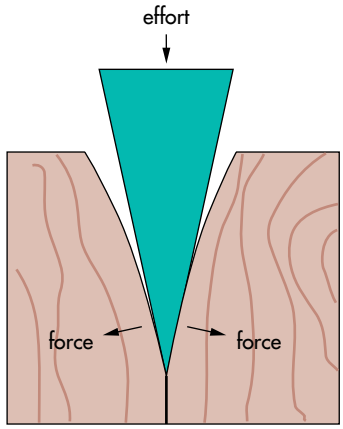

FIGURE 6.22
The two sideways forces act to split the wood.

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EXPERIMENT 2 THE TURN OF A SCREW

When you drive a screw into wood, the effort you apply is over a very large distance compared to the depth you have to sink the screw into the wood, which might be only 2 cm.

AIM

To compare the turn of a screw with the depth it is sunk.

MATERIALS

- A4 piece of paper
- scissors
- pencil

METHOD

1 Cut the paper into two triangular halves.

2 Measure the length of the longer side of the triangle. Call this length *T*.

3 Wrap the triangular piece of paper around a pencil.

4 Measure the length of the pencil.

Note: If the pencil were a screw, then *T* would be the distance you would have to turn to sink it a distance equal to the length of the pencil.

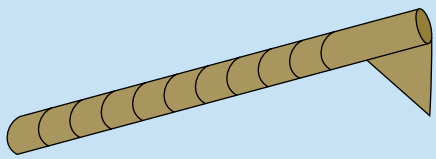

FIGURE 6.23
Wrap the paper around a pencil so that it looks like a screw thread, only flat.

The wedge

Axes and **wedges** can be used to split wood, an axe directly and a wedge by being struck with a sledge hammer.

Their shape is actually that of a double inclined plane. Figure 6.22 shows the effort and forces involved when an axe or a wedge is driven into wood.

The screw

Ramps, slopes and wedges are not the only examples of inclined planes. A **screw** is an inclined plane in a spiral shape. You can think of a screw as being an inclined plane that has been wrapped around a cylinder.

It is not all gain

When you push a barrel up a ramp, the ramp carries much of the weight of the barrel. If you lifted it straight up, you would have had to carry all the weight. However, you have to push the barrel a greater distance. The same applies to a screw whether it is part of a screw-clamp, a car jack or a wood screw.

Questions

1 Explain, in your own words, why an inclined plane is a machine.

2 Explain why meat mincers, axe blades and car jacks are examples of inclined planes. In your explanation, you should think about the distance that you would have had to move without the machine, and the distance you do have to move when using the machine.

3 Some corkscrews, axes with handles and staplers involve both levers and ramps.

a Describe how a corkscrew reduces the effort required to open a bottle of wine.

b Describe how an axe reduces the effort needed to chop up wood.

4 Explain why stairs and escalators perform the same function as a ramp.

5 Can you think of other examples of ramps being used to make work a little more convenient?

Extra worksheet

FACT FILE

In 70 AD, the Romans built a ramp in order to attack a Jewish fortress on the top of a 434 m high mountain near the Dead Sea.

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6.4

Wheels, axles and gears

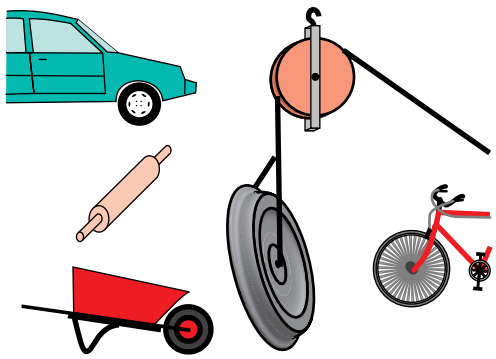

FIGURE 6.24
Wheels are an important part of everyday life.

FACT FILE

The Inca Empire of South America reached its peack in the 1500s. Despite having an advanced agricultural and road system the wheel was never adapted for transport.

Wheels—special types of levers

Wheels are used in almost every part of our society, from a child's pram and radio volume controls to the most sophisticated control mechanisms in transport, in mining, in manufacture and in entertainment. They make it much easier to move loads, whether they are large or small.

A wheel on an axle is a special kind of lever.

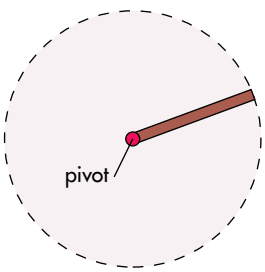

FIGURE 6.25
Imagine a rod pivoted at one end. The other end rotates about the pivot.

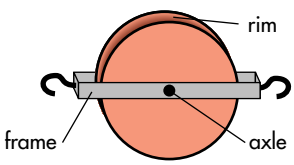

FIGURE 6.26
A single pulley. A wheel with a rim.

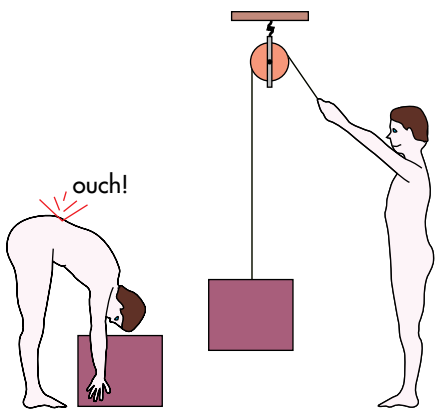

FIGURE 6.27
Many lifting tasks may cause back strain. Machines can reduce the strain.

The lever in this case is the wheel. The wheel revolves around the fulcrum which is in the centre of the axle.

Like all machines, wheels can make a job easier by amplifying the force we apply to a job.

A wheel can also be used to make it easier to lift things by changing the direction of the force.

Pulleys

A **pulley** is like a wheel with a rim. Like a wheel it acts like a lever with the axle as the pivot.

A single pulley alters the direction of a force, making many lifting jobs easier.

More complicated pulley systems not only alter the direction of force but can also alter the force needed.

Gears—wheels with teeth

Turning one wheel against another will make the second wheel turn, especially if the surfaces in contact are rough.

The trouble with wheels is that they sometimes tend to slip or stick. To make them work together smoothly, teeth are cut into them to make them into **gear wheels**.

Questions

1 Give two examples of how wheels can make life easier.

2 In what two ways can pulleys be used to making lifting heavy objects easier? Diagrams could help your explanation.

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EXPERIMENT 3 INVESTIGATING PULLEYS

Spreadsheet

AIM

To compare two pulley systems.

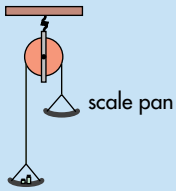

FIGURE 6.28

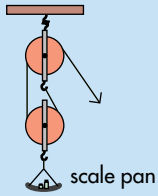

FIGURE 6.29

METHOD

1 Set up the pulley systems in Figures 6.28 and 6.29.

2 Find what mass is needed to just balance a 100 g mass in each case.

3 In each case, pull the mass on the end of the string a distance of 10 cm. Measure how far the other mass moves.

RESULTS AND DISCUSSION

Record your results in a table.

What do we gain from using a pulley system and what do we lose? How does increasing the number of pulleys used relate to the size of the load needed to balance the 100 g?

EXTENSION

Design a pulley system using three or four pulleys. Even if you do not set up the system, you can suggest what the gains and the losses are in using this system.

USING SCIENCE 3 CHANGING SPEED AND DIRECTION

Copy and label each diagram to show the direction of each driven wheel and whether the driven is faster, slower or the same speed as the driver. Explain your reasoning.

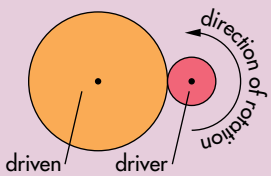

FIGURE 6.30

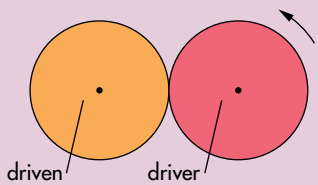

FIGURE 6.31

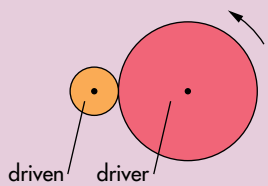

FIGURE 6.32

3 The gear arrangement of an egg beater ensures that the beater operates at high speed. It also changes the direction of movement. Explain, with diagrams, how it does this.

4 Figures 6.30–6.32 are the gearing arrangements in three different bicycles. Consider the distance moved by the bicycle wheel for one turn of the pedals in each.

a Which arrangement would you use to ride a bicycle very fast?

b Explain what the other arrangements would be suitable for.

c Which arrangement would you use for a bicycle with only one gear? Check your answers if you have access to a suitable bicycle.

Activity A

Using beer bottle tops, tacks and a piece of wood, make systems of gears that reverse the direction of motion. Simply nail the tops loosely so that they are touching each other. Draw your gears, showing how they work.

What other objects could you use to make larger or smaller gear wheels?

Extension worksheet

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6.5

Types of machines

FIGURE 6.33
Car engines consist of many simple machine components.

A **simple machine** does only one thing. There are six types of simple machines: the lever, the inclined plane, the wedge, the screw, the wheel and axle, and the pulley.

Complex machines are made up of two or more simple machines. There are many complex machines around you.

Engines are powered machines. They need an energy source, such as electricity or from burning fuel, to work.

FACT FILE

The largest bicycle made so far had a wheel diameter of over 3 m. Guinness Book of Records 1994

TABLE 6.1

Type of machine	What it does	Examples
Simple machine does only one thing	**either** makes it more convenient to do something **or** reduces the effort needed	A lever, an inclined plane, a wedge, a screw, a wheel and axle, and a pulley
Complex machine is made up of two or more simple machines	**either** makes it more convenient to do something or reduces the effort needed or both	A wheelbarrow consists of levers and wheels; a stapler consists of levers and wheels and wedges
Engines use a source of energy like electricity, gas, wood, coal, oil and petrol to perform some task	Same as complex machines	A petrol engine in a lawn mower, or an electric motor in a blender

USING SCIENCE 4 THE BICYCLE AND THE BODY

FIGURE 6.34
A bicycle with one gear arrangement.

Describe the three parts of the bicycle highlighted in Figure 6.34. Include in your description what sort of simple machines are used.

There are about 206 bones in the human body and many of these can be moved. The joints holding these bones allow movement. Just like in a bicycle, there are three types of joint in the human skeleton. Copy and complete the table comparing the bicycle to the human.

TABLE 6.2

Type of joint	Human example	Bicycle example
Hinge	elbow and knee joints	
Pivot—turn from side to side	head–neck joint	
Ball and socket—turn many different directions	shoulder joint	

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USING SCIENCE 5 YOUR SKELETON

The bones making up the human skeleton are hollow. They develop special strengthening near the joints where forces have the greatest effect. Hollow tubes are much stronger than solid ones of the same mass. Bicycles have hollow tubes in their frames for the same reason. Many other structures are made with hollow tubes, for example the chair you are sitting on.

AIM

To show that hollow tubes are stronger than solid ones of the same mass and material.

WHAT YOU NEED

- plastic straw
- thread
- small coins
- two glasses
- plastic lid

WHAT TO DO

1 Rest a straw on two glasses.
2 Hang a small pan made of a plastic lid from the centre of the straw.
3 Gradually add coins until the straw just bends.
4 Repeat the experiment using a flattened straw.
How do the two loads compare?

EXPERIMENT 4 THE ARM AS A LEVER

AIM

To make a model arm.

MATERIALS

- cardboard pieces
- rubber bands
- fasteners

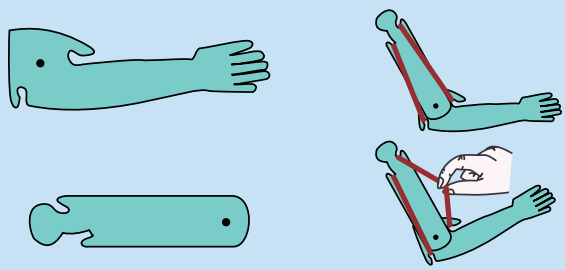

FIGURE 6.35

METHOD

1 Copy these shapes onto cardboard.
2 Cut them out.
3 Use paper fasteners to join the pieces.
4 Fix rubber bands onto the pieces as shown.
5 Operate the arm by pulling the rubber bands.
6 Use the cardboard model to show how the arm can be used as a lever by hanging a 100 g mass on the 'forearm' at various distances from the elbow.

DISCUSSION

1 When do the 'muscles' (rubber bands) in the upper arm have to do most work?
2 Where is the pivot in the arm?
3 Draw your own arm and label the fulcrum. Using arrows, show where the load and the effort are acting when you lift a bag with your forearm horizontal.

Questions

1 What is a simple machine? Give two examples.
2 What is a complex machine? Give two examples.
3 What is an engine? Is it a simple or complex machine? Explain.
4 Classify the following as either simple or complex machines:
a hammer b wheelbarrow c axe d bicycle e crane

Find out

Find out about a machine that is used to assist a person with a disability. For example, you might investigate devices that use a small movement of a part of the body such as the eyelid. Present a poster which shows how the machine works.

Web page

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6.6

Machines need energy

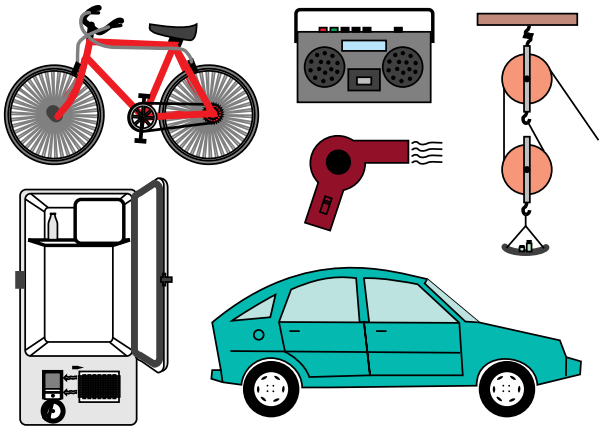


FIGURE 6.36
Imagine life without some of the machines we have around our home! Machines need energy to work.

Machines and energy

Something is said to have energy when it is capable of doing something, for example, cause a change. When you lever the lid off a coffee tin, your muscles supply the energy to cause the change.

USING SCIENCE 6 FORMS OF ENERGY

Energy comes in many forms. The tables show some of the different forms of energy, with examples.

TABLE 6.3

Type	Example
Static electricity	A rubbed balloon stuck to a ceiling
Gravitational	Anything lifted up, e.g. water in a dam, a hammer
Elastic	A stretched rubber band, a compressed spring, a hacksaw blade under tension
Nuclear	Energy stored in a nuclear pile
Magnetic	A magnet floating above another one in a beaker

TABLE 6.4

Type	Example
Motion	Anything moving
Light	Light from a globe, sunlight
Heat	Produced by rubbing, by a fire or light globe
Sound	Voices, music, thunder
Electricity	Electricity in wires

1 What do you think are the differences between the two tables?
2 What is common in each table?

EXPERIMENT 5 HUMAN-GENERATED ELECTRICITY

AIM
To convert your energy into electricity.

MATERIALS
- bicycle with a dynamo

METHOD
1 Turn the bicycle upside down.
2 Press the dynamo against the wheel and spin the wheel.

DISCUSSION
1 Where does the electrical energy come from? If your answer is the wheel, where does the wheel get its energy from?
2 Another way of making the wheel go round is to make use of the energy of a falling object fixed by a string to the wheel. Where does the energy to turn the wheel come from in this case? Think of other sources of energy that could be used.

eGlossary

Machines need energy to perform their functions. This energy can come from muscles or from sources such as petrol, gas, oil, coal, wood or electricity.

We can usually tell when a machine has energy because it is able to produce a change through a push, a pull or a twist.

Two kinds of energy

Energy can be stored in a battery, muscles or in petrol, or it can be immediately available in the form of running water, electric current or a moving marble on a ramp. All forms of energy are either one or the other—stored or movement.

Potential energy is the scientific name for *stored* energy. Potential energy is energy that is available for use. To be able to do things, it must be released by some process such as burning or releasing a coiled spring.

Kinetic energy is *motion* energy. Kinetic energy is in a form that can be used to do things immediately.

TABLE 6.5 Forms of energy.

Potential (stored energy)	Kinetic (motion) energy
Gravitational	Heat
Chemical	Light
Elastic	Sound
Magnetic	Motion
Static electricity	Electricity
Nuclear	

Efficiency

When we lift a mass up using a single pulley system, some of the energy is lost overcoming the **friction** between the wheel and its bearings. Even if we grease the bearing, there is still a little loss of energy through friction. Whenever a machine uses energy to do some work, some of that energy is always lost. **Efficiency** is a measure of how much energy a machine produces as useful work.

Questions

1a Identify the sources of energy for each of the household machines shown in Figure 6.36.

b What would happen if the machines in the drawing did not have energy?

2 What sort of energy is found in a battery? How can the energy in a battery be released, for example for use in a torch or TV remote control?

3 What sort of energy do a falling brick, electric current and light have?

4 A calculator uses electrical energy from a battery or a solar cell. Is the source of the electrical energy the same in each case?

5 Movement on ice is said to be virtually friction free, but why do we stop sliding even on ice?

6 A diesel engine is said to be 35% efficient compared to an electric motor which is 80% efficient. What do you think this means?

Think about

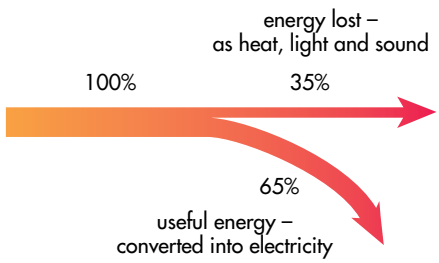


FIGURE 6.37

Figure 6.37 shows the energy flows in a power station. Draw similar diagrams showing the energy flows for:

1 a runner who only uses 15% of the energy from food to maintain movement

2 a car in which 10% of the energy is used to maintain movement, while 15% is lost in the gears and 75% is lost in heating up the engine.

FACT FILE

The energy in coal, oil and gas comes from sunlight that was absorbed by plants many millions of years ago. Heat and pressure slowly converted plants into coal, while oil and gas come from long dead plants and animals. The animals obtained their energy by eating plants.

Foundation worksheet 1

Internet assignment

eGlossary

HISTORY OF SCIENCE

Machines through the ages

FIGURE 6.38
An Egyptian shadoof. If a field needs 20 L of water per day for each square metre and the shadoof holds 10 L, how many times would it need to operate to irrigate a field of 450 m²?

Throughout history many cultures have contributed in some way to the development of the world as we know it today. Many of these contributions have been in the form of improved machines.

There are just six **simple machines**, with more **complex machines** being a combination of these simple machines. Some of these simple machines have been used for all of recorded history.

The ancient Egyptians made many advances in the use of simple machines. This is not surprising considering they were at the height of their power for over 3000 years. To build the pyramids during the Old Kingdom, 2686–2181 BC, they probably used huge **inclined planes** (ramps) to assist in getting the enormous boulders up to the building level. The boulders were broken up by using a very early type of chisel (double-sided **wedge**). Later on, during the New Kingdom, 1550–1069 BC, the Egyptians drew water from the Nile by using an application of a **lever** called a shadoof.

Between the Tigris and Euphrates Rivers, parts of which we now call Iraq, was an area known as Mesopotamia. It is believed that it was here that the **wheel** and **axle** was first used around 3500 BC. As with many inventions it had a military application—in this case, the chariot.

You have probably all seen a well with a bucket attached to a rope. This is really an application of a **pulley** and has been used in the Middle East since at least 800 BC.

Archimedes, one of the many famous Greek mathematicians, philosophers and scientists, wrote about the **screw** around the year 236 BC. In particular he described a water-lifting device which is now known as Archimedes' screw. It was actually an inclined plane twisted into the spiral of a screw.

FIGURE 6.39
(below left)
Archimedes' screw in action. It provided a continuous flow of water as long as the user had the energy to keep turning it!

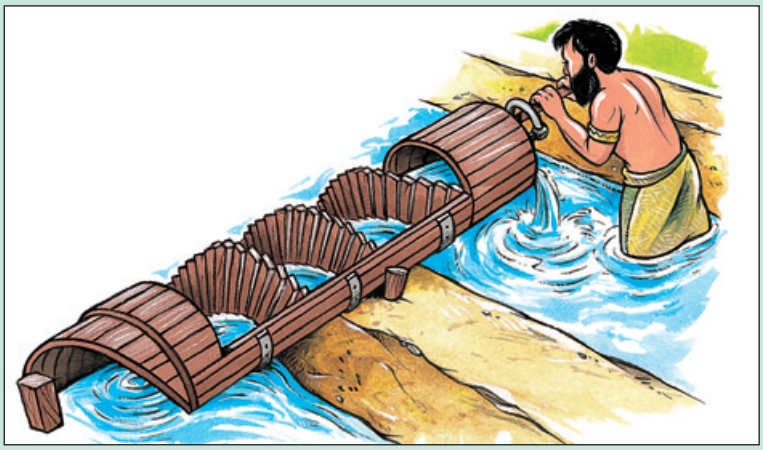
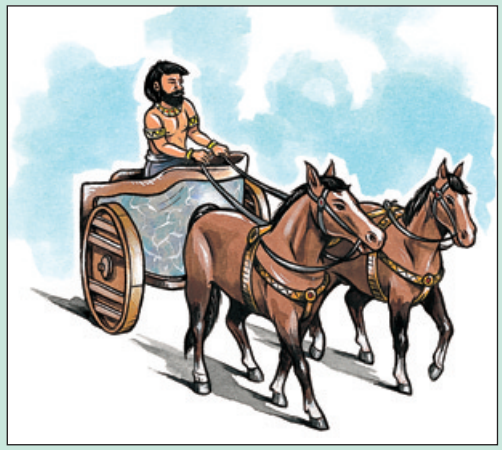


FIGURE 6.40
(below right)
A Mesopotamian chariot—notice the solid wheels, these were probably made from wood.



USING SCIENCE 7 MAKING A MACHINE

WHAT TO DO

Design and construct a machine that involves some combination of the simple machines studied in this chapter. Your machine could be a toy, or it could be used to make our lives easier and simpler!

A FEW HANDY HINTS

- Make it simple—the more complex a machine the longer it will take to build, let alone get to work!
- Decide on its purpose—if it is a present for someone make sure they would like it. If it is a machine for class demonstration make sure it is big enough for everyone to see it.
- Make a sketch—this will help to make your idea clearer to your teacher, as well as to you!

A FEW SUGGESTIONS

- Use Lego blocks a make a machine. This might be a crane that can lift a load, carry it somewhere and release it.
- Use bottle tops as gears to make a simple calculator that counts up to hundreds.
- Make a model of a simple irrigation system that includes a well and a means of transporting the water over some distance.
- Have a look at some of the fantastic inventions drawn by William Heath Robinson (see Figure 6.46 in the review section at the end of this chapter) or the impossible but beautiful drawings by M.C. Escher. Use these as inspirations for your own design!

Present your model to the class with a brief explanation of the simple machines involved.

Questions

1 Draw a simple diagram of the shadoof showing how it uses the principles of the lever. What sort of lever is it?

2 Many early simple machines were used to raise water; these include the lever, pulley and screw. Why do you think this was this such an important part of life in the ancient world?

Think about

What advantages do you think Archimedes' screw had compared to the shadoof and the well bucket? Which had the greatest potential to become a powered machine?

Activity

Draw a timeline that shows each of the simple machines mentioned on the opposite page. Add at least ten other machines to the timeline including the wheelbarrow, the printing press, the steam engine and the water mill. You could research this in the library or use the Internet.

Find out

1 In the past many advances in machines came from military needs. Today, space research provides similar advances. Find about some machines that have been produced or improved as a result of space research.

2 Trace the development of the bicycle by finding out what contribution each of the people listed made. Also include their country of origin: Baron Karl de Drais de Sauerbrun in 1818; Kirkpatrick Macmillan in 1839; Pierre and Ernest Michaux in 1861; James Starley in 1870; John Dunlop in 1888.

FACT FILE

Archimedes once used a compound pulley system to single-handedly pull a grounded galley up onto the beach.

Foundation worksheet 2

eGlossary

IMPLICATIONS FOR SOCIETY AND THE ENVIRONMENT

Machines and progress?

FACT FILE

Opposition to new machines has a long history. Workers in factories in France in the late 1800s threw their wooden clogs, called sabot, into their textile machines to break the machine. This is the origin of the word sabotage.

FIGURE 6.41
A Pentium III microprocessor. It can keep track of the computer operations coming from it and can provide identification information about the user.

From Stone Age to Bronze Age to Iron Age and beyond, technology has continued to advance. Many advances in technology have been welcomed because they have made life easier, safer or healthier. But do improvements in technology always lead to a better world?

Innovations in technology have resulted in weapons of mass destruction such as the machine gun, or even the atomic bomb. The technology that allowed mass production of items such as the car or woollen fabrics also caused mass unemployment and the resulting social problems. We have benefited greatly from the development of the computer, but this has also resulted in a world where our privacy is potentially threatened by the ease with which personal data can be obtained. George Orwell, in his novel *1984*, described a world where every action of every person is watched over by an all-seeing government. Could this be coming true?

Some groups today have turned away from modern technology. The Amish people in Pennsylvania, United States, believe that a simple lifestyle is important to their religion, and that modern devices make people lazy and greedy for more luxury. They still use horse-drawn carriages, don't use electricity, and some don't even use buttons. In Australia, as well as many other parts of the world, some people band together to live in self-sustaining communities which are often called 'communes'. Many of these communes only use very simple machines and do most of the work by hand because they believe that it is good for the environment and the soul.

Advances in technology have advantages and disadvantages. The development of robots in factories has meant that people don't have to do as many dangerous or

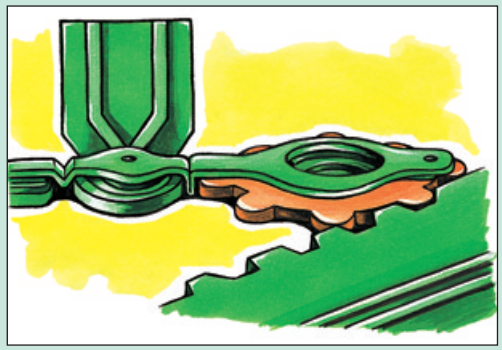


FIGURE 6.42
This silicon gear system is smaller than a human hair. It is an example of a nanomachine. Could nanomachines be the 21st century equivalent of the transistor and the silicon chip?

USING SCIENCE 8 CLASS DEBATE

As a class debate the topic: Advances in technology are always good for society.
Organise a class debate as follows.

1 Divide the class into two groups. One group will argue that advances in technology are always good for society (the affirmative side) and the other group will argue that advances in technology are not always good for society (the negative side).

2 Three students should be chosen from each side to be the speakers.

3 Each team may need to do some research to organise their arguments. Those students who are not speakers should form the group's 'research team'.

4 Before the debate, decide as a class how the debate should be judged. You could bring in a guest judge or set up a panel of students who are not speakers. Scoring should be based on criteria such as content and research, presentation, and strength of argument. The class should also agree on a strict time limit for each speaker.

5 During the debate, speakers from each side take turns in presenting their argument. Students who are not speakers can play the role of the audience and ask questions of the speakers, at the appropriate time.

boring jobs, and the increased efficiency has meant lower prices and higher quality because of higher accuracy. But the introduction of such technology has also led to job losses, and enormous amounts of money are needed to set up such techniques. Can you think of any other advantages or disadvantages?

Questions

1 Make a list of all the powered machines you use in a typical day.

2 In your list of machines for question 1 put a line through the ones you think you could do without.

3 For each of the remaining machines try to give a non-powered alternative (e.g. a bicycle instead of a car).

Activity

For each of the following machines make a list of the positive features and a list of the negative features associated with it:

- **mobile telephone**
- **'smart' bomb**
- **computer**
- **automobile**
- **microwave oven**
- **nanobots, which can kill all disease organisms and cancer cells in a person's body.**

Compare your list with that of at least one other student. Do you agree?

Think about

What aspects of today's technology could you not do without? Imagine you are setting up your own alternative-lifestyle commune. What features would it have? How would you describe it to other people who might like to join it?

FIGURE 6.43
Amish people working the fields. What advantages would there be to this lifestyle?

FACT FILE

Many of the scientists involved with the development of nuclear weapons later decided that they should be banned. These scientists included Enrico Fermi who led US development of the atomic bomb and Andrei Sakharov who led work in the Soviet Union.

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Using scientific language

1 Define a machine. List five simple machines. Now explain what is meant by saying a machine is simple.
2 What is the difference between a simple and a complex machine? Illustrate your answer with examples.
3 The crowbar has the pivot between the point where the load force acts and the point where the effort force acts. What type of lever is it? Explain the differences between the three types of lever, giving at least one example in each case.
4 What are the two types of energy? Explain the difference between them, giving examples.

Check your knowledge

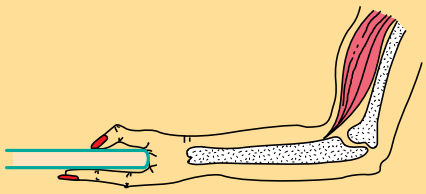


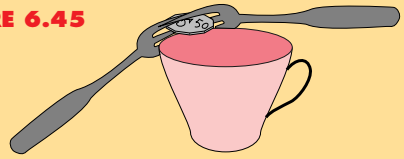
FIGURE 6.44

1 Copy and label the pivot with a *P* and the direction of the load force and effort force with *L* and *E* respectively.
Which class of lever is the arm?
To where would you move the book so that it exerts the minimum turning effect about the pivot, keeping the arm horizontal?
2 What sort of machine is a drill bit for making holes in wood? Explain.
3 What sorts of energy does a lemon have as it falls off a tree?
(Hint: think about the type of energy it had just before it fell.)

Apply your skills

1 Why are spanners for car wheel nuts made fairly short?
2 A rolling pin is a wheel on an axle. What do we gain by using it and what do we lose?
3 Some people always want to get something for nothing. They would like to have machines which are 100% efficient so there is no energy loss. Explain if this is possible.
4 How would you test the stability of a double decker bus with a large load on its top deck? How is the bus like a lever? Where should the centre of gravity of a bus be, even when it is fully loaded on the top deck and empty of passengers on the bottom deck?
5 Balance a 50 cent coin on the edge of a glass. Try it on its edge or on a flat face. It can be done by overlapping the prongs of two forks and slipping the coin between the prongs. Explain why you can balance the coin in this way. Include the terms 'turning effect' and 'centre of gravity' in your explanation.

FIGURE 6.45

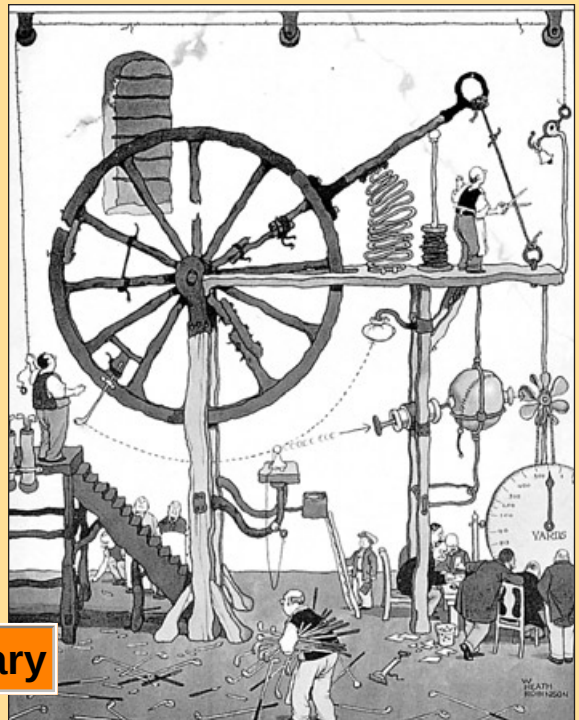


6 An egg beater would not be good for mixing dough. What sort of gear arrangement would a dough mixer use?
7 A particular steam engine is only 5% efficient. What does this statement mean? The energy to drive the steam engine comes from burning wood. 20% of the energy is used to overcome friction, 5% in producing noise, 60% in heating up the engine itself, and the rest is lost with steam that escapes. Draw a diagram to show the energy flows in the steam engine.

Challenge yourself

William Heath Robinson (1872–1944) was an artist whose name has passed into the English language. A 'Heath Robinson' machine is a comic mechanical fantasy. Draw your own Heath Robinson machine.

FIGURE 6.46
'Testing Golf Drivers' by William Heath Robinson.



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return to main menu

CHAPTER 7

REPRODUCTION

All living things reproduce. It is a basic characteristic of life. For most large animals reproduction involves two parents, one male and one female, who together produce offspring. Not any more! In 1997 scientists in Scotland produced a female sheep with only one parent. The parent sheep was in fact dead and its tissues had been frozen. The scientists took an udder cell from the frozen tissues and removed its nucleus. They then removed the nucleus from an egg cell of another sheep. The nucleus from the udder cell was injected into the egg cell. This egg cell was then implanted into a third sheep which acted as a surrogate mother. Five months later, Dolly, the first cloned sheep, was born. Dolly was an identical copy of the dead sheep whose tissues were used to supply the nucleus.

What was special about Dolly was that after 277 attempts the scientists produced a clone from the nucleus of an adult organism. Before this, it was thought that adult cells lost their ability to divide and develop into a new individual. Dolly has since reproduced herself in the old fashioned way, giving birth to a lamb called Bonnie in 1998.

In this chapter you will learn about reproduction in simple and complex organisms, as well as some of the ways science and technology are altering reproduction.

Web links

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- **Distinguish between sexual and asexual reproduction and give examples of each.**
- **Grow a plant from a cutting.**
- **Dissect and label the reproductive structures found in flowers.**
- **Compare and contrast reproduction in different types of mammals.**
- **Identify the major structures involved in human reproduction and give their functions.**
- **Explain the major steps involved in human reproduction, culminating in birth.**
- **Discuss how current research into cloning might affect peoples lives.**
- **Explain what is meant by reproductive technology and analyse some of the associated issues.**

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7.1

Reproduction

FIGURE 7.1
(above)
This animal is a zeedonk. Who are its parents?

FIGURE 7.2
(above right)
Although the child is the offspring of these two parents, and has some characteristics from each parent, she is unique.

No organism can live forever. All living things reproduce and pass on their characteristics to their offspring through reproduction. In this way the continuation of every species is ensured. A **species** is a single type of organism.

There are two types of reproduction in living organisms: sexual reproduction and asexual reproduction. Some organisms are capable of both.

Sexual reproduction

Sexual reproduction involves two individuals who combine their sex cells (called **gametes**) to form a new individual with characteristics of both parents. This union (or fusion) of gametes is called **fertilisation**. It results in the production of a unique individual. With the exception of identical twins no two organisms resulting from sexual reproduction have exactly the same features.

Can different species mate?

Sometimes two different species mate to produce an offspring such as a *liger* or a *geep*. These are known as **hybrids**.

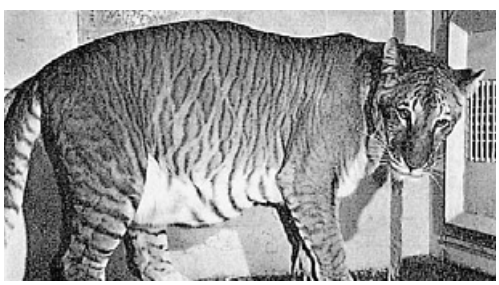

FIGURE 7.3
This animal is a liger. Who are its parents?

Hybrids are sometimes fertile, that is, able to produce offspring, but more often they are **sterile**. For example, a mule is a sterile hybrid of a donkey and a horse.

Asexual reproduction

Asexual reproduction requires only one parent and does not involve the union of gametes. It results in offspring who are the same as the parent.

There are five methods of asexual reproduction: budding, vegetative propagation, fragmentation, spores and fission.

Fission occurs in single-celled organisms such as bacteria and protozoa. In fission the parent cell simply divides into two equal parts, each of which develops into a new organism.

Fragmentation is like fission but occurs in multicellular organisms, particularly animals like worms.

Budding occurs in both single-celled and multicellular organisms. A bud grows off a parent to become a new individual.

Spores such as seen on the underside of fern fronds contain a number of reproductive cells which will grow into new

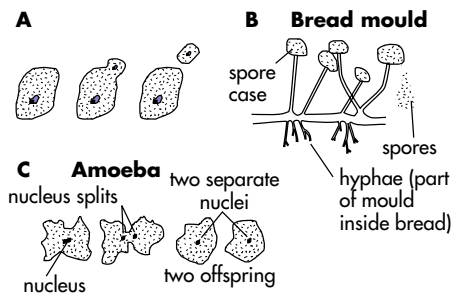

FIGURE 7.4
Which method of reproduction is shown in diagrams A, B and C?

Extension worksheet

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individuals if they meet suitable conditions.

Some flowering plants are able to reproduce asexually as well as sexually. They are able to reproduce by growing parts which separate from the parent plant to form new plants. The process is known as **vegetative propagation**. For example, strawberry plants send out a 'runner' which puts down roots and grows into a new strawberry plant. As only one parent is involved it is a form of asexual reproduction. Each strawberry plant will look like the next.

We can make plants reproduce asexually. A cutting taken from one plant and placed in the soil may grow roots from the cut end of the stem. The cut branch becomes a new plant.

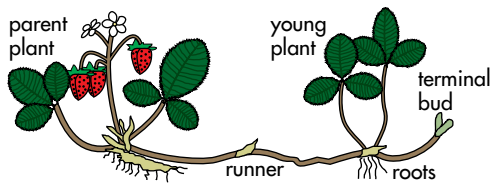

FIGURE 7.5
Strawberry plants can reproduce asexually by vegetative propagation.

EXPERIMENT 1 PLANT PROPAGATION

Spreadsheet

AIM
To grow a new plant from a cutting.

MATERIALS
- young fuchsia or geranium plants
- good quality cutting blade
- plant hormone rooting powder
- ice-cream stick
- rooting medium (e.g. commercial propagation mixture)
- 2 pots
- plastic bag
- wire

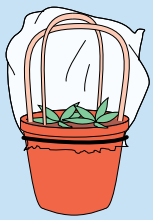

FIGURE 7.6
Propagating plants.

METHOD
1 Select side stems approximately 10 cm long with four leaf joints (nodes).
2 Cut stems from the parent plant close to and just below a leaf node.
3 Strip off the two bottom pairs of leaves.
4 Dip the cut end of the stem into the plant hormone rooting powder.
5 Using the ice-cream stick, create a hole in the rooting medium. Place the cut end of the stem in the hole.
6 Press the rooting medium around each cutting and cover the pot with a plastic bag stretched over two pieces of wire arranged crossways.
7 Keep the pot in a sheltered, shady place and plant out each cutting to a 10 cm pot when roots have developed (about 6 weeks).

RESULTS
Keep a weekly journal of any changes, including height, colour, new growth, amount of water needed.

DISCUSSION
1 Draw a graph showing changes in the height of your cutting over the growth period.
2 Was the plant you grew identical to its parent? Give reasons for your answer.

CONCLUSION
In this method of plant propagation, what type of reproduction is used?

Questions

1 What is reproduction? Why is it necessary?
2 Answer the question asked under Figure 7.4.
3 Design a classification key that distinguishes sexual from asexual reproduction.
4 Draw a table that summarises the main difference between the offspring of sexual and asexual reproduction.
5 What is a hybrid? What parents would be responsible for the following hybrid offspring: wallaroo, geep, hinny?

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7.2

Sexual reproduction in flowering plants

FIGURE 7.7
Many flowers, for example tulips, contain both male and female sex organs within one flower. Other flowers, such as pumpkin or zucchini blossoms, may contain either only male or only female sex organs.

Plants make their own food by using energy from the Sun in photosynthesis. Not only does this provide food for all other living things, it also produces oxygen which is vital for life on Earth. If plants were not able to reproduce, no life could exist on Earth.

Talk about
What sort of things might stop plants from reproducing? What effect would this have on life on Earth? Explain.

Angiosperms

Plants reproduce in many different ways. Most of the plants you are familiar with are **angiosperms** or flowering plants. These range from mighty eucalypts to small herbs and grasses. Flowers are the reproductive organs for these plants.

Some flowers contain only male gametes, **sperm**, which are contained in pollen on the **stamens**. Some flowers

EXPERIMENT 2 FLOWER PARTS

AIM
To dissect a flower and identify the major parts involved in reproduction.

MATERIALS
- flower (your teacher has many different samples, so choose one not being studied by another group)
- dissecting board
- dissecting instruments (scalpel, forceps, probe, scissors)

METHOD
1 Work with a partner.
2 Place the flower on the dissecting board.
Using your dissecting instruments, examine the flower in detail.
3 Identify the parts shown in Figure 7.8.
4 When you have completed your dissection, reconstruct your flower piece by piece by gluing it on a sheet of white paper.

RESULTS AND DISCUSSION
1 Using Figure 7.8 and the information provided, label your flower parts and give a short explanation of their function.
2 Compare your labelled reconstruction with that of other students.
3 What similarities and differences can you find? How do you explain these?

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contain only female gametes, **eggs**, which are stored in the **pistils**. Many flowers, however, contain both types of gametes.

Stamens are made up of two parts: the **anther** and the **filament**. The anther is a sac on the end of each filament which produces the **pollen**. Pistils are made up of three parts: the **stigma**, the **style** and the **ovary**. The ovary can be found at the base of the pistil and contains ovules in which the eggs are stored. The stigma is often feathery or sticky to enable it to capture pollen. The **receptacle** at the base of the flower supports all the reproductive parts.

Pollination

In order for a new plant to be formed the sperm must somehow meet the eggs. The transfer of pollen from the anther to the stigma is called **pollination**. **Self-pollination** is the transfer of pollen from the stamens to the pistils of the same plant. If the pollen is transferred from one plant to another then **cross-pollination** has occurred. Insects and other animals and the wind are responsible for taking the pollen from one plant to the next.

Following pollination, a pollen tube grows down the style towards the egg cells in the ovary. The pollen tube contains the sperm and one of these fertilises the egg. Fertilisation occurs in the ovary. Fertilised eggs grow into plant embryos. Structures develop around each plant embryo, forming **seeds**.

The ovary then enlarges to become a fruit. Fruit structures range from apples, oranges and grapes to structures such as gumnuts, banksia follicles and even wheat grains.

Most of the fruit we eat has been bred for increased fleshiness and flavour and fewer seeds.

Questions

1 Would the method of reproduction in tulips and pumpkin blossoms be sexual or asexual? Explain.

2 Shape, colour and scent are important for flowers that contain only male or only female gametes.

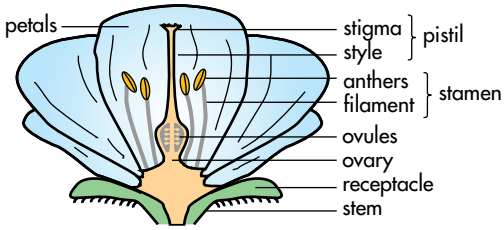


FIGURE 7.8
The reproductive organs of a flower.

Explain why this is so.

3 Some flowers have stamens that extend far beyond their petals. Suggest how this could be both an advantage and a disadvantage to the plant.

4 Would you get much variety in plants if self-pollination took place most of the time?

5 A primary school student has brought you a flower and asked you to identify it as male or female. How will you respond?

Think about

Seeds must somehow scatter over a large area to increase their chances of successful growth.

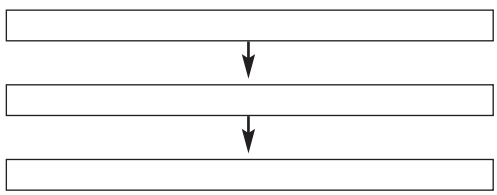
1 Why do you think this is true?

2 Suggest as many ways as you can in which plants may achieve this.

Activity

Using the information in the previous paragraphs, draw a flow chart showing the stages of fertilisation in flowering plants.

A flow chart looks like this:



In the first box you could write:

Pollen sticks to stigma of flower

Continue from here.

FACT FILE

The largest flower in the world is called Rafflesia. This rare flower, which grows in Malaysia, is more than 1 m in diameter and smells like rotting meat.

Word process

FIGURE 7.9
Rafflesia, the world's smelliest plant!

7.3

Sexual reproduction in animals

FIGURE 7.10 (above)
I am a female web-building spider. I am bigger and much more colourful than the males. I remain stationary and let the males come to me.

FIGURE 7.11 (right)
I am a male lyrebird. I have a spectacular tail. I build a mound on which I dance, showing off my beautiful feathers. I even bend my tail right down over my head. As well as dancing I call to my mate.

FIGURE 7.13 (far right)
I am an oyster living in Holland. Ten days after the full moon between 26 June and 10 July I release my sperm into the water to mix with the eggs from the females.

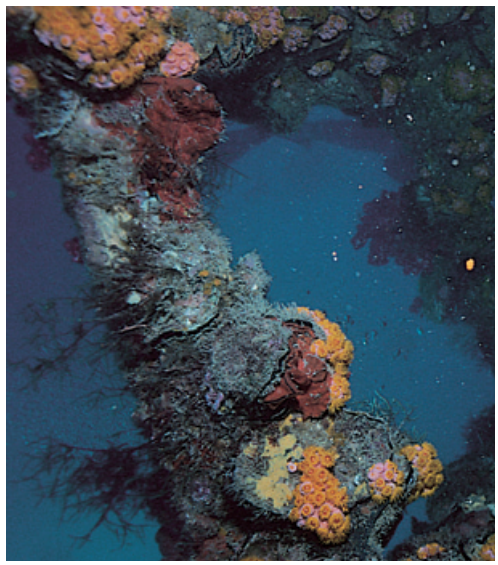
Talk about
Courtship behaviour helps animals find a suitable mate. Discuss examples of courtship behaviour among animals including humans.

The mating game

Once animals of a species are ready to reproduce the two sexes must meet so that mating can occur. This is not as simple as it sounds. Members of a species must mate with the opposite sex of the same species at the correct time, otherwise no offspring will be produced. If not enough offspring are produced the species becomes extinct.

As in sexual reproduction in plants, in most animals male gametes, sperm, fertilise female gametes, eggs. For many animals that live or breed in water this process is **external**, that is, it occurs outside the female's body. Fish and amphibians are two groups who reproduce in this way. The female releases her unfertilised eggs into the water. The male releases his sperm into the water surrounding the eggs. Which of the animals from the five autobiographies is an amphibian?

FIGURE 7.12
I am a male fiddler crab. One of my claws is enormous and brightly coloured. I wave my claw, beckon with it and tap the ground from time to time.



Three other animal groups also lay eggs—birds, reptiles and monotremes. All lay eggs with a tough outer shell. How then do the sperm reach the inside of the egg once it is laid? They don't! Both birds and reptiles use **internal fertilisation**. What does this mean?

All animal sperm can swim in water, but in internal fertilisation the sperm must be introduced into the female where they swim to the egg. The process of introducing the sperm into the female's body is called **copulation**, and is the natural reproductive behaviour of all land-living animals. Even snails, spiders and insects rely on this method of reproduction. This means that when the egg is laid it is already fertilised and will develop into a new organism of that species.

Extra worksheet

FACT FILE

Snails are hermaphrodites because each is both sexes—male and female. Every adult produces both types of sex cells.

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Reproduction in mammals

What do a whale and a kangaroo have in common? An ape and a possum? A bat and a tiger? An echidna and a dolphin? They are all **mammals**.

Mammals are classified into three groups, based on their method of reproduction.

In all cases—monotremes, marsupials and placentals—internal fertilisation occurs through the male and female of the species copulating. For fertilisation to be successful there must be a male sperm and female egg present at the same time.

Mating generally occurs at a time that enables the young to be born when there is increased likelihood of sufficient food and appropriate temperatures. Humans are one of the few animal species that are continuously fertile.

Monotremes

The **monotremes**, like all mammals, have fur and suckle their young. However, they lay eggs like reptiles. The only two monotremes in Australia are the echidna and platypus. They both lay their eggs on land (although the platypus spends a great deal of time in the water). The yolk of the egg provides the developing embryo with the nutrients it needs until it hatches.

Marsupials

Kangaroos, koalas, possums and wombats are all **marsupials**—the second group of mammals. Marsupials give birth to live young after a very short **gestation period**. This means the young are not fully developed when born and must be protected by means of a pouch. For example, the adult red kangaroo, standing at almost 2 m, gives birth to a blind, naked offspring 2.5 cm long after a gestation period of only 33 days. This newborn marsupial crawls up the mother's fur and into the pouch where it is suckled for the next 8 months. During this time the joey develops and grows.

Placental mammals

The third group of mammals includes humans. They are the **placentals**. Placental mammals have a long gestation period after which they give birth to offspring who are well advanced in their development. The offspring are able to remain inside the female until this stage is reached because the placenta allows the mother's and embryo's blood supply to come into close contact with each other. In this way the mother can pass nutrients and oxygen to the offspring, and the offspring can pass wastes back to the mother for removal.

Questions

1 Explain the following terms: external fertilisation, copulation, internal fertilisation.

2 Why is it important that members of the same species are able to recognise each other as the same species?

3 a Summarise the differences between monotremes, marsupials and placental mammals.

b Can one be distinguished from the others on the basis of a single characteristic?

4 Look at Figures 7.10–7.13.

a What type of reproduction is involved in each case?

b What signals does each organism use to attract its mate?

5 Some mammals produce gametes only at certain times of the year or when certain temperatures are reached. Why do you think this is so?

Think about

How do echidnas manage to copulate without spiking each other?

Find out

How long is the gestation period for a human, a seal, a giraffe and an elephant?

FIGURE 7.14
Echidna hatching.

FIGURE 7.15
Many animals have defence mechanisms to protect themselves from predators. The echidna has spines that protect it from other animals. They are particularly effective when it rolls into a ball. These spines or quills are composed of modified hair.

FACT FILE

Temperature decides the sex of a crocodile's egg. Cooler incubation temperatures produce more females whereas warmer incubation temperatures produce more males.

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IMPLICATIONS FOR SOCIETY AND THE ENVIRONMENT

ARKS 'n' SPARKS

Sometimes animals have difficulty breeding in the wild. There may be insufficient numbers, for example because of predators or disease, their habitat may have been destroyed, or they may not have enough food. Zoos and sanctuaries are concerned about the threat to some of the world's species as these changes occur. They have

USING SCIENCE 1 KEPT IN THE ZOO

You are the keeper of endangered species at Alderly Edge Zoo. A flightless bird called the green spatquok is under threat. Its habitat has been destroyed and it is near extinction. You must design a captive breeding program to keep this species alive. You currently have one female green spatquok in your zoo. Listed below are ten items considered essential to the survival of the green spatquok in captivity. Your task is rank the items in order of importance with number 1 indicating the item most necessary for breeding in captivity. Rank these items independently of anyone else by copying the following information into your workbook and recording *your* numbers in the column headed 'Individual Rating'.

Now that you have listed the items in order of importance, you must reach agreement with the people who will be working with you. In groups of four, compare your lists. The group must reach consensus (agreement) as to the order of importance so that the program can begin. There are two rules in this discussion—you are not allowed to force anyone to change their opinion, and you are not allowed to trade off, i.e. you're not allowed to bargain such as 'OK, we'll make yours number 3 if mine can be number 4.' Your group must use reasonable discussion to determine the order, each person having the opportunity to put a point of view. Good luck! Record the group's answers in the column labelled 'Team Rating'. Then compare the two columns by recording the difference between the number you gave an item and the number the team allocated. For example, if you gave 'nesting material' a 7 and the team gave it a 4, the difference would be 3. You may have discovered in your discussion that all the items are important to some degree.

Individual rating		Team rating	Difference
	fully enclosed aviary		
	water		
	suitable male of same species		
	enough space for pair and offspring		
	possible mates from other species		
	special diet		
	natural habitat		
	incubator (machine used to hatch eggs)		
	nesting material		
	second female of same species		
	animal nursery		

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therefore set up captive breeding programs where they try to increase the chance of successful reproduction.

Zoos and sanctuaries have discovered that the most important things affecting the success of a captive breeding program are:

- suitable mates
- sufficient space to keep the pair and their offspring
- an environment as close as possible to the species' natural habitat
- an understanding of the animal's diet and mating behaviours.

Zoos and sanctuaries have to share their resources and information. A good zoo needs to be part breeding centre and part sanctuary for non-breeding animals. Many zoos try hard to copy a species' natural environment, but ultimately conditions in captivity are different. By splitting captive populations among several zoos around the world, the zoos aim to maintain the variety within a species so that if animals can be returned to the wild they have the ability to cope with their natural environment.

It is important to keep track of animals moving around the world as part of captive breeding programs. More than 300 zoos are currently using ARKS (Animal Recording Keeping System) to keep track of over 100 000 animals in 32 countries. SPARKS (the Single Population Analysis and Record Keeping System) also allows zoos to set up breeding programs that achieve the ideal distribution of a species' population.

FIGURE 7.16
Meet the long-footed potoroo.

VENUE: Healesville Sanctuary, Victoria
STATUS: Victoria's most endangered mammal species
NATURAL HABITAT: Scattered areas of East Gippsland and the border of south-eastern New South Wales
CAUSE OF DECLINE: Logging
NATURAL DIET: Underground fungi
BREEDING: Every 37 days between May and November
NUMBER IN CAPTIVITY: 4 adults and 18 young

Questions

1 What is a captive breeding program?

2 What things are most important if a captive breeding program is to be successful?

Think about

Write a paragraph expressing your point of view in response to this statement:
'Captive breeding should be seen as a support, not a substitute, for conservation efforts in the wild.'

Find out

Taronga and Western Plains zoos are just two locations in New South Wales where captive breeding programs are in operation. Both native and non-native endangered species are involved. Use your library and the Internet to research these programs. Which captive breeding programs have been successful enough to return endangered species back into the wild? Prepare a written or oral report to present your findings to your class.

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7.5

Human reproduction. 1

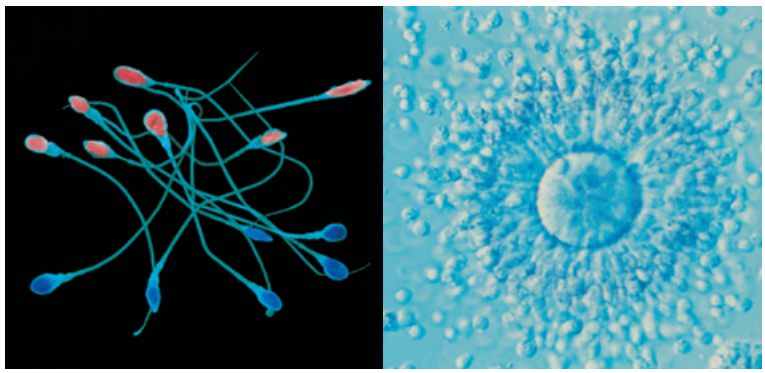

FIGURE 7.17
Human gametes, sperm and ovum.

Humans, like all other mammals, reproduce sexually. Two parents, one male and one female, each produce sex cells or **gametes**. The male gametes are sperm and the female gametes are eggs or ova (singular = **ovum**). Gametes contain half the number of chromosomes found in ordinary body cells. To produce a new individual, a sperm must join with an ovum. This is called **fertilisation**. **Copulation**, which is also called sexual intercourse in humans, takes place first to bring the sperm close enough to the ovum for fertilisation to occur. The new cell formed at fertilisation is called a **zygote**. This cell has a full set of chromosomes; therefore, it has all the necessary information to grow. The zygote will grow and develop becoming an embryo inside the body of the female parent.

The male reproductive system

Sperm cells are produced by the **testes**, which are located in a loose pocket of skin called the scrotum. Each testis consists of thousands of tiny coiled tubes which produce the sperm cells. These tiny tubes join to form the epididymis at the base of the testis. Sperm cells are stored within the epididymis which in turn connects to the sperm duct or vas deferens. The two sperm ducts, one from each testis, join into the urethra as it leaves the bladder. The urethra passes through the penis. In humans the urethra carries urine from the bladder as well as sperm cells to the outside.

There are three important glands located along the urethra which produce components of **seminal fluid**. Seminal fluid plus sperm cells is called **semen**. Seminal fluid nourishes and activates sperm cells. Semen is released from the penis during sexual intercourse and is deposited in the vagina of the female. A single ejaculation can release millions of sperm cells.

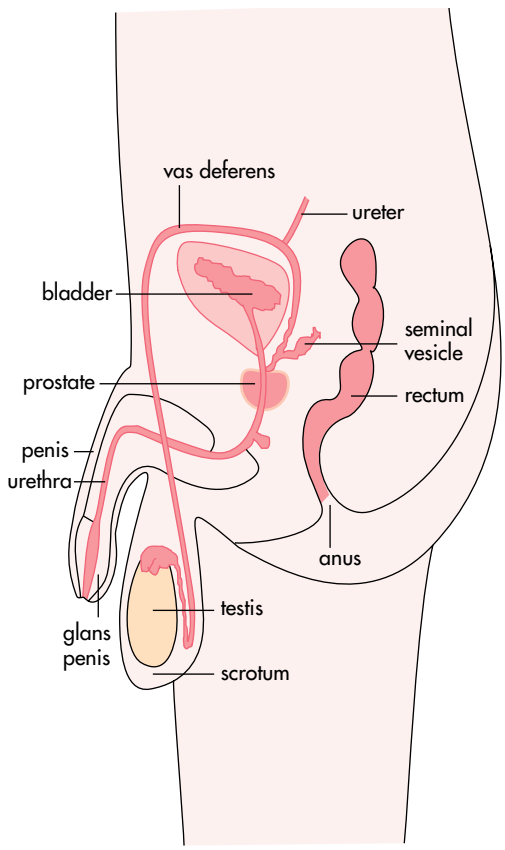

FIGURE 7.18 (right)
The male reproductive system.

FACT FILE

In Australian males, cancer of the prostate gland is the second highest cause of death by cancer, after lung cancer.

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USING SCIENCE 2 OBSERVING REPRODUCTIVE CELLS AND TISSUES

WHAT YOU NEED

- monocular microscope
- lamp
- prepared slides of sperm cells, ova, testis, ovary

WHAT YOU DO

1 Set up the microscope and lamp.

2 Place the prepared slide on the stage. Focus the microscope.

3 Draw a detailed diagram of the tissue. Label your diagram with the name and magnification used.

Puberty

At about the age of 11–14 years a boy's reproductive organs become mature. His testes produce the male sex hormone **testosterone**. The testes also start to make sperm cells. A number of other changes take place at the same time. The penis, testes and scrotum grow larger. Males may have a ejaculation of semen while sleeping. This is called a wet dream and is perfectly normal.

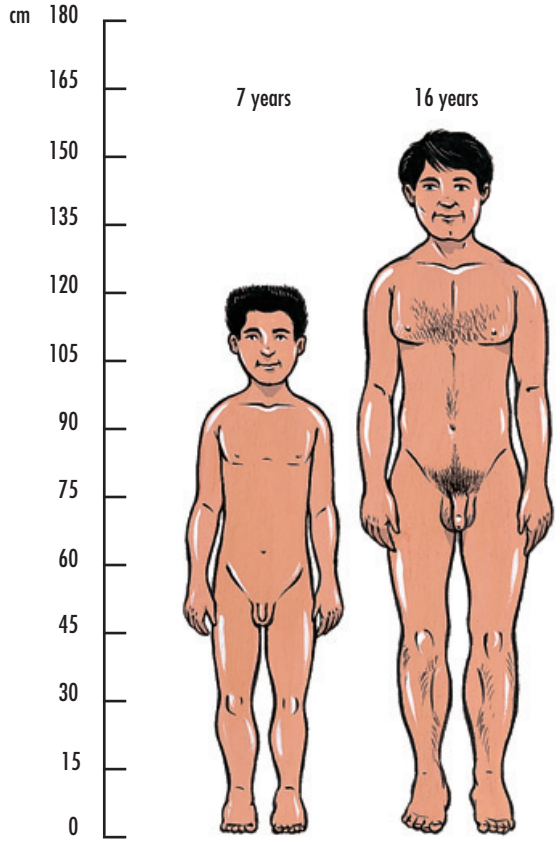

FIGURE 7.19
The changes that take place at puberty are referred to as secondary sexual characteristics.

Questions

1 What is a gamete? How do gametes differ from other body cells?

2 How do sperm cells differ from ova in their structure?

3 Which parts of the male reproductive system:

a produce gametes?

b store gametes?

c transfer gametes?

4 List the structures, in the correct order, through which sperm cells must pass from when they are formed in the testes to the time they leave the urethra.

5 Describe other changes (not already mentioned) that happen to males during puberty.

Find out

1 Why do the testes hang outside the main part of the body?

2 The three glands which produce components of seminal fluid are the prostate gland, the seminal vesicle and Cowper's gland. Find out what is produced by each of these glands and how each component aids the survival of sperm cell and fertilisation.

FACT FILE

A human male can produce up to 86 million sperm cells each day. This would be enough to populate the world in less than 12 weeks.

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7.6

Human reproduction. 2

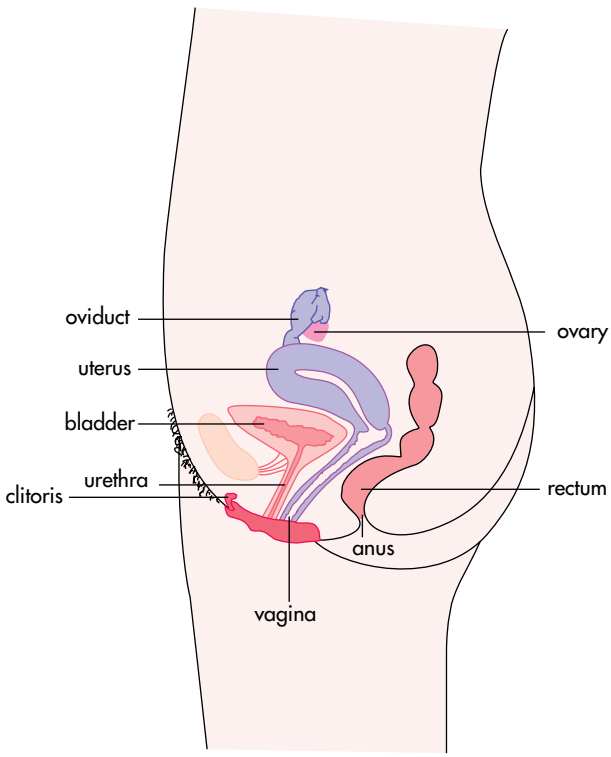

FIGURE 7.20
The female reproductive system.

The female reproductive system

Unlike the male's, a female's reproductive organs are located inside her body. The ovaries are the specialised organs which produce the eggs or ova. The ova produced by a woman are present in an immature form at birth. Each **ovary** will contain around 200 000 eggs. Whereas the testes can continually produce sperm cells, the ovaries usually release only one ovum approximately every 28 days. This is called **ovulation**. Each ovary usually alternates production of ova so that only one is released at a time. Occasionally both ovaries will produce ova at the same time. If both are fertilised then non-identical twins may develop.

The funnel of the **oviduct** catches the ovum and channels it down the tube to the **uterus**. If fertilisation has taken place during the journey, the embryo will implant in the uterus and develop and grow. If not fertilised, the ovum will be discharged through the **cervix** and **vagina**. The cervix is a ring of muscle separating the uterus and vagina. The vagina has three functions. It is:

1 the place where sperm are deposited by the male
2 the passageway for menstrual flow to leave the uterus
3 the passageway for the birth of a baby.

Puberty

At about the age of 10–14 years a girl's ovaries start to produce the female sex hormone **oestrogen**. At the same time her ovaries start to release ova. A number of other changes take place as well. A girl's breasts grow larger, her hips widen and hair grows in her armpits and pubic region. The uterus and the vagina also increase in size.

Once puberty is reached females begin their **menstrual cycle**. The arrival of a girl's first **period** is known as **menarche** and indicates that her reproductive system has reached maturity. Figure 7.21 shows what happens throughout a typical 28 day cycle. This cycle is controlled by chemical messengers called **hormones**.

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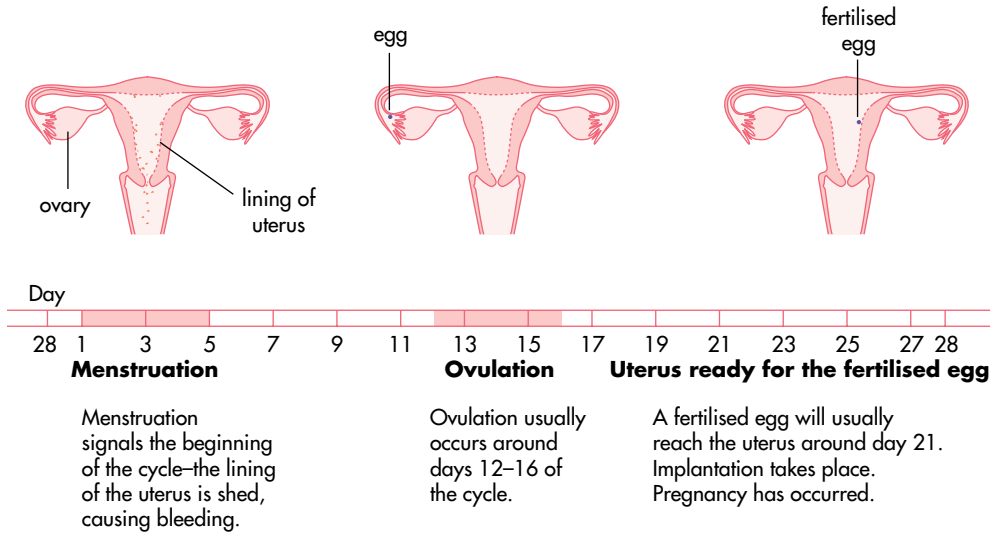

FIGURE 7.21
The menstrual cycle.

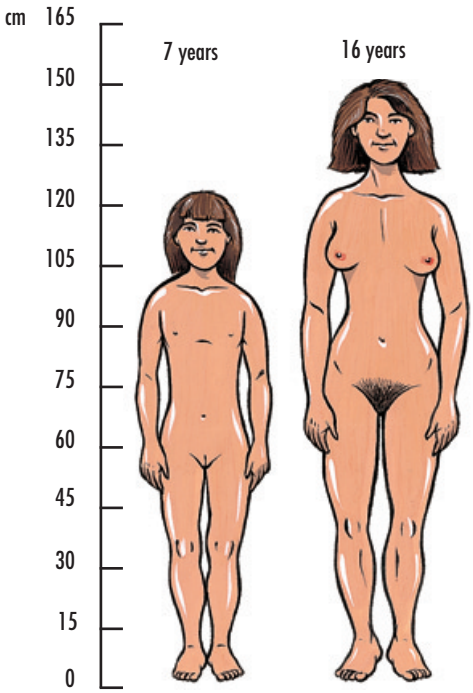

FIGURE 7.22
Female secondary sex characteristics.

Questions

1 Where are ova made?
2 How often are ova released?
3 Which are there more of, ova or sperm cells?
4 Describe how the female sex hormone oestrogen affects the body.
5 What happens to an ovum that is not fertilised?
6 How does the uterus prepare for a fertilised ovum?

Find out

What role do the sex hormones oestrogen and progesterone play in the menstrual cycle.

FACT FILE

Around the age of 50 years, women stop ovulating. Menstruation also ceases. This change is called menopause. Men continue to produce sperm cells well into old age.

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7.7

Pregnancy and birth

FIGURE 7.23
Although millions of sperm are released into the female's reproductive tract during intercourse, only one will fertilise the egg.

Fertilisation

Look back at the calendar showing the **menstrual cycle** (Figure 7.21). It is important to note that 28 days is the *average* length of this cycle. For some women it will be longer and for others it will be shorter. Timing for all other dates including **ovulation** and menstrual bleeding can vary, and even be missed altogether.

If fertilisation takes place, days 15–28 will be vastly different.

There are some important things to note about fertilisation in humans. The small amount of semen released at ejaculation (about 4 mL) contains millions of sperm. Many of these are killed by the acid in the female's vagina or simply are not strong enough or mobile enough to swim through the cervix. The ones that make it through the uterus swim into the oviduct in which the egg is travelling. Only one of these sperm, however, will fertilise the egg. Once the head of one sperm has broken through the egg's membrane, a barrier is set up stopping any other sperm from entering.

How then are twins formed? **Identical twins** develop from a single ovum and sperm. Sometimes when the cell divides in two following fertilisation, it splits completely, and two embryos are implanted in the uterus lining. Because the two embryos have come from the one sperm and one ovum their chromosomes will be identical, giving them identical characteristics. **Fraternal** (non-identical) **twins** develop from the simultaneous release of two ova, which are fertilised by two different sperm. Different sperm and ovum means different chromosomes and therefore non-identical characteristics.

FACT FILE

Identical twins are not identical in every way. They have different fingerprints.

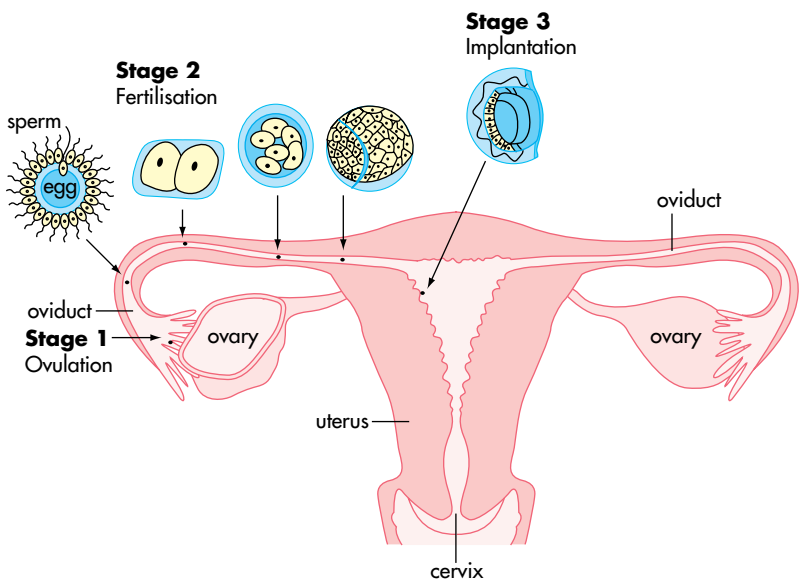

FIGURE 7.24
Steps in reproduction.

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The growing embryo

The dividing cells rapidly take on different forms, becoming nerve cells, blood cells, muscle cells, etc. These form tissues which in turn join with other tissues to become organs like the eye, heart and brain. By 6 weeks the **embryo** (about the size of a fingernail) has a pumping heart and a brain. It is protected by a sac of fluid called the **amniotic sac**. An **umbilical cord** joins the embryo to the **placenta**. The embryo's blood circulates through the placenta, picking up food and oxygen from the mother's blood and releasing carbon dioxide and other wastes into the mother's blood.

Two months after fertilisation the embryo is now termed a **foetus**, and it has a face, limbs, fingers and toes. At 22–24 weeks the foetus' eyelids open and the hands begin to grip. At this stage the foetus looks very like a newborn baby, and begins to grow in length and put on weight.

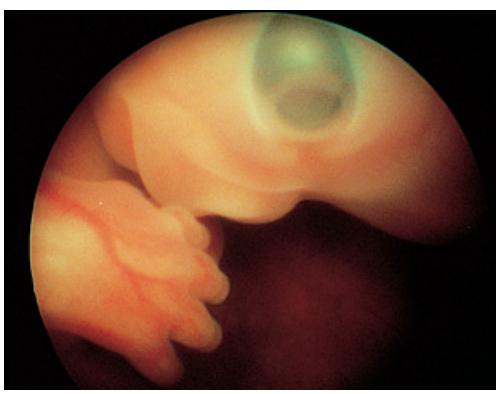

FIGURE 7.25
Human embryo 8 weeks after fertilisation.

Birth

The normal position of the baby just before birth is upside down with its head resting just above the cervix. The cervix needs to open sufficiently to let the baby's head pass through into the vagina. At about this time the amniotic sac bursts and the **amnion fluid** escapes. This is commonly referred to as the 'waters breaking' and signifies the first stage of **labour**.

The muscles in the uterus wall contract rhythmically throughout the birth. They press the baby's head through the cervix into the vagina, where the mother assists by pushing down with her abdominal muscles. The second stage of labour lasts from the time that the cervix is fully stretched until the baby is completely born.

As soon as it is born the baby has its air passages cleaned of fluid and begins to breathe on its own. It is still attached to the placenta by the umbilical cord, which is clamped and then cut, releasing the baby from its mother. However, the placenta is still inside the mother's uterus. The expulsion of the placenta is called the **afterbirth** and is the third and final stage of labour.

Questions

1a Why is it not totally accurate to say that ovulation takes place on day 14 of the menstrual cycle?

b What does this suggest about when fertilisation can occur?

2 Explain the difference between identical and non-identical twins.

3 Look at the sequence of events in Figure 7.24. Write an explanation of what is happening at ovulation, fertilisation and implantation.

4 When is the developing embryo termed a foetus?

5 Why can the umbilical cord be cut after birth?

Think about

Imagine you are in the birthing and delivery suite of a hospital. You are about to become either a mother or a father. Draw a cartoon strip illustrating the events leading up to the birth of your first child. Remember there are three stages of labour and different things happen in each of these. Don't forget to let us know what sex your baby is and what you've named 'it'.

FACT FILE

Some viruses and drugs can pass across the placenta into the baby's bloodstream. HIV is one such virus.

Multimedia

Foundation worksheet 1

Internet assignment

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CURRENT ISSUES, RESEARCH AND DEVELOPMENT

Reproductive technology

FIGURE 7.26
Dr Carl Wood, leader of the IVF team.

Australia has pioneered many of the techniques currently used in assisting humans to reproduce. In late 1970 the Melbourne in-vitro fertilisation (**IVF**) team was established by Carl Wood.

Teams involved in the test-tube baby programmes do not make life; they assist in creation by using God's materials—the sperm of the husband, the egg of the wife, and the brains and skills of scientists, doctors and many others.

Carl Wood and Ann Westmore, from Susan Downie, *Baby Making: The Technology and the Ethics*, page 161

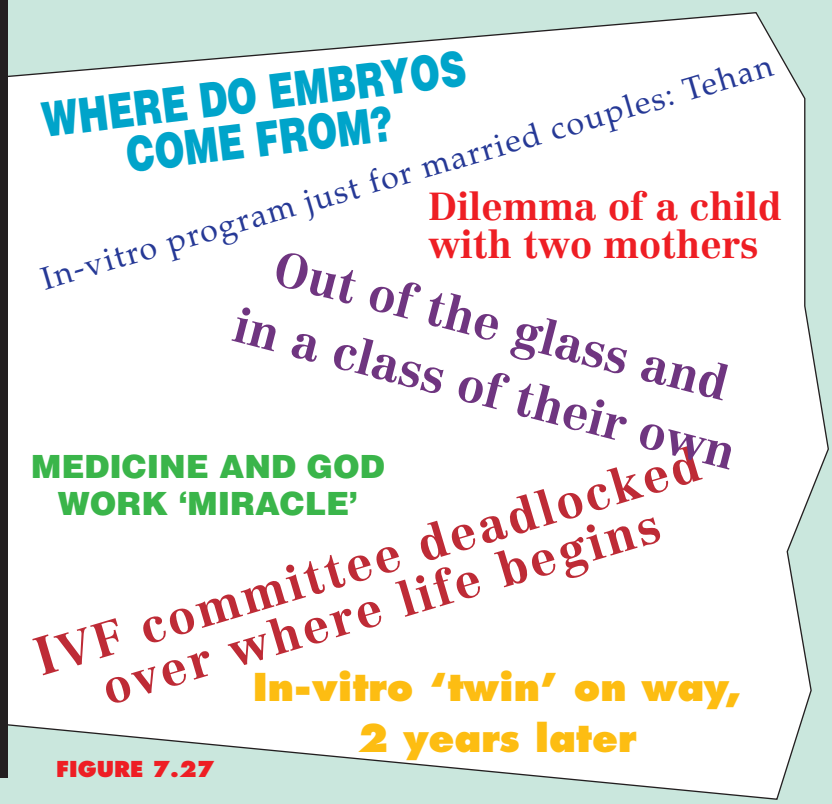

FIGURE 7.27

By early 1973 this team had achieved two early IVF pregnancies but it wasn't until June 1980 that Australia's first (and the world's third) IVF baby, a girl, was born in Melbourne. Following this success, Melbourne saw the birth of the first IVF quadruplets as well as the first donor embryo baby in January 1984. In March 1984, the first frozen embryo baby was born, also in Melbourne. Then controversy ... In June 1984 a couple died, leaving behind two frozen embryos. To whom did they belong? To the couple's estate? To the laboratory? To the government? Despite this dilemma the progress continued and technology developed further. In November 1984 Melbourne welcomed the first donor egg baby, followed, in July 1986, by frozen egg twins, born in South Australia. And so it continues ...

What is IVF and what do all these terms mean?

There are two common methods of helping women reproduce—IVF and **GIFT**. At first these techniques were used by couples having trouble conceiving children because of a blockage in the woman's oviduct. A mature ovum from the ovary is collected and either transferred to the uterine side of the blockage where it can be fertilised by the sperm after copulation—GIFT—or transferred to a suitable medium and fertilised with sperm from the father in a glass dish—IVF. If fertilisation takes place in vitro ('in glass') the embryo is transferred to the uterus in the hope that implantation will occur.

In order to improve the chances of this technology working, the woman is often treated with drugs which cause superovulation, that is, the maturation of several ova at the same time. With more than one ovum available for fertilisation there is the possibility that more than one embryo will

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be formed. In the early days of the technique, several of these were transferred into the female to increase the likelihood of implantation, which resulted in many multiple births. Embryos that are not transferred may be thrown away or frozen for future use. Embryos that were frozen at the same time can be thawed out at different times, creating 'twins' born 2 to 3 years apart.

Technology has progressed even further with the availability of donor 'banks', that is, donated sperm and/or ova. **Surrogacy** involves a second woman receiving the embryo, proceeding through the pregnancy, and then giving the baby to a couple who could not have a child.

One of the most controversial issues has been deciding who are the legal parents of a child born to a surrogate mother who received a donor egg and donor sperm then gave the child to an infertile couple. Such a child would have six 'parents' or participants.

Susan Downie, *Baby Making: The Technology and the Ethics*, page 6

FIGURE 7.28
Alice and Maggie Kirkman.

TABLE 7.1

Issue	Your opinion
Infertile couples in overcrowded countries should not be assisted to reproduce.	
Millions of dollars are spent on both research and providing the technology to infertile couples while children are living in poverty.	
People are not generally concerned with farmers who breed larger than normal sheep but many are worried that the same technology could be used to produce a race of super-humans.	
Sperm and ova donors and mothers who are willing to act as surrogates should be allowed to do so for profit.	
Donor sperm and ova should only be collected from people of exceptional athletic ability or intelligence.	
Frozen embryos should be used in a variety of ways, e.g. raised in order to provide blood for transfusions and organs for transplant; sold to infertile couples; used for research.	
People should be able to choose the sex of their baby.	
People have a right to know who their biological parents are.	

Think about

AI (artificial insemination) and AID (artificial insemination by donor) are more common than IVF or GIFT. Do these raise the same issues? Why are they debated less often?

Activity

Do not discuss the following issues with anyone. Copy Table 7.1 into your workbook and write your own opinions.
Now, in a group of six, choose one of the issues for debate. Divide your group into two teams of three and toss a coin to decide which team will take the positive and which the negative. Research your topic by using as wide a variety of resources as possible, e.g. newspaper and magazine articles, journals, reference books, discussion papers, Internet. Debate the topic in front of the class, whose job it is to decide the team with the most thoroughly researched and best presented arguments.

Foundation worksheet 2

FACT FILE

Alice Kirkman has an interesting story to tell. She was born in 1988 as a result of an egg taken from her mother Maggie. The egg was fertilised by a sperm from an anonymous donor (not Maggie's husband) by using the IVF technique. The embryo which formed was implanted into Maggie's sister's uterus. Maggie's sister Linda was Alice's 'surrogate' mother. Who are Alice's parents?

7.9

Cloning around

FIGURE 7.31
Send in the clones. The original mouse on top with some of her clones.

Talk about
Why would scientists want to be able to clone animals and plants? Do you think scientists should be allowed to clone people? Would you like to meet a clone of yourself?

FIGURE 7.29
Identical twins.

A **clone** is an exact copy or duplicate of another living organism. Clones are produced by asexual or 'one parent' reproduction, so the clone has exactly the same genetic make-up as it parent. Clones are common in simple plants and animals. However, in more complex organisms which reproduce sexually the only way to produce a clone naturally would be to have identical twins.

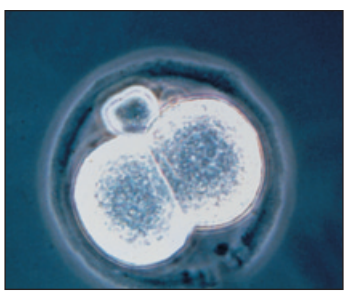
Experimenting with cloning

Scientists have been trying to clone larger animals since the 1960s. The first experiments were carried out on frogs. The scientists took cells from the intestines of tadpoles, removed the cell nuclei and injected them into egg cells (whose nuclei had been destroyed). Healthy, fertile frogs were produced. However, when the scientists tried to use cells from an adult frog, only a few of the clones grew into tadpoles and none survived into adulthood. Many experiments followed on mice, sheep and cows with similar results.

Scientists became convinced that cloning was not possible with adult cells. The only cells which could be successfully cloned were the unspecialised cells formed when a fertilised egg cell begins to divide early in an embryo's development. Once an embryo begins to grow further its cells begin to specialise. The cells are programmed by their genetic code to become muscle cells, bone cells, nerve cells and so on. Until recently scientists believed that it was not possible to make a cell go back to its unspecialised state and behave like a newly fertilised egg cell.

However, several researchers have produced clones by using techniques that have proved successful for some specialised cells. Since Dolly the cloned sheep was born in 1997, French researchers have cloned a calf from a muscle cell, and a Japanese scientist, Ryozo Yanagimachi, working in Hawaii, has cloned over 50 mice.

Yanagimachi used a cumulus cell, which is a body cell that surrounds and nourishes a ripening egg cell. He called his cloned mouse Cumulina. Cumulina has herself been cloned several times, producing three generations of identical mice all perfectly normal and healthy.



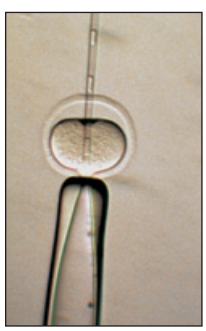


FIGURE 7.30
Technique for removing nucleus from an egg cell and inserting it into an adult cell.

Why clone?

What benefits could come from cloning? This is the topic of much discussion as scientists and governments decide on what controls should exist on cloning research. Sections of the community see cloning as an aid to farming and medical research. For some time, large numbers of plants with desirable characteristics have been cloned by using tissue culture techniques. Animals which are resistant to disease or produce large yields of milk could also be 'carbon-copied' to give a boost to agricultural industries. However, scientists who are genetically engineering animals to carry human genes see cloning as a means of mass producing life-saving drugs. In the future it may even be possible to use cloning to produce replacement human organs for transplantation.

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USING SCIENCE 3 THE ETHICS OF CLONING

Read the article and answer the questions below.

1 Use a dictionary to help write down the meaning of the following words as used in the article: progeny, gestation, pharmaceutical, derived, ethical, inhibition, diversity.

2 What is Dr Dawson's opinion on the cloning of people?

3 Suggest a reason why 'it would be dangerous to have a national flock of merino sheep that were all identical'.

4 Some governments have passed laws to ban experiments on human cloning. Do you think this is a good idea?

Scientists deny wolf in sheep's cloning

By science correspondent GRAEME LEECH

THE science that cloned a Scottish sheep is unlikely to be used to clone a human being—even though it is technically possible.

The creation at Edinburgh's Roslin Institute of a sheep that is an identical twin of its mother has raised fears the technique could be used to create a perfect copy of a living human.

'Scientists have been able to clone a human for 15 years, but who would really want to,' asked a geneticist at Monash University's Institute of Reproduction and Development, Dr Karen Dawson.

'There is a bit more to people than just genetics.'

This included differing environmental conditions, for example, 'You'd need a parallel environment to reproduce the same person,' Dr Dawson said.

The main benefit to be derived from the technique of cloning an animal from adult cells rather than embryo cells would be the potential for production of pharmaceuticals, Dr Dawson said.

The chairman of the Gene Therapy Committee of the National Health and Medical Research Council, Dr Ron Trent, said there were no obvious reasons for cloning a human.

Australian scientists say that laboratories involved in the complex science of cloning an entire animal, or investigating the genetic manipulation of an animal's biology, know the type and progress of research elsewhere.

This is an effective brake on any laboratory prepared to ignore the ethical inhibitions on cloning a human, according to Dr Kevin Ward of CSIRO's division of animal production at Prospect in western Sydney.

Dr Ward said the scientists who cloned the Scottish sheep, named Dolly, had visited labs in Australia to exchange information.

'One laboratory by itself can't do it all,' he said.

Dr Ward's lab is working on modifying the genetic properties of sheep to improve meat and wool quality.

It is close to changing the biochemistry of the animal so that it can gain more benefit from common nutrients.

Another CSIRO development could prevent blow-fly strike in sheep by making them produce their own insecticide, Dr Ward said.

Once a sheep had been genetically modified, it passed on the characteristic to all its progeny.

The British development would allow the rapid multiplication of gene types—at least halving the gestation period.

Cloning would never displace conventional breeding because of the need to maintain genetic diversity.

'It would be dangerous to have a national flock of merinos that were all identical,' Dr Ward said.

Other than controls within individual universities and research institutions, the supervision of genetic research in Australia is in the hands of Dr Trent's Gene Therapy Committee and the Genetic Manipulation Advisory Committee. There is no law in Australia preventing the cloning of humans.

The Australian
25 February 1997, p. 3

Questions

1 What is a clone?

2 Why is it so difficult to clone animals from adult cells?

3 List three possible benefits of cloning.

Think about

Is the cloning of animals for human use ethical? Give three reasons to support your opinion.

Activity

Collect media stories about cloning. Look in newspapers, science magazines and on the Internet. Write a short comment on each article saying if it is for or against cloning.

Find out

There are many science fiction stories as well as movies such as 'Jurassic Park' and the 'Boys from Brazil' which make predictions about the use of cloning. Select a short story or view a movie and write a report about how cloning technology is portrayed.

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Using scientific language

Copy the paragraph and complete it by inserting the correct response for each missing word.

At birth girls have all the __ they will require throughout their life (plus thousands more!). Until __ these __ are immature. Somewhere between the ages of ten and sixteen, all females begin their reproductive cycle. They develop __ sexual characteristics such as growth of pubic hair and breasts. Their first menstruation is called __ and means that an ovum has matured and is ready for release. Approximately every __ days females will experience a menstrual cycle involving many __ (chemicals that regulate the functioning of the reproductive cycle), __ (the release of the ovum) and __ (the shedding of the __ lining).

Check your knowledge

1 Draw a classification key showing the different methods of reproduction.

2 Describe the difference between:
- a pollination and fertilisation
- b internal and external fertilisation
- c asexual and sexual reproduction
- d sperm and ova.

3 Copy the following and number the statements from 1 to 9 in order of occurrence during a typical menstrual cycle.
__ blood vessels broken as lining is shed
__ follicle produces oestrogen to repair lining
__ egg will die if not fertilised
__ menstrual flow begins
__ progesterone produced to stop the lining from thickening any further
__ primary follicle in ovary starts to grow and egg matures
__ egg released into oviduct
__ when a certain level of oestrogen is reached, brain releases luteinising hormone, causing one follicle to burst
__ surface layer of cells begin to die as arteries constrict

4 Draw a flower that has been dissected vertically through the centre. Draw and label all its reproductive parts.

5 Explain the differences between the three types of mammalian reproduction.

6 Describe the three stages of labour.

Apply your skills

1 Suggest a reason for each of the following statements.
- a Only one sperm can fertilise an egg.
- b Monotremes lay eggs.
- c Flowering plants usually cross-pollinate.
- d The testes are located outside the body.
- e Many animals display courting and mating behaviours.

2 Figure 7.32 shows nine different organisms. Identify each and explain their likely method of reproduction.

FIGURE 7.32
In this diagram, you should locate nine different organisms in their natural environment.

Challenge yourself

Sometimes problems occur in pregnancy. Research one of the following or a topic approved by your teacher.
- Breech births
- Thalidomide
- Smoking and pregnancy
- Alcohol and pregnancy
- Down syndrome
- AIDS and pregnancy
- Miscarriage
- Stress and pregnancy
- Rubella and pregnancy
- Premature babies

Present your findings to the rest of the class. You may choose to use a poster, oral presentation, model, etc.

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return to main menu

CHAPTER 8

GENETICS

Genetically modified food (GMF) is a subject you are sure to hear more about in the future. Potatoes, apples, tomatoes, grains, soybeans and corn are just a few of the foods which have been genetically modified overseas and in Australia.

Changing the genetic make-up of our food is not new. Plant and animal breeders have been altering the inherited characteristics of the things we eat for thousands of years. What has changed is the process. The genetic make-up of plants and animals can be changed slowly by crossing different varieties and selecting individuals with the desired characteristics. Genetic engineering is a rapid way of altering the genetic make-up of a plant or animal. It involves the direct manipulation of the genetic code inside cells. It is even possible to take genetic code from one organism and insert it into the cells of a completely different organism. The 'mixtures' produced are called transgenic organisms. By use of genetic engineering techniques, scientists have inserted genetic code from bacteria into corn plants to make them produce their own insecticide. Human genetic information has been inserted into pig cells, which increases the growth of the pigs. Human genetic code has also been inserted into bacteria, which causes the bacteria to manufacture human insulin, a life-saving chemical for diabetics.

In this chapter you will learn about DNA, the chemical which carries the genetic code and how scientists are developing new techniques to control inheritance.

Web links

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- **Trace the discovery of the structure of DNA and the understanding of its role in inheritance.**
- **Make a model of DNA and use this to explain its structure and ability to replicate.**
- **Explain that characteristics are determined by genes, which are subunits of chromosomes.**
- **Describe two types of cell division: mitosis and meiosis.**
- **Predict the outcomes in crosses involving dominant/recessive and intermediate inheritance.**
- **Explain the role of X and Y chromosomes in sex determination and sex-linked inheritance.**
- **Evaluate the issues arising from genetic engineering.**
- **Explain how scientific concepts are applied to genetic counselling.**

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8.1

The double helix

FIGURE 8.1
What do all these people have in common? What differences can you see? What determines these similarities and differences?

Talk about

How many instructions would it take to build a living human being? Around 10 000 or 100 000 or 1 000 000 instructions? How could you write these instructions to fit inside a single fertilised egg cell?

FACT FILE

Nobel Prizes are not awarded posthumously. Rosalind Franklin may well have been included for her contribution but she had died of cancer, probably brought on by her research.

Scientists rarely carry out their research in complete isolation. Scientific journals and conferences provide opportunities for scientists to share their findings and have their work built upon.

Gregor Mendel was an Austrian monk who experimented with pea plants. In 1865, he hypothesised that there were 'units of heredity' that provided the instructions that directed the growth and development of organisms. These units of **heredity** were later to be termed **genes**.

In 1869 a chemical known as **deoxyribonucleic acid** or **DNA** was first discovered. It was not until 1944 that it was shown that genes were composed of DNA. The race to discover the structure of DNA spanned two countries. In the early 1950s, Linus Pauling in the USA and Rosalind Franklin and Maurice Wilkins in England were using **X-ray crystallography** to determine the shape of the DNA molecule. Basing their work on the research of these three scientists, James Watson and Francis Crick, also in England, started to build scale models of DNA in the hope that a convincing structure would emerge.

Watson and Crick had access to much of Franklin's and Wilkin's X-ray work and, combining the information with that of some 1951 research of an American, Erwin Chargaff, they discovered the shape known as the double helix. Their findings were first published in 1953 and were to earn Watson, Crick and Wilkins the Nobel Prize in 1962.

Watson and Crick showed that DNA is a long, thread-like molecule similar in shape to a rope ladder twisted into a spiral pattern—the **double helix**. The sides of the ladder are made of alternating sugar and phosphate units. The rungs of the ladder are made of paired chemical units or bases. There are four different bases: adenine, guanine, cytosine and thymine (usually referred to as A, G, C and T). It was found

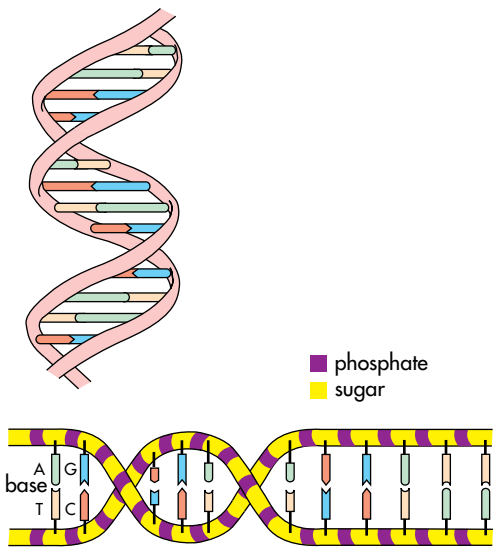

FIGURE 8.2
The DNA molecule 'unwound' to show the relative positions of the bases and sugar and phosphate units.

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that adenine only paired with thymine and guanine only paired with cytosine. These four bases compose the basic structure of the DNA of all living organisms—the difference between the DNA of different organisms is the number and arrangement of the bases.

Because it is composed of only four different bases it would be easy to think that there must be very little difference between different molecules of DNA. However, a single molecule of DNA may be made of hundreds of thousands of bases and so there is an extraordinary number of different combinations of base sequences that can make up the individual DNA molecules.

The discovery of the structure of DNA has led to an enormous amount of research and new knowledge. This forms the basis of genetic engineering, in which experimental work began in the 1970s. Since the work of Watson and Crick, scientists now understand the way in which DNA is able replicate itself, how it is able to determine the characteristics of organisms, and how this information can be used to alter and create **genes**.

FACT FILE

To write the code for the 100 000 genes which contain all the instructions for a human would produce a sequence of letters over 8000 km long.

USING SCIENCE 1 A MODEL OF DNA

WHAT YOU NEED

24 paper clips in four different colours, strips of paper, coloured pencils.

WHAT YOU DO

1 Use paper clips to form the bases of your molecule (one colour for each of the four bases) and strips of paper to form the sides.
2 Make your model with 12 rungs of any base order.
3 Draw the positions of the sugar and phosphate molecules on the paper strips.
4 Compare your model to the others made in your class. Are any exactly the same as yours? How are they different?
5 Look carefully at the order of bases in a number of models. What does this imply about the structure of DNA?

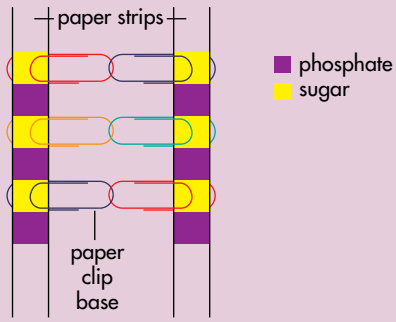

FIGURE 8.3
Construction of the paper clip DNA molecule.

EXTENSION

Make up one-half of the helix using a paper strip and paper clips and ask your partner to complete the molecule using the appropriate bases.

Questions

1 Describe the basic structure of DNA. Include a description of the make-up of the 'rungs' and 'sides' of the 'ladder'.
2 Draw and label a diagram of a double helix in your workbook.
3 Why do we describe DNA as being a double helix?

Think about

Can Watson and Crick be given complete credit for the discovery of the structure of DNA? What does this tell you about the nature of scientific research?

Find out

Use your school library to draw a time line of events from the discovery of DNA in 1869 to the present day, indicating the major stages in the increase in knowledge.

Web page

FIGURE 8.4
James Watson and Francis Crick with their model of the double helix DNA molecule.

8.2

Genes and chromosomes

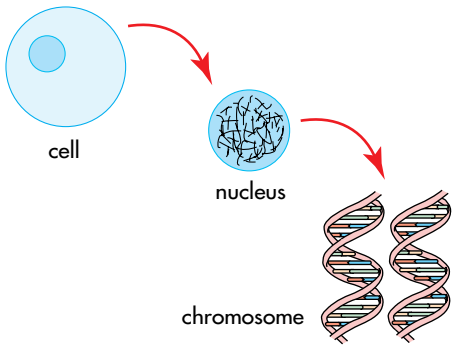

FIGURE 8.5
All cells with a nucleus contain genetic material.

FIGURE 8.6
(above right) An electromicrograph of the sex chromosomes of a male.

Chromosomes are long, thin, thread-like structures made of DNA and are found in the nucleus of cells. The cells in the human body each contain 46 chromosomes (in 23 pairs) with the exception of the sperm and egg cells which only contain 23 chromosomes (or one of each pair). When a cell contains pairs of chromosomes we say that it has a diploid number of chromosomes. When a cell only has one of each pair we say that it has a haploid number of chromosomes.

FACT FILE

If all the chromosomes in a single human cell were unwound and connected end to end they would measure 1 m in length. The 1 m length of human DNA would contain around 3 billion base pairs.

Carrying the message

The bases making up the DNA of a chromosome carry encoded messages or genes that determine the characteristics of an organism. Each chromosome may carry over a thousand genes. It is now possible to explain how genes are able to provide the instructions necessary to produce different characteristics.

Amino acids are the building blocks for proteins such as enzymes and hormones which control the various functions and characteristics of living organisms. For instance one protein may cause you to have red hair, a different protein will cause a tree to grow roots, and a third will cause a cell to produce mucus. The genes are codes which enable an organism to produce these particular proteins. One gene may code for the precise sequence of more than 1000 amino acids that makes up the specific protein that controls a certain characteristic or function.

The bases on the DNA molecule work in threes to code for different amino acids. Thus the combination AAT will code for one amino acid, CGC will code for a different amino acid, TCC for another, and so on. There are 20 different amino acids to code for; Table 8.1 lists the base combinations and the amino acids they produce. A particular code will always specify the same amino acid, whether in a plant cell, an animal cell or a bacterium.

If one amino acid in a particular protein is deleted, added or altered in any way, then a different protein molecule, with a different function, could be produced. For example, changing the third base in Figure 8.8 from AAC to AAG would code for a different amino acid—phenylalanine instead of leucine.

When additions, deletions or alterations of bases in a gene occur spontaneously they are known as **mutations**. The abnormal proteins which form can produce different characteristics in individual cells. Mutations occur randomly within a population and any gene can undergo mutation at any time. The rate at which mutations occur increases if an organism is subjected to radiation, UV light, high temperatures and many chemicals (e.g. formaldehyde, some pesticides and some drugs).

Most mutations are harmful to an organism. However, it is possible that a mutation may cause a more advantageous characteristic to be produced than the one for which it was originally coded.

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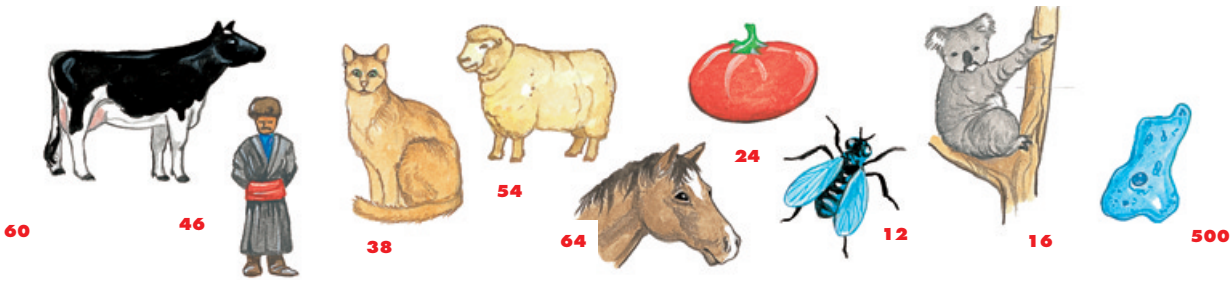

FIGURE 8.7 Different organisms will have different numbers of chromosomes. For instance, humans have a diploid number of 46, horses 64, cows 60, sheep 54, tomatoes 24, blowflies 12, koalas 16, cats 38 and some amoebae may have over 500. The chromosomes carry different types and numbers of genes as well.

TABLE 8.1 Triplet codes and the amino acids they produce. Note that some amino acids are produced by more than one triplet code.

DNA coding strand	Amino acid	DNA coding strand	Amino acid	DNA coding strand	Amino acid	DNA coding strand	Amino acid
AAA AAG	phenylalanine	AGA AGG AGT AGC	serine	ATA ATG	tyrosine	ACA ACG	cysteine
AAT AAC	leucine			ATT ATC	*stop*	ACT	*stop*
						ACC	tryptophan
GAA GAG GAT GAC	leucine	GGA GGG GGT GGC	proline	GTA GTG	histidine	GCA GCG GCT GCC	arginine
				GTT GTC	glutamine		
TAA TAG TAT	isoleucine	TGA TGG TGT TGC	threonine	TTA TTG	asparagine	TCA TCG	serine
TAC	methionine/*start*			TTT TTC	lysine	TCT TCC	arginine
CAA CAG CAT CAC	valine	CGA CGG CGT CGC	alanine	CTA CTG	aspartic acid	CCA CCG CCT CCC	glycine
				CTT CTC	glutamic acid		

CGA	AGA	AAC	TCA	GCA	←	base chain
alanine	serine	leucine	serine	arginine	←	amino acids

FIGURE 8.8 A series of bases will code for a particular amino acid.

USING SCIENCE 2 CODE FOR LIFE

Use Table 8.1 to translate the following code for a protein molecule into the correct sequence of amino acids:

AAT ACG TTA CGC GGG TTC CCA
GAT CAT CCT GGG

1 If the sixth and twelfth bases were changed to T, what would be the effect on the protein molecule?
2 What would happen if the eighth base was deleted?
3 What would happen if a G was inserted between the sixteenth and seventeenth bases?

Questions

1 Is there a connection between the complexity of an organism and the number of chromosomes its cells contain? Explain.
2 Name six types of cells which contain a diploid number of chromosomes.
3 What type of cells contain a haploid number of chromosomes?
4 What is a mutation? What types of things can increase the rate of mutations?

Think about

1 Draw a flow chart to explain how a particular gene could cause you to have brown eyes.
2 What could be the effect of a mutation in a skin cell compared to cells that produce eggs or sperm?

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CURRENT ISSUES, RESEARCH AND DEVELOPMENT

Genetic engineering

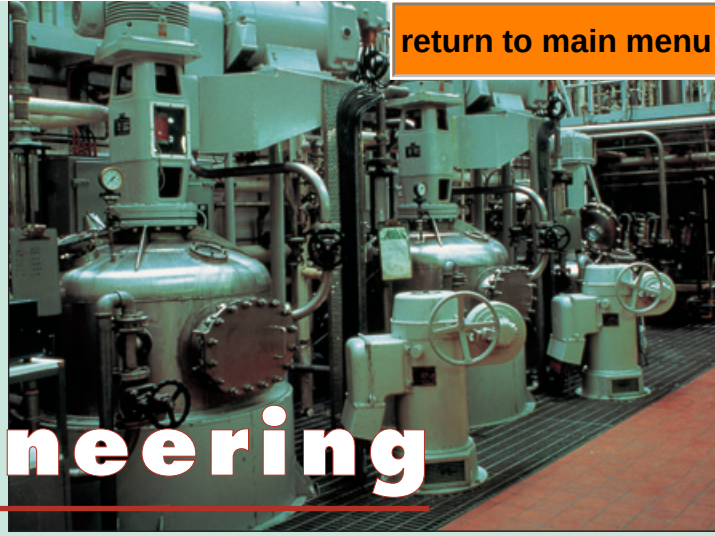

FIGURE 8.9
Large vats containing genetically modified bacteria produce a supply of valuable drugs and vaccines.

Gene technology

Genetic engineering involves altering DNA in the laboratory. Basically, a piece of DNA is cut out of one chromosome and stuck into the chromosome of another cell. The new DNA then codes for a different protein which will end up changing a characteristic of an organism. This technique is called recombinant DNA technology.

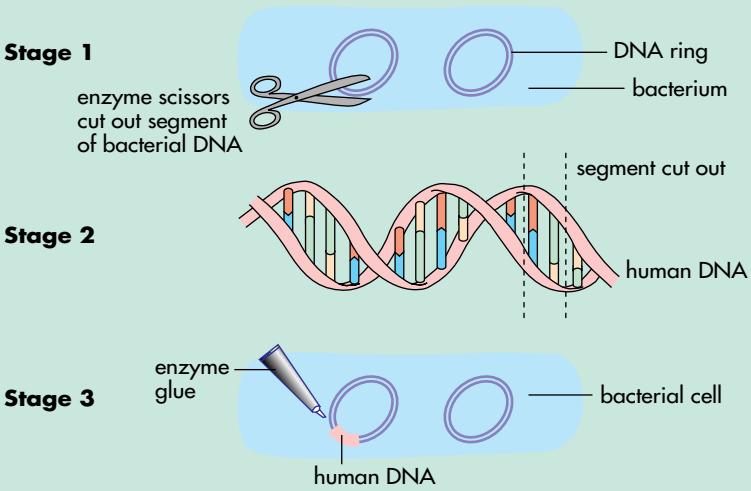

FIGURE 8.10
Recombinant DNA technology. Cutting and gluing genes!

Genetic engineering techniques have also been developed to duplicate selected parts of a DNA molecule. The enzyme polymerase, in the polymerase chain reaction (PCR), causes segments of DNA to make exact copies of themselves. This allows large quantities of genetic code to be made for such purposes as testing embryos for genetic diseases, detecting the presence of the AIDS virus and identifying criminals.

Gene shears is another gene technology. This is a new technique being developed by Australian scientists and has the potential to alter plant and animal cells to make them resistant to viral diseases.

Using genetic engineering

The applications of genetic engineering are limitless. Currently, the technology is used to manufacture proteins, improve crops and livestock and screen for inherited diseases. However, genetic engineering also has the potential to produce building materials, plastics, fibres and new sources of energy as well as to cure genetic diseases.

Manufacturing proteins

Each gene is responsible for making a specific protein. Genes for human proteins can be inserted into the DNA of bacterial cells. The bacterial cells will then manufacture the human protein in large quantities. Genetically modified bacteria are manufacturing proteins such as insulin for diabetics, blood-clotting Factor 8 for haemophiliacs and human growth hormone. In the same way bacteria can be modified to produce vaccines for humans and farm animals.

Improving crops and livestock

A wide range of genetically modified plants is currently being produced. Plants such as soybeans have been engineered to withstand herbicides that are used to kill weeds. This means the weeds can be sprayed without damaging crops. Other plants such as corn have been engineered to produce their own insecticide to resist

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the attack of insect pests. Genetic engineering can also change the characteristics of a crop to produce potatoes that don't turn brown when you cut them, capsicums with a higher vitamin content and tomatoes which ripen more slowly.

This means that the tomatoes can be picked at a later stage when they are tastier. It also makes it easier to transport them over long distances.

Turnips and potatoes are currently being experimented upon to reduce their susceptibility to frost. The 'anti-freeze' gene from flounder (a type of fish) has been isolated and inserted into the vegetables, resulting in a frost-resistant plant.

The genetic engineering of livestock has not proceeded as rapidly as in plants. An Australian company has patented a transgenic pig which contains an extra gene to make it grow faster. The transgenic pigs are more efficient because they use less food but develop more meat with less body fat. However, the company has so far not been allowed to sell the transgenic pigs without Federal Government approval. The gene which makes the pigs grow faster includes some human genetic code which may cause consumers to reject this product.

Screening tests

Research into DNA has enabled scientists to find the genes responsible for certain hereditary diseases such as cystic fibrosis, muscular dystrophy, haemophilia and Huntington's disease. Gene technology allows simple screening tests for parents. Mothers with a higher risk can have tests to see if an embryo has faulty genes. They can then choose whether to continue with a pregnancy.

FIGURE 8.11
Cotton leaves with and without a gene for insect resistance.

Other applications

Genetic engineering techniques may also be used to produce a supply of suitable transplantable organs. **Transgenic** pigs carrying human genes have been developed. The pigs' organs can be transplanted into humans with less chance of rejection. This technique can save lives because there is frequently a shortage of donor organs available for transplantation.

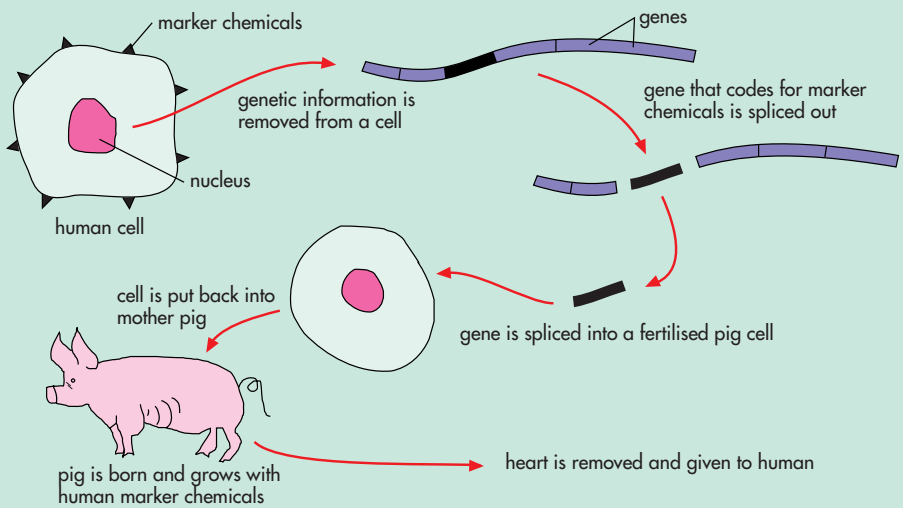

FIGURE 8.12
A body will reject foreign cells because they have different marker chemicals (proteins) on the outside of their cell membranes. If human marker chemicals can be genetically engineered onto pig cells then transplanted pig organs will not be rejected.

FACT FILE

In the movie 'Jurassic Park', prehistoric dinosaurs are brought back to life by genetic engineering. Dinosaur DNA is collected from ancient blood-sucking insects preserved in amber. Could this be possible? Probably not, is the current answer. DNA decomposes over time, so even if fossilised insects did contain dinosaur blood it would be extremely unlikely that the DNA would be intact.

Genetic engineering (cont.)

FIGURE 8.13
A range of products which can contain genetically modified ingredients.

FACT FILE

In a recent survey, 81% of people questioned said that genetically modified food should be labelled, and 58% indicated that if it was labelled they would not buy it.

Talk about

If scientists insert animal genes into plants such as turnips and potatoes, does this mean that vegetarians would not want to eat those plants?

How far should we go?

There are always problems when scientific breakthroughs and new technologies have the potential to greatly change our lives. People are often concerned that the new technology will be misused or have long-term negative effects. Genetic engineering is no exception. The application of genetic engineering to food is a controversial issue. Environmental, health, consumer and animal welfare groups have all expressed concerns. Sections of the community have called for the labelling of all genetically modified foods, so that consumers would have the ability to choose. However, producers and manufacturers say this would be both impractical and expensive.

USING SCIENCE 3 FOOD FOR THOUGHT

1 Write down your initial opinions about genetically modified food.
2 Form a group and discuss your ideas. Together, brainstorm a list of benefits and problems associated with genetically modified food.
3 Read the following extracts and add ideas to your list.
4 Research the topic further using library and Internet sources and record any other ideas.
5 For each problem associated with genetically modified food discuss whether it may be solved by science in the future.
6 Some problems do not have a scientific solution. Governments and society must find solutions. What regulations do you think should exist to control genetically modified foods?
7 What are your opinions now? After weighing up the benefits and the problems of genetically modified food, state your position on the subject.

EXTENSION

Design your own survey on genetically modified food and use it to conduct a survey of opinions in your school and community. Write a report of your findings.

Researchers are developing beans and grains with much higher levels of protein—no small thing for parts of the world where meat is scarce. Plants with clever genes are being developed to produce bigger harvests even on poor soils in overpopulated countries.
Adapted from 'Brave new farm', James Walsh, *Time*, 11 January 1999.

Some consumers are concerned that genes will be inserted into food which will increase the production of toxic and allergenic substances.
Adapted from 'Transgenic foods: the case for caution', Natasha Stott Despoja, *Australian Science*, Vol. 19, No. 8, 1998.

Biotechnology companies will want farmers to grow plants with herbicide, pesticide and viral resistance genes. These genes will spread from the crops to weeds through wind blown pollen. This will create super weeds.
Adapted from 'Apocalypse when?', Jeremy Rifken, *New Scientist*, No. 2158, 31 October 1998.

People have false fears about genetically modified foods. Many nightmares predicted for genetically engineered crops have already happened. It's just that not many people noticed or cared because they were caused by conventional breeding, not genetic manipulation.
Adapted from 'Strange fruit', Phil Cohen, *New Scientist*, No. 2158, 31 October 1998.

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USING SCIENCE 4 DEBATING THE ISSUE

'Super pigs' row likely

Major churches face internal conflict over the morality of animal organs being transplanted into humans. The breeding of 'super pigs' for a harvest of transplantable hearts, lungs and kidneys has prompted the vexed question of how far medical science can advance before the global community says enough is enough.

Churches were surprised by the news of the special-purpose breeding of the pigs in England. What was once the stuff of science fiction has suddenly become a reality, raising a host of moral questions.

A Catholic spokesperson said the church needed to further explore the implications. 'Different people in the church probably have different ideas. Some people may see it as an act of charity (in saving human life) but I'm not really too sure,' the spokesperson said.

'There is a certain repugnance that medical science has gone so far as to erode the process of nature. How far can medical science go? Catholics believe that one must take proportionate steps to preserve human life. Some people argue that human heart transplants are disproportionate.

'The idea of transplanting animal organs into humans takes the debate a step further. We are talking here about the implications of medical science. These are areas that are still being explored by the church.'

The 'super pigs' have 'human hearts' after scientists successfully introduced human genes into fertilised sows' eggs. As a result, the pigs' hearts, when transplanted, are expected to be readily accepted by the human immune system.

This has also paved the way for the 'farming' of pigs' lungs and kidneys, although there is a question over whether the process will work with livers.

The pigs are being kept at a secret location in England to thwart animal rights activists who have attacked scientists' homes. The scientists have called for a public debate of ethical questions.

Adapted from a report in the *Advertiser*, 15 March 1993.

Read the article '"Super pigs" row likely', then answer the following questions.

1 What organs can be obtained from the 'super pigs'?

2 Why might people object to the practice?

3 Why are the scientists keeping the pigs at a secret location?

4 Would you accept a heart from a specially bred pig to stay alive? Should you be allowed to?

5 Should pigs and other animals be specially farmed to provide organs for humans?

Organise a class debate on this issue.

Questions

1 List the basic steps involved in genetic engineering.

2 Why might agricultural scientists want to use genetic engineering in preference to traditional breeding methods of selecting characteristics?

3 What does the term 'controversial' mean? Why is GMF a controversial issue?

Think about

1 It is possible that in the future parents would be able to go to a 'Gene Bank' and select the characteristics that they would want their child to have. What advantages and disadvantages would you see in this situation?

2 Imagine you were able to alter genes and modify any food. Write a description of a new food you would create by using the genes of any plant or animal. Describe the features of your designer food such as taste, texture, colour etc.

Activity

A scientist identifies a gene that codes for a growth hormone that will increase the yield of wheat in arid environments. An application is made to patent the gene. This would mean that no other scientist or plant breeder could use the gene without permission. Imagine that you have to debate the issues involved in approving the patent. Outline the reasons for and against granting such a patent.

Foundation worksheet 1

Internet assignment

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8.4

Multiplying and dividing

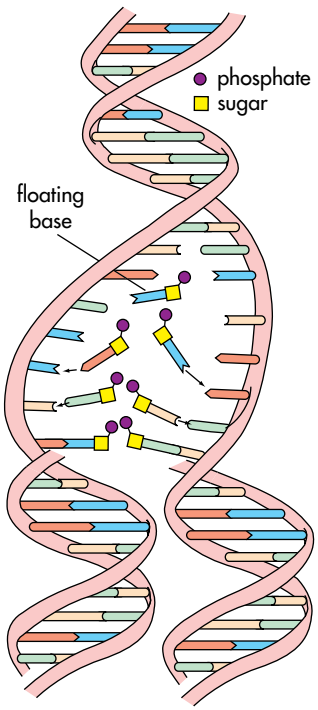

FIGURE 8.14
The 'unzipping' process that leads to the production of new DNA strands.

Animation

Foundation worksheet 2

Talk about

The human body consists of billions of cells yet it began as a single cell. Where did this first cell come from and how did it develop into billions of cells?

Cells have the ability to replicate the DNA they contain. To do this the bases of the double helix separate from each other, in much the same way as a zip opens, leaving the bases exposed. Within each nucleus there are bases floating freely. These floating bases pair with the exposed bases and also attach themselves to the sugar and phosphate molecules of the adjacent base, causing two identical double helices to form.

The amount of DNA has doubled inside the nucleus and the cell is now able to divide. Cell division can occur in one of two ways: by mitosis or by meiosis.

Mitosis

Mitosis is the process that most cells use to divide. It is the process whereby a single cell divides to form two identical daughter cells. These daughter cells carry the same diploid number of chromosomes and exactly the same genetic information as the parent cell.

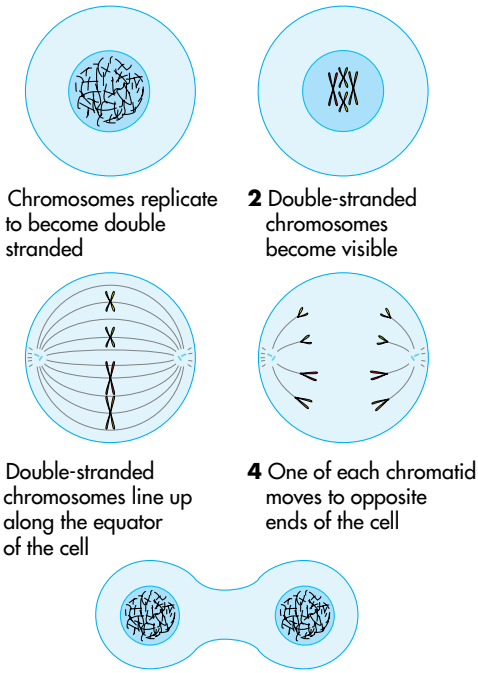

FIGURE 8.15
The stages of mitosis.

The DNA within the nucleus has already duplicated itself when the cell starts to divide. The chromosomes shorten and thicken, becoming visible under a microscope. Each chromosome can now be seen to consist of two strands attached at only one point, the **centromere**. Each chromosome has made a copy of itself and now consists of two identical strands or **chromatids**.

The process is precise. The 'double-stranded' chromosomes line up down the middle of the cell and split into their two chromatids. The chromatids move to opposite ends of the cell and when one of each chromatid reaches its destination, the cell divides into two identical cells, each with the same number of chromosomes as its parent.

USING SCIENCE 5 IDENTIFYING THE STAGES OF MITOSIS

WHAT YOU NEED

Prepared slides of onion root tips, microscope.

WHAT YOU DO

1 Using the microscope, identify onion root cells that are undergoing the various stages of mitosis.

2 Select four cells in different stages of mitosis. Draw them in the correct sequence for mitotic division, using the information above to label them clearly.

Multimedia

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Meiosis

Animation

Meiosis is a special type of cell division used only for the production of **gametes**—sperm or egg cells. When gametes unite at the point of fertilisation their nuclei combine. If two human gametes each had 46 chromosomes and were to combine at fertilisation they would produce a single cell (or **zygote**) with 92 chromosomes. This cell would no longer be the cell of a human. Hence, it is important for the gametes to be produced with only a haploid number of chromosomes so that the zygote that is produced has the correct diploid number of chromosomes.

Meiosis is similar to mitosis except that there are two divisions with four haploid cells being produced. The first division involves the separation of the pairs of chromosomes and the second division the separation of the chromatids.

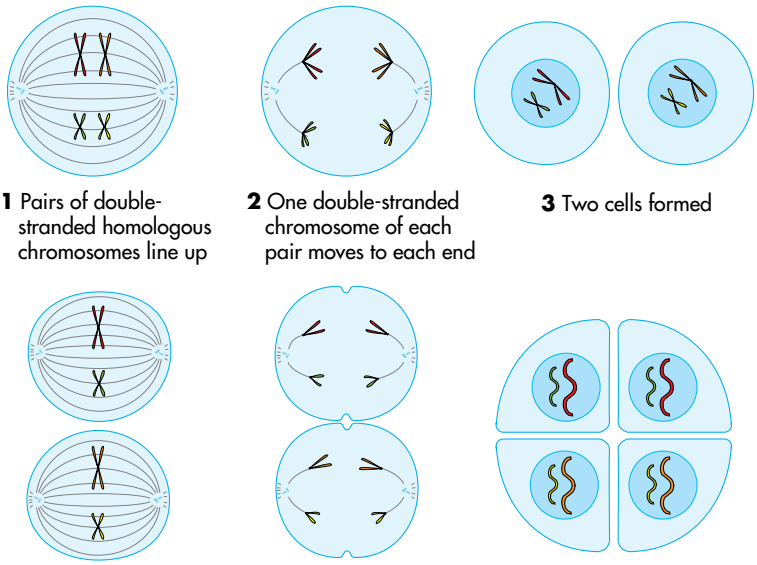

FIGURE 8.16
The stages of meiosis.

USING SCIENCE 6 MODELLING MITOSIS AND MEIOSIS

AIM
To model mitosis and meiosis.

WHAT YOU NEED
4 pipe cleaners

WHAT YOU DO
1 Use the pipe cleaners to represent the chromatids of two chromosomes.
2 Twist two pipe cleaners around each other to represent one chromosome after replication. Use the other two pipe cleaners to form the second chromosome (Figure 8.17).
3 Model mitosis by manipulating the pipe cleaners.
4 Using the same four pipe cleaners, model meiosis.
5 Draw the basic movements of the pipe cleaners to show mitosis and meiosis.

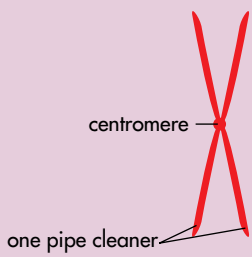

FIGURE 8.17
'Pipe cleaner' chromosome after replication, showing two chromatids.

Questions

1 Why is it essential for the chromosomes to replicate before cell division occurs?
2 Why is it important for the 'double-stranded' chromosomes to line up in single file down the centre of the cell prior to its splitting in two during mitosis?
3 If a cell started off with 28 chromosomes prior to going through mitosis, how many chromosomes would it end with?
4 Why is it important for the cells that are produced to be identical to the parent cell in mitosis but not in meiosis?

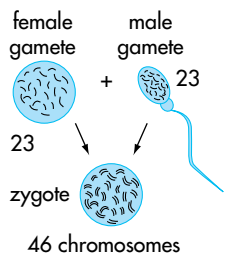

FIGURE 8.18
Haploid gametes fuse to form a diploid zygote.

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8.5

Characteristics and inheritance

FACT FILE

Before the work of Gregor Mendel very little scientific investigation into heredity had taken place. Early folklore suggested that characteristics could be passed on through the 'mixing of blood'. People believed that children inherited a blend of characteristics from their parents. This idea led to common terms such as 'blood lines', 'blood relative' and 'royal blood'. We know now that blood is not the hereditary material and these terms have no biological meaning.

After the fusion of a human egg and sperm has taken place, the chromosomes in the **zygote** now exist as 23 pairs. The members of each pair are said to be homologous. **Homologous** chromosomes have the same genes for particular characteristics at the same location on the chromosome.

Each new cell has two genes for each characteristic, one from each chromosome of a homologous pair.

Talk about

If we all have the same genes on our chromosomes, why don't we all look the same?

Gregor Mendel carried out experiments with pea plants to answer this question. He worked with pure-breeding red- and white-flowered pea plants. **Pure breeding** meant he could produce only red-flowered offspring from red-flowered parents and white-flowered offspring from white-flowered parents.

Mendel decided to use the pollen from red-flowered plants to cross-pollinate the white-flowered plants. All of the flowers

FIGURE 8.19
Gregor Mendel.

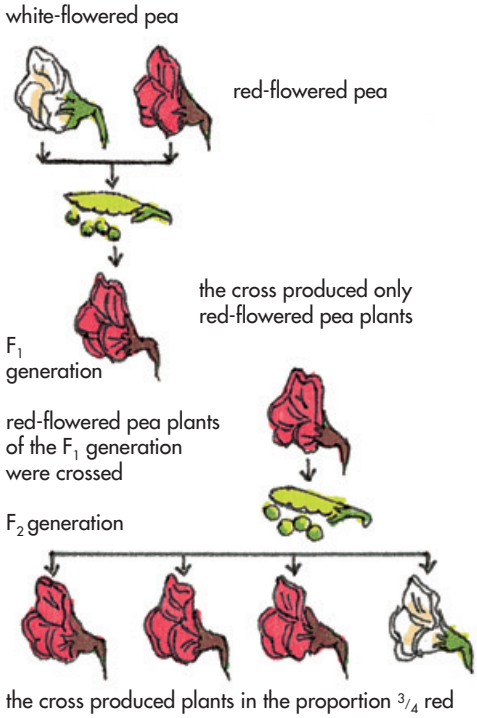

FIGURE 8.20
The results of some of Mendel's experiments.

that resulted from this cross were red. It seemed to Mendel that red somehow appeared stronger than white. He called red the **dominant character** and white the **recessive character**.

Mendel then cross-pollinated these new red flowers. Each set of crosses that he performed produced both white and red flowers in roughly the proportions of $\frac{3}{4}$ red flowers and $\frac{1}{4}$ white flowers.

It seems that genes work in pairs to determine which characteristic is shown or expressed. **Geneticists** would say that the gene with which Mendel was experimenting was pea flower colour and that the different forms of the gene (i.e for red flowers and white flowers) are the **alleles** for that gene.

The allele that produces a red flower is dominant so we can call it R. The allele for white flowers is recessive and we call it r. We use the capital and lowercase forms of

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the same letter, R and r, to show we are talking about the same gene. An RR combination of alleles will produce a red flower. Rr will also produce a red flower and only rr will produce a white flower.

The RR and rr combinations of alleles, where the offspring has identical alleles, are known as **homozygous**. The Rr combination of alleles is known as **heterozygous**.

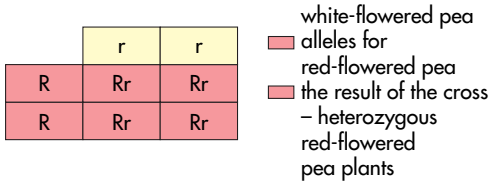

FIGURE 8.21
This diagram, called a punnet square, shows us all the possibilities for the offspring that may result. We do not assume that the offspring will appear exactly in this order and in this exact ratio, but it does tell us what we could expect to occur.

USING SCIENCE 7 ALTERNATIVE CHARACTERISTICS

Spreadsheet

Select some easily observed physical characteristics of students in your school, e.g. left-handed versus right-handed, ability to roll tongue, eye colour, widow's peak, attached earlobes. Add to this list if you can. Survey as many students as you can to find out the proportions of these characteristics in the school population. Show your results in a table and as bar graphs.

1 How could you determine if the characteristic is inherited or not?

2 Is it possible to decide whether any of these characteristics are dominant or recessive based on your findings? What else might you need to find out?

Question

Eye colour is determined by a gene with two major alleles, brown and blue. Brown (B) is dominant to blue (b). Using the same format as that shown in Figure 8.21, show how we might determine which offspring would result in the following crosses:

a a homozygous brown-eyed parent mating with a blue-eyed parent

b a heterozygous brown-eyed parent mating with a blue-eyed parent

c a heterozygous brown-eyed parent mating with a heterozygous brown-eyed parent.

Think about

1 In a particular herd, the sheep had either black wool or white wool. A farmer carried out a breeding experiment to determine which colour was dominant and which recessive. Two white sheep were crossed and white and black sheep were produced. Two black sheep were crossed and only black sheep were produced. Which allele of the wool colour gene is dominant? Which is recessive? Draw the cross between the two white sheep in a diagrammatic form to show the ratio of white and black offspring you would expect.

2 Brown hair in humans is dominant to blond hair. If you have blond hair but both of your parents have brown hair, what alleles must your parents have?

3 You are blond but your partner has brown hair. Your partner's mother, however, had blond hair. If you were to have children with your partner, what ratio of offspring would you expect?

Find out

You might like to do some research on Mendel and include some of the more unusual aspects of his life. Present your findings in a short talk to the class.

FACT FILE

Huntington's disease is a degenerative brain disease that causes a gradual loss of the control of movement and thinking capacity, and eventually death. Huntington's disease is caused by a dominant allele. Be careful not to think that dominant somehow means better or stronger or even more numerous.

Extension worksheet

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8.6

Intermediate inheritance

FIGURE 8.22
What would you expect the offspring of these two chickens to look like?
How sure are you?

It would be very straightforward if all genes were inherited as simple dominant and/or recessive processes. However, the determination of our physical characteristics is much more complex than that. There are three other forms of inheritance that we must also take into account. These are **incomplete dominance**, **co-dominance** and **sex-linkage**.

Incomplete dominance

Some genes do not have dominant and recessive alleles. Instead, they have alleles that provide a 'blending' of the different alleles. The homozygous form always breeds as you would expect, but the heterozygous form is a mixture of the two alleles.

Let us look at the example in domestic chickens where black feathers are incompletely dominant to white feathers. The heterozygous chicken has blue-grey feathers.

FIGURE 8.23
Intermediate inheritance in snapdragons.

If a homozygous black-feathered chicken (BB) mates with a homozygous white-feathered chicken (WW), all of the offspring would appear to be blue-grey feathered (BW). Note we used the letters B and W rather than B and b or W and w: this reminds us that neither allele is dominant to the other.

We call the first generation produced the **F_1 generation**.

	W	W
B	BW (blue/grey)	BW (blue/grey)
B	BW (blue/grey)	BW (blue/grey)

FIGURE 8.24
A diagrammatic representation of the cross between black- and white-feathered chickens.

If two blue-grey chickens were to mate and produce a second generation (the F_2 generation), we would find offspring in the ratio of $\frac{1}{4}$ black, $\frac{1}{4}$ white and $\frac{1}{2}$ blue-grey.

	B	W
B	BB (black)	BW (blue/grey)
W	BW (blue/grey)	WW (white)

FIGURE 8.25
A diagrammatic representation of the cross between two blue-grey feathered chickens.

FACT FILE

Up to 40% of your DNA has no known coding function. This DNA is often referred to as 'junk' DNA, although many geneticists suspect it does serve some purpose.

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We call the terms BB, BW and WW the **genotype** of the chickens. This means the type of genes that the chickens possess for that characteristic. The colour black, white or blue-grey is the **phenotype**, or physical appearance, of the chicken.

Co-dominance

There are some cases where both alleles in the heterozygous offspring are expressed in the phenotype. Blood grouping is an example. There are three alleles for the blood group gene, and blood groups A, B, O and AB exist. The heterozygous form AB has alleles that produce both A and B blood, but type AB has characteristics of both A and B blood—it is not a blend of the characteristics. The inheritance of blood groups is further complicated because both A and B are dominant to O. Hence, the O blood group will only occur in a homozygous individual.

TABLE 8.2 The inheritance of blood groups in humans.

Genotype	Blood group (phenotype)
AA	A
AO	A
BB	B
BO	B
AB	AB
OO	O

Questions

1 In snapdragons, red flower colour is incompletely dominant to white flower colour. A red snapdragon is crossed with a white flower and only pink flowers result. These pink flowers are then crossed with each other to produce a second generation. Draw these two generations of crosses, starting with the red and white parents and finishing with the F_2.

a What ratio of phenotypes would you expect at each generation?

b If 100 plants were produced in the F_2 generation, how many would you expect of each colour, pink, red and white?

2 The litter resulting from the mating of two short-tailed cats contains three kittens without tails, two with long tails and six with short tails.

a What would be the simplest way of explaining the inheritance of tail length in these cats?

b What would be the genotypes of the kittens? (Use your own letters.)

3 Would it be possible for a person with blood group A to mate with a person of blood group B and produce a child with blood group O? Show how you worked this out.

Think about

In all of the examples and questions that we have looked at to date, we have shown a direct link between genotype and phenotype, i.e. BB produces brown eyes in humans, BW produces blue-grey feathers in chickens, RW produces pink flowers in snapdragons. Do you think there are any other factors which affect an organism's phenotype? Give examples if you can. (Hint: Some hydrangea plants will produce blue flowers if they are grown in acidic soil and pink flowers if they are grown in basic soil. Identical twins brought up in different countries may appear to have differences in their phenotypes.)

Extra worksheet

FACT FILE

Some characteristics can be controlled by a large number of genes. In these cases the pattern of inheritance is very complex. Human eye colour, skin colour, height and intelligence are examples of traits controlled by multiple alleles. Within the human species these traits show continuous variation. For example, people are not either short or tall. All possible intermediate heights exist within a population. Up to five pairs of genes may determine human height.

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Sex determination

return to main menu

FIGURE 8.26
In humans, sex is determined by just two chromosomes. Females are XX and males XY. It is thought that a gene on the Y chromosome acts as a switch to cause an XY embryo to become a male.

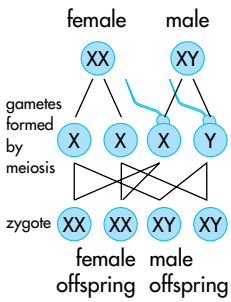

FIGURE 8.27
Male gametes with X or Y chromosomes determine the sex of the offspring.

FACT FILE

In birds the sex chromosomes are the same (ZZ) in the male and different (ZW) in the female. In some other organisms sex is determined by a single gene.

Of the 46 chromosomes that are found in each of your body cells, two of these play an important part in determining which sex you are. These are referred to as the X and the Y chromosomes. Females have two X chromosomes in each of their cells and males have an X and a Y. These chromosomes can be seen clearly in Figure 8.31.

All of the eggs produced by a female during meiosis will have 23 chromosomes, one of which is an X. Half of a male's sperm cells have an **X chromosome** and the other half have a **Y chromosome**. If a sperm containing an X chromosome fertilises an egg, the offspring will be a female (XX). If a sperm carrying a Y chromosome fertilises an egg the offspring will be a male (XY).

Talk about

Henry VIII was famous for having six wives. Of his six wives, only one bore him a son. Many of the other wives did not have any children or they had daughters, so he had them beheaded or he divorced them. He blamed his wives for not giving him the sons that he wanted. What advice would you give him today about where the blame lay?

Sex linkage

There are some genes that are found on the X chromosome and not on the Y chromosome. These are called the **sex-linked genes**. As there is no homologous gene on the Y chromosome, these genes can operate as a single allele in males.

Colour blindness is caused by a sex-linked recessive gene. This means that it is a gene found on the X chromosome and it is recessive to the dominant normal colour vision. If a woman with normal colour vision mated with a man with colour blindness, her daughters would all be heterozygous for colour blindness but would have normal vision. If her daughters were to mate with males with normal vision, none of their daughters would be colour blind but half of their sons would be expected to be colour blind. See Figure 8.28.

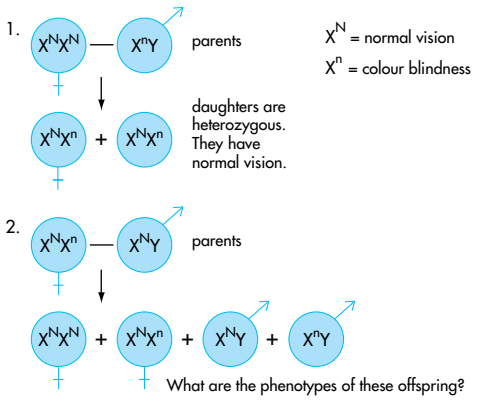

FIGURE 8.28
The inheritance of colour blindness in humans. Would it be possible to have colour-blind females?

Questions

1 Explain the process that leads to sex determination in humans.
2 What do we mean by sex-linked characteristics?
3 Why would you expect to find more colour-blind males than females in the world?

Think about

1 If a woman, heterozygous for colour blindness, marries a colour-blind male, what possible genotypes and phenotypes would you expect in their sons and daughters?
2 Haemophilia is also a recessive sex-linked condition. It is a disease

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EXPERIMENT 1 VEGETABLE PEOPLE

Word process

AIM

To model inheritance by building vegetable people.

MATERIALS

- potato
- onion
- sultanas
- peas (fresh)
- celery sticks—long and short
- carrot slices
- parsnip slices
- toothpicks

METHOD

1 There are two sexes of vegetable people—potato and onion—and each has body parts with appearances (phenotypes) which are expressions of dominant or recessive genes. Table 8.3 is a summary of the gene combinations (genotypes) of the body parts and the phenotypes they produce.

TABLE 8.3 Phenotypes and genotypes of vegetable people.

Characteristic	Phenotype	Genotype	Responsible genes
Eyes	sultana	BB or Bb	B (dominant)
	pea	bb	b (recessive)
Arms	long celery stick	LL or Ll	L (dominant)
	short celery stick	ll	l (recessive)
Feet	carrot slices	CC or Cc	C (dominant)
	parsnip slices	cc	c (recessive)

Table 8.4 shows the genotype of Ma Potato and Pa Onion.

TABLE 8.4

Parent	Sex chromosomes	Eye genes	Arm genes	Feet genes
Ma Potato	XX	bb	ll	Cc
Pa Onion	XY	Bb	LL	Cc

In your workbook write out Table 8.4 and for each genotype add the phenotype for the characteristic.

2 On small pieces of paper write each sex chromosome and gene for Ma Potato and Pa Onion's characteristics. This should give you four 'gene' piles each for Ma and Pa.

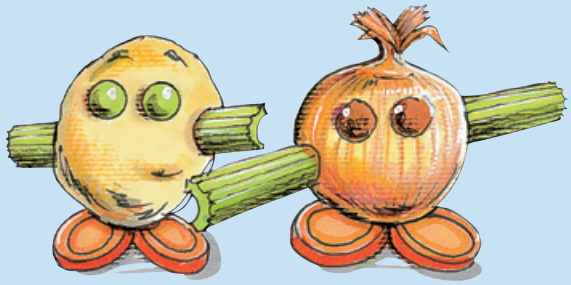

FIGURE 8.29

3 Randomly select one sex chromosome and then one of each gene from Ma's four piles of genes then do the same for Pa's genes. These are the genes of your vegetable child. Write out the genotype and phenotype of your vegetable child.

4 Construct your vegetable child from the materials provided.

5 Look at the vegetable children made by other members of the class and record the phenotype and genotype of at least four different examples.

6 All groups should now contribute to a set of class results. Each group should randomly select a pair of sex chromosomes from their two gene piles to determine the frequency of occurrence of potato and onion children. Then do the same for each of the other characteristics, remembering that there are three possible genotypes for each. Record these in a table.

DISCUSSION

1 What is the probability of a vegetable child being a potato or an onion? Compare this with the results of the class random selection.

2 Draw a punnet square to determine the probability of each genotype for the three characteristics. Compare this with the results of your class random selection.

EXTENSION

If you wanted to select genes for a second and a third vegetable child, would you need to do anything to your gene piles? Explain.

that prevents the affected person's blood from clotting. Generally, only males can have haemophilia, because female foetuses with haemophilia will abort naturally before birth. Explain why you think this might be the case.

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8.8

Chromosomal abnormalities

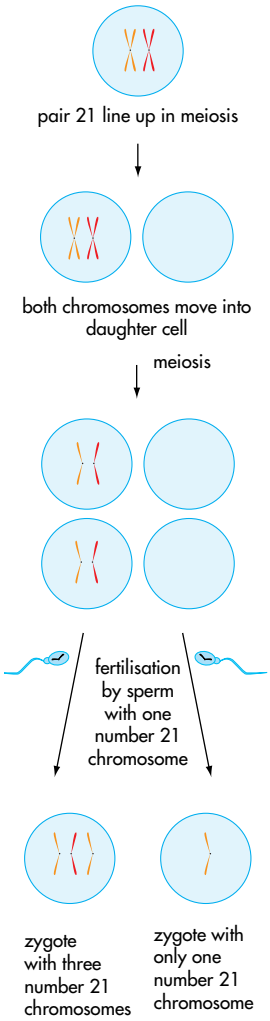

FIGURE 8.30
A cell is shown undergoing non-disjunction. Notice the effect on the number of chromosomes in the resultant cells after telophase is complete.

Karyotypes

It is possible to isolate some cells of a developing foetus and photograph the chromosomes in the cells. The photograph is then cut up so individual chromosomes can be arranged into homologous pairs according to their size. The pairs are numbered for ease of identification. The **karyotype** which is produced can provide information about the developing foetus.

Chromosomal defects

In all, about 2200 chromosomal variations have been described by scientists. Some are the result of a loss of part of a chromosome, a rare occurrence, but most **chromosomal abnormalities** are caused by changes in the chromosome number, either an increase or a decrease.

Non-disjunction

Before the chromosomes separate during meiosis they form pairs down the middle of the cell in preparation for pair separation. One member of each pair should go to each end of the cell. For some reason, in certain conditions, a pair may not separate properly and both chromosomes go to the one end. This non-separation, called non-disjunction, results in gametes that have too many or too few chromosomes.

Down syndrome

Down syndrome is a common disorder affecting about one in every 600 births, particularly the first-born children of mothers over 40. It was the first disorder found to be due to chromosomal abnormality. Before antibiotics, most of

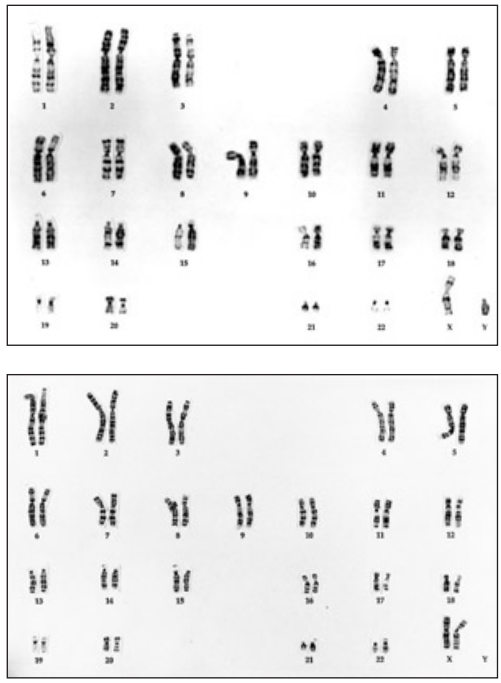

FIGURE 8.31
The karyotypes of a normal male and a normal female. What is the main difference that you can see between these two karyotypes?

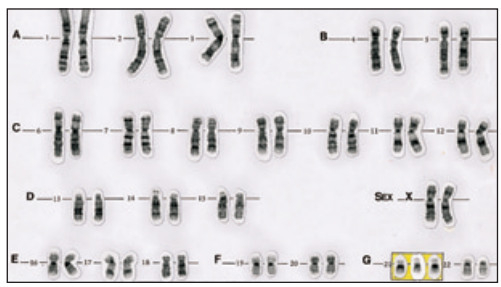

FIGURE 8.32
The karyotype of a person with Down syndrome. Can you determine which chromosome exists in threes rather than as a pair?

FIGURE 8.33
People with Down syndrome have a characteristic appearance.

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these children died in infancy from pneumonia, but now about half have a much greater chance of surviving.

In 1959 it was discovered that children with Down syndrome have an extra chromosome in their cells (47 instead of 46).

Klinefelter's syndrome

About one in every 500 males suffers from a disorder known as Klinefelter's syndrome. These people carry three sex chromosomes XXY instead of XY. Sufferers tend to have severe acne and an awkward build and do not achieve sexual maturity without hormone treatment. Even with treatment they remain sterile.

Turner's syndrome

Women who suffer from Turner's syndrome have only one sex chromosome (XO), so have a total of 45 chromosomes in each cell. This occurs in about one in 2500 births. They fail to achieve sexual maturity without hormone treatment and are sterile because they do not develop ovaries. Embryos with only a Y chromosome fail to develop and are aborted naturally.

'Super' males

Males with XYY chromosomes appear normal but are taller than average and tend to be less intelligent and more prone to aggressive behaviour. They can have children with a normal chromosome count. This condition occurs in about one in every 3300 live male births.

'Super' females

Women with an XXX chromosome count are sexually under-developed and are sterile. This condition occurs in about one in every 800 births.

Questions

1 What is a karyotype?

2 How are chromosomes organised in the normal male and normal female karyotype?

3a How many number 8 chromosomes do you have?

b How many number 8 chromosomes did you inherit from your mother?

c How many number 8 chromosomes did you inherit from your father?

4 Explain the term non-disjunction.

5 Complete the table below:

Syndrome	Chromosomes	Egg	Sperm	Frequency
Down	47	24	23	1 in 600
Klinefelter's	XXY	XX		
Turner's			0	
'Super' male		X		
'Super' female			X	

Think about

1 Examine the karyotype of the person with Down syndrome, Figure 8.32.

a Which chromosome is affected?

b What sex is this person with Down syndrome? How do you know?

2 At a chromosomal level, what is the difference between a male with Klinefelter's syndrome and a 'super' male?

3 Why do some people have their foetuses tested to produce a karyotype? Can you think of any objections to the procedure?

4 Do you think it would be possible to 'cure' someone suffering from a condition caused by a chromosomal abnormality?

Find out

Research one of the chromosomal abnormalities listed in this section. Prepare a report which includes causes of the condition, characteristics associated with the condition, and programs and treatments to assist sufferers of the condition.

FACT FILE

When older women have a baby they face a higher risk of chromosomal abnormalities such as Down syndrome occurring. It is possible to find out whether an embryo has a genetic problem by using a technique called amniocentesis. This involves using a needle to collect fluid from the amniotic sac in which the embryo is growing. The fluid contains cells from the embryo which can then be examined.

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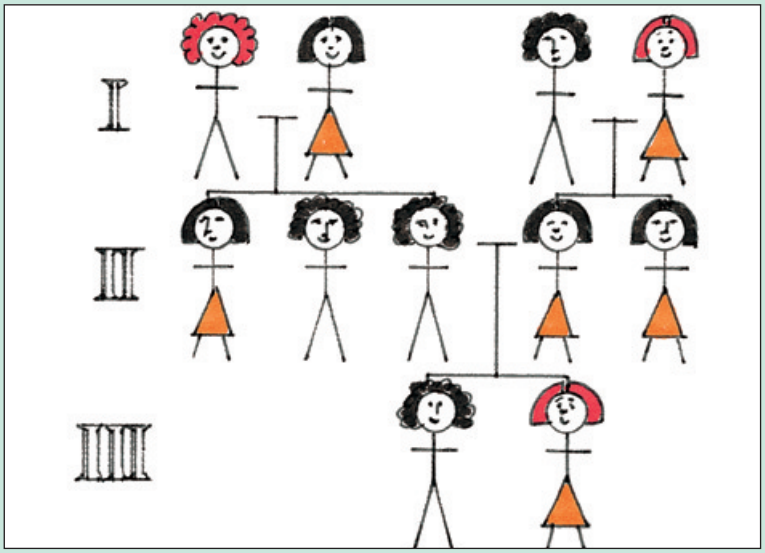
APPLICATIONS AND USES OF SCIENCE

Genetic counselling

Pedigrees

FIGURE 8.34
A 'stick-figure' diagram of inheritance.



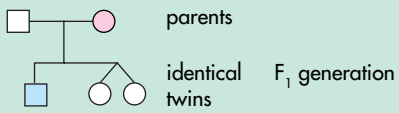


FIGURE 8.35
The symbols used in a pedigree.

As we have read in this chapter, genetically inherited characteristics are passed from one generation to another. A **pedigree** is a convenient and useful method of showing which individuals exhibit or carry characteristics in successive generations.

The method of displaying the information in Figure 8.34 is cumbersome, so a system of symbols is used.

Pedigrees and their use in genetic counselling

Many diseases and conditions are **inherited**. Cystic fibrosis, haemophilia, Huntington's disease and thalassaemia are examples of disorders which are passed from one generation to the next.

Cystic fibrosis causes a build up of thick mucus in the lungs. Sufferers may not survive beyond their teens. Thalassaemia is a serious form of anaemia which can be fatal. Haemophilia and Huntington's disease have already been mentioned.

It is possible, using the knowledge of a number of generations in a family, to

USING SCIENCE 8 PEDIGREES

Look at Figure 8.34. Draw a pedigree of the three generations of this family using the correct symbols. For each generation (labelled as I, II, and III), suggest the genotype of each individual and give an indication as to your confidence in this suggestion (e.g. very sure, sure, not sure). You will need to decide on appropriate symbols for the alleles for brown and red hair before commencing.

EXTENSION

If the couple in generation III had another child, what would be the chance of it being:

a brown haired
b red haired
c a boy with red hair?

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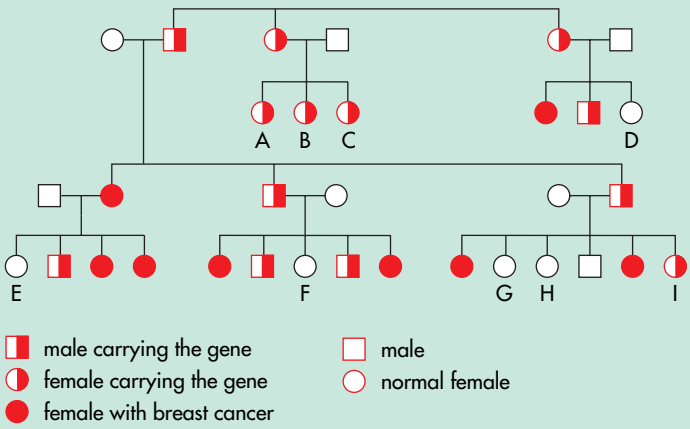

Individuals A–I would be counselled to ensure early detection of breast cancer

FIGURE 8.36
Pedigree of the incidence of breast cancer in a particular family.

determine the probability of having a child with a particular condition or the probability of a family member being a carrier of a particular gene. In cases such as these a couple wishing to have children can be advised as to the risk of their children suffering from the disease or being a carrier of it.

Specially trained counsellors work with potential parents to advise them about the options that they have if they are at risk of producing children with a genetic disease. On the basis of this information a couple may choose not to have children, to terminate a pregnancy, or make preparations for bringing up a child with a genetic disease.

Talk about

The characteristics of Huntington's disease do not usually appear until the affected person is in their 30s, often after they have already had children and hence potentially passed the gene on to them. Using family pedigrees, **genetic counsellors** can determine if a parent is likely to be at risk of producing a child with the Huntington's gene. Further tests can then be carried out to determine if a foetus has the DNA associated with the gene. If you were a genetic counsellor, what advice would you give to a person who is carrying a foetus that has been shown to be likely to have the Huntington's gene? Remember that the sufferer lives a perfectly normal and healthy life until the symptoms eventually start to appear.

Pedigrees can also be used to determine the likelihood of developing a disease later in life. Figure 8.36 shows the incidence of a particular mutant gene known to be a factor in the development of one form of breast cancer. The information obtained from this would enable women with this pedigree to be counselled so that early detection of any cancer would be possible. Women with a long family history of breast cancer could have the option of a mastectomy before the cancer could develop.

Questions

1 What advantages are there in using pedigrees?
2 What symbols are used in making pedigrees?
3 What is the role of a genetic counsellor?

Activity

Choose an easily observed characteristic that is found in members of your family. Investigate as many members of your family as you can to determine the presence or absence of this particular characteristic. Choose as many generations of your family as you can. Assign appropriate symbols and draw a pedigree showing the prevalence of this characteristic. Can you work out whether this is a dominant or a recessive characteristic? Show your working.

FACT FILE

Around the year 2003, one of the most monumental scientific investigations ever undertaken will be complete. It is called the Human Genome Project. It began in 1990 with a team of scientists slowly reading and documenting the 3 billion chemical codes which make up the human genome. The purpose of this project is to reduce or find cures for some of the 5000 genetic diseases which are known today.

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Using scientific language

Write the term which is described by each of the meanings given below:

a pairs of chromosomes containing genes for a particular characteristic which are located in identical positions
b alternative forms of the same gene
c having a single set of chromosomes
d a pattern of inheritance found with genes carried on the X chromosome
e having two different alleles for one gene
f the appearance or characteristics of an organism resulting from the interaction of its genotype and the environment
g a sudden and permanent change in the genetic material of an individual.

Check your knowledge

1 What is a karyotype?
2 Explain the term 'mutation'. What consequences can result from mutation?
3 How many chromosomes are present in a normal human:
a skin cell b muscle cell
c ovum d sperm.
4 What are sex-linked genes?
5 Describe and/or draw the basic structure of a DNA molecule.
6 Write down the matching strand of DNA to the following single strand:
AAC TTG CGT AGC TAG CTC TTA TAA CGC GGC
7 Copy and complete the following table showing the differences between mitosis and meiosis.

	Mitosis	Meiosis
Number of divisions	one	
Chromosomes in daughter cells		haploid
Number of daughter cells		
Cells using the process	'body' cells	

Apply your skills

1 In guinea pigs, black coat colour is dominant to brown coat colour.
a Select an appropriate symbol to represent the alleles for black and brown coat colour.
b What colour offspring would you expect from the following crosses? (Include a punnet square for each cross.)
i homozygous black × homozygous brown
ii heterozygous black × heterozygous black
iii heterozygous black × homozygous brown

2 A particular variety of radish is known to have three different shapes: long, short and oval. These shapes are known to be determined by a single pair of genes.
a Draw a punnet square to show the proportion of genotypes and phenotypes in the F1 generation if a homozygous long radish (LL) is crossed with a homozygous short radish (SS).
b Two radishes of the F1 generation are crossed. Draw a punnet square for this cross, showing the proportion of genotypes and phenotypes in the F2 generation.
c What type of inheritance is demonstrated in these radishes?

Challenge yourself

In pea plants, the height of the stem is controlled by a pair of alleles. The allele for tallness (T) is dominant over the allele for dwarfism (t). The flowers of a single tall plant (195 cm) were self-pollinated and the height variation in the plants produced is shown in Figure 8.37.

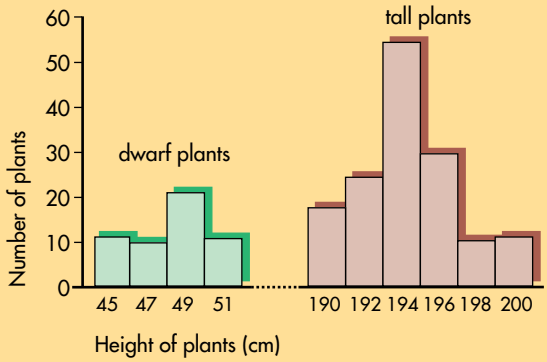

FIGURE 8.37

1 What is the meaning of the term self-pollinating?
Use the information shown in the chart to:
2 Work out the total number of tall plants and the total number of dwarf plants.
3 Work out the genotype of the parent plant. Give one reason to support your answer.
4 Draw a punnet square to show this cross and work out the ratio of genotypes in the F1 generation.
5 Consider the variation in height of the dwarf plants. Give one possible reason for this difference.
6 Design an experiment to show that some individual tall plants of the F1 generation are homozygous for tallness.

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return to main menu

CHAPTER 9

FIGHTING DISEASE

There is a war going on. It's a war between microbes and humans. Disease-causing microbes have been extremely effective at controlling human populations. Until the 20th century, that is! Humans have fought back with a deadly array of weapons. Antibiotics, antiseptics, disinfectants and vaccines are all used against the enemy. Some microbes such as the smallpox virus have almost been wiped off the face of the planet. It only survives in high-security laboratories.

By the end of the 1960s some scientists thought that it was only a matter of time before all disease-causing microbes would be defeated. However, at the end of the 20th century a whole range of deadly new bacteria and viruses have emerged. Ebola, Mad Cow, Legionnaires, HIV and Lassa fever are just a few causing concern.

Some diseases previously thought to be under control have re-emerged and some microbes have become resistant to the very antibiotics which were used to destroy them.

Scientific research is constantly looking for new methods of controlling the microbial menace.

In this chapter you will learn about the causes of disease and the ways the body tries to defend itself.

Web links

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Recognise that abnormal cell function may result in disease.
- Describe how a range of micro-organisms can cause illness in humans.
- Grow bacteria and fungi.
- Describe how the body's defence mechanisms respond to disease.
- Relate the scientific processes involved in the development of antibiotics.
- Explain how disease spread through food can be controlled and prevented.
- Discuss some of the issues associated with using vaccination to control disease.
- Discuss current health issues in Aboriginal communities.

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9.1

In sickness and in health

FIGURE 9.1
(above left)
Good personal hygiene involves washing your hands after using the toilet and before handling food.

FIGURE 9.2
(above right)
Regular exercise is essential to avoid many lifestyle diseases such as obesity and Type 2 diabetes.

Talk about

What causes disease and ill health? List the things you think can cause ill health. Compare your list with other students' lists.

Health is a state of physical and mental well-being. This state is achieved when all cells, tissues and organs are functioning normally. Often this state of well-being is taken for granted until we experience ill health or disease.

Disease can be defined as any condition in which the body or parts of the body do not function properly. There are many causes of disease. Some diseases are caused by **micro-organisms** such as **bacteria, viruses, fungi** and **protozoa**. The micro-organisms which damage or kill body cells are called **pathogens**. Pathogens can spread from one individual or host to another; hence, these diseases are described as **infectious**. Infectious diseases include influenza, diphtheria, thrush and malaria.

Many diseases are not transmitted from one person to another. These **non-infectious** diseases are caused by a wide range of factors. Causes of non-infectious diseases can be grouped in the following way:

- genetic or congenital—these occur when abnormal genes are inherited from parents
- age—as people get older some organs function less well
- environmental factors—exposure to various forms of radiation, poisons and pollutants
- lifestyle—poor diet, stress, drugs.

Unfortunately there are some illnesses that we do not know the exact cause of. An example is multiple sclerosis (MS), which is a nerve disease. We know how it affects the body, but not how it begins.

FIGURE 9.3
A diet rich in fruit, vegetables and grains but low in animal fats can reduce the risk of some diseases.

Staying healthy

While we cannot select the genes we inherit from our parents or stop the ageing process, we can make a conscious effort to maintain good health. By eating a balanced diet, taking regular exercise, practising good personal hygiene and avoiding environmental hazards we can reduce the chances of disease. Maintaining a healthy lifestyle helps our body's defence system to fight off pathogenic organisms and avoid disease.

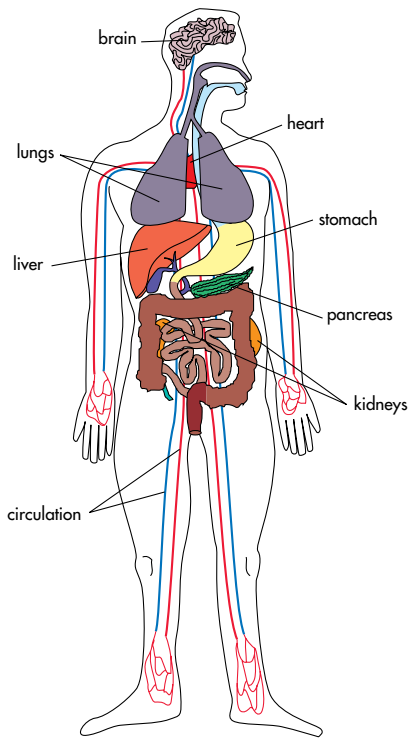

FIGURE 9.4
All areas of the body can be affected by disease.

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USING SCIENCE 1 FACTORS THAT CAUSE ILLNESS

1 Rule up a table with the five main causes of illness as listed above.
2 Under each heading list as many examples as you can of illnesses that are associated with the organs and systems shown in Figure 9.4.
3 Draw an outline of a human body similar to Figure 9.4. Enter illnesses from your list on the diagram.
4 Compare your lists with others in the class.
5 Make a list of any diseases that you think of that there is not yet a cure for.

Questions

1 Explain how you could tell if someone was suffering from a disease.
2 List five factors that can cause disease.
3 Why do you think people are encouraged to:
a wash their hands before eating?
b not drink alcohol to excess?
c run cars on unleaded petrol?
d wear sunscreen?

Think about

1 In the Middle Ages (1100–1500), some popular 'cures' were:
- **headache: hang the beak of a bird around your neck**
- **toothache: put goat's cheese on your head**
- **black plague: build a totem pole in the town to scare off the disease.**

Why do you think that people believed in these cures?
2 Why can't many diseases be cured? Give at least three reasons.
3 In what ways does everyone in Australia benefit from people having healthier lifestyles?
4 Suggest ways in which Australians could be encouraged to lead healthier lives.
5 From the lists of diseases you have made, write down which ones are preventable, and how they can be prevented.

Activities

What kills Australians?
Look at Figure 9.5 which shows the causes of deaths in Australia.

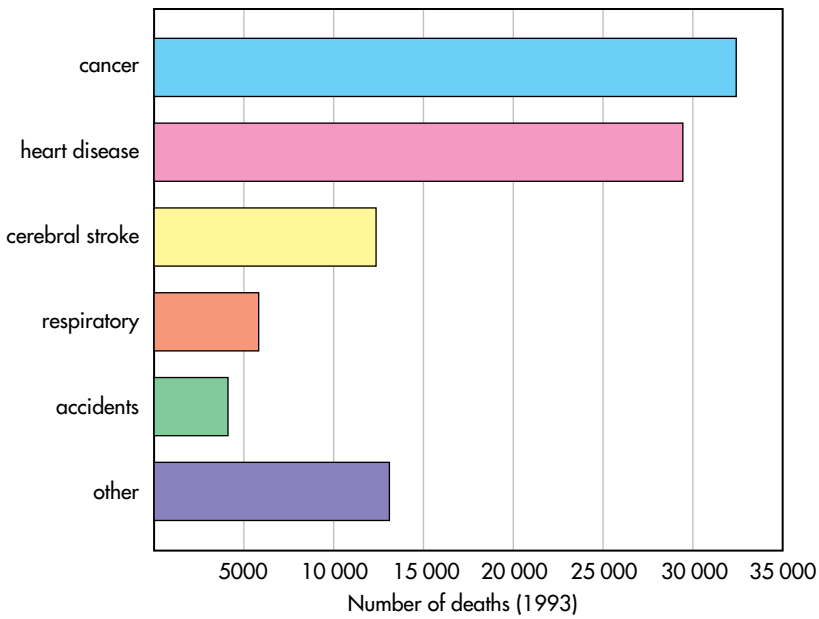

FIGURE 9.5
Causes of deaths in Australia in 1993.

1 Which diseases do you think could be reduced by:
a not smoking?
b being fitter?
c having a better diet?
d using a sunscreen?
2 In which category of causes would you place these diseases:
a heart attack?
b pneumonia?
c AIDS?
d strokes?
e deaths causes by drugs?

FACT FILE

In 1918, just after World War 1, influenza (the 'flu') killed an estimated 22 000 000 people world wide.

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9.2

Infectious diseases

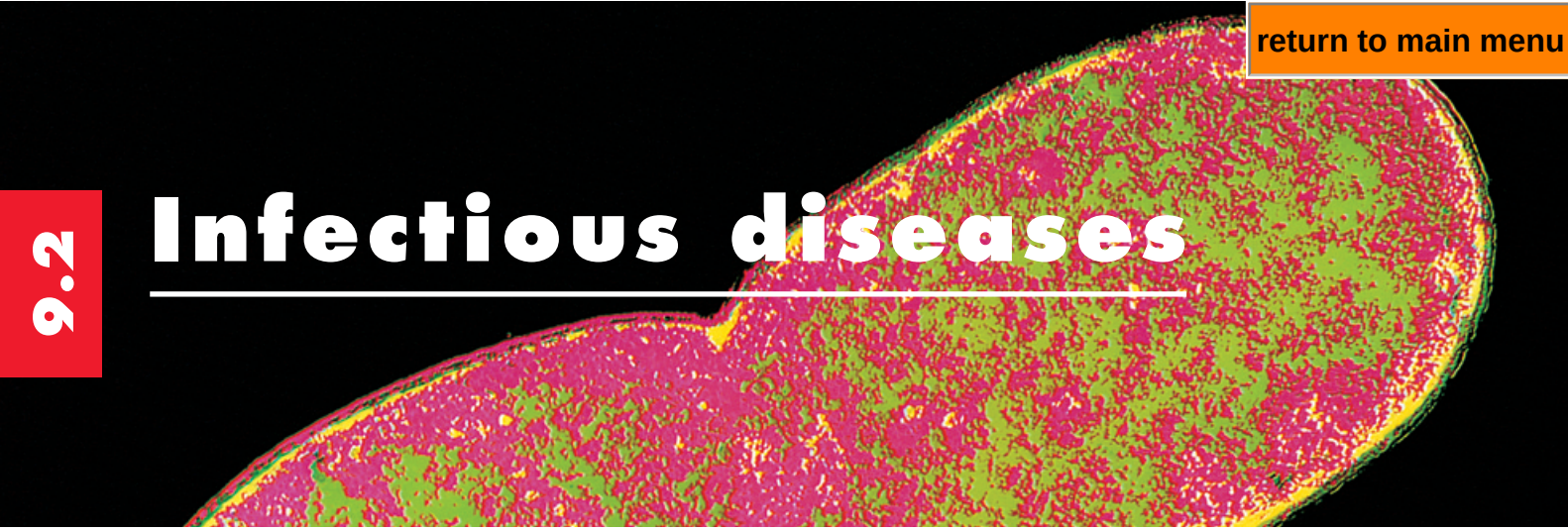

FIGURE 9.6
Escherichia coli dividing. Bacteria are small simple cells. They lack internal structures including a proper nucleus.

FACT FILE

Micro-organisms a hundred times smaller than viruses have been discovered. They are called prions and consist of protein with little or no nucleic acid. They are thought to cause diseases of the nervous system in humans.

Bacteria and viruses are responsible for most of the diseases which attack people and domestic animals. These micro-organisms harm us in two ways. They attack and destroy our cells and they release poisonous substances called **toxins**.

Micro-organisms can enter the body in a number of ways. Once inside they quickly multiply; this is called the **incubation period**. Several days or weeks may pass before the signs or symptoms of the disease appear. Sometimes people become infected without developing any **symptoms**. However, they may still pass the disease on to others. These individuals are called carriers. Micro-organisms can enter our bodies in a number of ways as shown in Figure 9.7.

Bacteria

Bacteria are simple single-celled organisms. Bacteria inhabit almost every place on Earth. They are very small, averaging 1–5 µm (micrometres) across. They can be found in water, in the air, in soil, on most surfaces, as well as on and in the bodies of animals, including us. Most do not cause disease.

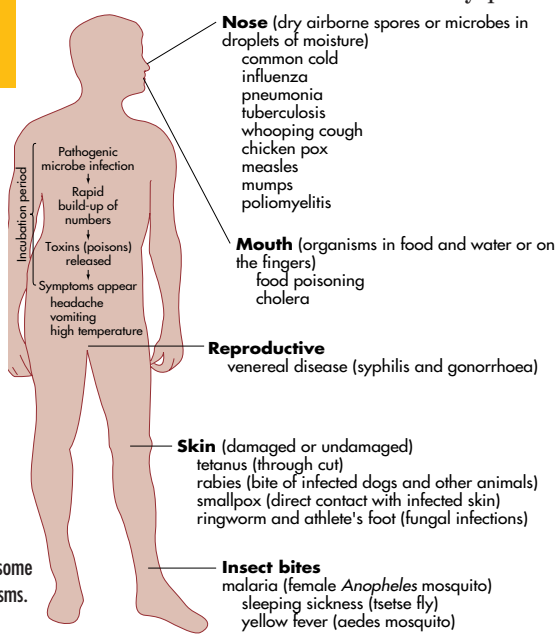

FIGURE 9.7
Entry points on the body for some disease-causing micro-organisms.

Once inside the body, bacteria can multiply very quickly. They do this by dividing in two (**binary fission**), and under ideal conditions some can do this every 20 minutes or so. This would mean just one bacterium could become 1 million in under 7 hours!

Viruses

Viruses are extremely small. Their size ranges from 0.01 to 0.3 µm across.

Viruses are not living organisms in the true sense of the word—they do not eat, drink, produce wastes, or do any of the other things living organisms do. Their structure is non-cellular (i.e. they are not made up of cells). They are simply a piece of nucleic acid (DNA or RNA) wrapped in a protein coat.

Viruses cannot even reproduce on their own! They can enter the body in the same ways as bacteria. When a virus enters a cell, it reproduces by forcing the cell to make more viruses. The cell then bursts, releasing the viruses to infect other cells.

Questions

1 What is a pathogen?

2 For each way that pathogens can enter the body, suggest at least one way this could be prevented.

3 How do bacteria reproduce? What could be an advantage of this method?

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EXPERIMENT 1 GROWING BACTERIA

Spreadsheet

When bacteria reproduce they form colonies. A colony may contain millions of bacteria. Different types of bacteria form colonies with different appearances—this helps microbiologists to identify the type of bacteria.

Warning: Treat all bacteria as potentially dangerous. Do not open the lids of the agar plates once you have grown the bacteria. Your teacher will dispose of the plates correctly.

AIM

To show that bacteria are common in our environment, including on our skin.

MATERIALS

- agar plates with nutrient agar
- permanent marking pen

METHOD

1 Take an agar plate, and mark the base as shown in Figure 9.8.

2 Run your finger down the side of the agar marked 'unwashed'. Be careful not to gouge into the agar.

3 Now wash your finger with soap and water and repeat the process on the 'washed' side.

4 Seal the plates with sticky tape and put them in an incubator set at 25–30°C overnight.

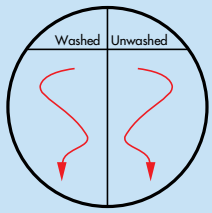

FIGURE 9.8
Preparing the agar plate.

RESULTS

1 Count the number of different types of colonies on each side, and the total number of colonies. Bacterial colonies tend to be smooth and round. You may also have some moulds: these tend to be more fluffy in appearance.

2 Describe the appearance of one or two of the colonies.

DISCUSSION

1 Explain your results.

2 Are there any bacteria on the 'washed' side? How could this have occurred?

3 Why is it important to wash your hands after going to the toilet and before eating?

EXTENSION

Design and carry out an experiment to test the effectiveness of a known disinfectant.

4 Why are viruses not considered to be living?

5 Describe the structure of a virus.

6 On average, how much bigger is a bacterium than a virus? How much bigger is a human cell than a virus?

7 How do pathogens cause the symptoms of a disease?

Find out

Mary Mallon, known as Typhoid Mary, was a famous carrier of typhoid fever. Find out how she spread the disease and how many people she infected.

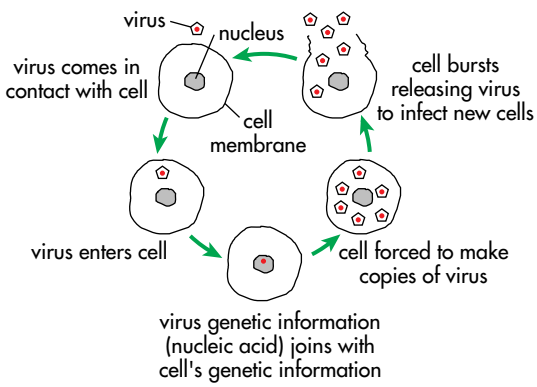

FIGURE 9.9
Virus infecting a cell.

Extra worksheet

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9.3

More pathogens!

FACT FILE

It has been estimated that by the time a person first shows signs of malarial fever, there are about 150 million protozoans in their blood.

Some **fungi** and **protozoans** (single-celled 'animals') can also cause diseases.

Fungi

Most fungi cause no problems for humans; in fact, we extract antibiotics from some, and eat others.

One of the more common fungal infections is **ringworm** (tinea). Tinea also causes **athlete's foot**. Ringworm can be caught from contact with infected people or animals such as cats and dogs. It is called ringworm because it grows in a ring pattern. Athlete's foot occurs between the toes, and can be spread by people sharing bathrooms. The fungi make spores which can be easily spread.

FIGURE 9.10
Ringworm.

Protozoans

Protozoans are small single-celled animal-like organisms which are common in water and moist soil.

They generally reproduce by dividing in two, but some can reproduce sexually. Most are not harmful to humans, but some are pathogenic. Many have a complex life cycle.

Diseases caused by these organisms are relatively few in Australia owing to sanitation, and our climate. For example, amoebic dysentery is spread through contaminated drinking water. It causes severe pain and diarrhoea. This is why it is important to only drink boiled or bottled water in some countries where there is not a guaranteed clean water supply.

Protozoan diseases tend to be more common in tropical climates. Sleeping sickness is most prevalent in Africa, and is spread by the bite of the tsetse fly.

The best known protozoan disease is malaria. This disease is a major world health problem. It is estimated that about 100 million people world wide have malaria. Malaria is spread from human to human by the *Anopheles* mosquito. The female mosquito bites an infected person and sucks up blood. The protozoan breeds in the salivary glands of the mosquito. When the mosquito bites a person it first injects saliva to stop the blood from clotting. In this way the protozoans are injected into another person. Malaria protozoans breed in a person's red blood cells. Infected cells then burst simultaneously, releasing millions of new protozoans to invade more cells. The disease is often fatal—partly because so many red blood cells are destroyed that the blood cannot carry enough oxygen around the body.

Symptoms include bouts of shivering, fever, sweating and fatigue. Travellers to countries where malaria occurs should take necessary medication.

There has been a massive world-wide effort to control malaria, the effort being directed at the mosquito rather than the protozoan.

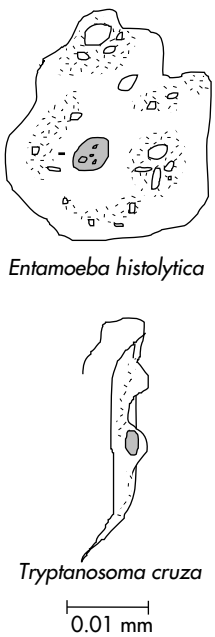

FIGURE 9.11
Some pathogenic protozoans that cause amoebic dysentery and sleeping sickness.

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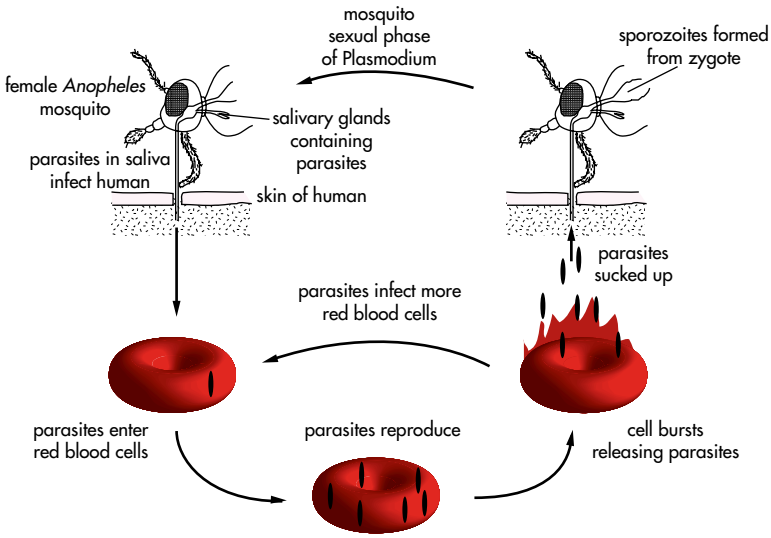

FIGURE 9.12
The life cycle of the malarial protozoan.

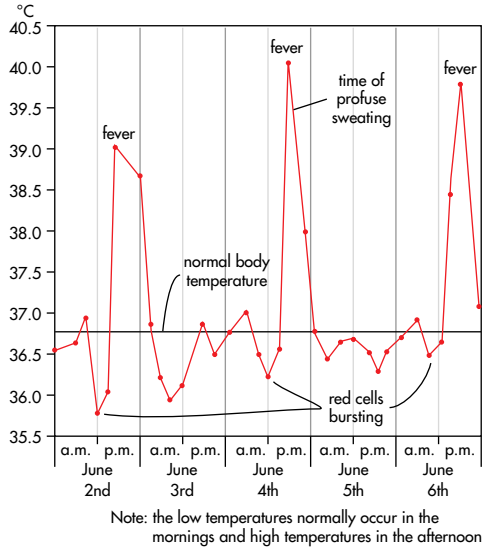

FIGURE 9.13
The temperature of someone suffering from malaria.

EXPERIMENT 2 GROWING FUNGI FROM SPORES

AIM
To show how fungi can easily be spread by spores.

MATERIALS
- agar plate with nutrient agar; or fresh bread
- a piece of mouldy bread or fruit
- small paint brush

METHOD
1 Transfer some spores from the mouldy sample to the agar plate or fresh bread either by direct contact, or by using a fine art paint brush.
2 Seal the agar plate (if used) with sticky tape, label it, and place the bread or agar plate in an incubator or warm place for 24–48 hours.

Warning: Do not open the plate unless directed to by your teacher!

RESULTS
1 Describe the appearance of any fungus (mould) that grows.
2 Under supervision you can look at the fungus under a dissecting microscope, or make a slide and look at it under a monocular microscope.

DISCUSSION
1 What did the experiment show?
2 What are some other ways that spores could be spread?

Questions

1 Why is it incorrect to call a ringworm a worm?
2 How can tinea be spread between family members?
3 How could you prevent the spread of tinea?
4 Why are travellers to certain countries advised to drink only unopened bottled water?

Think about

1 An organism which carries a parasite from host to host is called a vector. What are the vectors for sleeping sickness and malaria?
2 What precautions would you take if you were travelling in a country where malaria occurred?
3 Why do you think we do not have malaria in Australia?

Web page

Foundation worksheet 1

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9.4

Defence systems

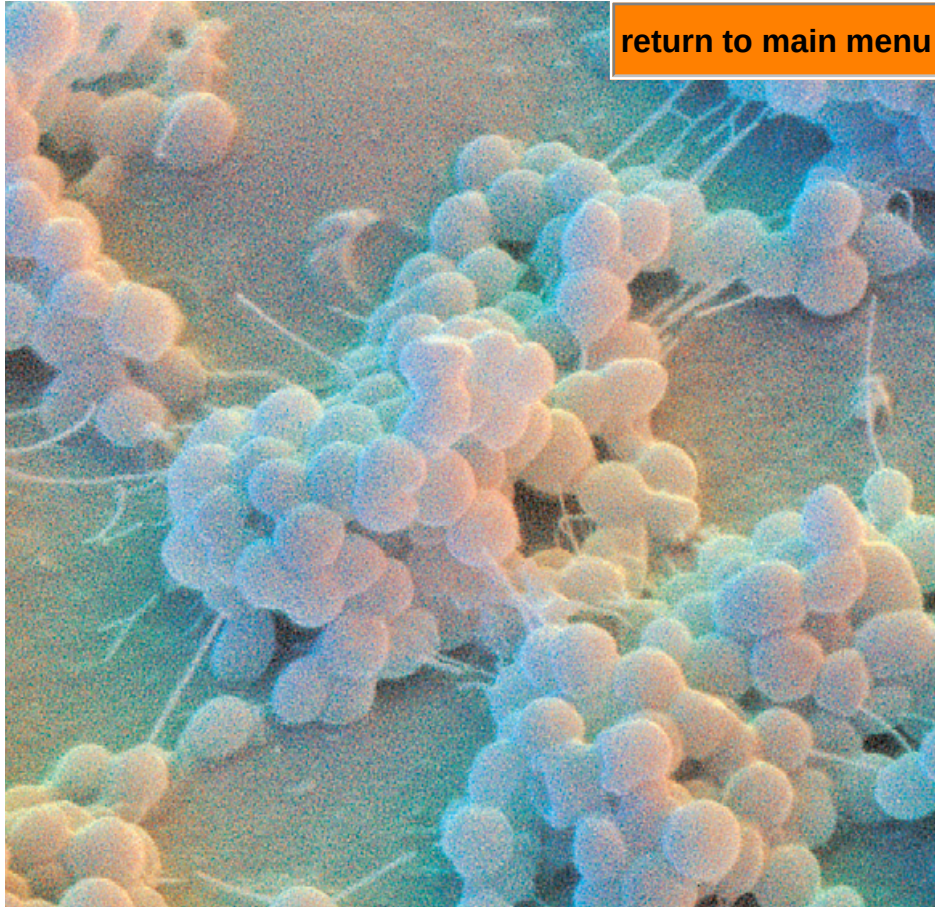

FIGURE 9.14
The bacteria that live on our skin.

Our bodies are under constant attack by micro-organisms and other foreign substances. However, we seldom become ill. The body can defend itself against disease in a number of ways.

Putting up the barriers

The skin and other body surfaces are natural barriers which prevent the entry of many disease-causing agents. The skin produces sweat which contains fatty acids and lactic acid and this acidic secretion kills many micro-oganisms. Some non-pathogenic bacteria that can tolerate the acidic conditions live in the pores and produce chemicals which inhibit the growth of other organisms.

The internal surfaces of the nose, throat and windpipe produce sticky **mucus** which traps airborne particles before they reach the lungs. Nasal hairs and cilia lining the air passages also filter particles and move them out of the respiratory system.

The tear ducts and salivary glands produce chemicals which destroy micro-organisms. The stomach produces hydrochloric acid. This kills most pathogens before they can enter the blood stream. The acidic environment of the vagina also inhibits the growth of invading organisms.

The immune system

In times of stress or accidents foreign particles can pass through the barriers of the skin and linings. When this happens the body's immune system responds.

Travelling around the bloodstream are millions of **phagocytes** (a type of white blood cell). If the body becomes infected, phagocytes will engulf and destroy any invaders. The site of the battle between the phagocytes and the invaders soon becomes red, swollen and painful. This response is called inflammation. Phagocytes continue to attack the invaders until they are destroyed or the phagocytes themselves die. The mixture of dead phagocytes, body cells and invaders appears as pus at the infection site.

If the pathogens are not destroyed they will soon spread into the **lymphatic system**. The lymphatic system consists of lymph vessels and lymph nodes. The lymph nodes, spleen and thymus filter fluids containing

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pathogens and destroy them. They also produce lymphocytes. Lymphocytes offer another line of defence because they make antibodies. **Antibodies** stick to antigens (proteins on the surface of foreign particles) and bring about the destruction of the invader in two ways.

1 They destroy the pathogen or its toxin.
2 They cause the pathogens or toxin to clump together. The clump is then engulfed and destroyed by lymphocytes.

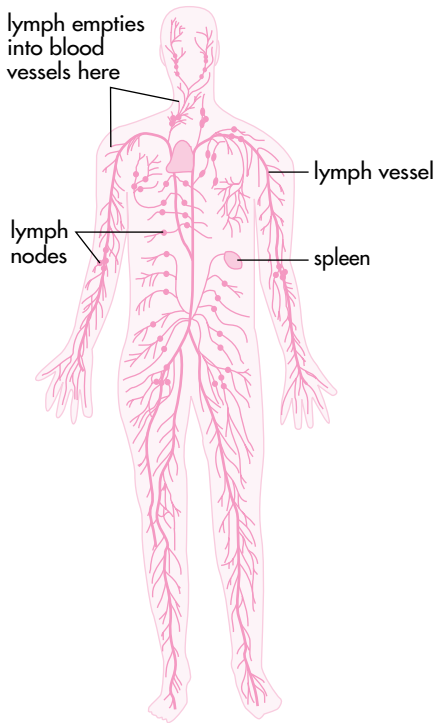

FIGURE 9.15
The human lymphatic system.

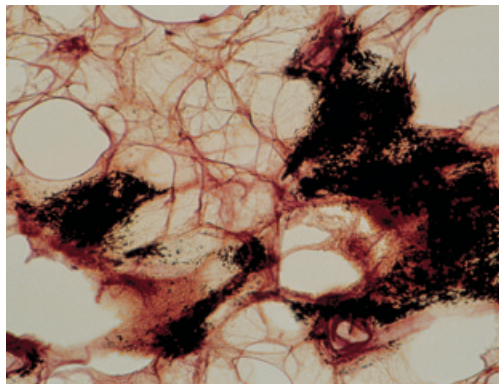

FIGURE 9.16
Killer cells destroying invaders.

Questions

1 Write a paragraph explaining how the body defends itself. Use the following key terms: barrier, secretion, mucus, inhibit, engulf.
2 Name three locations where the body produces acid which inhibits the growth of pathogens.
3 What is the difference between antigens and antibodies?
4 Pus in a wound can be seen as a good sign. What does the appearance of pus tell us?

Think about

The body is considered to have three lines of defence in the war against disease. What are they?

Find out

1 Disease can occur in the body's immune system. Select one of the following diseases, research it and write a report:
a leukemia
b AIDS
c allergy.
2 Australia has produced some well-respected immunologists such as Sir Frank Macfarlane Burnet, Sir Gustav Nossal and Peter Doherty. What is an immunologist and why are these men so highly regarded?

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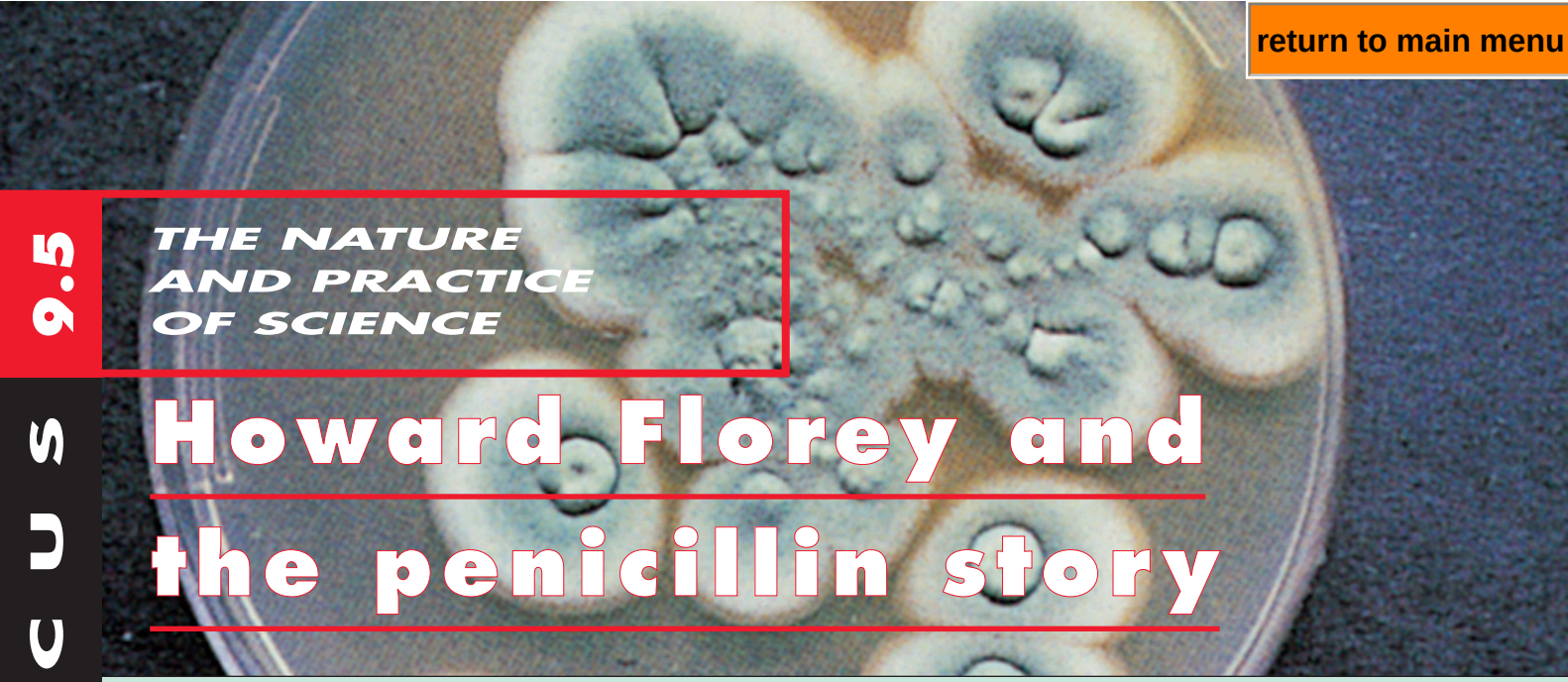

THE NATURE AND PRACTICE OF SCIENCE

Howard Florey and the penicillin story

FIGURE 9.17
Penicillium mould.

FACT FILE

Howard Florey was born in Adelaide in 1898. At school he was an excellent student. In 1917 he began studying medicine at the University of Adelaide. He won a scholarship in 1921 and left for Oxford University in England to begin a research career investigating naturally occurring antibacterial agents (substances which could kill bacteria). After reading an article written by Alexander Fleming, Florey began to investigate a mould called *Penicillium*.

Early observations of moulds

For thousands of years people have observed the disease-destroying abilities of moulds without understanding how they worked. Louis Pasteur in the 1870s observed that mould inhibited the growth of anthrax bacteria. Joseph Lister observed that samples of urine contaminated with mould didn't allow bacteria to grow. Then in 1928, Alexander Fleming observed that bacteria growing in a petri dish were killed by mould. The mould had accidentally contaminated the petri dish when the lid had been left off. Fleming **hypothesised** that the mould produced a chemical which could kill the bacteria. He called the chemical penicillin. No further interest was shown in **penicillin** until Howard Florey decided to investigate its action.

FIGURE 9.18
A child with facial infections treated with penicillin in 1942 showed a remarkable recovery.

In 1938, Florey gathered a team of scientists to work together on *Penicillium*. Today scientists often work in teams but at that time it was unusual. The team had several problems to solve.

- How could they grow the mould in large quantities?
- How could they extract the antibacterial chemical called penicillin from the mould and purify it.
- How did penicillin work and was it safe to use on patients?

Once the first sample of penicillin was produced they began to test it on living things. One member of the team, Ernst Chain, tested penicillin on eight mice injected with a deadly bacteria. Four of the eight mice were given penicillin. These mice survive but the four untreated mice soon died. Then in 1941, a man suffering from severe blood poisoning was injected with penicillin. At first he showed signs of recovery; however, the team didn't have enough penicillin to keep up the treatment and he died. Other patients, particularly children, showed remarkable recoveries.

Florey realised that penicillin was needed in large quantities. In the middle of World War II he travelled to America in a blacked-out aeroplane to arrange for large-scale production. At first the penicillin was only available for the military and it was used to save the lives of many soldiers.

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Then in 1944, Australian began manufacturing penicillin and was the first country to make it available to all its people.

The **antibiotic** penicillin was a miraculous drug that has saved millions of lives since its development. Howard Florey, Ernst Chain and Alexander Fleming all received the Noble Prize in 1945 for their work on penicillin, one of the greatest medical advances of the 20th century.

Modern antibiotics, antiseptics and disinfectants

The first antibiotics were produced from micro-organisms, mainly fungi, but many of the commonly used antibiotics are now synthetically manufactured. Different antibiotics have different actions, but most either stop the bacteria dividing by attacking their cell walls or disrupt their internal functioning.

Today there are many different antibiotics available to treat the range of infections. Antibiotics can be taken in tablet form or given in injections. Unfortunately, however, some types of bacteria have become resistant to certain antibiotics, and so the search for new drugs is always on.

An **antiseptic** is a substance which can destroy bacteria, and which can be used on the body. It cannot be consumed or injected like an antibiotic. A **disinfectant** is similar to an antiseptic, but is used on surfaces other than the body.

Questions

1 Explain the term 'antimicrobial agent'.

2 What are antibiotics and how do they work?

3 What three problems did Florey have to solve before the drug penicillin was used to treat patients.

4 Why did Dr Chain treat four of the eight mice with penicillin and leave four untreated?

5 What advantages would scientists gain by working in a team?

6 What important contribution did Alexander Fleming make?

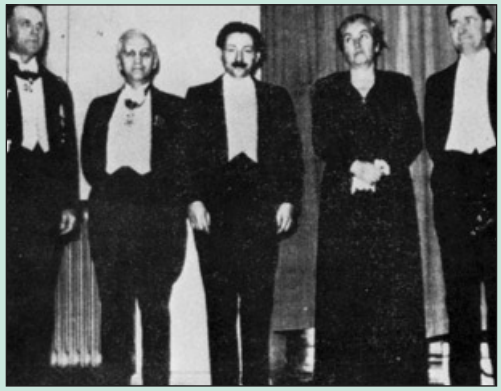

FIGURE 9.19
From left to right: Sir Alexander Fleming, Dr Ernst Chain and Sir Howard Florey at the Nobel Prize Festival in 1945.

FIGURE 9.20
Antibiotics, antiseptics and disinfectants are agents used to destroy bacteria in different situations.

7 Explain the difference between antibiotics and antiseptics.

Think about

1 Why are biologists continually trying to find new antibiotics?

2 Doctors advise that patients should take the complete course of antibiotics prescribed. This often means taking the treatment after the symptoms disappear. Why do they give this advice?

3 Antibiotics are not effective against viral diseases such as influenza and measles. Explain why, using your knowledge of viruses.

Find out

What contribution did Joseph Lister make to medicine?

Activity

Antibiotic discs are small circles of filter paper soaked in antibiotic solutions. They can be used to test the effectiveness of the antibiotic against bacteria growing in petri dishes. Your teacher may place a disc in a petri dish of bacteria for you to observe.

Foundation worksheet 2

FACT FILE

Penicillin is made by the fungus *Penicillium*. This is often observed as a grey-green mould growing on fruits such as oranges. By preventing bacteria from growing nearby, the fungus does not have to compete for food and space.

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9.6

Food alert

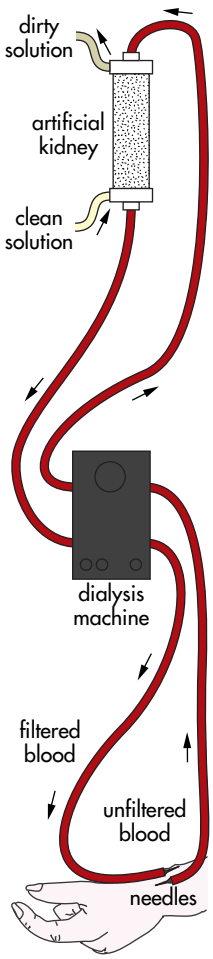

FIGURE 9.21
During dialysis blood is pumped through a dialysis machine, cleaned in an artificial kidney and returned to the body. To clean all of the blood takes up to 3 h.

Talk about
What causes food poisoning? What happens to people who have it? How can it be prevented?

Haemolytic Uraemic Syndrome (UHS)

In January 1995 a number of children in South Australia became seriously ill with a mystery disease.

It was up to a team of doctors and scientists to work out the cause of the illness and to find a way of treating it.

First, they had to find out if there was something in common with all the sufferers—where had they been, what had they come in contact with, and what had they eaten? It seemed that they had all eaten a particular brand of a **cured meat** called mettwurst. Samples of the mettwurst were tested to see if any bacteria could be grown from it. One problem was that most of the mettwurst had already been eaten!

A strain of *Escherichia coli* (*E. coli*), a bacterium common in human and animal intestines, was found, and was grown in large numbers in a liquid nutrient solution.

Once the bacterium was isolated and identified, its story could be pieced together. After the bacterium has been eaten (only a few need to be consumed), it goes through to the large intestine, where it sticks to the inside wall. At first it causes diarrhoea, which contains some blood because the intestine wall is slightly damaged. Here the bacteria breed, and produce a toxin which enters the bloodstream and moves throughout the body, affecting cells of the kidneys, and those that line blood vessels.

Its effect on blood vessels results in blood clots forming—these can block the small blood vessels in the kidney, and sometimes the brain, causing further damage.

The main effect seen in the children was kidney failure. As a result, they had to have **dialysis** treatment, sometimes daily. If the blood is not filtered regularly to remove the body's waste substances, then a person will become seriously ill, and eventually die. Unfortunately one child did die from this disease, which is called Haemolytic Uraemic Syndrome.

The next problem was how to treat the disease. Antibiotics were ruled out as they kill bacteria by causing them to burst, and this would release more toxins. There is also, at the moment, no way of removing or deactivating the toxin. Fortunately intestinal bacteria are eventually eliminated in the faeces, and so the cause of the disease will eventually pass. It was decided therefore to treat the effects of the disease (i.e. dialysis) until the bacteria were eliminated from the body. Luckily, the kidneys of the majority of children will eventually heal and return to working order.

The outbreak has since passed, but we can be sure that in the future there will be outbreaks of new or rare diseases, once again sending a team of scientists into action.

Questions

1 Why was it important for the source of the disease to be identified quickly?
2 What could be meant by a strain of a bacterium?
3 Draw a flow chart to show how the bacterium was identified.
4 Why couldn't antibiotics be used to treat the patients?
5 List the people involved in identifying and treating the disease. Write down how each would contribute.

Think about

Our drinking water is regularly checked to see if *E. coli* bacteria are present. How would they find their way into the water supply?

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Food poisoning

The incidence of food poisoning is on the increase. There are about 2 million cases in Australia each year. Pathogenic bacteria and viruses contaminate food and cause nausea, vomiting, diarrhoea and upset stomachs. In severe cases death may result, particularly in children, the elderly or people with other illnesses.

FIGURE 9.23
How safe is your lunch?

Causes of food poisoning

From 1980 to 1995, 128 food-poisoning outbreaks were reported. This resulted in 5952 individual cases and six deaths.

The causes of these food poisonings included:

- inadequate cooking (food should be heated to a high temperature until thoroughly cooked)
- inappropriate storage (food should be frozen or refrigerated at very low temperatures)
- contamination from infected water, soil and animal wastes (via unclean utensils, surfaces and storage containers)
- poor personal hygiene by food handlers (failure to wash hands, cover hair, cuts and coughs).

FIGURE 9.22
Processed meat and meat products have been involved in several food-poisoning outbreaks.

FACT FILE

In a recent study into food safety, 40 families were asked to prepare sandwiches as they would for school lunches. The sandwiches were collected between 8 and 9 a.m. and stored at 26°C until 12.30 p.m. They were then chilled and tested. The results showed that one in seven breached food safety standards.

TABLE 9.1 Some causes and effects of food poisoning.

Cause	Usual food contaminated	Symptoms	Incubation period
Campylobacter jejuni	Raw chicken, raw milk, untreated water	Diarrhoea, fever, stomach pains	2–7 days
Staphlylococcus aureus	Meat and meat products, eggs, milk and dairy products	Nausea, vomiting, diarrhoea, cramping	1–6 h
Salmonella species	Poultry, fish, milk, cream, peanut butter, chocolate	Stomach pains, nausea, watery diarrhoea	8–72 h
Listeria monocytogenes	Milk, cheese, raw vegetables, smoked mussels	Fever, chills, backache, headache, diarrhoea, vomiting	Average 30 days
Clostridium botulinum	Home preserves, canned food, soft cheeses	Dizziness, blurred vision, difficulty swallowing and breathing, death	18–36 h

Questions

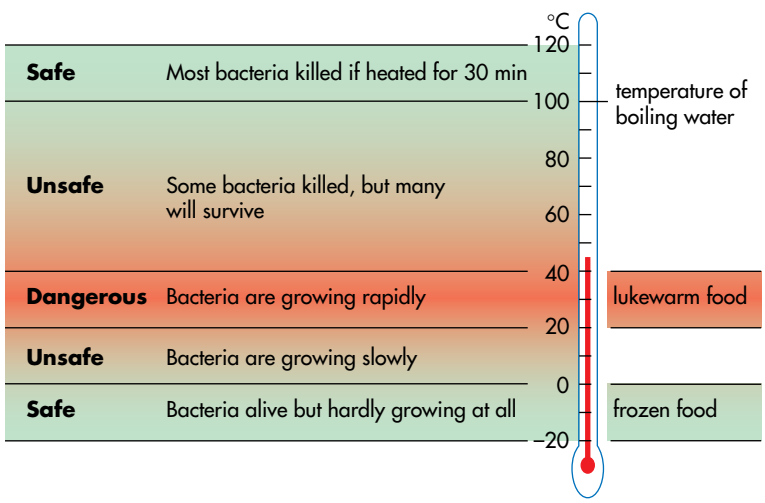
1 Examine Figure 9.24 and use it to explain why inadequate cooking and storage can lead to food poisoning.

2 People eat more processed foods and take-away meals now than in the past. How could this lead to more cases of food poisoning?

Activity

Design a pamphlet on ways to avoid food poisoning at home. Include advise for eating out.

FIGURE 9.24



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Vaccination

FACT FILE

The first vaccinations of sorts were practised by the Chinese thousands of years ago. They treated people (with some success) with scrapings from the scabs of others who had survived smallpox, a deadly disease.

Immunity

Talk about

Why do we only ever catch some diseases such as chicken pox or measles once in our life and are then immune to subsequent infections?

When we are exposed to a pathogen such as a bacterium or virus, the immune system recognises antigens on the surface of the pathogen. **Lymphocytes** then make large numbers of **antibodies** to destroy the pathogen. After the infection, some of the lymphocytes remain in the body along with the antibodies. This provides a level of protection against re-infection by the same pathogen. We say a person has developed natural immunity to that disease. The immunity may last several months or years.

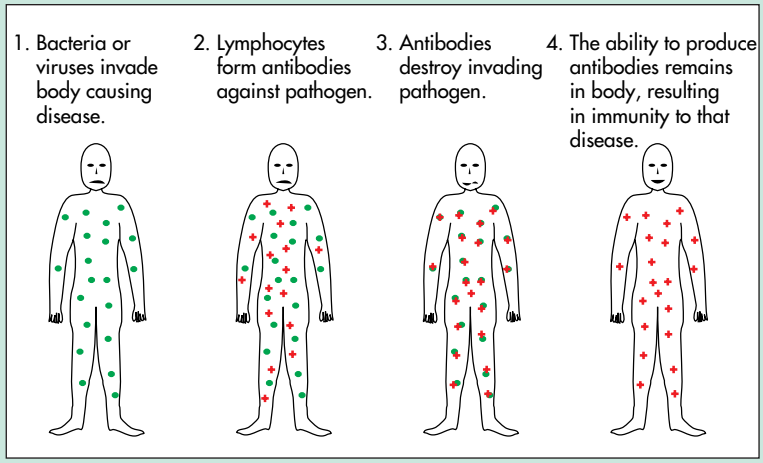

FIGURE 9.25
Antibodies are formed against invading micro-organisms. As well as destroying the pathogen, they give immunity to subsequent infections.

Vaccination

Allowing for **natural immunity** to occur is acceptable for non-life-threatening diseases, but not for serious diseases such as poliomyelitis, tetanus and others. This is where vaccinations are important. **Vaccination** or **immunisation** is the process of administering vaccines either by injection or by swallowing. A **vaccine** commonly contains a weakened form of a pathogen which will cause antibodies to be formed in the body but which gives rise to only very mild disease symptoms. These antibodies now give immunity against the serious form of the disease.

TABLE 9.2 Australian Standard Vaccination Schedule.

Age	Disease	Vaccine
2 months	Diphtheria, tetanus, pertussis Poliomyelitis Hib	DTPa or DTPw OPV-Sabin vaccine Hib vaccine (HbOC or PRP-OMP)
4 months	Diphtheria, tetanus, pertussis Poliomyelitis Hib	DTPa or DTPw OPV-Sabin vaccine Hib vaccine (HbOC or PRP-OMP)
6 months	Diphtheria, tetanus, pertussis Poliomyelitis Hib (HbOC schedule only)	DTPa or DTPw OPV-Sabin vaccine Hib vaccine (HbOC)
12 months	Measles, mumps, rubella Hib (PRP-OMP schedule only)	MMR Hib vaccine (PRP-OMC)
18 months	Diphtheria, tetanus, pertussis Hib (HbOC schedule only)	DTPa or DTPw Hib vaccine (HbOC)
Prior to school entry: 4–5 years	Diphtheria, tetanus, pertussis Poliomyelitis	DTPa or DTPw OPV-Sabin vaccine
10–16 years	Measles, mumps, rubella Hepatitis B (1st dose)	MMR HBV
1 month later	Hepatitis B (2nd dose)	HBV
6 months after 1st dose	Hepatitis B (3rd dose)	HBV
Prior to leaving school: 15–19 years	Diphtheria, tetanus Poliomyelitis	Td (ADT) OPV-Sabin vaccine
Every 10 years	Diphtheria, tetanus	Td (ADT)
Post-partum for non-immune women	Rubella	Rubella vaccine or MMR
Over 50 years	Pneumococcal infections	Pneumococcal vaccine (every 5 years)
(Aboriginal and Torres Strait Islander people)	Influenza	Influenza vaccine (annual)
Over 65 years	Pneumococcal infections Influenza	Pneumococcal vaccine (every 5 years) Influenza vaccine (annual)

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FIGURE 9.26
Chickenpox. A vaccine for this disease is now available.

FIGURE 9.27
Vaccinating a child.

Young children especially are at risk of getting a disease and passing it on to others if they are not vaccinated. Vaccinations are not compulsory anywhere in Australia.

Vaccination rates in Australia fell to a low 52% in 1997. This is one of the lowest vaccination rates in the developed world. At the same time the incidence of some infectious diseases are increasing. Vaccination programs in the past dramatically reduced the incidence of diseases like whooping cough but now it has reappeared. In 1996 there were 1007 reported cases of whooping cough, also known as pertussis, in Victoria, with the death of a 10-week-old baby. In New South Wales 975 cases occurred in the same year with two 7-week-old babies dying from the infection.

WHY CAN'T WE CURE THE COMMON COLD?

This is a very common question asked, especially by people who are currently suffering with a cold.

Since viruses are not really living, it is hard to kill them. Antibiotics are not effective against viruses, and so will not cure colds and flu. There are a few anti-viral drugs, but because they can have harmful side-effects they are only used for very serious diseases.

Some viruses can be vaccinated against, for example polio, smallpox, rubella and hepatitis. When you have a viral disease, the body's immune system learns to recognise certain chemical markers (antigens) on the surface of the virus. This takes a little time. Vaccines make the body's immune system act as if you have had the disease, and so it is ready for any real invasion.

Unfortunately in the case of colds, flu and HIV, the chemical markers keep changing. Some of the viruses made by infected cells have altered antigens, and so the body cannot recognise them again. If you have two doses of the flu in a year, then it is due to two different viruses. Being vaccinated in advance against one type of the flu will not prevent you from catching the other.

FACT FILE

The first true vaccination was performed by Edward Jenner around 1790. He deliberately gave people cowpox, a relatively harmless disease. The people then became immune to cowpox and smallpox. Jenner did not know how the vaccination worked—the explanation came 100 years later. Smallpox is now eliminated from the world.

Extension worksheet

Internet assignment

USING SCIENCE 1 SHOULD IMMUNISATION BE COMPULSORY?

Despite what many people see as the benefits of immunisation, not all parents choose to have their children or themselves fully immunised.

1 In a group, make a list of reasons why people should be immunised.

2 Make a list of reasons why some people are not immunised.

3 Should immunisations be compulsory? Make lists of possible arguments for and against compulsory immunisations. This issue could be debated by the class.

4 Write an article for a newspaper or design a pamphlet urging people to get immunised.

EXTENSION

Multimedia

1 Some diseases cannot yet be vaccinated against. Can you find out which ones, and why?

2 Vaccinations can work in two different ways—some are called passive, while others are active. Find out the difference between the two.

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return to main menu

Controlling disease

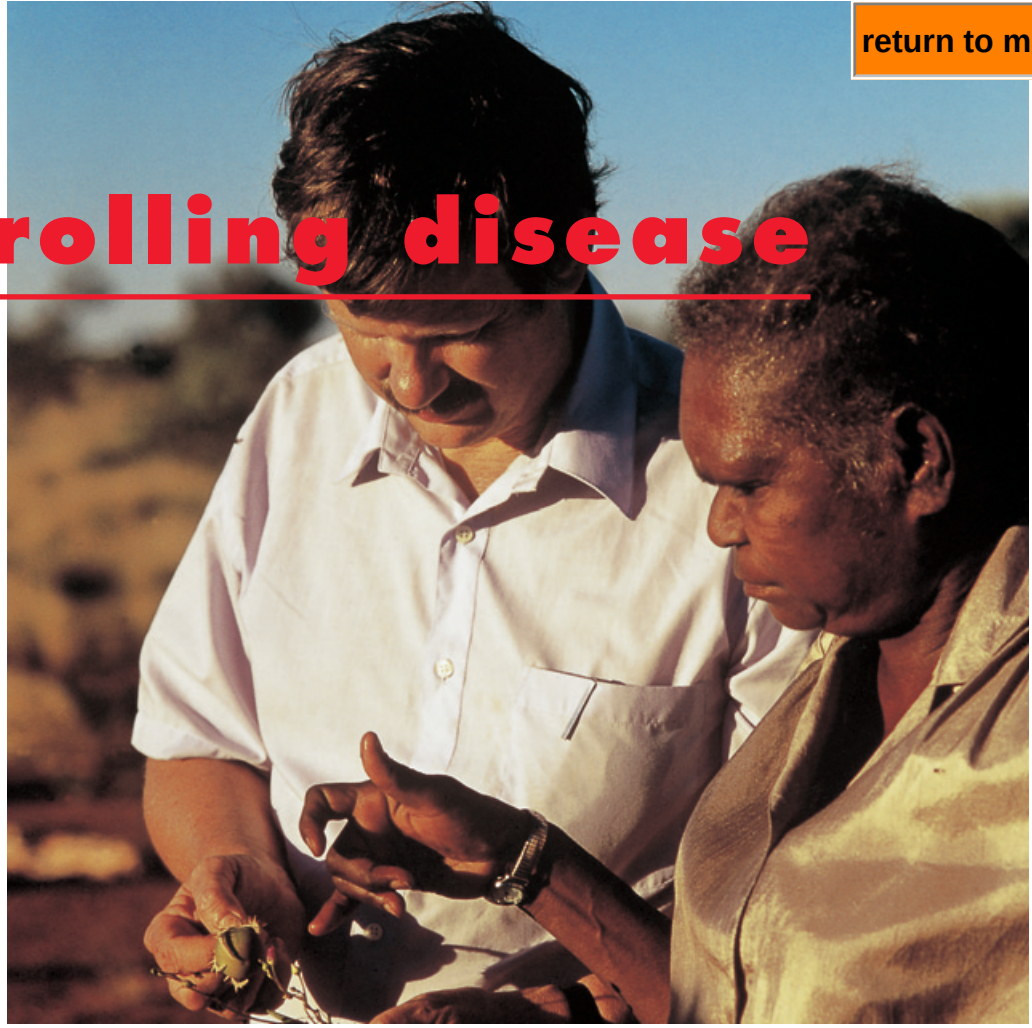

FIGURE 9.28
Traditional native foods and medicines are the subject of research by CSIRO scientist Chris Harwood, who believes that bush foods can be used to help struggling communities in central West Africa. Aboriginal elders of the Warilpiri people are sharing their invaluable knowledge of edible *Acacia* (wattle) species with scientists. The *Acacia* seeds can be used to make a flour which is rich in nutritional fibre. One species contains 23% crude protein, 53% dietary fibre and 11% fat. *Acacia* trees can grow in poor soils with extremely low rainfall. They can be used in times of drought when other food crops have failed and so may help reduce famine.

Great advances have been made in reducing the impact of disease during the 20th century. Life expectancy for Australians has increased by 20 years. This improvement has been brought about by a number of factors, including better nutrition, access to clean water, more effective sanitation (sewage and garbage) treatment, increased knowledge of the causes of disease and improved disease treatment and prevention methods. Some factors such as a healthy diet and vaccination help the body defend itself against disease. Other factors such as water treatment with chlorine, sewage treatment and garbage disposal help prevent the spread of micro-organisms. Public health education programs provide information to reduce the incidence of disease. The 'safe sex' education program helped slow the spread of HIV in Australia.

Unfortunately, not all Australians enjoy the same benefits of medical and public health advances. Aboriginal Australians on average experience greater health problems than non-Aboriginal Australians.

History and social disadvantage

Before 1788 Aboriginal people lived a hunter–gatherer lifestyle which provided a nutritious diet of native foods and plenty of exercise. Many infectious diseases in other parts of the world were not present on the Australian continent. Aboriginal people practised their own form of health care using bush medicines.

Today, many Aboriginal Australians live in poverty, due to lack of employment and education opportunities. Poor living conditions, poor food and inadequate access to health services are responsible for the high incidence of both infectious and non-infectious diseases.

The solution to this problem is very complex. Health problems can only be solved when the causes of social disadvantage are addressed.

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When life is 17 years shorter

By ANITA CATALANO

ADULT Aboriginal people die at a younger age than the rest of the Australian population and so do their babies. They develop a higher rate of health problems such as heart disease, diabetes and sexually transmitted diseases.

A report released by the Federal Governmet last year, Australia's Health, found that Aboriginal men and women lived an average of 17 years less than the non-Aboriginal population.

The life expectancy for an Aboriginal man was 57 years compared with 74 for a non-Aboriginal man. For an Aboriginal woman, the life expectancy was about 62 years compared with 80 for non-Aboriginal women.

Infant mortality was four times higher than in the non-Aboriginal population. About 27 Aboriginal babies per 1000 live births die each year compared with seven per 1000 in the non-Aboriginal population.

A study by the West Australian Health Department backed up the national figures. It found that Aborigines in Western Australia were dying at twice the rate of the rest of the community.

The study of 2500 deaths in Western Australia between 1983 and 1989 found that Aboriginal men as young as 30 were being killed by heart disease caused by Western influences such as alcohol, tobacco, dietary changes and a more sedentary lifestyle.

National figures from the Australian Institute of Health and Welfare show Aborigines die at four times the rate of the general population. They show that 237 per 100 000 Aborigines die of cardiovascular disease compared with 100 per 100 000 of the non-Aboriginal population.

A study by the Menzies School of Health Research in Darwin found that Hepatitis B cases in Aborigines were 42 per 100 000 compared with four per 100 000 in the non-Aboriginal community.

Two separate studies from researchers in Darwin found that malnutrition in Aboriginal children was more than 10 times the national average, while 8 per cent of Aboriginal children suffered from middle-ear disease.

Aboriginal women also face higher-than-average health problems. A report by the National Health and Medical Research Council found that deaths among pregnant and postnatal Aborigines was more than eight times that of the rest of the population.

It found that the Aboriginal maternal mortality rate was 12.7 per 100 000 of total pregnancies last year compared with 11.8 per 100 000 three years earlier.

The Australian
20 February 1995, page 5

1 Form a group, read the article and discuss what impact it has on you.

2 The writer, Anita Catalano, uses many statistics comparing Aboriginal health with non-Aboriginal health without making any comments or giving any explanations. What is the purpose of this style of reporting?

3 The same statistics could be displayed in table or graph form. Design a table that compares life expectancy for Aboriginal and non-Aboriginal Australians. Design a graph that compares the incidence of cardiovascular disease, hepatitis B and maternal mortality between the two groups.

4 Discuss this statement, 'All Australians have the same right to good health'.

Questions

1 What is 'life expectancy'?

2 How would life expectancy be calculated?

Think about

In tropical areas of Australia, public education programs advise people to keep their homes and yards free of stagnant water. How could this help stop the spread of disease?

Find out

Australia has strict quarantine laws to prevent infectious diseases being brought into the country. Animals from overseas must be isolated to make sure they are not carrying diseases such as rabies. Find out how long animals from other countries must be kept in quarantine and research the cause and symptoms of rabies.

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Using scientific language

Copy and complete the following:
Bacteria and _____ are micro-organisms which can cause disease. Any organism which causes disease is called a _____ . Further examples of these are fungi and _____ .
_____ are chemicals which can be swallowed or injected to combat bacteria. They are not effective against viruses.
_____ can help prevent disease by stimulating the body to make antibodies against disease-causing micro-organisms. This results in _____ to that particular disease.

Check your knowledge

1 What is meant by an infection? Give an example.
2 Name five ways that bacteria and viruses can enter the body.
3 Bacteria reproduce by binary fission. What does this mean?
4 Match the diseases in Column B with the cause in Column A:

TABLE 9.3

A. Cause	B. Disease
Infection	Lung cancer
Genetic	Influenza
Ageing	Lead poisoning
Environmental	Alzheimer's disease
Lifestyle	Haemophilia

Apply your skills

1 Prevention is better than cure. Explain what this means in scientific terms, with an example.
2 Is the purpose of vaccination to cure or prevent a particular disease? Explain your answer.
3 Give examples where diseases have been controlled by:
   a better public hygiene
   b acting responsibly in personal habits
   c vaccinations.
4 Our skin is home to large populations of many types of bacteria. How could having them there help stop infections from other bacteria and fungi?
5 True or false?
   a Viruses need oxygen to survive.
   b Viruses do not need nutrients.
   c Viruses are very similar to bacteria.
   d Viruses only reproduce inside other cells.
   e Viruses are killed by antibiotics.
6 Figure 9.29 shows the distribution of microbe colonies on the surface of a nutrient agar plate that has been incubated for 48 h. Two types of 'patches' grew, both being obvious microbe colonies.

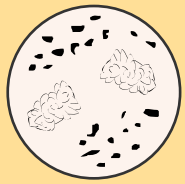

FIGURE 9.29
Agar plate with bacterial and fungal colonies.

   a What type of microbe colonies would the large patches be?
   b What type of microbe colonies would the small patches be?
   c Suggest why the small patches do not grow close to the large ones.

Challenge yourself

1 Some people think that we catch more colds in the winter because of the weather and temperature.
What do you have to say about this?
2 Table 9.4 shows recorded cases of hepatitis types A and B in South Australia.

TABLE 9.4

YEAR	1984	1985	1986	1987	1988
Hepatitis A	68	141	510	145	101
Hepatitis B	199	182	122	76	43

Hepatitis A is highly infectious and caused by eating or drinking contaminated goods. The organism is excreted in the faeces. Hepatitis B is a blood-borne organism and is generally transferred by exchange of body fluids (e.g. sharing needles).
   a Plot this data as line graphs.
   b True or false?
      i The trend of hepatitis A is a steady increase.
      ii The data allows us to confidently predict the number of expected cases for the next year or two.
      iii The data for hepatitis A in 1986 would have been unexpectedly high.
      iv The steady fall in hepatitis B could be related to AIDS education and prevention.
      v The data shows that hepatitis B is a more serious disease than the A type.
      vi Hepatitis A can be prevented by good personal hygiene.
      vii Hepatitis A would be more prevalent in drug users.

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CHAPTER 10

ENERGY AND MATTER THROUGH ECOSYSTEMS

When we eat, what substances do we and other animals actually consume? Where do these substances originally come from? Apart from the 'naturally occurring' range of proteins, lipids, carbohydrates, vitamins and minerals, animals also consume a large number of different industrial chemicals and pesticides. The number of new chemicals which have been released into the environment in last few decades is estimated to be in the thousands. More are being added every year.

We need to know the sources of these substances and their effects on living things. To do this we need to understand how all substances move through the environment and find their way into organisms. Do they then accumulate in an organism or are they recycled?

Scientists have long recognised the problems caused by many of these substances, as predicted by Rachael Carson in her landmark book *The Silent Spring* in 1956. Many are now researching ways to reduce the entry of chemicals into our environment. Biological control and genetic engineering are two tactics being used to solve part of the problem.

Conservation groups are also concerned about the problems caused by the release of these substances.

In this chapter you will investigate how ecosystems function, and begin to examine the impact humans have had on them.

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- Distinguish between biotic and abiotic features of an environment.
- Explain the role of living things in the cycling of matter and flow of energy.
- Compare the roles of producers, consumers and decomposers in ecosystems.
- Describe the importance of cycles of materials in ecosystems.
- Describe some impacts of human activities on ecosystems.
- Discuss issues associated with conservation of the environment.
- Discuss the positive and negative impacts of the use of pesticides and other chemicals.
- Describe some recent scientific contributions made by Australian scientists to improving the environment.

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Web links

10.1

The biosphere—where life exists

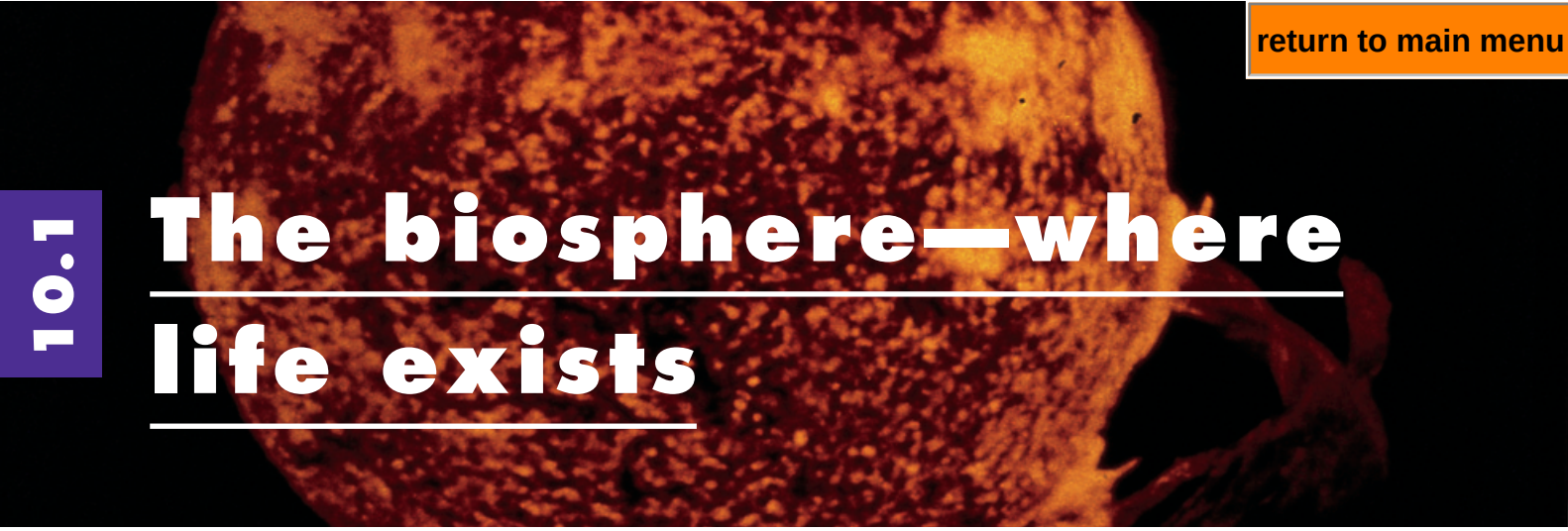

FIGURE 10.1
The Sun—a limitless supply of energy for life in the biosphere.

FACT FILE

The Sun provides more energy for Earth in 1 h than all our other sources of energy do in 1 year. Although we say the Sun is a limitless supply of energy, it will in fact run out of its hydrogen fuel—in 5000 million years or so!

The **biosphere** is the place where all life as we know it exists. It comprises the Earth and its atmosphere. Beyond the atmosphere there is space. Our biosphere is isolated in space. So what enters or leaves our biosphere? In terms of matter, not much. Gravity prevents the atmosphere and substances from leaving, and meteorites represent the only arrival of matter. We say that the biosphere is a closed system, and all the matter contained within it is finite. Non-renewable resources can run out, and unwanted wastes can accumulate.

What can enter (and leave) the biosphere is energy. Through nuclear **fission** the Sun provides us with a limitless supply of light and heat energy. Much of the energy is reflected back into space or absorbed by the atmosphere. The estimated 15% which reaches the Earth's surface can be trapped by plants. Green plants have the ability to trap light energy, and convert it into chemical energy in order to make a simple sugar, glucose. They do this in a process called photosynthesis:

$$\text{carbon dioxide} + \text{water} \xrightarrow{\text{light energy}} \text{glucose} + \text{oxygen}$$

Fortunately for animals, this also removes carbon dioxide from the atmosphere and adds oxygen to it. The fate of the glucose is equally vital. Plants can convert it to other organic compounds such as more complex carbohydrates, lipids, proteins and vitamins. These then form the basis for all food on the planet, through many varied food chains.

That is not the end of the story. When plants or animals die or produce wastes, these are recycled back through the biosphere. Remember, the supply of energy is limitless, not the matter to make up the chemical compounds. Gases are also recycled through the biosphere. Just think, you could have atoms as part of you that were once part of a dinosaur, a tree, a bacterium or even one of the gases of the atmosphere.

USING SCIENCE 1 ANALYSING ENVIRONMENTS

Word process

1 Work in a small group. Each group needs a piece of A3 paper and access to library references.
2 Rule a line down the middle of the page and head one side 'Plant' and the other 'Animal'.
3 Choose a particular native Australian plant and animal, and write their names under the headings.
4 Briefly describe where they live (e.g. river, reef, forest). You may need to do a little research but keep this part simple.
5 Now list the biotic and abiotic environmental factors for each.
6 Present your findings to the class and compare yours with others.
7 Were there any common factors which most groups listed? Explain.

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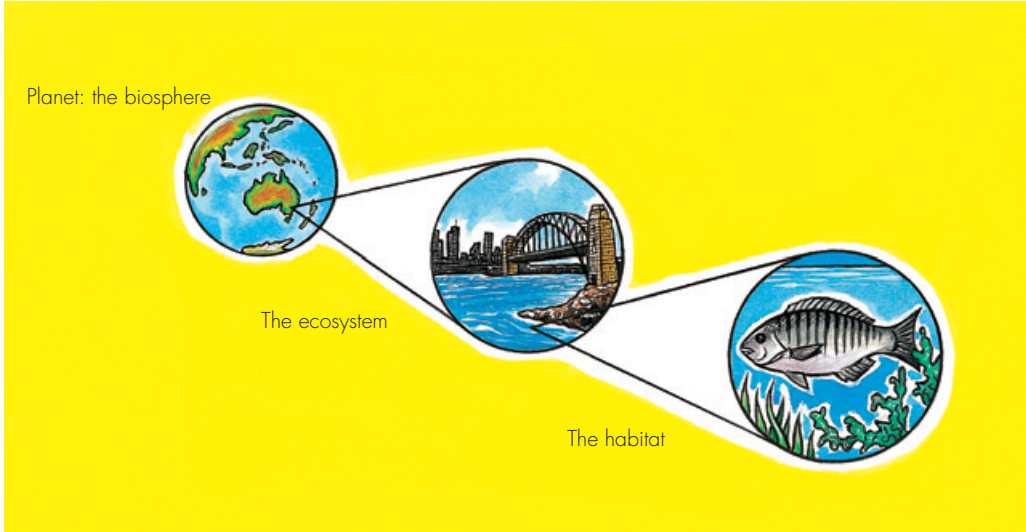

FIGURE 10.2
The relationship between the biosphere, an ecosystem and a habitat.

USING SCIENCE 2 ENVIRONMENTAL ISSUES

1 Work in a small group with a piece of A3 paper.
2 List as many environmental issues as you can.
3 Classify each as a 'local' or 'global' problem.
4 Compile a class list.
5 What could be meant by the slogan 'Act locally, think globally'?
6 You may like to keep a record of this activity and review it after completing this chapter.

Biotic and abiotic

The biosphere is made up of many **ecosystems**. In an ecosystem organisms interact with each other and their environment in a balanced way. Ecosystems can be further divided into habitats and environments.

A **habitat** is the place where an organism lives. For example, a fish's habitat may be in a section of the harbour. An organism's environment includes both its physical (**abiotic**) and living (**biotic**) surroundings. The fish's abiotic environment includes coastal currents, tides, the temperature, water salinity, and chemicals added by human activities. Its biotic environment includes the organisms on which it feeds, or those which compete for the food and space (these may be its own or different species), and those which may feed on it. All these environmental factors influence the survival of the fish.

The broad area of science which deals with the study of organisms in their environments is called ecology. A better understanding of ecology is needed for humans to deal with many of the ecological issues that have arisen because of our impact on ecosystems.

FIGURE 10.3
Through photosynthesis plants convert light energy into chemical energy in the form of food. This energy forms the basis for virtually all food chains on Earth.

Questions

1 What is the difference between the biosphere, an ecosystem and a habitat?
2 Why is the biosphere regarded as a closed system? Explain how a carbon atom found in carbon dioxide in the atmosphere could become part of your body.
3 Explain why photosynthesis is essential to life.
4 List three biotic and three abiotic parts in the environment of a kangaroo.

Think about

1 The greenhouse effect is partly due to excess carbon dioxide in the atmosphere. How might the removal of large areas of forests increase this effect?
2 How do you think energy from the Sun enters an aquatic ecosystem?

Find out

The Earth's ozone layer is being depleted. Research the ozone 'story' and find out what measures are being taken to remedy the situation.

Interactive

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10.2

Producers versus consumers

FACT FILE

Light energy can be used by green plants and chemical (food) energy can be used by animals and plants, but the escaping heat energy cannot be recycled by any living organism. It is therefore lost from the biosphere.

Recall that an **ecosystem** is a system formed by living things interacting with each other and their non-living surroundings. The living things are **interdependent**.

Talk about

What does interdependent mean? Give three ways that organisms could be interdependent.

Ecosystems include both living and non-living components. An ecosystem is **terrestrial** if it occurs on land and **aquatic** if found in water. Virtually all ecosystems regardless of their position on Earth are supported by the energy from the Sun.

The flow of energy through an ecosystem occurs in only one direction. This one-way flow begins with solar energy that is converted into chemical energy by photosynthetic organisms. Animals rely completely on these green plants for their energy. Green plants produce the chemical energy (food energy) which is consumed by animals. Thus we call plants **producers**, and most animals **consumers**. The links between the producers and the consumers where energy is transferred from one organism to another is known as a food chain.

The terms 'motion energy', 'light energy', 'chemical energy', 'nuclear energy' and 'heat energy' can be used to describe the transfer and conversion of energy in a food chain. Some chemical energy must be used to power metabolism in a living organism, including cellular respiration. However, very little of the energy stored within an organism is actually used for the growth and development of the organism that eats it. A large amount of energy is lost to the environment as heat.

Figure 10.5 shows the Sun's light energy being converted into chemical energy in the producer. This chemical energy can be used by the plant or it can be stored, in which case it is ready for use by the **first-order consumer**, the grasshopper. When the grasshopper eats the grass some of the stored chemical energy in the grass is passed on as food for the consumer. Similarly the grasshopper loses heat energy owing to its higher level of activity and higher body temperature. When the

FIGURE 10.4
A river red gum is a typical producer as it manufactures its own food via photosynthesis. A wombat is a typical consumer as it must take in energy in the form of food.

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second-order consumer, the lizard, eats the grasshopper, it too receives some of the chemical energy stored in the food chain. More energy is lost as heat. And so too the **third-order consumer**, the eagle, receives some energy and loses some. If along the way either the producer or one of the consumers dies, its energy will be transferred to the bacteria that will decompose it, again with a heat loss.

From each **trophic level**—feeding level in a food chain—to the next, only 10% of the chemical potential (stored) energy is passed on. In quantitative terms the grass would store 100% of the chemical energy; the grasshopper would receive 10% of this; the lizard a further 10% (now we're down to $\frac{1}{100}$ of the original) and the eagle (third-order consumer) would only receive $\frac{1}{1000}$ of the energy originally available in the producer. When an organism operates at more than one trophic level a food web will be formed.

Questions

1 What do the arrows in a food chain indicate?

2 Use the information supplied in the following graph to answer questions a–d.

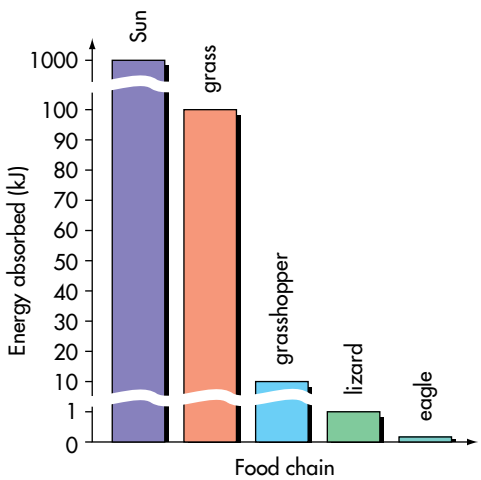
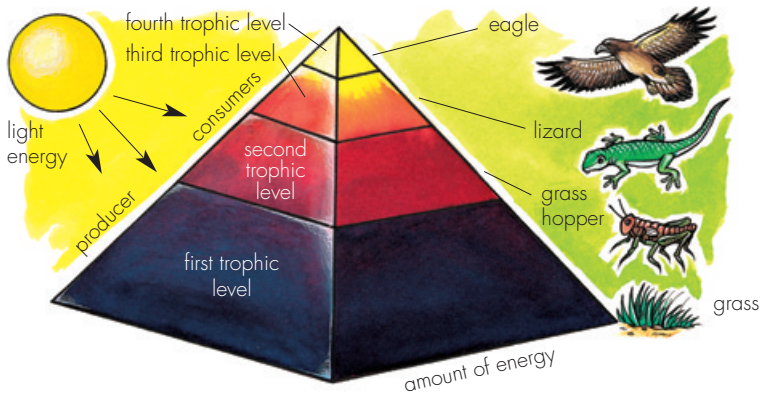


FIGURE 10.6

a What percentage of energy is passed on through each level of the food chain?

b Which organism can be classified as a second-order consumer? Explain.



c What do you think happens to the lost units of light energy depicted in your graph?

d If a fourth-order consumer entered the food chain, how much energy would it receive from its food source?

3 Draw a flow chart showing how the 'energy' terms—motion, light, chemical, nuclear and heat—form a 'chain'.

FIGURE 10.5
Energy is lost as heat as it flows through a food chain. The food pyramid indicates the amount of energy passed from one trophic level to the next.

Think about

1 Compare the two sets of teeth in Figure 10.7.

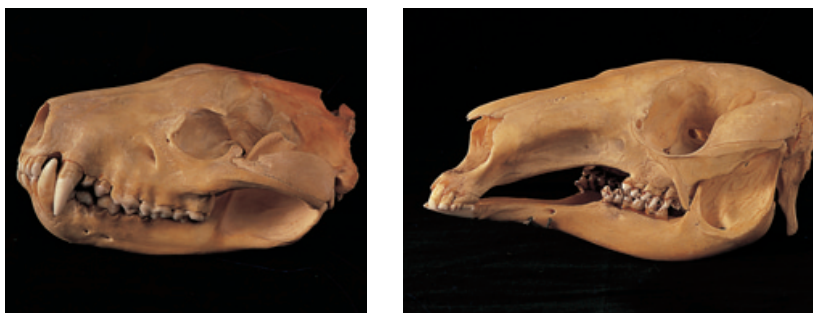


FIGURE 10.7

Which belongs to a first-order consumer, which to a second-order or third-order consumer? Explain your answer.

2 Do you think the heat energy escaping from food chains could be harnessed? Explain your answer.

3 Why do you think a compost heap gives off heat?

Extra worksheet

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10.3

Caught in a web

FACT FILE

Geologists have provided other scientists with a method of identifying who eats whom without actually being on the scene. Geologists use stable isotope tracing to analyse rocks but now it can be used to analyse the proportions of carbon isotopes in tissues of living things. (Some atoms can occur with differing numbers of neutrons in their nucleus. These different forms are called isotopes.)

Plants trap about 1% of the light energy from the Sun which falls on their leaves. Consumers make use of this energy in daily activities. They only manage to convert about 10% of it to stored energy ready for the next consumer.

Talk about

For the following food chain draw a graph showing the transfer of stored chemical energy from one organism to the next.

algae → microscopic animals → minnow → trout → human

Make an inference as to why humans (top-order consumers) generally eat three meals per day involving a wide variety of foods.

More than 50% of the energy obtained in photosynthesis is used immediately in the plant's own **cellular respiration**. Energy not used in respiration can be stored in the tissues. However, at any time this energy may enter one of the two main food chains—those involving grazing or browsing, or those involving decay. The small percentage of energy found in each chain is essential to the existence of the biosphere and humans.

The food chain in the Talk About identifies only one possibility for the human consumer. Many more food chains could be drawn showing how energy flows from one organism to the next but they would not show any interrelationships between one food chain and another. To do this we must draw a **food web**.

FIGURE 10.8
Fungi are decomposers. They gain energy by breaking down the remains of organisms. They do this by releasing digestive enzymes onto the remains then taking up the digested nutrients.

The easiest method is to identify levels in the food web as shown in Figure 10.9.

At any stage of a food web an animal or plant may die—its stored energy is lost to the consumers relying on it. However, **decomposers** and **scavengers** rely on the energy from these dead organisms for their own survival. Decomposers are responsible for breaking down the dead remains of the organism, thereby obtaining the chemical energy they require. Scavengers feed on the remains of dead animals, thereby taking up that organism's stored energy.

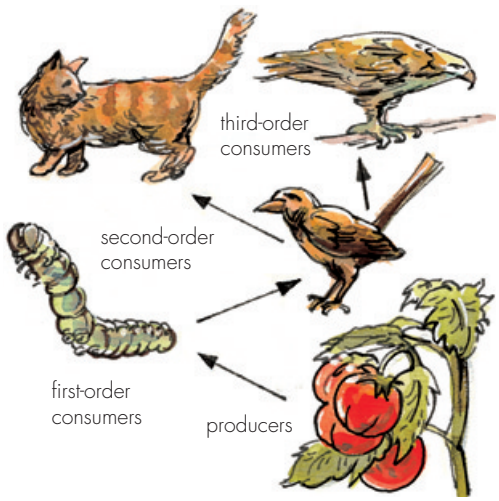

FIGURE 10.9
Levels in a food web.

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USING SCIENCE 3 DESIGNER FOOD WEBS

Work in a group of three. Your teacher will give you a set of arrows and drawings depicting different organisms. Using the information outlined in Table 10.1, construct a food web showing how energy flows within the web. Remember → means 'energy flows to'.

Once you have constructed your food web paste it on a piece of A3 paper, then answer the following questions.

1 Which organism is a fourth-order consumer?

2 With what other organisms does this fourth-order consumer have to share its energy source?

3 Are there any decomposers in this food web? If not, add some!

4 What organisms, if any, can be regarded as both second and third (or third and fourth) order consumers? Explain.

TABLE 10.1

Organism	Description—is eaten by . . .
Flies	Spider and field mouse
Eucalypt leaves	Koala, moth larvae and greenfly
Leaf litter	Woodlice and earthworms
Koala	Dingo and fox
Woodlice	Kookaburra and magpie
Greenfly	Ladybirds and blue wren
Moth larvae	Blue wren and kookaburra
Ladybirds	Spider
Earthworms	Magpie and kookaburra
Spider	Kookaburra
Field mouse	Dingo, fox and tawny owl
Magpie	Tawny owl and grass snake
Blue wren	Tawny owl

Biomass

Biomass is the total dry weight of organisms in the area. When you look back at the pyramid drawn in Figure 10.5 you can identify trophic levels. Most biomass pyramids are like this one—wide at the bottom and narrow at the top. The transfer between trophic levels indicates how energy can be lost when converted to biomass. In some aquatic ecosystems, however, the pyramid is inverted. The biomass of the producers is smaller than that of the consumers. Can you suggest a reason for this?

Phytoplankton—producers in an aquatic ecosystem—reproduce rapidly but are almost immediately consumed by **zooplankton**—the consumers. As zooplankton live longer their numbers at any given time are greater than those of the producers. Hence, an inverted pyramid.

Questions

1 How does a food web differ from a food chain?

2 How is energy lost from a food web?

3 Explain the difference between decomposers and scavengers.

FIGURE 10.10
A scavenger gains energy by feeding on the remains of dead animals.

4 What is biomass? What is a biomass pyramid?

Think about

What happens to the other 99% of light energy not trapped by photosynthesis?

Activity

Write a list of all the things you've eaten today. What animals or plants did your food originally come from? Draw a food web showing the relationships between the food you've eaten today and yourself.

Interactive

You might like to review the interactive module 'Food webs' from Heinemann Interactive eScience 2.

Foundation worksheet 1

Internet assignment

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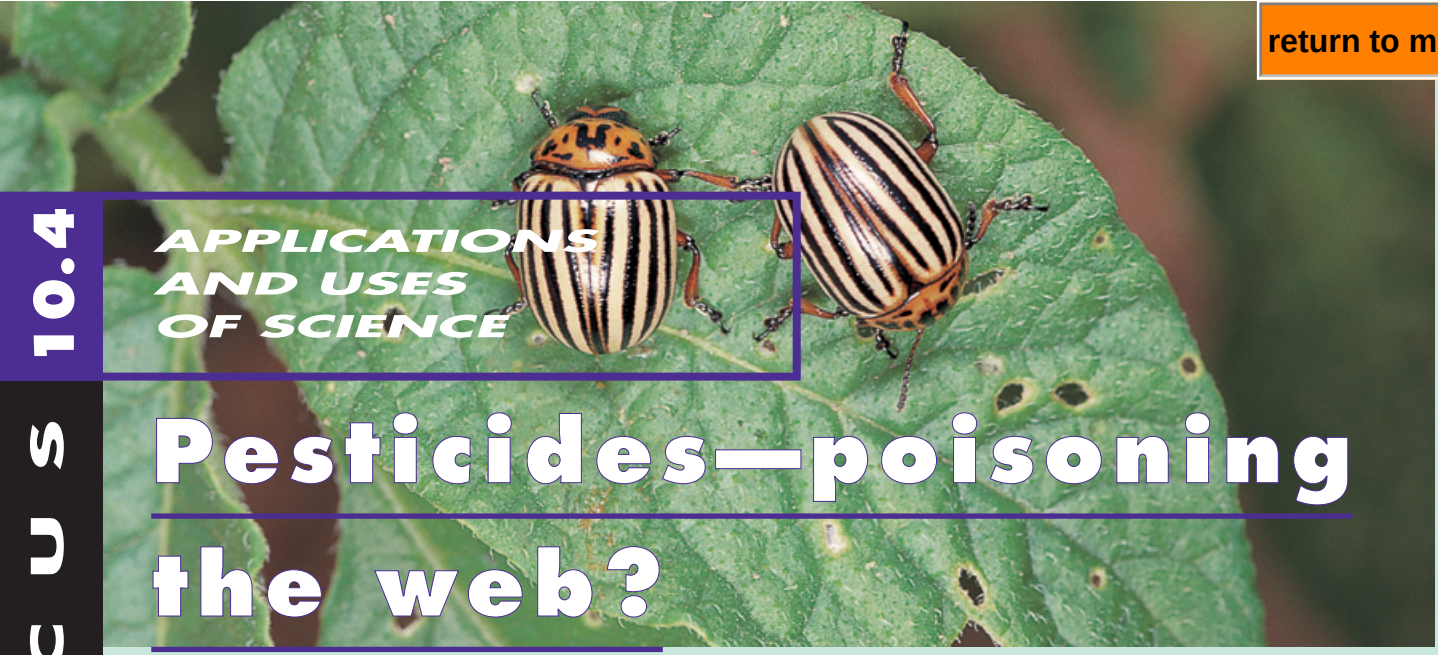

APPLICATIONS AND USES OF SCIENCE

Pesticides—poisoning the web?

FIGURE 10.11
The Colorado beetle, a crop pest, was controlled by the introduction of the pesticide 'Paris green' in 1867.

The people of the world need to be fed and clothed, and farmers and agricultural companies wish to make a profit. For these things to happen, we need to produce healthy crops at a reasonable cost. These crops are also an excellent food source for many organisms which compete with humans for food. We often call these organisms 'pests'.

Pesticides are used to control insects, fungi, mammals and weeds. Some pesticides such as **DDT** have also been used to kill insects that cause massive health problems (e.g. malaria). Generally they are sprayed over a large area from light aeroplanes. This may create a range of environmental problems.

- Most are non-specific, meaning that they may kill non-pest species as well, or even worse they may kill a predator of the pest species!
- Pesticides often find their way into the soil from where they can be washed into waterways.
- Many of these substances are very chemically stable, meaning that they are not easily broken down into harmless substances by organisms such as soil bacteria, or in the bodies of animals that ingest them. One group are called 'Persistent Organic Pollutants' or 'POPs' and include organochlorine insecticides such as DDT (see Figure 10.12).

One unfortunate consequence of pesticide use is **'bio-accumulation'** or 'biomagnification'. This means that the concentration of pesticide increases in the bodies of organisms as it moves through the **food chain**. How does this work? Say, for instance, that a pesticide was sprayed in the concentration of 1 ppm (part per million) onto a crop. The crop may be eaten

FACT FILE

Bio-accumulation also occurs with heavy metals (such as arsenic, lead, cadmium and mercury), and industrial chemicals such as PCBs and dioxin. Each of these is very toxic to humans.

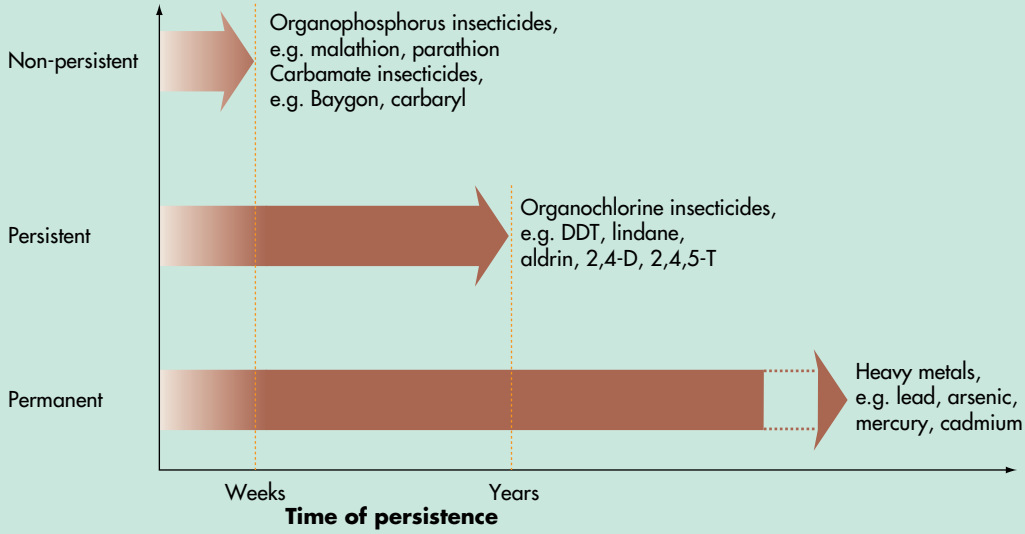

FIGURE 10.12
Some chemicals will last a few weeks in the environment, others will last a few weeks.

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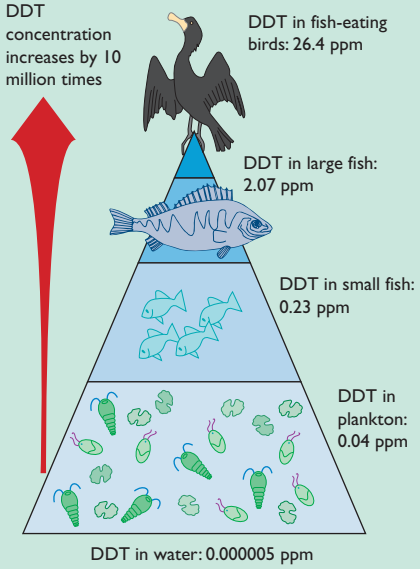
by some insects which the **insecticide** does not directly kill. One insect may eat a large number of plant cells, and so the level builds up in it. These insects may be eaten in large numbers by small mammals, and so the level builds up in them too. Predatory birds may eat a number of these mammals. Now the concentration in the birds' bodies will be considerably higher than 1 ppm (see Figure 10.13). Bio-magnification of DDT and other pesticides has been blamed for causing the death of many birds.

Humans are not exempt from the effects of bio-accumulation—after all we are a 'top-order consumer' in many food chains. Many pesticides are fat soluble, and they accumulate in fatty tissue and are not excreted. Here they can build up until they reach a toxic level. Each of us would have traces of many industrial chemicals and pesticides in our bodies; hopefully they will not reach toxic levels.



FACT FILE

Some bio-accumulates mimic naturally occurring hormones, and disrupt the immune system, behaviour and reproduction in many animal species.

FIGURE 10.13
DDT passes along food chains to high level carnivores. Because the amount of biomass decreases at each level in the food chain, DDT is concentrated.

USING SCIENCE 4 PESTICIDE DATABASE

Using any resources available to you, research a particular pesticide. You can present your findings as a poster, an essay, a media report or a talk. You may wish to consider the following points.

- Why and how is the pesticide used?
- When was it first used?
- Does the pesticide have any short- or long-term effects on the environment?
- Is there any evidence of it bio-accumulating?
- Has it been linked with any health problems in animals (including humans)?
- Have there been any steps taken to control the use of it?
- What could be the alternatives (if any) to its use?

USING SCIENCE 5 BIOLOGICAL CONTROL

Present a short report on one of the following.

- Find out how the CSIRO has developed a genetically altered cotton plant which makes its own pesticide. How could this benefit the environment?
- Find out how the 'sterile male' method is used to control fly populations (e.g. fruit fly).
- Find out how the CSIRO has found a way of immunising cattle against ticks.
- The story of the cane toads is one of a biological control methods that failed. Explain why, and examine what lessons we have learnt from this.

Find out

There is a naturally occurring insecticide called pyrethrum. What can you find out about it? Is it any less harmful than manufactured insecticides.

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10.5

Decomposition

FIGURE 10.14
Detritus!

Detritus is the term commonly given to the dead organic matter found in an ecosystem. On land it is fairly easy to establish just what detritus consists of—fallen leaves, twigs, dead animals, and so on. In an aquatic environment we look for similar signs. Here detritus is made up of plants, for example, algae, that once grew on the edge or the bottom of the aquatic system but have died. Aquatic detritus is also made up of the remains of aquatic animals. Wherever we find detritus there is a mass of food waiting to be eaten.

We have mentioned previously that scavengers feed on the remains of dead animals and plants. Also known as the **detritivores**, scavengers play an important role in the cycling of matter. Detritivores, for example, the common earthworm, physically break up the detritus into smaller particles ready for further action by the **decomposers** such as bacteria and fungi.

Although the detritivore obtains some energy from the detritus, most of its energy is supplied through the consumption of the decomposers attached to the dead plant or animal. These decomposers, bacteria (which are microscopic in size) and fungi, are very numerous. They are truly the last link in the food web.

USING SCIENCE 6 BREAKING IT DOWN

Examine the two food chains illustrated in Figures 10.15 and 10.16.

1 In which ecosystem do both the detritivore and decomposer food chains start with dead organic matter?

2 Where does the detritus come from in Figure 10.16?

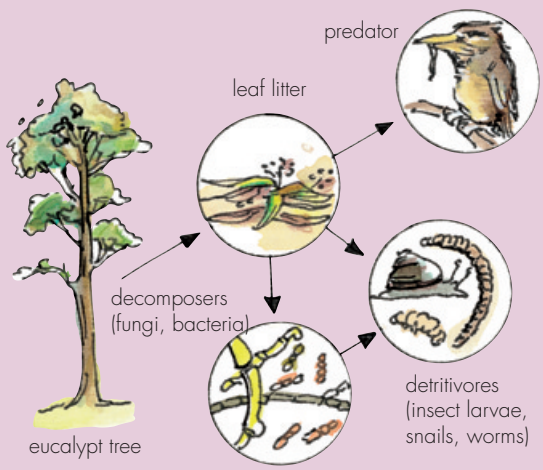

FIGURE 10.15
A terrestrial ecosystem.

3 An inference can be made from these two food chains that 'the detritivores in Figure 10.16 obtain greater energy from their food source than those in Figure 10.15.' What does this mean? Do you agree with the inference? Why, or why not?

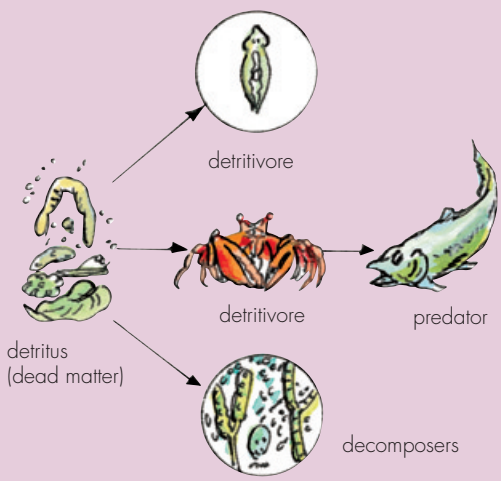

FIGURE 10.16
An aquatic ecosystem.

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EXPERIMENT 1 DECOMPOSERS ON THE GO

Datalogging

AIM

To grow micro-organisms, e.g. bacteria, from a soil sample.

MATERIALS

- sample of topsoil containing organic matter
- 4 petri dishes with lids
- 40 mL of milk (approximately)
- 2 × 100 mL beakers
- crucible
- Bunsen burner, tripod and heatproof mat
- pipe-clay triangle
- gauze mat
- matches
- stirring rod
- spatula
- marking pencil
- narrow masking tape
- incubator (25°C)

METHOD

1 Set up the apparatus as shown in Figure 10.17.

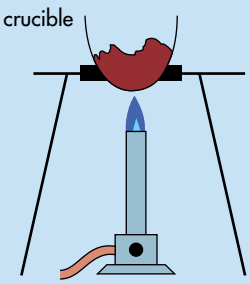

FIGURE 10.17

2 Place about two spatulas of soil in the crucible and heat for 15 minutes. Stir occasionally with the stirring rod so that it does not overheat. This will sterilise the soil. Cover and allow it to cool.

3 Pour half the milk into one of the beakers and heat gently until it begins to boil. This will sterilise the milk. Cover and allow it to cool.

4 Label the petri dishes 1 to 4 and add the mixtures specified below:
Dish 1: one spatula of sterilised soil plus 10 mL of sterilised milk.
Dish 2: one spatula of sterilised soil plus 10 mL of unsterilised milk.
Dish 3: one spatula of unsterilised soil plus 10 mL of sterilised milk.
Dish 4: one spatula of unsterilised soil plus 10 mL of unsterilised milk.

5 Seal the lids of the petri dishes by sticking masking tape around the sides. Do not open them again from this point on.

6 Swirl the petri dishes gently to mix the soil and milk.

7 Incubate the dishes at 25°C for 4 days.

8 Do not open the petri dishes when you remove them from the incubator.

RESULTS

Draw labelled sketches of each of the petri dishes, showing any growths that have occurred. Note their size, shape and colour.

DISCUSSION

1 Millions of micro-organisms are found in just 1 g of soil. Were there any dishes that showed no growth of micro-organisms in your experiment? What happened to the original micro-organisms in the soil?

2 Which dish contained the greatest area of growth? Suggest reasons for this.

3 Why is topsoil used in this experiment?

4 Can you be sure that the micro-organisms you have grown are bacteria? What else could they be?

5 Can you conclude what conditions are required for the growth of decomposers?

Although decomposers do 'feed', as their name suggests their action more closely resembles the breakdown of a food source. These organisms secrete digestive enzymes over the dead organic matter which then breaks down to form soluble organic molecules such as glucose. In completing this process, the decomposers release carbon dioxide and water into their environment as well as sulfates, nitrites and nitrates, all of which are very important to plants.

Questions

1 How did you supply the micro-organisms grown in the petri dish with the energy they required?

2 Give five examples of detritus.

3 Why are decomposers known as the last link in the food web?

4 Suggest why bone and wood decompose only very slowly.

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10.6

Cycling matter

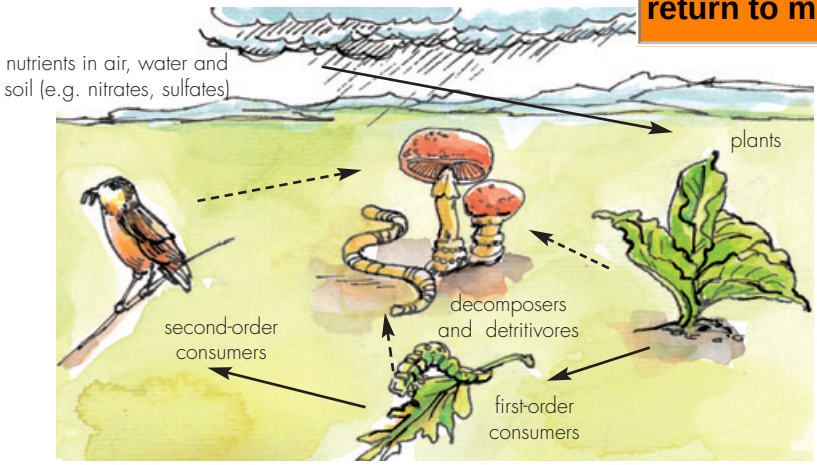

FIGURE 10.18
Nutrients are cycled through an ecosystem.

The energy in an ecosystem is renewable. The Sun continuously supplies energy to the system even though much of this energy is lost as heat. Nutrients on the other hand have no continuous source and so must be cycled through the ecosystem.

Every ecosystem provides its organisms with nutrients. Living organisms feed on each other, and the materials that make up dead organisms are returned to the ecosystem by the action of decomposers. Decomposition involves the breakdown of **organic material**, releasing the required nutrients to the air, water and soil. Two important elements that are recycled are carbon and nitrogen.

Carbon is a chemical element occurring in all living organisms. It is found in proteins, carbohydrates, lipids and many other organic compounds. The carbon dioxide in the atmosphere provides plants with their carbon through photosynthesis. The carbon is converted into a carbohydrate (glucose). As already discussed this glucose is used for many things within the plant—starch for storage, cellulose for cell walls and so on. Animals obtain their carbon through consumption of the plants or by eating other animals that eat the plants.

The carbon is in the food chain but the atmosphere does not have an endless supply of carbon dioxide. Therefore the carbon must be put back—recycled. How does a plant or an animal return carbon to the atmosphere?

Remember that living organisms respire, that is, they use oxygen to release energy in their cells. Two by-products of this cellular respiration are water and carbon dioxide. Thus the carbon is returned to the atmosphere. Animals may also recycle carbon by excreting carbon in the soil.

When an organism dies, decomposers release the carbon back into the soil and atmosphere by breaking down the organism's organic compounds. Other means of releasing carbon into the atmosphere come from the burning of fossil fuels. Coal, gas and petroleum are formed from the remains of plants and animals. As such they hold a store of carbon just waiting to be released into the atmosphere as carbon dioxide.

The **carbon cycle** appears quite simple when compared to the second essential nutrient, nitrogen. Nitrogen is present in the proteins manufactured by all living organisms. Its cycle is dependent on the action of certain kinds of bacteria. Plants are not capable of absorbing nitrogen from the atmosphere as they can carbon. They must obtain their nitrogen from the soil—the habitat of the bacteria.

When a plant or animal dies nitrogen is released into the soil as ammonia (a nitrogen compound, NH_3). Bacteria in the soil use this ammonia as their source of

USING SCIENCE 7 BUNDLING

Using the information above, select the three most important points in relation to the cycling of carbon. Write each piece of information on a small slip of paper.

Join up with a partner. Lay your three pieces of information out on the table. If you and your partner have both written the same piece of information, remove one of the slips of paper. Bundle together all the information you've kept and join with another pair. Repeat the 'sifting' process until the four of you have a complete set of researched notes. Organise and stick all your slips of paper onto a piece of poster paper for display in the classroom.

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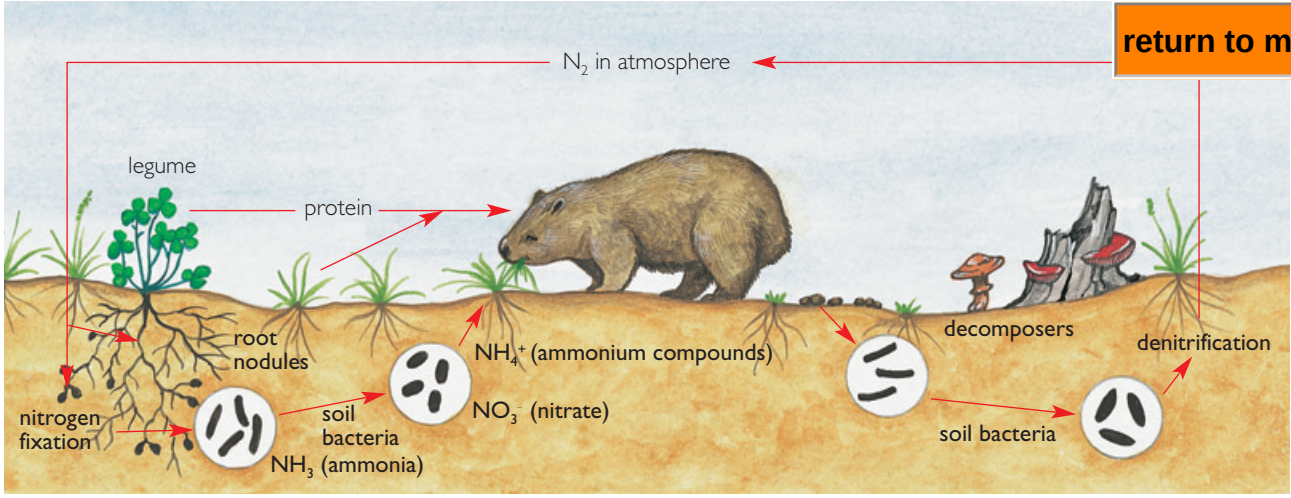


FIGURE 10.19
The nitrogen cycle.

energy. In using this energy they release nitrates. Plants take up both ammonia and nitrates through their roots, with nitrate being more easily absorbed.

A second group of bacteria called **nitrogen-fixing bacteria** can absorb nitrogen as a gas from the air spaces in the soil. They then build this gas into ammonia which is converted to nitrates by yet more bacteria.

Rain water can assist in the **nitrogen cycle**. Nitrates are very soluble, that is, they dissolve easily in water. As the rain water passes through the soil it dissolves the nitrates, removing them or sending them deeper into the soil.

And then there is another group of bacteria . . . These bacteria obtain their energy by breaking down the nitrates into nitrogen gas which escapes from the soil into the atmosphere.

Questions

1a List the role each of the three groups of bacteria plays in the nitrogen cycle.

b Why is nitrogen important for plants and animals?

2 Matter is cycled through an ecosystem. Explain how the energy in an ecosystem is renewable.

Think about

How would the nitrogen cycle operate in an aquatic environment?

Activity

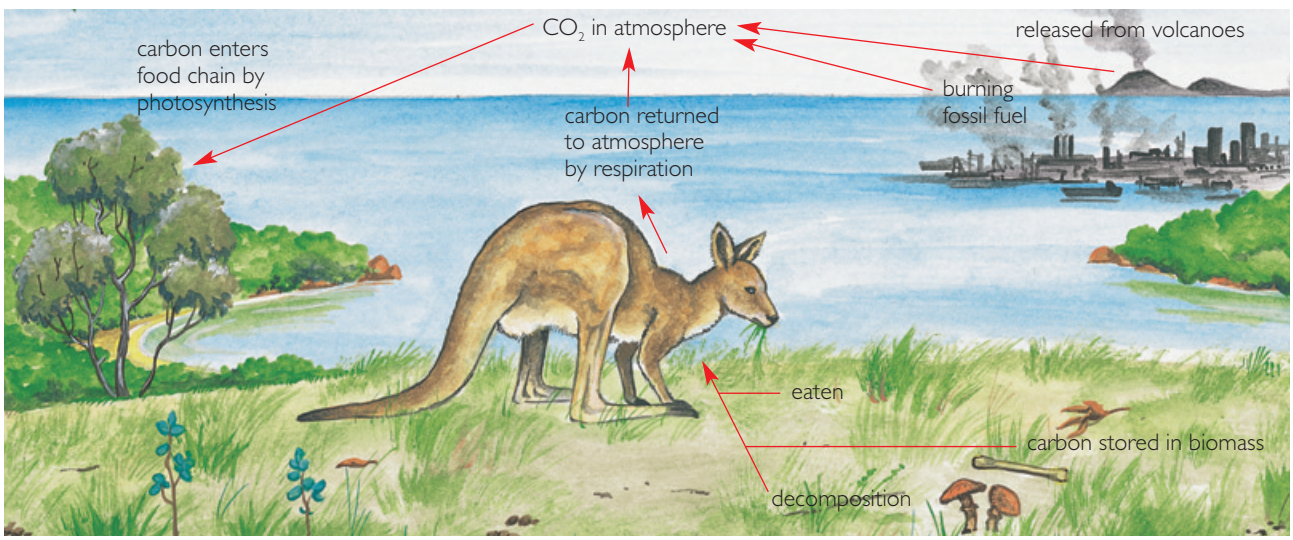
Draw a diagram illustrating the carbon cycle in an ecosystem. Remember to include all levels of the ecosystem or food web.

Find out

In agriculture most of the crop is usually removed before it decomposes. How do farmers maintain the nitrogen in their soil?

Extension worksheet

FIGURE 10.20
The carbon cycle. The element carbon is cycled through the biosphere.



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CURRENT ISSUES, RESEARCH AND DEVELOPMENT

Vermi—what?

FIGURE 10.21
Worms in action.

Vermiculture is relatively new on the scene of large scale recycling, but it has been occurring in nature for millions of years. It involves using worms to break down common organic wastes into **vermicast**—worm excrement.

Worms have tremendous appetites and can consume and excrete their own body weight in organic material in a single day. Common wastes which can be consumed by worms include food scraps, manure, garden litter, paper products, sewage sludge and crop wastes. The vermicast makes a perfect, natural plant food because the nutrients are immediately available to the soil. They are also more environmentally friendly than many fertilisers. Another appealing feature of the system is that it is relatively cheap to establish and maintain. It can also be used on a small scale at home or on an industrial scale.

The City of Melbourne in conjunction with Waste Organic Recycling Management Systems Australia Pty Ltd (WORMS) trialled a program involving worms and waste material from the Queen Victoria Market. Their aim was to test the feasibility of using worms on a commercial basis to help reduce the amount of waste being sent to tips.

The first problem encountered by the program was the contamination of the waste! Not only was there organic matter, that is, fruit, vegetables, meat and fish, but there were also substances that are **inorganic**, such as glass. Naturally, worms cannot feed on this latter form of waste. Their first trial held from June 1993 to March in 1994 was therefore not a great success. Only 13.75% of the organic matter was converted by the worms but they did recover an estimated 68 000 worms!

FIGURE 10.22
Windrows used in vermiculture.

In the second trial conducted at Werribee Disposal Facility, 32% of organic matter was converted. The expected conversion time was 12 weeks but in fact it took 16 weeks. The expected conversion rate was 45%. Even so it can be seen that improvement was made in this second trial. The general improvements came in the area of aeration and drainage—two essential components to the activity of the worms.

The reason stated for the lower than expected conversion rate in this second trial was low nitrogen levels in the waste. With your knowledge of the nitrogen cycle, why do you think this was an important factor?

The actual process used in the 'worm factory' can be outlined in a flow diagram.

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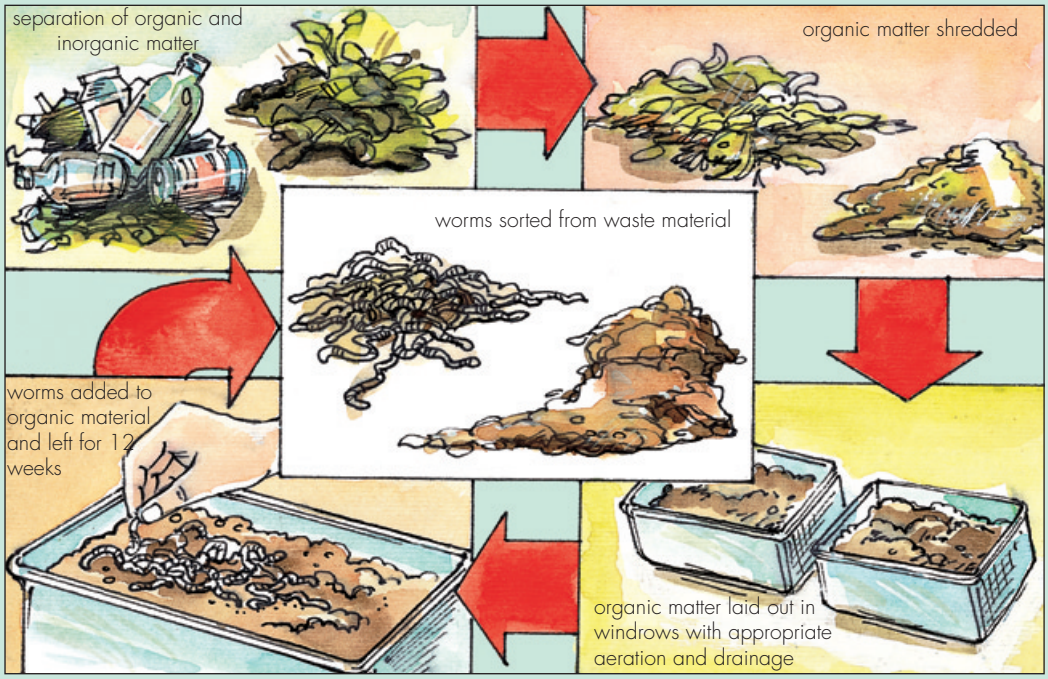

FIGURE 10.23
A flow diagram for a worm factory.

Foundation worksheet 2

USING SCIENCE 8 A WORM FACTORY OF YOUR OWN

Your class has established a restaurant business. However, the Council has told you that you must cut down the amount of waste material you are placing in the Council bins. You are already responsible recyclers of glass, paper and plastic so your only problem is what to do with all the waste food materials. You decide to hire hundreds of employees to assist you in your campaign but these are not your ordinary employees. These ones are slimy and have no backbone . . . You are entering the world of vermiculture. You and your hundreds of employees are going to spend the next 12 weeks experimenting with the conditions required to break down a quantity of organic waste material into smaller molecules which will then be more easily decomposed once it reaches the tip or is used to fertilise school gardens!

As a class design a worm factory. You will need to consider the size of the container—make it fairly small or else the process will take too long—and the amount of organic waste—this is the waste food materials. What types of foods will decompose most quickly? Perhaps you could run a preliminary test on different foods to determine this. The number of worms will also need to be considered as will aeration and drainage. Your vermiculture **windrow** needs to be well ventilated and moist but not damp. Look back at the nitrogen process outlined earlier. How will the level of nitrogen affect your experiment? Finally, where should you position your factory—remember worms don't like direct sunlight! You may like to set up two factories and vary the conditions to find the best possible vermiculture situation. This will be a long process so be patient. And another question—where will you get the worms from?

When you have conducted your trial you must write a report beginning with a summary outlining the decision-making process you used, a summary of the method, and results. Next write an aim (the purpose of your experiment), list your materials, and write the method in step-by-step format. You may like to include some diagrams. Maintain regular checks so as to record detailed results. Determine what it is that you are looking for—remember the aim of your experiment. Including a summary of your findings might be a good idea. Tables? Graphs? These usually come under the heading Discussion. And last but not least—the conclusion. Try to make your report as informative as possible—maybe the Council or your school will end up using it!

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10.8

The water cycle

FIGURE 10.24
The Earth consists of three basic parts: the lithosphere—the solid ground on which we stand; the hydrosphere—the water found in oceans, lakes and rivers; and the atmosphere—home to the gases which sustain life.

FACT FILE

The human body is about 65% water.

FIGURE 10.25
Pollution from industrial waste is a problem for our waterways.

Talk about
List as many ways as you can why water is important for living things.

Water can exist in solid, liquid or gaseous form and a balance must be maintained between all three parts of the Earth and between all three forms of water. Thus evolved the **water cycle**.

Of course there are many factors which influence the cycle on a day to day basis. Can you list what factors affect the amount of rainfall a certain area will receive?

Evaporation from the oceans provides most of the water vapour. As a result, the air over the oceans becomes loaded with vapour. Winds move this air over the land until it reaches mountain barriers which cause it to rise. As the air rises, it cools and its density increases. This means that it can hold less water and reaches **saturation** (or a temperature known as **dew point**). Further increases in altitude result in the water vapour condensing to form precipitation or rain.

What happens to the water after it has fallen as rain? Some of it makes it way back to the oceans via rivers. Some is stored by humans in dams and reservoirs. A certain amount finds its way into ground water (e.g. the Sub-Artesian Basin) and may then rejoin the cycle at various points. Lastly, some is used by plants and animals. Water cycles through living things. Plants absorb it from the ground and lose much of this as vapour through their leaves in a process called transpiration. Animals drink it or obtain it in their food and lose it either by evaporation (e.g. sweating) or when they excrete.

Human impact on the water cycle

Humans, like all living organisms, need water, not only for drinking, but also for a range of other activities including irrigation, washing, industry, and even flushing toilets. Our demands are great, but they need to be balanced with the supply. We must also ensure that we do not harm our water supplies. Not just for our own sake but for that of all living things.

Firstly let us examine the balance. The overuse of water for irrigation especially along the Murray–Darling Basin causes changes in salinity, resulting in land loss by dry-land salinity, and river salting. Scientists are working to 'intercept' the salt as it flows along the river. Planting more

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USING SCIENCE 9 MODELLING THE WATER CYCLE

Water moves through the biosphere in a cycle (Figure 10.25). Your task is to build a model depicting the changes in state the water undergoes in its cycle, the method by which it is cycled, and the environment through which it travels. Be creative in your use of objects to symbolise each part of the cycle.

trees along the river has also been shown to reduce the problem. Factories often use water for cooling. This means that the water returned to the waterway can often be warmer than is desirable for the organisms that live in it.

Another obvious problem for waterways is pollution. This not only occurs in rivers and dams, but also in the oceans. What things pollute the water?

- Nutrients, such as fertilisers—an excessive amount of these in the water can cause excessive growth of algae, called blooms. This can upset the oxygen balance in the water and some also produce toxins. The process of excess nutrient build-up is called eutrophication.
- Toxins—these can include pesticides (see section 10.4), heavy metals and industrial wastes. Many aquatic creatures can bio-accumulate toxins to alarming levels.
- Oil—a particularly unpleasant pollutant, which is not only toxic, but also coats animals and threatens their lives.
- Detergents—can affect the life cycles of some animals, and contribute to eutrophication.
- Physical pollutants—such as litter, especially plastics which can entangle animals.

Questions

1 Which part of the water cycle represents water in its liquid form? In its gaseous form?

2 How does water which has fallen as rain make its way back to the oceans? What things could be added to the water as this happens?

3 How might the following affect the water cycle?

a using more water for agriculture

b building factories next to rivers

c washing cars in the street so that the water goes down the stormwater drain

d using too much fertiliser on crops

4 Can you suggest a reason why inland New South Wales and Queensland have experienced such severe droughts?

5 How is water cycled through living organisms? Draw a diagram illustrating your answer.

Find out

Many natural disasters arise from interruptions in the water cycle. Using either floods or droughts as your example, explain how the cycle can be interrupted resulting in a disaster. Make sure you include the impact on living things.

FACT FILE

In the hydrosphere, 97% of the water is found in the oceans. Of the remaining 3%, 75% is ice, 22% is found under the ground and only 2% (of that 3%) is actually on the surface in streams or lakes.

FIGURE 10.26
Drought is a natural interruption to the water cycle.

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10.9

Conserving the environment

FIGURE 10.27
The Great Barrier Reef—a delicate ecosystem which could so easily be damaged.

So what if a species becomes extinct, or a rainforest disappears to make way for farmland or developments? Is saving them just too difficult in a world with an expanding population and all its needs?

In section 10.1 you may have made a list of the environmental problems we face. This could have been very long and maybe even a little depressing. By understanding the cause of some of these problems, we humans should be able to find solutions. We cannot 'give up' on the planet. It is the only one we have!

Conservation

Conservation is the protection and careful management of our natural resources. Conservation or national parks are one means of protecting some of the great variety of living organisms (referred to as the **biodiversity**) in Australia. One only has to look at a list of threatened or endangered species to realise that urgent action is required to save many of our native animals and plants.

What has caused this loss of biodiversity? In Australia there are two main causes:

- loss of habitat—this may result from farming, mining, building cities or climatic changes
- introduced species—these may compete with native species for food (e.g. rabbits), or directly kill wildlife (e.g. cats, foxes). Introduced plants can also overrun natural vegetation.

In other parts of the world hunting and poaching of some animals (e.g. rhinoceros and tigers) has been an additional problem. In Australia the yellow-footed rock wallaby was shot to near extinction as sport.

The clearing of forests for wood and developments is also a controversial issue. Each year Australia exports about 5 million tonnes of woodchips to make paper and disposable products. In addition we use about 1 million tonnes of paper per year.

Ecologists now talk of **sustainable development**. This was defined by The World Commission on Environment and Development in 1987 as 'development which meets the needs of the present without compromising the ability of future generations to meet their own needs'. From such definitions come the ideas of the 3Rs—'reduce, recycle, reuse'. Consider your own lifestyle. How well does it fit in with the definition?

FACT FILE

Biologists have listed over 100 animal and more than 200 plant species that are endangered in Australia, that is, at risk of extinction. Many more arelisted as vulnerable and rare.

Internet assignment

Does extinction really matter?

Does it really matter if a few species that we have never really heard much about become extinct? The answer should be 'yes' for a number of reasons. Each organism has a role in an ecosystem, for example a link in a food chain. If it is lost then that role may not be performed and many other species may be affected. Many organisms are of great value to humans, such as a source of medicines or food, or to keep another organism's numbers in check. We have yet to catalogue the vast variety of life on the planet and its potential uses. Who knows what useful compounds as yet remain undiscovered. The final reason is an ethical one. Do we have the right to

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cause the elimination of another species? How would future generations view us if we allowed this to happen?

What can be done?

Individuals can have an effect in many ways: by deciding what they will and won't buy, by applying the rules of the 3Rs, by becoming aware and using their vote wisely.

Of course there are many issues involved in conservation. These often involve jobs and other economic aspects. For example, the woodchipping industry employs many people and the economies of some towns rely on the money generated from it. Mining companies are always looking for new resources. They also employ many people and contribute money to the economy by way of taxes and royalties.

Our challenge is to balance these aspects and to leave the planet no worse off for future generations.

FIGURE 10.28
Woodchipping employs people and puts money into the economy, but is it ecologically sustainable?

USING SCIENCE 10 ACTING LOCALLY

Web page

Carry out a 'Personal Environmental Impact Statement'. List ways in which your lifestyle has both a negative and positive impact on the environment.

You could do this covering all your activities over a 24 h period. Maybe give yourself a 'score' or an 'environmental impact rating' based on a set of standards decided by the class. Things you could consider include use of transport, electricity, recycling. Make a list of things you could do to reduce the 'negative' and increase the 'positive' impacts.

Is there anything that your class or community can do?

Questions

1 What is meant by 'biodiversity'?
2 What human activities have caused animals and plants to become endangered?
3 List the reasons why it is important to conserve endangered species.
4 In your own words explain what is meant by 'sustainable development'.

Activities

1 Find out what is meant by 'World Heritage Area'. Where are such areas in Australia?
2 Find out what steps are being taken by scientists to conserve a particular animal or plant.

FACT FILE

Since European settlement at least 41 bird and mammal species and more than 100 plant species have become extinct in Australia.

FIGURE 10.29
Australia has a poor record on species extinctions. What can be done?

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Using scientific language

Match one of the words in the following list with each of the clues: biosphere, producer, consumer, detritus, ecosystem, abiotic, bio-accumulate, habitat, sustainable, salinity, transpiration, photosynthesis, aquatic, organic, decomposer, biotic. You may use a word more than once.

1 A food web in its non-living surroundings.
2 Pesticides increase in concentration along a food chain.
3 This organism uses the Sun's energy to make organic matter.
4 Substance produced by living organisms.
5 Relies on other plants or animals for its energy.
6 The Earth and its atmosphere.
7 Non-living part of an environment.
8 Using no more of a natural resource than can replaced.
9 Living part of an environment.
10 The place where an organism lives.
11 A earthworm is an example.
12 Dead organic matter.
13 Salt concentration of a waterway.
14 Water loss through the leaves of a plant.
15 Animals and plants in a river or an ocean.
16 Using light energy to make organic matter.

Check your knowledge

1 What does the term 'producer' mean?
2 In what form do cells require their energy? Draw a flow chart showing the changes undergone by energy from the Sun before the cells can use it.
3 Briefly describe the nitrogen cycle.
4 Look at Figure 10.30. Construct a table showing the producers, first-order consumers, second-order consumers and decomposers.

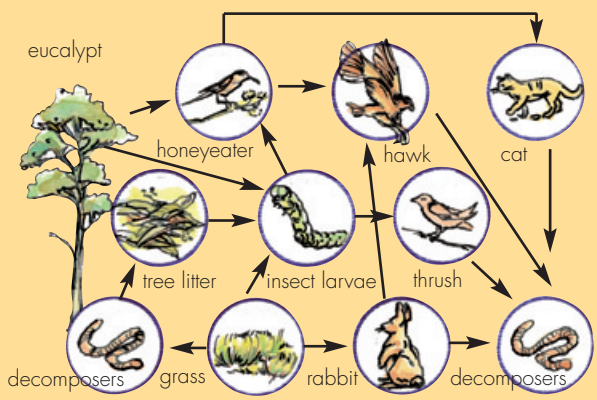

FIGURE 10.30

5 What is the difference between decomposers, detritivores and scavengers? Give examples.
6 Glucose can be converted to many different forms depending on the needs of the organism. Describe three such conversions and the purpose they serve.
7 Explain how water cycles between living organisms.
8 A food chain does not only show who eats whom. What is its other function?

Apply your skills

1 'Throughout a food web energy is constantly lost yet a constant flow of energy is maintained.' Explain.
2 Why are changes of state important in the water cycle?
3 Look at Figure 10.31. Predict what would happen if the number of grasshoppers decreased.

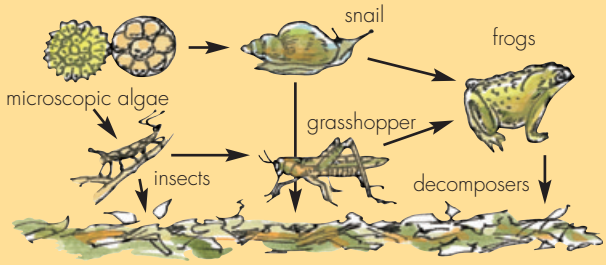

FIGURE 10.31

4 The pyramid in Figure 10.32 shows the changes in energy levels within a food web. Label each layer with the appropriate group of organisms. What do the arrows indicate?

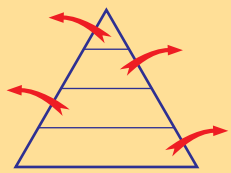

FIGURE 10.32

5 How does carbon enter a food chain?
6 Sulfur and nitrogen are essential nutrients for plants yet they can also be responsible for their destruction. Explain how this is possible.

Challenge yourself

Energy can be harnessed through efficient use of the biomass. Dairy farmers use it to power their milking machines. Design a poster depicting how biomass can provide electrical energy to drive a generator.

eTester

return to main menu

CHAPTER 11

DYNAMIC EARTH

The restless unpredictable nature of our world is evident every time we watch the television or read the newspaper and learn about another natural disaster. The tsunami which struck Papua New Guinea on 17 July 1998 is just one example where the ever-changing Earth has caused devastation and death. This tsunami was caused by an undersea earthquake approximately 30 km off the coast in the Bismarck Sea. The earthquake caused a large mass of water to rise up and spread out, forming waves. Within 30 minutes a series of three waves struck the Papua New Guinea coastline in quick succession. The third wave was at least 10 m high. Three villages were completely destroyed and ten others received extensive damage. Over 2500 people lost their lives.

The north-west coast of Papua New Guinea has a long history of tsunamis. The same region was affected by a larger tsunami in 1935. Scientists expect that the region will continue to be affected by tsunamis, earthquakes and volcanic eruptions in the future.

In this chapter you will learn about the dynamic nature of the Earth's crust and how scientists attempt to explain and predict natural disasters.

Web links

CHAPTER OUTCOMES

Successful completion of this chapter will allow you to:

- **Model the interior of the Earth, showing the relative thicknesses of the different layers from crust to core.**
- **Describe the characteristics of earthquakes and volcanoes and their effect on the surface of the Earth.**
- **Build a model of a volcano and describe its internal structure.**
- **Relate cooling rates to the formation of different igneous rocks.**
- **Describe some careers which require training in geology.**
- **Describe folding and faulting and associated landforms.**
- **Discuss the theories of continental drift and plate tectonics, and the evidence supporting these theories.**
- **Describe how scientists use devices to monitor changes in the Earth's crust and atmosphere to provide warnings of potential natural disasters.**

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11.1

Windows into the Earth

FIGURE 11.1
The surface of the Earth is familiar to most of us but what is the Earth like inside?

When you look out your bedroom window you see only a tiny part of the world. An Inuit child looking out a frosty window in northern Canada would see a very different view. Each window gives only a small amount of information, but as more and more 'windows' are added together, the view of the world becomes more complete.

Talk about
Why do we need the 'views from many windows' to get a complete view of the world?

High temperatures and pressures have so far prevented scientists from drilling more than about 10 km into the Earth. Because of this, we have to rely on the Earth's 'natural windows' to provide us with information about the planet's interior. These windows are as yet very small, but through them we are steadily increasing our knowledge. Although nobody has seen the inside of the Earth, we have a good idea what it looks like. What are some of these 'windows'?

Earthquakes provide one window into the Earth. Scientists called **seismologists** who study earthquakes tell us that the inside of the Earth has several layers, some of which are more solid than others. Volcanoes are another kind of 'window'. **Vulcanologists** are able to study material from the upper part of the mantle when it comes to the surface during a volcanic eruption. **Meteorites** are thought to be lumps of 'left-over building material' from the time of the formation of the planets. Some scientists believe that these rocks consist of similar chemicals to those inside the Earth. We also know that the Earth has a magnetic field around it, and this offers another clue. Many scientists suggest that the magnetic field is caused by a layer of liquid iron inside the Earth—most probably the outer core.

FACT FILE

A group of geophysicists has used supercomputers to combine millions of seismological observations from 3000 surface monitoring stations to form a single overall picture of the Earth's outer core. The molten outer core appears to be studded with mountains and riddled with lower-density fluid that forms ocean-like regions. There may even be a bizarre form of rain: showers of iron particles that fall onto the core.

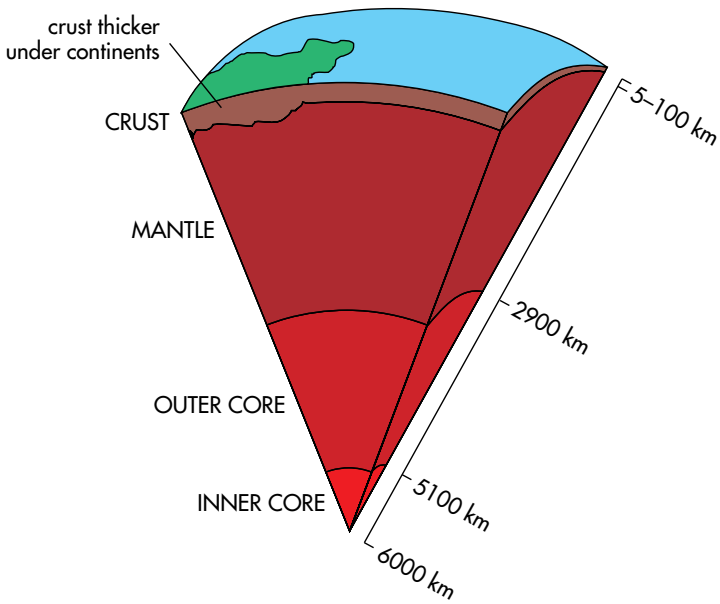

FIGURE 11.2
The layers inside the Earth.

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USING SCIENCE 1 THE LAYERS OF THE EARTH

Work in five groups to make a rough scale model of the interior of the Earth. You will need to work in a large outdoor area such as on the school oval.

WHAT YOU NEED

- 5 large pieces of stiff cardboard (poster size)
- 5 position markers (witches' hats or buckets)
- 5 thick marking pens

WHAT TO DO

1 Each group chooses one of the layers and makes a very large, highly visible sign for it. For example:
TOP OF CRUST
TOP OF MANTLE
TOP OF OUTER CORE
TOP OF INNER CORE
CENTRE OF EARTH

2 Take the signs, position markers and one book per group outside to a large open area.

3 Make a starting point at one end of the area. The 'TOP OF CRUST' group stays here, holding up their sign and placing the marker on the ground at their feet.

4 The 'TOP OF MANTLE' group stands in front of the first group, as close as possible. They hold up their sign and put their marker at their feet.

5 The other groups take 30 large paces from 'TOP OF MANTLE'. The 'TOP OF OUTER CORE' group stays here with their sign and marker.

6 The remaining two groups take 20 large paces and 'TOP OF INNER CORE' stays here.

7 The last group goes to the 'CENTRE OF THE EARTH', a further 15 paces away.

8 Everyone should hold up their signs and look at the model they have made. Leave the signs and markers in place, and go back to the beginning.

POINTS TO CONSIDER

1 The people at the top of the crust would be most affected by earthquakes. Notice how thin the crust is: humans have not even drilled through this layer!

2 The group at the top of the mantle were standing where volcanic eruptions start.

3 The people of the outer core were probably standing in the liquid iron and nickel that are thought to generate the Earth's magnetic field.

4 The group at the inner core were so squashed by the weight of everything else above that they were rock solid.

5 At the centre of the Earth, nearly 6000 km down, the last group was sizzling at over 4000°C.

EXTENSION

Find out more about the different layers of the Earth and produce your own large, colourful wall chart. For each layer, include information such as temperatures and what the layer is made of. You might even like to make part of the wall chart three-dimensional, moveable, or perhaps 'pop-up'.

Questions

1 What does a vulcanologist study?

2a List four natural occurrences that can give scientists clues to the interior of the Earth.

b Explain how each of these clues adds to the 'big picture'.

3a Which is the thickest layer of the Earth?

b Which is the thinnest?

Think about

A famous man, Will Durant, once said:
'Civilisation exists by geological consent, subject to change without notice.'
Discuss this statement with a friend, your group, or the whole class.
What do you think he meant? What 'changes without notice' could be brought about by activity within the crust and mantle?

eGlossary

11.2

Earthquakes and tsunamis

FIGURE 11.3
In 1946 a young Hawaiian girl stood in disbelief at the sight from her clifftop home above Hilo Bay. As if drawn by some giant vacuum cleaner, the sea water below her was sucked away leaving countless stranded fish flapping about on the exposed sea floor. As if in slow motion, a huge wall of grey water 30 m high surged back to the shore. It swamped the town below her, killing 96 people and leaving the whole area in ruins.

Talk about

Imagine that a 30 m wave was 30 seconds away from your home. What would you do?

The event described in the caption was a **tsunami** (Japanese for 'harbour wave'). An earthquake under the ocean is the most common cause of tsunamis, but the biggest and most destructive tsunamis yet recorded have been started by underwater volcanic eruptions.

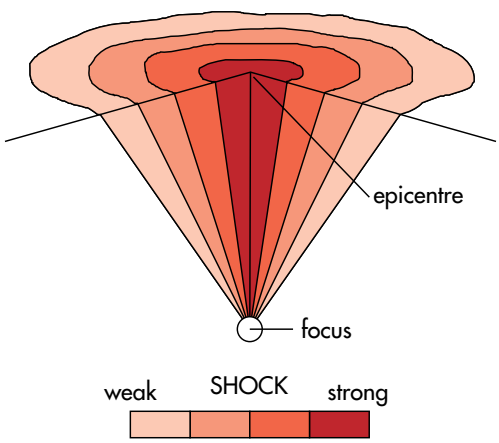


FIGURE 11.4
When an earthquake occurs, shock waves travel in all directions from the focus, reaching the epicentre first.

Movements of the Earth's crust are called **earthquakes**. Earthquakes start in the solid crust and upper mantle of the Earth, at depths between 70 and 700 km. Parts of the Earth's crust are under enormous pressure. When this pressure becomes too much, the rocks may move or break, sending out shock waves called **seismic waves** in every direction. This results in ground movement which can cause great damage.

The point at which an earthquake starts is called the **focus**. The point on the Earth's surface directly above the focus is called the **epicentre**. The strongest shaking is usually felt at the epicentre.

Measuring the strength of earthquakes

About 150 000 earthquakes are recorded each year. **Seismographs**, machines that measure the amount of ground shaking, produce **seismograms**, graphical records of the size and duration of the earthquake. Computers can be used to find the exact location and depth of the focus.

The strength of an earthquake is often described in terms of numbers on the Richter scale, as shown in Table 11.1.

eGlossary

TABLE 11.1 The Richter scale.

Richter scale measurement	Earthquake effects
3	Felt slightly in areas near epicentre. No damage.
4	Felt by most people up to several kilometres from earthquake. Some objects upset.
5	Strongly felt. Some damage to weak buildings.
6	Moderately destructive. Some severe damage to weak buildings.
7	Major earthquake. Destruction of many buildings.
8	Very destructive. Total destruction of structures.

Each increase of 1 on the Richter scale corresponds to a ten-fold increase in energy. For example, an earthquake measuring 6 on the Richter scale is ten times more powerful than an earthquake measuring 5.

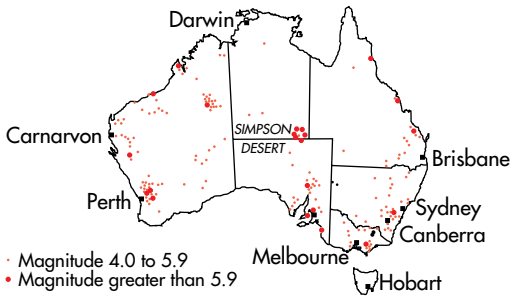


FIGURE 11.5
Known earthquake sites in Australia.

Australia has very few large earthquakes. Those earthquakes that do occur usually don't cause much damage because our country is so lightly populated. The Newcastle earthquake of 1989 was an exception. Twelve people lost their lives, and damage to buildings in this heavily populated area was extensive. Yet the earthquake itself was only moderate, measuring 5.6 on the Richter scale.

Extra worksheet

USING SCIENCE 2 MAKE A MODEL SEISMOGRAPH

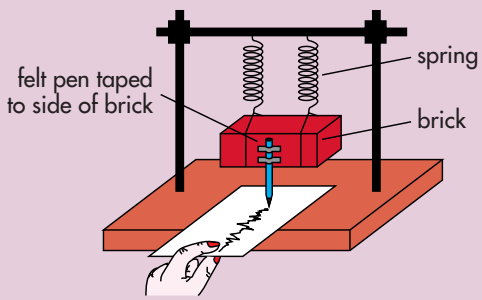


FIGURE 11.6
A model seismograph.

WHAT TO DO

1 Use Figure 11.6 as a guide to designing and building your own seismograph. Try out different materials until it works.

2 Test your seismograph by bumping the table or jumping on the floor while another student pulls the paper past the pen. See what other tests you can come up with.

3 Develop your own scale for measuring the strength of your bumps and thumps.

Questions

1a What causes tsunamis?

b Tsunamis are often called 'tidal waves'. Is this an accurate description? Explain your answer.

2 Use the map in Figure 11.5 to answer the following questions.

a Where have the biggest earthquakes in Australia occurred?

b Would these earthquakes have caused a lot of damage? Why?

c Are there any capital cities that could claim to be 'earthquake-free'? Which one(s)?

3 If the tsunami that hit Hawaii in 1946 had travelled 3200 km in 5 h, how fast did it travel (in km/h)?

4 Use the Richter scale (Table 11.1) to suggest what damage you would have expected in the Newcastle earthquake.

FACT FILE

The greatest death toll due to an earthquake occurred in China in 1556 when approximately 830 000 people were killed.

Multimedia

eGlossary

11.3

Volcanoes

FIGURE 11.7 (above) Volcanic eruptions on the island of Hawaii produce vast amounts of lava which cover large areas of land, but rarely harm people.

FIGURE 11.8 (far right) As magma rushes to the surface of the Earth, the size of the gas bubbles increases.

FACT FILE

The Western District of Victoria is dotted with volcanic cones and crater lakes. Though there is no sign of volcanic activity, some scientists believe that these volcanoes are too young to be considered extinct.

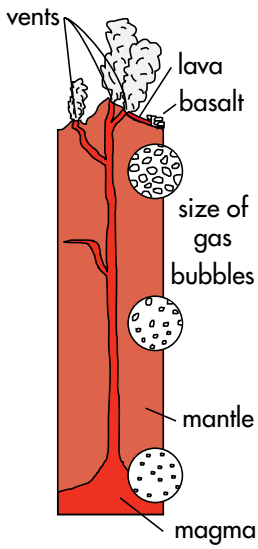
Next time you are about to open a fresh bottle of soft drink, give it a shake first. Watch carefully to see what happens as you slowly unscrew the cap, releasing the pressure in the bottle. Hundreds of bubbles suddenly appear where there were few before, and they rapidly get bigger as they race towards the top. When the cap is removed completely, the soft drink gushes out of the bottle (and all over your hands).

Now consider a volcanic eruption. Under tremendous pressure, pockets of molten material (called **magma**) in the upper mantle start moving upwards. Gases dissolved in the magma form bubbles that grow bigger and bigger as the magma rushes to the surface. With spectacular force, the gas-driven mixture bursts out of the narrow vent, and a new **volcanic eruption** is born.

Talk about

How is a volcanic eruption:
a similar to opening a bottle of shaken soft drink?
b different from opening a bottle of shaken soft drink?

The magma emerges white hot at temperatures of 1000°C or more. As this liquid loses its gases to the surrounding



atmosphere it becomes lava, and flows along the ground away from the vent. As it cools the lava changes colour from yellow through orange and red, finally hardening into a shiny black rock called basalt. Often the gas escape holes are preserved in the cooling lava.

If the magma is very sticky to start with, or the vent is blocked, the gases may cause enormous pressure to build up just below the surface. A catastrophic explosion may result as hot gases and shattered rock in the form of ash surge up and outwards at

USING SCIENCE 3 HOW HOT IS IT?

Using Figure 11.7 as a guide, draw a full-page colour sketch of a volcanic eruption. Use the table opposite to label each part of the eruption with its approximate temperature.

Temperature	Colour
Above 1150°C	white
1000–1150°C	yellow
900–1000°C	orange
500–900°C	red
Less than 500°C	black

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EXPERIMENT 1 MAKING A MODEL VOLCANIC ERUPTION

AIM
To observe how gases cause liquid material to erupt.

MATERIALS
- lots of newspaper
- large piece of thin card (poster size)
- masking tape
- scissors
- cup-shaped piece of egg carton
- red and yellow food dyes
- cornflour
- water in bottle or beaker
- glass stirring rod
- 2 Alka Seltzer tablets (or similar)

METHOD
1 Make a volcanic cone using the cardboard and masking tape. Make sure it can stand by itself.
2 Cover the work area with several layers of newspaper. Sit the volcano in the middle of the newspaper.
3 Cut enough off the tip of the cone so that the eggcup fits in snugly. This will act as the volcanic crater. Secure the rim of the crater to the cone all the way around with tape.
4 Half fill the crater with cornflour. Gradually add small amounts of water and stir until the mixture is a thin paste.
5 Add 3 drops of each food dye.
6 Drop in the two Alka Seltzer tablets and watch!

DISCUSSION
1 Draw a diagram of your volcano, showing the shape and extent of your 'lava flows'.
2 Let your volcanic eruption stand for a while. What has happened to the bubbles of gas? What has happened to the lava?

EXTENSION
Using the same cardboard model, try another eruption using vinegar and baking soda … and don't forget the food dye.

very high speeds. This type of eruption causes the most deaths because people do not have time to get out of the way of the deadly cloud (called a **nuee ardent** or 'glowing cloud').

Not all volcanic regions in the world are erupting all the time. If lava or ash are currently erupting, we say the volcano is **active**. If there is a long period of calm (years or centuries), the volcano may be classed as **dormant**. When there has been no sign of activity for many thousands of years, the volcano may be declared **extinct**.

Scientists classify volcanoes into two main types, based on they way they form and the nature of their eruptions. The two main types are called composite volcanoes and shield volcanoes.

Questions
1 What role does gas perform in a volcanic eruption?
2 What is the difference between an extinct volcano and a dormant one?
3 What could cause a volcano to become extinct?
4 Why does basalt often have many holes in it?

Think about
Imagine you are a pool of magma in the mantle. Describe your journey to the surface, out into the cool air. Don't forget to describe the changes in your temperature, pressure and gases as you go.

Find out
Consider the following volcanoes: Mt Vesuvius, Mauna Loa, Mt St Helens, Etna, Surtsey, Kilauea, Krakatoa, Fuji. For each find out:
a their location
b when they last erupted
c whether they are a composite or shield volcano.

Word process

Internet assignment

FACT FILE
Several members of the solar system have been shaped by volcanic activity. The Moon, Mars, Venus and Io, one of Jupiter's moons, all show evidence of volcanism. In fact the largest volcano is on Mars. It is around 24 km high and is called Olympus Mons.

Extension worksheet

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11.4

Rocks made of crystals

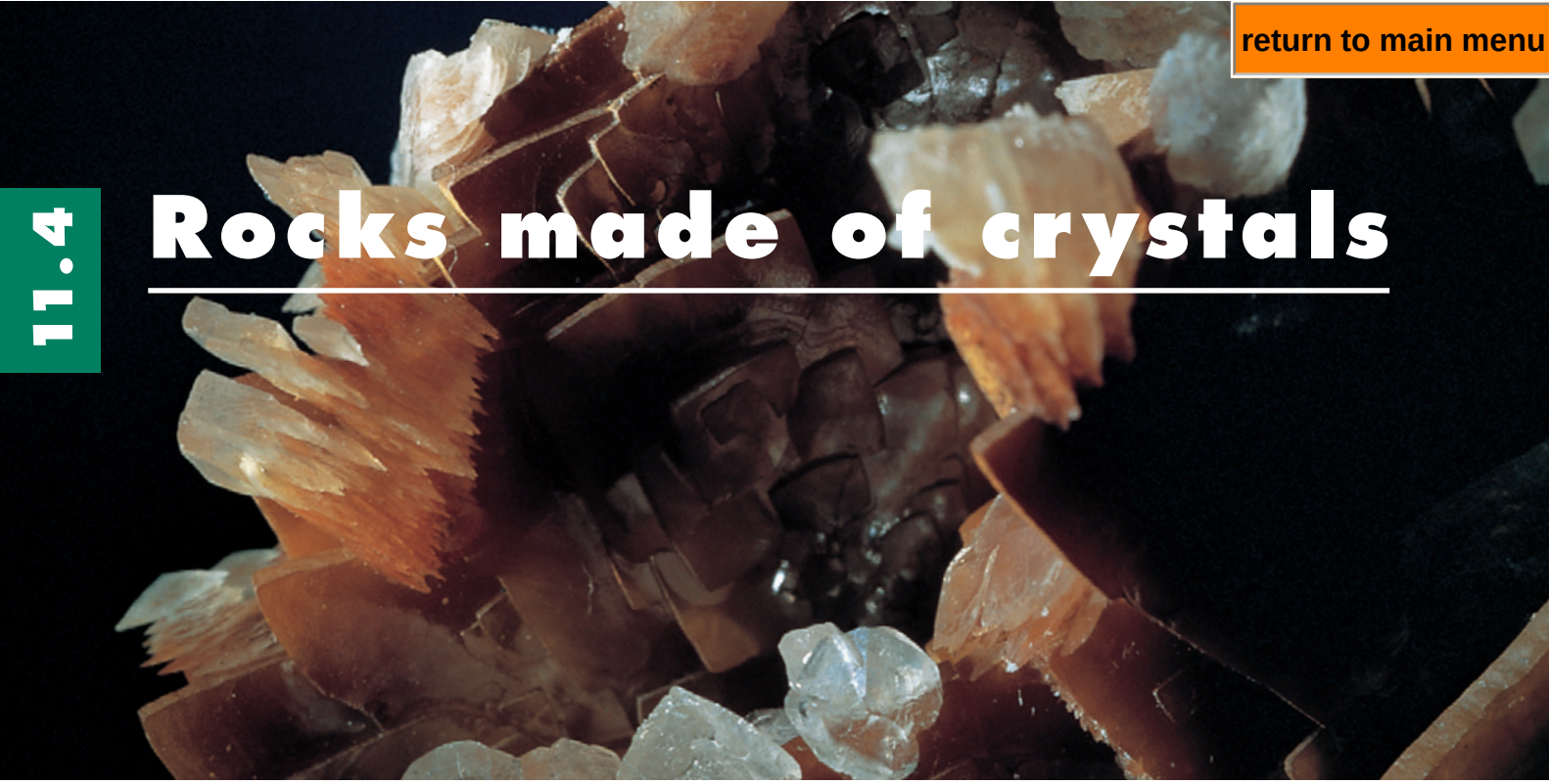

FIGURE 11.9
These are quartz crystals.

FACT FILE

Lava may cool to form basalt in a matter of hours or days (thick flows may take a year). Magma forming granite deep in the Earth's crust may take a million years to cool.

Talk about

What do you think of when you hear the word 'crystal'? Do you think of magic? Or of sparkling, glassy jewels with beautiful colours? You might imagine them set into a crown, or in a treasure-filled chest. In a small group, brainstorm a list of words that you associate with crystals. How does your group's list compare with the lists of other groups?

A **crystal** is a solid chemical substance with flat edges or 'faces'. The faces of natural crystals may be so perfectly arranged that they look as if they had been made by humans.

One way that natural crystals form is from molten rock, or magma. As the magma cools, certain chemicals cluster together into minerals. A **mineral** is a naturally occurring substance with a distinct chemical composition. For instance, the mineral quartz is made when silicon and oxygen combine.

As the minerals separate out of the cooling liquid magma, they grow into crystal shapes. The first mineral to 'crystallise' can form a distinct crystal shape with regularly arranged faces because there is only liquid around it. As more and more crystals form, they 'grow into' each other, so later crystals are less well formed. When the magma has finished cooling, the crystals all fit together like a jigsaw puzzle.

Basalt and granite are two examples of rocks formed from hot, molten magma. They are known as **igneous** ('born of fire') rocks.

FIGURE 11.10
Basalt rocks are hard, igneous rocks.

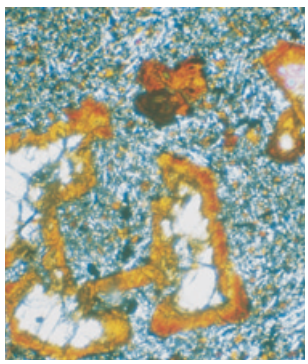

FIGURE 11.11
The crystals in basalt are microscopic.

FIGURE 11.12
Granite rocks are examples of igneous rocks.

FIGURE 11.13
Granite is formed from large crystals.

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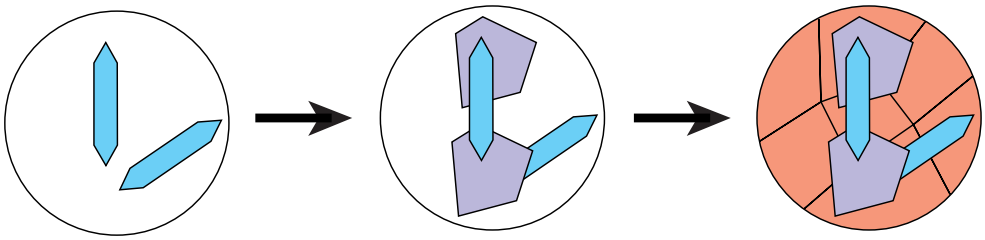

FIGURE 11.14
The stages of crystal formation.
1 The first mineral starts forming in molten magma.
2 The second mineral forms.
3 The last mineral to form fills any remaining space.

Magma that reaches the surface of the Earth through a volcano cools very quickly. The lava that flows out hardens to form rocks such as basalt. The crystals in basalt are usually so small that you need a microscope to see them.

Some magma does not reach the surface. Magma bodies that remain 10 km or so beneath the Earth's surface cool very slowly. **Granite** is formed in this way. Granite boulders are only exposed at the surface after the overlying rock is worn away over millions of years. The mineral crystals in granite are large enough to be seen without a microscope.

Foundation worksheet 1

EXPERIMENT 2 COOLING RATE AND CRYSTAL SIZE

AIM
To discover the effect that cooling rate has on the size of crystals formed from a molten substance.

MATERIALS
- 3 clean test-tubes, test-tube holder and test-tube rack
- spatula
- beaker containing ice cubes
- Bunsen burner
- hand lens
- saturated solution of potassium nitrate (freshly made)
- potassium nitrate
- safety glasses

METHOD
1 Pour about 5 mL of potassium nitrate solution into a test-tube.
2 Add one spatula of potassium nitrate to the test-tube.
3 Slowly heat the test-tube over the Bunsen burner flame until the solution begins to boil.
4 Pour half of the hot solution into each of the remaining test-tubes.
5 Put one test-tube in the beaker of ice to cool quickly.
6 Put the other test-tube in the test-tube rack to cool slowly.

DISCUSSION
1 How did the sizes of the crystals on the walls of the two test-tubes compare?
2 Can you make a connection between crystal size and cooling rate?
3 Look at Figures 11.11 and 11.13. How can you tell that the granite cooled slowly and the basalt cooled quickly?

Questions

1a What is a crystal?
b What is a mineral?
2 Can a substance be both a mineral and a crystal? Explain.
3 Why don't all crystals in a rock cooled from magma have perfect shapes?
4 What is the main difference in the crystals formed in basalt and those formed in granite? What causes this?

eGlossary

11.5

The tortured crust

FIGURE 11.15
Uluru is the eroded remnant of a vast block of sandstone which has been squeezed over millions of years. What we see now is one part of a giant fold.

FIGURE 11.17
The movement of crustal blocks on fault lines can cause total devastation in populated areas. Whether rocks fold or fault depends on four main factors: the types of rocks, the direction of the force (pulling or pushing), how big the force is, and how rapidly it is applied. **Mudstone**, which is often soft, will fold easily, but **sandstone** is easily fractured. If a squeezing force is applied slowly any rock type can be folded. When a massive force is applied in the dramatic few minutes of an earthquake, all of the rocks near the centre of the earthquake will break, no matter what type they are.

In Australia's dry heart stands Uluru, a rock formation made of steeply tilted layers of sandstone deposited by massive rivers millions of years ago. The layers were once horizontal.

Carving up the landscape of the United States west coast into long thin strips are many great scars.

The blocks of crust are sliding past each other. Sometimes they creep slowly, but sometimes they lurch suddenly, producing devastating earthquakes.

FIGURE 11.16
The San Andreas Fault, a huge **fault** which cuts through San Francisco and Los Angeles, is a fracture between one block of crust which is moving south and another which is moving north.

Questions

1 Name the two types of structures we observe in rocks which tell us that great forces within the crust of the Earth have existed in the past.

2 List four factors which determine whether rock masses will bend or break. Do an illustration which shows that you understand these four factors.

3 If Uluru is part of a fold, which part of the fold could it have been? Draw a picture showing the rest of the fold to aid your explanation.

Think about

Would a reverse fault result from pressure (pushing) or tension (pulling)? What about a normal fault? Could a strike-slip fault result from either pressure or tension?

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USING SCIENCE 5 MODELLING FAULTS AND FOLDS

During this activity you will investigate the effects of pressure and tension (pushes and pulls) on layers of material which represent layers of rock.

WHAT YOU NEED

An old telephone book, coloured pencils or textas, Playdoh or Plasticine of three different colours, 10 cm piece of nylon fishing line, 2 ring washers.

WHAT YOU DO

1 Look at one end of the telephone book. Imagine that the stacked pages represent layers of rock. Mark several coloured layers along the edge to make the layered effect more pronounced.

2 Push the binding with one hand and the opposite edge with the other to make folds in your layers.

FIGURE 11.18
Push opposite edges of the book.

3 Use Figure 11.19 to identify the parts of the **folds** which you have made with your telephone book pages.

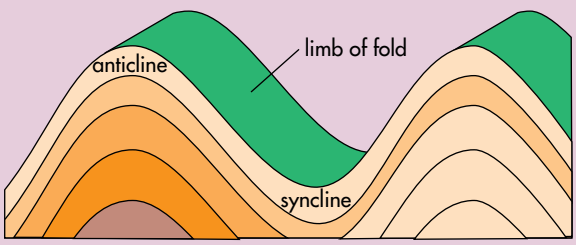

FIGURE 11.19
Upward-bending folds are called **anticlines**. The downward-bending ones are **synclines**. The straight parts in between are called **limbs of the fold**.

4 Make a block of Plasticine or Playdoh with three different horizontal layers in it.

5 Tie one washer to each end of the fishing line and use this as a cutting wire to cut down through the layered block at an angle.

6 Move the blocks along the 'fault line' to create the 'normal fault' shown in Figure 11.20. Stick the blocks in their new positions with a little pressure.

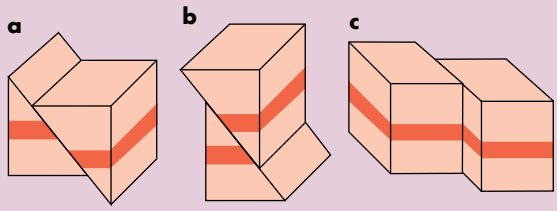

FIGURE 11.20
Types of faults.
a A **normal fault** results when one block slips down along the fault line.
b A **reverse fault** occurs when one block slides up along the fault.
c A **strike-slip fault** is one where the blocks slide past each other horizontally.

7 Imagine the higher block was eroded down to the level of the lower block. Simulate this with your cutter by slicing off the top of the higher block horizontally.

8 Imagine this new ground surface was being mapped by geologists. What would they find as they walked across the fault line?

9 Look at the fault line from the side of the block. Imagine this was a railway cutting. What would you see as you passed the fault line in the train?

10 Repeat the fault-simulation process for the other two fault types shown in Figure 11.20.

EXTENSION

Make Playdoh models of **anticlines** and **synclines**. 'Erode' the tops off with your cutter and explain to your partner what geologists walking around on top would see. Could the sequence of rocks they walk across give them a clue as to whether they had an anticline or a syncline under their feet? Explain.

Find out

1 Use a library to find out how Uluru actually formed, from the first sediments deposited to the agents of erosion now working to wear it down. Make a flow chart from your research.

2 How do people living on or near the San Andreas Fault cope with 'living on the fault line'? Locate some magazine articles, newspaper cuttings or videos about this. Discuss how you would feel.

Web page

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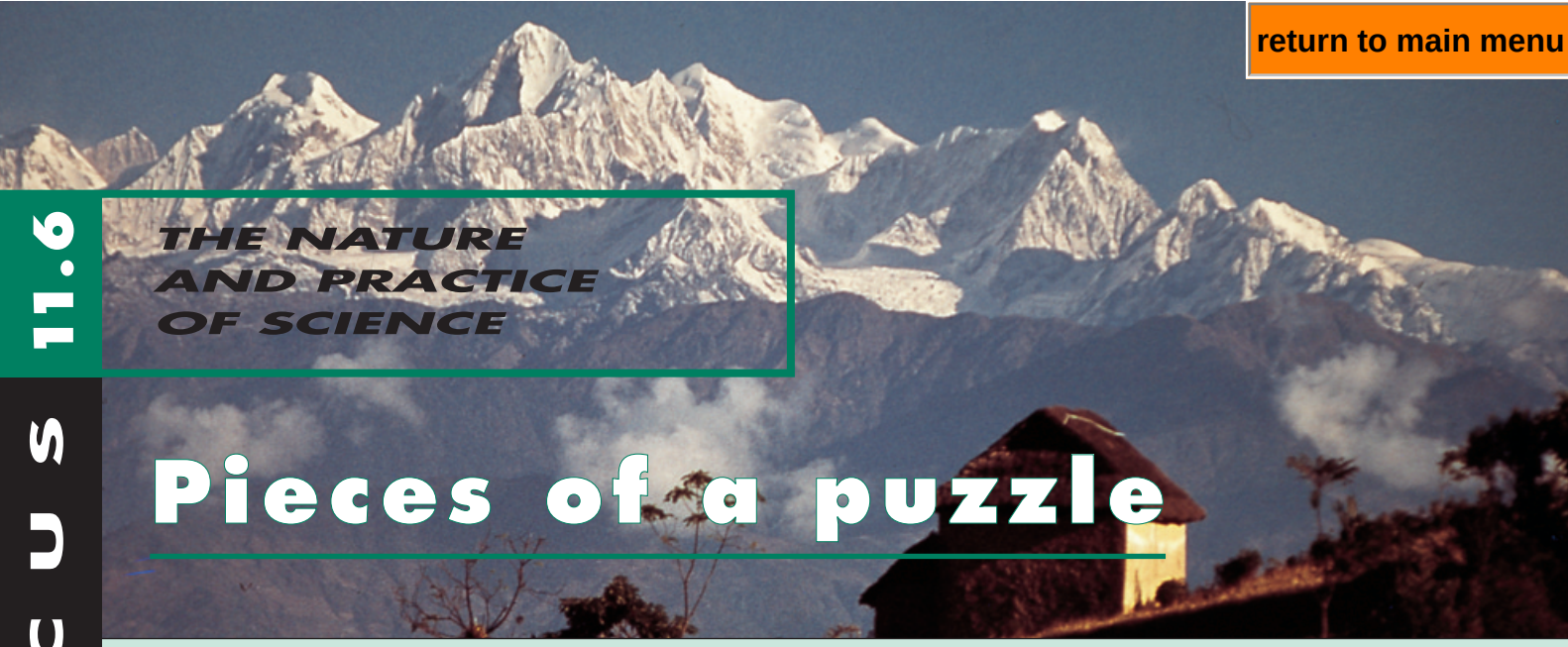


THE NATURE AND PRACTICE OF SCIENCE

Pieces of a puzzle

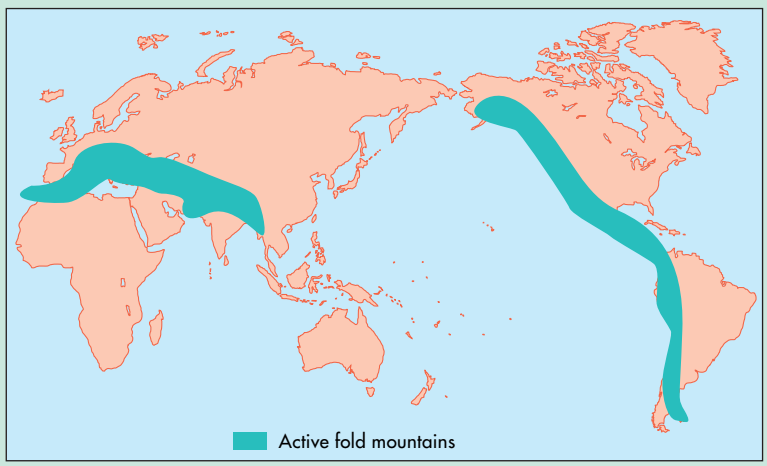
FIGURE 11.21
The world's highest mountain ranges are called **fold mountain belts** because they consist of huge masses of folded and faulted rock in long narrow zones. On a world scale, these mountain chains can be seen to cross countries and continents.

Once you start looking at road cuttings and other places where rocks are exposed, **folds** and **faults** appear to be all around us. To see the most intensely folded and faulted rocks in the world, you would need to visit the highest mountains on Earth. These mountains form long chains and include the Himalayas in Asia, and the continuous line of mountains which runs down the western edges of North and South America.

Why are these mountains where they are, and what has happened to form them? The most widely held belief among geologists today is that these mountains formed when two land masses collided and their edges crumpled up.

This theory of continental collisions may sound bizarre, but scientists have been gathering evidence for the movement of pieces of Earth's crust for centuries. The theory began with Alfred Wegener, a German meteorologist, who in 1912 noticed that continents such as South America and Africa could be pieced together like a giant jigsaw puzzle. He proposed that these continents had once been joined, and had since 'drifted apart'. The **continental drift theory**, as it became known, was not very popular at first, but since Wegener's time scientists all around the world have contributed to the growing body of supporting evidence.

FIGURE 11.22
Fold mountain belts occur right across the Earth. The ones shown on this map are still being pushed up today.



The major clues in this geological detective story come from many different branches of science.

1 **Geologists** have mapped large areas of similar rocks on continents which are now widely separated. When the continents are fitted together in certain ways these rocks match up across the joins.

2 **Palaeontologists** (people who study fossils) have found many fossil land plants and animals of certain species which are shared by continents now separated by vast oceans.

3 **Biologists** have shown links between present-day plant and animal groups which are native to continents separated by oceans.

4 **Geologists** have also studied rocks of certain ages which show evidence of ice cover at certain times during Earth's history.

All of this evidence, and much more, has been used to build up a picture of the way continents have changed their positions in the past. That the continents have drifted is no longer in doubt. They have also been torn apart and pushed back together in new configurations many times over.

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USING SCIENCE 6 THE CONTINENTAL JIGSAW

In this activity you will assemble the ancient continent of **Gondwana**. This is the name used for the landmass which is thought to have consisted of Australia, India, South America, Africa and Antarctica, before they all broke apart about 80 million years ago.

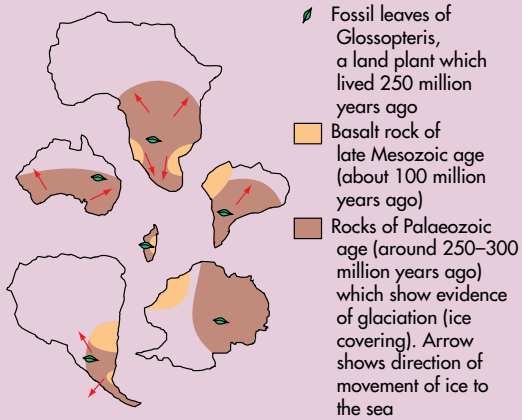

FIGURE 11.23
Enlarge these continents. Cut them out and put the information on them.

WHAT YOU NEED

Your own large, hand-drawn copies or enlarged photocopies of the continent shapes shown in Figure 11.23.

WHAT YOU DO

1 Use only the coastlines as a guide first. Match the continents up like a jigsaw until you are happy that they fit together reasonably well.

2 Now look at the other geological and biological information. Do these match up as well? Make adjustments to your 'jigsaw' until these pieces of information make sense too.

DISCUSSION

1 Could more than one configuration be possible?

2 Which pieces of information do you think were the most reliable for helping to match up the continents? Which information might be unreliable?

3 Why are there still some gaps when the coastlines have all been fitted together?

Questions

1 What two features of the Himalayas suggest that they formed from upward crumpling when two land masses collided?

2 Look at Figure 11.22. On which continents are fold mountains still forming today? Which continents are not experiencing fold mountain activity?

3 What was the first piece of evidence put forward to support the theory of continental drift?

4 Why do you think people might have had trouble accepting the idea of continental drift at first?

Find out

Use your atlas to find the major mountain chains of the world. Compare this to Figure 11.22 so that you can match up the names of the mountain ranges which are still being built. You may also be able to discover some major mountain ranges which are no longer forming, or which were formed in other ways.

Animation

Foundation worksheet 2

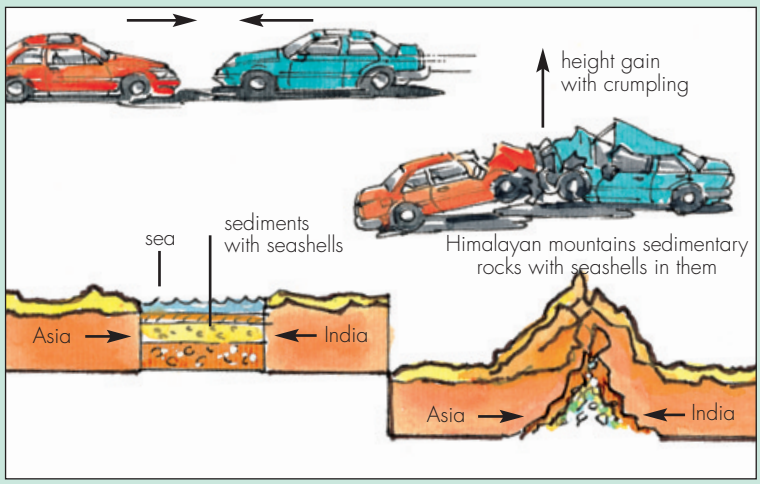

FIGURE 11.24
When two cars collide the crumpling results in shortening in a horizontal direction and a vertical height gain. The Himalayas are thought to have formed when two continents squeezed together, folding upwards the seafloor sediments which had been between them.

Adding weight to this theory is the fact that sedimentary rocks in the highest mountains in the world, the Himalayas, contain fossils shells of creatures which were once on the sea floor.

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Completing the puzzle

FIGURE 11.25
The earthquake that triggered the 1998 Papua New Guinea tsunami was caused by the slow-moving Australian plate, which is travelling north, ramming into the Pacific plate, which is moving north-west. The Pacific plate is being forced down beneath the Australian plate off the Papua New Guinea coast.

FACT FILE

Australian geologist Samuel Carey played an important role in gathering evidence to support the theory of continental drift. While working in New Guinea he made observations that convinced him that continents moved horizontally. Carey traced the position of mountain ranges and island arcs, showing how they could be connected to form continuous belts around the globe.

One of the reasons people found it difficult to accept the theory of continental drift was because it is hard to imagine something the size of a continent floating around the world. Only with the technology of recent decades have scientists been able to explore new evidence which might explain how the continents are able to move.

The first clue is to look on a world scale at where earthquakes and volcanoes are currently active. Like fold mountains, these other major disruptions in the Earth's crust occur in distinct zones (see Figure 11.30).

The earthquake and volcano zones mark the edge of pieces of the Earth's crust known as **plates**. These plates of crust are a bit like the cracked shell of a hard-boiled egg. Like the rubbery egg-white under the shell, the Earth has a slightly squashy layer under the crust, called the **mantle**.

Scientists think that movements in this lower layer cause the crustal plates to move, and at the edges of the pieces there is a lot of activity.

Not all of the plate edges experience the same type of activity, though. In some places the plates are moving apart from each other. At these plate edges **magma** (molten rock) from the mantle wells up and pushes the plates further apart. Scientists in miniature submarines have actually observed this process in action along the Mid-Atlantic Ridge, a huge fracture which divides the Atlantic Ocean into two plates. In fact it is spreading along this fracture

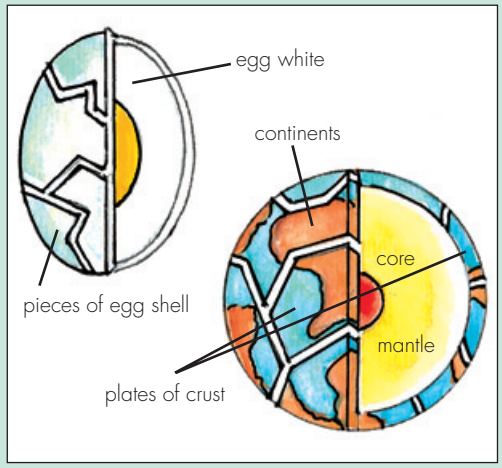

FIGURE 11.26
The two outer layers of the Earth are a bit like those of a cracked hard-boiled egg.

which provides the answer to Alfred Wegener's question about the snug fit of the coastlines of Africa and South America.

While some plate edges are pulling apart, others are pushing together. We have already seen that continents which push together can create huge belts of fold mountains. This usually happens when continents riding on the plates come into contact. But this isn't the whole story. Some plate edges, rather than simply crumpling upwards when they collide, actually override other plates. This process is called **subduction**.

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USING SCIENCE 7 MAKING A MODEL OF THE EARTH'S CRUST

In this activity you will make a model of the plate tectonic processes shown in Figure 11.27.

The major processes of the plate tectonic model of our Earth are subduction, seafloor spreading, fold mountain building and the horizontal sliding of plates past one another.

You will be making the model from cardboard, paper and other readily available materials. You will need to design the model on paper first, and show how you can actually make it move. Some sort of winding apparatus might be involved. Show your design and list of materials to your teacher before you start. Try focusing on one or two of the processes first. If you are really ambitious, you could include all four in your design.

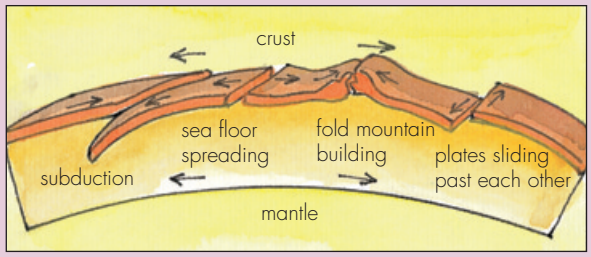

FIGURE 11.27

In other parts of the world the plates slide past each other horizontally. This is what seems to be happening along the San Andreas Fault (see Figure 11.16).

All of these movements and types of activity at the edges of crustal plates are explained by the new theory of **plate tectonics**. Tectonic comes from the Greek word 'tektonikos' which means 'built'. These tectonic processes are constantly building, demolishing and rebuilding the surface of our planet.

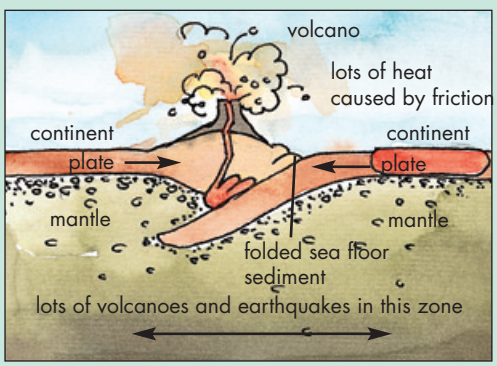

FIGURE 11.28
One plate is pushed over the top of another. The friction and heat produced during this grinding process cause a lot of violent earthquake and volcanic activity.

Questions

1 Name three geological features of the Earth's crust which occur in distinct zones.

2 Explain the difference in the movements of plates in subduction zones to the movement in seafloor spreading zones.

3 Subduction and fold mountain building both occur when plates push into one another. In what way are these processes different?

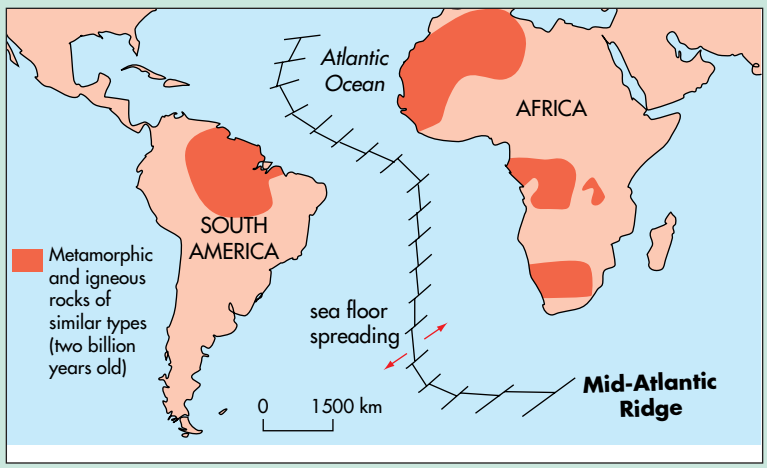

FIGURE 11.29
South America and Africa were broken apart when a crack in the Earth's crust allowed magma from the mantle to push up. This spreading has continued, resulting in the two continents moving further apart as new sea floor is created from the cooling magma.

Find out

1 How did the work of Fred Vine and Drummond Matthews provide evidence for seafloor spreading along mid-ocean ridges?

2 How do the people in countries on or near plate edges cope with their tectonically active environment? Compile an information file of magazine and newspaper articles about recent events in some of these areas.

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11.8

Coping with a restless Earth

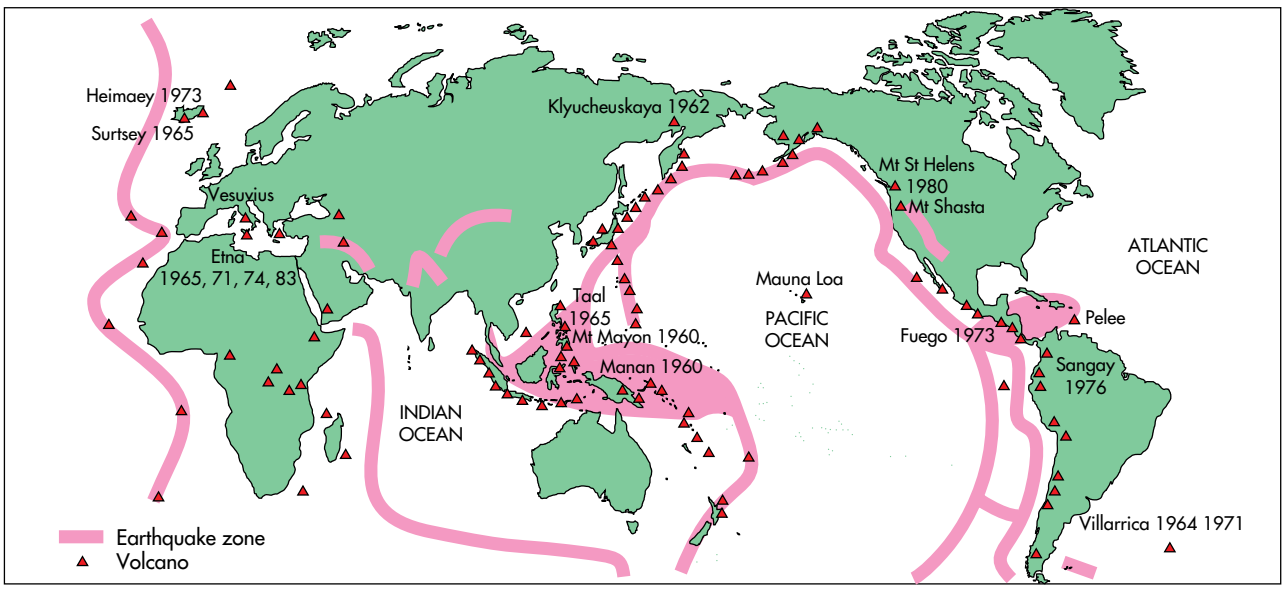

FIGURE 11.30
Recent volcanic eruptions and earthquakes are not spread evenly across the Earth but are concentrated into zones. Can you find a continent which is not currently a volcano or earthquake region?

FACT FILE

In 79 ad the people of Pompeii failed to recognise the warning signs of a volcanic eruption. The wells in the town dried up and small tremors were felt for days beforehand. Even when smoke came out of Mt Vesuvius it was looked upon more with curiosity than alarm.

Talk about

The rim of land around the Pacific Ocean is often called the 'Ring of Fire'. Why do you think it is called this?

Perhaps the most useful piece of information for preventing huge losses of life in a natural disaster is knowing where a particular disaster is likely to strike. Most earthquakes and volcanoes are located in distinct zones. These zones correspond to weaknesses in the Earth's crust, rather like cracks in a hard-boiled egg that has just been tapped all over ready for peeling (see Figure 11.26).

Scientific instruments can now detect changes in the activity of particular volcanoes, reducing the risk to life. Scientists can use seismographs and **tiltmeters** to measure small earth tremors and the bulging of the ground that occurs as magma rises towards the surface.

Satellites with special heat-sensitive cameras can measure rises in ground temperature, and gas samplers can pick up increases in sulfurous gases, another sign of an impending eruption. Some of the most recent volcanic eruptions in Hawaii have

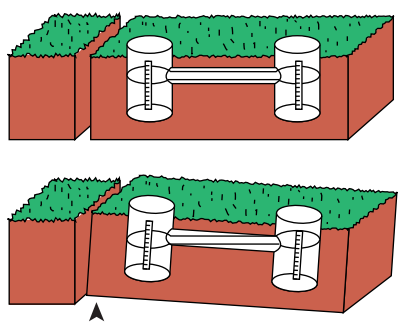

FIGURE 11.31
A tiltmeter is an instrument that measures the amount of tilt in the ground caused by magma pushing up from underground and making the sides of a volcano swell. This can help scientists predict where an eruption will occur.

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USING SCIENCE 8 PREPARING A DATA GRID

Using any reliable resources available to you, make a colourful, informative poster about one of the four natural phenomena mentioned on these two pages. As well as words, your poster could include tables, graphs, photographs and diagrams.

A good way to gather information in note form, and to keep track of the resources you use at the same time, is to use a data grid. An example is shown opposite.

The words 'Who', 'What', 'Where', 'When' and 'Why' are often a good start as headings for your grid.

Once you have filled most boxes in the grid with short notes, you can use the grid to do your poster. This method helps you to put information into your own words.

Source of information	Where are tsunamis likely to occur?	Famous examples	What can be done to prepare for tsunamis?	Pictures
Earth magazine, July 1993	80% hit coasts of Pacific Ocean	Hawaii: Hilo Bay, 1946 Java: from eruption of Krakatoa, 1883	Seismographs detect underwater earthquakes. Coastal cities can be warned.	Good diagram p.6.
Encyclopedia of World Events, volume 13	All around Pacific and other coastal areas	Indonesian islands from eruption of Krakatoa. Over 30 000 killed		Photo of man being swept away p. 1137.

FIGURE 11.32
An example of a data grid on the topic of tsunamis.

been predicted hours, or even days, in advance.

Earthquakes give far fewer warnings. Chinese scientists claim to have noticed changes in certain gases contained in groundwater before some earthquakes, as well as strange behaviour amongst wild and domestic animals. Tilting, sliding and bulging of the ground, detectable with extremely sensitive instruments, are possible warning signs in some areas. The best defence against earthquakes is to build structures that will withstand the shock waves. Much research is being done into building designs and materials suitable for earthquake-prone areas.

Fortunately, the underwater earthquakes that often trigger tsunamis can be detected by seismographs all over the world. In many cases this has given people in coastal areas time to get to higher ground. Severe weather conditions such as cyclones are probably the most easily monitored of all. Satellites, ships at sea, weather balloons and automatic weather stations on remote islands all help to chart the course of cyclones at sea days, or even weeks, before they hit land.

Questions

1 What is the first step in preventing great losses due to natural disasters?

2 Which statement below best describes the distribution of earthquakes and volcanoes around the world?

a Earthquakes and volcanoes are always found in the same place.

b Earthquakes and volcanoes are usually found in well-defined zones.

c Earthquakes and volcanoes are evenly distributed over the planet.

3 Use the map in Figure 11.30 to answer the following questions.

a Name five countries that are likely to suffer earthquakes frequently.

b Name one place where volcanoes are active, but major earthquakes are rare.

4 Arrange the four natural disasters mentioned in order of difficulty to predict (from hardest to easiest). Explain your answer.

FACT FILE

The 1946 tsunami that devastated the coast of Hawaii led to the setting up of the Pacific Tsunami Warning System based in Honolulu. Here scientists use computers and satellites to keep a round-the-clock watch on the region. If a tsunami is confirmed, warnings are transmitted to the countries which may be affected.

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Using scientific language

1 Make a crossword for a friend to help them revise the work in this chapter. You will have to write the clues yourself. Try to use some of the words in this list.

tiltmeter	seismogram	volcano
crystal	tsunami	granite
lava	magma	Pompeii
mantle	core	ash
subduction	basalt	Gondwana
fold mountain		

2 Explain how the two terms in each pair are related, then explain how they are different.
- a mineral and crystal
- b seismograph and siesmogram
- c fold and fult
- d continential drift and plate tectonics

Check your knowledge

1 List four natural phenomena scientists study to find clues about what is inside the earth.

2 What is a tsunami? Name two events that can cause a tsunami.

3 a Where in the earth do earthquakes start?
- b What effects do they have at the surface of the earth?

4 Complete the following sentences.
- a Molten material in the mantle that still contains dissolved gas is called _____ .
- b As molten material erupts from a volcano and loses its gases it is called _____.
- c _____ is a type of rock formed from hardened lava.

5 Explain the differences between volcanoes that are active, dormant and extinct.

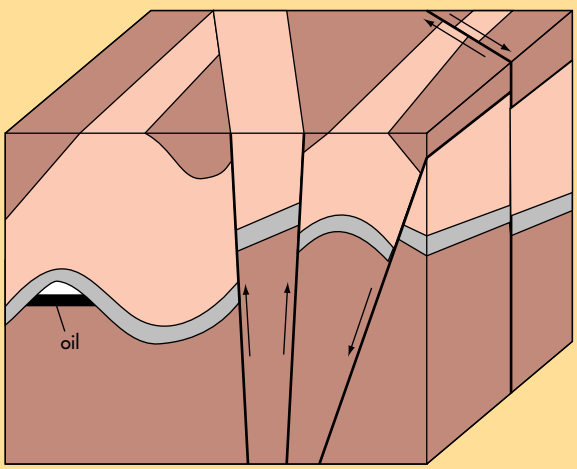

FIGURE 11.33

6 Copy Figure 11.33. Use arrows and labels to indicate these features: anticline, syncline, normal fault, reverse fault, strike-slip fault, limb of fold, axis of fold.

Applying your skills

1 a What are the two main reasons why we have not drilled very far into the Earth?
- b Can you think of other reasons why humans would have difficulty getting to the mantle?

2 Tsunamis are often called 'tidal waves'. Explain why this term is misleading.

3 Study the following picture of a granite boulder at Wilsons Promontory in Victoria.

FIGURE 11.34

It has some extremely large crystals, set in a background of much smaller crystals. This gives a confusing picture of the cooling rate history of the rock. Try to explain what might have happened.

Challenge yourself

1 In many countries people live in active volcanic areas. Find out how the people in Indonesia, New Zealand or Hawaii benefit from living near active volcanoes.

2 Humans can be affected severely by changes in the Earth's crust. Research a recent natural disaster and report on its effect on human lives. You could consider:
- how many deaths occurred
- how much damage was done
- how people coped
- where people went
- how it felt to live in the disaster area at the time.

Include in your report copies of at least three newspaper clippings written at the time, and comment on your feelings about each item.

3 Find out about the models and theories of the formation of mountains and continents that were popular before the theory of continental drift. Detail the evidence that led to them being rejected.

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GLOSSARY

abiotic non-living components of an environment 185
absorb to take in; light which is absorbed is neither transmitted nor reflected 99
accommodation ability of the lens in the eye to change shape in order to obtain a clear image of near objects; lack of accommodation is where the lens of the eye becomes stiff and cannot change shape sufficiently to focus on near objects 94
accuracy how close a measured result is to the actual result 4
acid a substance with a sour taste that reacts with metals to form hydrogen gas; turns blue litmus red 36
activity series (reactivity series) arrangement of metals in order of their ease of reaction with various substances such as oxygen, water and acids; aids in the understanding of properties of metals and enables predictions to be made about a metal's behaviour 56
additive mixing formation of secondary colours by mixing two primary colours, e.g. red + green gives yellow 100
afterbirth expulsion of the placenta following birth 137
algae plant-like organisms usually found in water 40
algicide inhibitors chemicals used to prevent algal growth 41
alleles alternative forms of a gene 154
alpha particles radiation consisting of positively charged particles (essentially the same as a helium nucleus) 24
alternating current (AC) current that constantly, rapidly changes direction in a circuit 81
amino acids subunits of proteins; there are 20 different amino acids 146
ammeter device for measuring electric current 70
amnion fluid fluid inside the amniotic sac surrounding the developing foetus 137
amniotic sac protective sac of fluid surrounding a developing foetus 137
ampere unit of electric current (A) 70
angiosperm flowering plant 126
anion negative ion, e.g. Cl^-, O^{2-} 35
antacid tablet tablet containing a base which is used to neutralise excess stomach acidity 54
anther structure on the tip of the stamen where pollen is found 127
antibiotics chemicals which can be introduced into the body to destroy bacteria 175
antibodies proteins which are active against foreign micro-organisms or substances 173
anticline upward bending fold 215
antiseptic chemical which can be used on the body to destroy bacteria 175
aperture diameter of adjustable diaphragm which controls the amount of light entering a camera 102
aquatic living in water (freshwater or marine) 186
asexual reproduction one-parent reproduction which does not involve the union of gametes 124
athlete's foot fungal infection which occurs between the toes 170
atom small particle that makes up all matter; composed of protons, electrons and neutrons 24
atomic number number of protons an atom has in its nucleus 30
axe machine to help split wood; a double inclined plane 111

bacteria simple, single-celled organisms which lack a true nucleus; most are harmless, some beneficial and some cause disease 40
basalt igneous rock which has cooled rapidly 210
base a substance with a soapy feel that neutralises acids; turns red litmus blue 36
bi-focal spectacles with lenses that are in two sections of different shapes and hence strengths 95
binary fission reproduction of bacteria by splitting in two 168
bio-accumulation (biomagnification) the increase in concentration of a chemical as it progresses along a food chain 190
biodiversity the variety of living things in a community or the biosphere 200
biologist scientist who studies living things 20, 216
biomass total mass of organic matter in an ecosystem 83, 189
biosphere region at the surface of the Earth that links all ecosystems; all living things exist within the biosphere 184
biotic living components of an environment 185
budding asexual reproduction where the new individual buds or grows directly off the parent 124

carbon cycle cycle of carbon-containing compounds through the biosphere 194
carbonates substances that react with acids to form carbon dioxide 54
catalyst substance that speeds up the rate of a chemical reaction but is itself unaffected 63
cation positive ion, e.g. Na^+, Mg^{2+} 35
cellular respiration see respiration 52

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centromere point on a chromosome at which chromatids are held together; point of attachment to spindle fibres 152
cervix opening between the vagina and uterus 34
charge electrons are said to be negatively charged and protons positively charged; their charges are equal but opposite; like charges attract, unlike charges repel. The charge of an ion is determined by how many electrons it has lost or gained to attain a stable outer shell 34, 60, 70
chemical change a change in which a new substance or substances are formed, e.g. combustion 46
chemical formula group of atomic symbols that shows the kinds and number (or ratio) of atoms in a substance 35
chemical reaction reaction in which a chemical change occurs 46
chromatids identical strands of a replicated chromosome held together at the centromere 152
chromosomal abnormality abnormal phenotype in an individual caused by an extra or missing chromosome; caused by an abnormal meiosis in one of the individual's parents 160
chromosomes long thread-like bodies of DNA found in the nuclei of cells 146
circuit a continuous path along which an electric current can flow 66
class 1 lever lever where the pivot is between the load and the effort 109
class 2 lever lever where the load is between the effort and the pivot 109
class 3 lever lever where the effort is between the load and the pivot 109
clone exact copy or duplicate of another living thing produced by asexual or one-parent reproduction 140
colour blind any condition where colour is not accurately perceived; the most common is red-green colour blindness 95
colour film film with three distinct layers, each one being sensitive to one of the primary colours 102
column graph graph showing information as columns or bars; allows comparison of measurements 10
combination reaction see synthesis reaction 52
complementary colours a primary and secondary colour that give white light when mixed, e.g. green + magenta = white 101
complex machine machine made up of two or more simple machines 114
components parts of circuit, e.g. globe, switch 66
compound a substance composed of two or more different types of atoms; can be broken down into simpler substances 28
concave lens lens that is thinner in the middle than at edges 90
concave (mirror) a mirror with a surface that curves inwards 86
concentration a measure of how much substance is dissolved in a solution 62
conductor a substance that will allow an electric current to flow through it 66
consumer organism that gets its energy by eating other organisms 186
contact lens small lens worn directly on the cornea 95
contaminants unwanted impurities in a substance 40
continental drift theory theory suggesting that the continents are slowly moving over the surface of the Earth 214
controlled experiment experiment with an identical set-up (the control) to the experimental set-up with the exception of the factor (the variable) being tested 2
converge come together 86
convex lens lens that is thicker in the middle than at edges 90
convex (mirror) a mirror with a surface that curves outwards 86
copulation mating process where male's sperm are introduced into the female's body 127
cornea thick curved transparent material at the front of the eye; most of the focusing of light on to the retina occurs here 95
coulomb unit of charge; one coulomb (C) is equal to the total charge of 6.25×10^{18} electrons 70
counterbalancing a technique in which a controlled experiment is done twice with the same subjects. On the second time through, the control group becomes the experimental group and the experimental group becomes the control group 4
covalent bonding force of attraction between two atoms as a result of sharing electrons 33
cross-pollination transfer of pollen from anther to stigma in different flowers 127
crystal solid substance in which the atoms are arranged in such a way as to produce definite shapes 210
cured meat uncooked fermented meat used in salami and other smallgoods 176
current see electric currents 90
cyan blue-green secondary colour formed by mixing blue and green light 100

data observations made during an experiment 10
DDT chlorinated hydrocarbon pesticide which accumulates in the tissues of organisms and becomes more concentrated up the food chain 190

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decomposers micro-organisms such as bacteria and fungi that break down the remains and waste products of plants and animals, thus returning important nutrients to an ecosystem 188
decomposition reaction reaction where one substance breaks down to simpler substances 52
deoxyribonucleic acid (DNA) makes up the chromosomes in cells; contains the heredity units or genes 144
dependent variable the variable that the experimenter measures but has no control over 2
detritivore consumer that eats detritus 192
detritus small particles of dead organic material such as leaves, twigs and animal remains 192
dialysis removal of wastes from the blood 176
diffraction the slight spreading of a light beam into a pattern of light and dark bands when it passes through a narrow slit 97
direct current (DC) current that always flows in the same direction in a circuit 81
disease any state in which the body does not function normally 166
disinfectant chemical which is used on surfaces other than the body to destroy bacteria 175
dispersion separation of light into its constituent colours, e.g. white light can be split into a spectrum by passing it through a prism 98
diverge spread out; light rays diverge after passing through a concave lens 86
doctor qualified person who practises medicine, involved in preventing and treating illness and injury 8
dominant character phenotype which is expressed in a heterozygous individual 154
dormant volcano a volcano that has not erupted for many years 209
double helix double spiral; the DNA molecule is a double helix 144
double replacement reaction reaction where two compounds react to form two new compounds 52
Down syndrome human chromosomal disorder caused by an extra copy of chromosome number 160

ecosystem system formed by living things interacting with each other and their non-living surroundings; the living things within an ecosystem are interdependent 185
efficiency measure of how well a machine converts the energy put into to it into useful work 117
effort the force applied to a lever to overcome the load 111
egg (ovum) female gamete; organisms or parts of organisms that produce eggs are called female 127
electric current a flow of charged particles (*I*) 70
electric motor energy converter which changes electrical energy into kinetic energy 77
electrical conductivity ability to conduct an electric current 30
electricity the energy that results from the movement of charged particles 66
electrolysis decomposition of a substance using electrical energy 37, 42
electrolyte soluble compound which forms a solution able to conduct an electric current 36
electromagnetic radiation waves which can pass through a vacuum, includes visible light, ultraviolet and infrared rays, microwaves, X-rays, gamma rays and radio waves 97
electromagnetic spectrum electromagnetic radiation of all possible wavelengths; includes visible light, ultraviolet and infrared rays, microwaves, X-rays, gamma rays and radio waves; see also electromagnetic radiation 97
electron fast-moving negatively charged particle that surrounds the nucleus of an atom 24
electron shell region surrounding the nucleus which can hold a set number of electrons 30
electroplating process of coating one metal with a thin layer of another metal by electrolysis 42
element a substance composed of only one type of atom; it cannot be broken down into simpler substances by chemical reactions 28
embryo early stage of newly developing organism from sexual reproduction 137
endothermic a chemical reaction that absorbs heat energy as it proceeds 60
energy the ability to do work or cause change 60
engine a powered machine 114
enthalpy (heat content) amount of energy stored within a substance; denoted *H* 60
epicentre position on the Earth's surface directly above the focus of an earthquake 206
equation method of showing reactants involved and products formed in a chemical reaction 46
exothermic a chemical reaction that releases heat energy as it proceeds 60
experiment a carefully planned activity designed to answer a question or test a hypothesis 2
external fertilisation union of gametes outside the female body 128
extinct volcano a volcano that has not erupted for thousands of years 209
extrapolating a process of estimating a value beyond the range of a set of measurements 12

F_1 generation first filial generation; offspring resulting from a cross of the parental generation 156

fault break in sedimentary rock caused by movement of the Earth's crust 213
fertilisation union of gametes 124
filament stalk-like structure which houses the anther 127
filter transparent material that allows only certain wavelengths of light to pass through 100
first-order consumer consumer that eats a producer 186
fission asexual reproduction via the splitting of single cells 124
fission splitting of an atomic nucleus 184
focal length distance from the centre of a convex lens to its focal point 90
focus (focal point) point where light rays intersect after passing through a convex lens 90
focus point at which an earthquake begins 206
foetus developing embryo after 8 weeks 137
fold bend in sedimentary rock caused by movements of the Earth's crust 213
food chain organisms linked together by their feeding habits in a single series 190
force a pull or push; can change the way things move; measured in newtons (N) 108
fossil fuels fuels formed from the remains of living things buried millions of years ago; oil, coal and natural gas 82
fragmentation asexual reproduction where new individuals grow from pieces of the parent organism 124
fraternal twins twins formed as a result of two separate fertilisations of sperm and eggs 134
friction force which acts against a moving object 117
fungi organisms not considered plants because of their lack of leaves, roots or chlorophyll. They are either parasites, living on other plants or animals, or saprotrophs, existing on dead matter 166

gametes sex cells; male gametes are called sperm and female gametes are called ova or eggs 124, 152
gear wheel wheel with teeth which can turn another gear wheel 112
generator energy converter that changes kinetic energy into electrical energy 77
genes segment of a chromosome (i.e. a sequence of DNA) which controls a particular characteristic of an organism 144
genetic counsellor person expert in giving advice about disorders present in a family or the chance of these being passed on to offspring 170
genetic engineering altering the DNA of an organism in the hope of producing a desirable characteristic; can involve inserting genes from one organism into the DNA of another 148
geneticists scientists who study heredity 154
genotype genetic make-up of an organism 157
geologist scientist who studies the Earth, its origin, history and structure and composition 21, 214
gestation period period of time from fertilisation to birth 129
gold-plated cheaper metal coated with gold via electroplating 42
Gondwana ancient supercontinent existing many millions of years ago; consisting of the present-day landmasses of Australia, India, South America, Africa and Antarctica 215
granite igneous rock formed from magma that has cooled slowly beneath the Earth's surface 211

habitat the place where an organism lives 185
heredity transmission of characteristics from parents to offspring via genes 144
heterozygous refers to an individual with different alleles for a given gene 155
homologous refers to chromosome pairs which carry the same genes (although these may be the same or different alleles); humans have a total of 46 chromosomes as 23 homologous pairs 154
homozygous refers to an individual with identical alleles for a given gene 155
hormone chemical messenger carried by the bloodstream 134
hybrid organism produced via a mating between two different species; normally infertile 124

identical twins twins formed from a single egg and sperm 136
igneous rock rock formed by the cooling of molten material from within the Earth 210
image formed by a lens or mirror; observable but not actually there 86
immunisation vaccination 178
incident ray light ray which strikes the boundary between two materials 88
inclined plane machine which decreases the effort required to overcome a load by increasing the distance over which the effort acts 110
incomplete dominance phenotype in which the heterozygote is different from (often intermediate between) either homozygote 156
incubation period time between micro-organism entering an organism and symptoms of disease appearing 168
independent variable the variable that is changed by the experimenter, usually in a regular way 2

indicator a substance that indicates a change, often by changing colour itself, that would otherwise be difficult to observe directly 38
infectious disease caused by a micro-organism 166
inherited passed from parents to offspring via genes 162
inorganic matter simple non-organic substances such as water, minerals, carbon dioxide 196
insecticides pesticide active against insects 191
insulator a substance that will not allow an electric current to flow easily through it 66
interdependent dependent on each other in some way 186
internal fertilisation union of gametes inside the female's body 128
interpolating a process of estimating a value between two measurements 12
inverted upside down 91
ion atom or group of atoms that has lost or gained electrons and is therefore charged, e.g. Na^+,Cl^- 24
ionic bonding electrical attraction that holds together two oppositely charged ions 28, 32
ionic compound compound made up of positive and negative ions in a set ratio 34
IVF in-vitro fertilisation 138

karyotype the number, size and shape of the chromosomes in an individual or cell 160
kinetic energy energy possessed by a moving object 117

labour contraction of the walls of the uterus eventually resulting in birth 137
lack of accommodation (presbyopia) see accommodation 94
lattice ordered arrangement of particles in three dimensions; an ionic lattice is made up of positive and negative ions in a set ratio 28
law of reflection reflected rays leave a mirror at the same angle as they hit it 86
lens transparent material such as glass with a shaped surface 90
lever simple machine consisting of a rigid rod pivoting on a point 106
limbs of the fold straight parts between synclines and anticlines 213
line graph graph which has measurements plotted as points on a set of axes; these are then connected by a line 10
litmus paper acid–base indicator which is red in acids and blue in bases 38
load the force that a lever is used to overcome 106
long sight (hypermetropia) eye defect where a person can see far objects but not close ones; the eyeball is too short 94
loudspeaker energy converter that changes electrical energy into sound energy 80
lymphatic system body system consisting of vessels and lymph nodes transporting a clear fluid called lymph around the body; filters out harmful substances and bacteria from the body 172
lymphocyte type of white blood cell which makes antibodies 173, 178

machine device that changes the size and/or direction of a force 106
magenta purplish secondary colour formed by mixing blue and red light 100
magma molten rock below the Earth's crust 208, 216
magnetic effect causing a magnetic field 78
magnetic field a region in which a magnetic object will experience a force, e.g. around a magnet or a wire carrying an electric current 79
magnified image image that appears larger than the object that it comes from 91
malleability ability to be beaten and worked without shattering 30
mammals class of endothermic vertebrates; have hair and feed their young on milk, e.g. humans, dogs, dolphins 129
mantle solid layer between the Earth's crust and magma; although solid it is thought that it does flow slowly 216
marsupial mammal where the young is born after a very short gestation period and continues its development in a pouch 129
matter anything that takes up space and can be weighed; material 24
meiosis cell division which results in daughter cells having half the number of chromosomes as the parent cell; gamete formation 153
menarche arrival of a girl's first period 134
menstrual cycle female reproductive cycle 134
metallic bonding force of attraction between positive metal ions and the free electrons between them 32
metallic lattice regular arrangement of positive metal ions surrounded by electrons that are free to move 32
meteorite an object that has reached the Earth's surface from space 204
micro-organism (microbe) organism that can only be seen clearly with a microscope, e.g. bacteria, algae 166
microphone energy converter that changes sound energy into electrical energy 80

mineral generally inorganic, naturally occurring chemical compounds 210
mitosis nuclear division in which each chromosome is replicated and passed to two daughter cells; used generally as cell division to produce two identical daughter cells 152
molecule group of two or more atoms bound together; there is a set number of atoms in a molecule 24
monotremes egg-laying mammals, e.g. platypus 129
mucus lubricating and protective fluid secreted by membranes in many different parts of the body 172
mudstone sedimentary rock formed by compaction of grains that cannot be seen or felt 212
mutation spontaneous change in the DNA structure of an organism 146

natural immunity immunity to a disease following that disease 178
negative first image formed on light-sensitive photographic paper in which light and dark areas are reversed 102
neutral neither acidic nor basic; neutral substances have pH of 7 38
neutralisation reaction of an acid with a base to form a salt and water 54
neutron uncharged particle found in the nucleus of an atom 24
nitrogen cycle cycle of nitrogen-containing compounds through the biosphere 195
nitrogen-fixing bacteria bacteria that convert nitrogen gas to ammonia. This is then converted to nitrate which is taken up by the roots of plants 195
noble gases (group VIII) elements with a full outer shell 31
non-electrolyte a compound which does not conduct an electric current in solution 36
non-infectious disease not transmitted from one person to another, caused by other factors 166
normal line at right angles to a surface 82
normal fault results when one block slips down the fault line or plane of breakage 213
nucleus found at the core of an atom; contains protons and usually neutrons 24
nuee ardent deadly cloud resulting from an explosive volcanic eruption 209

oestrogen female reproductive hormone that causes the uterus to thicken 134
ohm unit of electrical resistance (Ω) 73
Ohm's law in a metallic conductor the ratio of voltage to current is constant; the constant is called the resistance: $V/I = R$ 75
organic material complex carbon-containing compounds such as protein and carbohydrate 194
outer shell see valence shell 31
ovary organ in sexually reproducing organisms where the egg cells (ova) are formed 127, 134
oviduct (fallopian tube) tube which links ovary to uterus; fertilisation occurs here 134
ovulation release of ovum from an ovary 134
ovum (egg) female gamete 132
oxidation chemical reaction in which oxygen is added to a substance or hydrogen is removed from it; any reaction when a material loses electrons 57

palaeontologist scientist who studies fossils 214
parallel circuit an electric circuit in which current divides and some passes through each component 69
pathogen disease-causing organism 166
pattern a trend in data which is repeated over time 12
pedigree description or diagram representing the occurrence of a characteristic over a number of generations 162
penicillin antibiotic produced from *Penicillium* fungi or mould 174
period period of menstrual flow 134
periodic table table of the known elements arranged according to their atomic numbers and electron structure 30
pesticide any substance used to destroy or control pests 190
pH scale a number scale of 0 to 14 which indicates how acidic or basic something is 38
phagocyte type of white blood cell that can engulf and destroy bacteria or other invading micro-organisms 172
phenotype characteristics expressed by an individual; determined by the genotype and influenced by environmental conditions 157
photograph permanent image or picture formed on light-sensitive paper 102
photon particle or package of energy in which light and other forms of electromagnetic waves are emitted. It has both particle and wave properties, no charge and no mass 97
physical change a change in which no new substance is formed, e.g. ice melting 46
physics study of natural laws and properties of matter and energy 21
phytoplankton tiny photosynthetic organisms found in the plankton of an aquatic ecosystem 189
pivot (fulcrum) the point at which a lever pivots or pushes 106
placebo a non-acting substitute given to a subject in a blind or double-blind experiment 5

placenta an organ that attaches the developing embryo or foetus to the uterus, found in placental mammals 137
placentals mammals which have a relatively long gestation period during which the foetus is nourished via a placenta 129
plane mirror flat mirror 86
plate tectonics theory which suggests that the Earth's crust is made up of solid plates moving relative to each other 217
plates pieces of the Earth's crust which are moving 216
pollen structures formed on the anthers of flowering plants which contain sperm 127
pollination transfer of pollen from anther to stigma in flowering plants 127
polyatomic ion (radical) a group of atoms bonded together with an overall positive or negative charge 35
pool chlorination addition of chlorine compound to swimming pools to prevent the growth of algae and other micro-organisms 40
potential difference (voltage) electrical energy lost across a component or gained across a cell; measured in volts (V) 72
potential energy stored energy 117
power station site where electricity is generated 81
precipitate condensation of water resulting in rain (or dew); (chemistry): a solid formed when two solutions are mixed 46
precision the smallest division in a scale on a measuring instrument 4
primary colours red, green and blue; when added together form white light 100
print image created by exposing the negative to light; light and dark areas now correspond to the original situation; a photograph 102
producer organism that can manufacture its own food by photosynthesis; all plants are producers 186
product new substance produced in a chemical reaction 46
proton positively charged particle found in the nucleus of an atom 24
protozoa single-celled animal in water and moist soil 166
pulley wheel with a rim 112
pure breeding population of organisms which display the same characteristic over many generations; homozygous organism 154
pure spectrum spectrum formed by first passing light through a convex lens and then passing it through a prism 92
radioactive elements which are radioactive have unstable nuclei which break down into simpler nuclei with the emission of alpha and beta particles and gamma radiation 24
ramp machine to help lift objects 110
rate a measure of how fast something changes with time 62
reactants substances that react to form new substances in a chemical reaction. Reactants are written on the left-hand side and products are written on the right-hand side 46
reactivity measure of how likely a substance is to become involved in a chemical reaction 57
real image image that can be projected on to a surface 92
receptacle base of the flower; supports all reproductive structures 127
recessive character phenotype which is only expressed in a homozygous individual 154
redox reactions oxidation and reduction reactions 57
reduction chemical reaction in which oxygen is removed from a substance or hydrogen is added to it; any reaction when a material gains electrons 57
reflection rays or beams of light bouncing off a smooth surface; sound will also bounce off surfaces 86
refracted ray light ray that exits from one material into another, e.g. from glass to air 88
refraction change in direction of a wave due to change in speed as it passes from one medium to another, e.g. a light ray bends as it passes from air to glass 88
resistance (*R*) opposition to the flow of electric current through a conductor, measured in ohms (Ω); measure of voltage required to drive a certain sized current through a component 73
respiration oxidation ('burning') of glucose in cells to release energy; an exothermic reaction 188
retina light-sensitive cells on the interior surface of the eye 94
reverse fault results when one block slides up the fault line or plane of breakage 213
ringworm fungal infection of the skin; grows in a circular pattern 170

salt compound usually consisting of metal and non-metal atoms bonded together to form a crystal lattice of positive and negative ions 36
sandstone sedimentary rock formed by compaction and cementation of grains that can be seen and felt 212
saturation (dew point) state where the air can hold no more water vapour 198
scavenger consumer that feeds on dead animals 188

scientific model simple representation of a complex thing or idea 24
screw machine with a spiral-shaped inclined plane 111
secondary colours yellow, cyan and magenta; formed by mixing two primary colours 100
second-order consumer consumer that eats a first-order consumer 187
sector graph circle graph which is divided into percentages or fractions 10
seed protective structure surrounding embryonic plant 127
seismic wave shock waves from an earthquake 206
seismogram graphical records of the size and duration of an earthquake 206
seismograph device for measuring movements in the Earth's crust 206
seismologist scientist who studies earthquakes 204
self-pollination transfer of pollen from anther to stigma in the same flower 127
semen mixture of seminal fluid plus sperm cells 132
seminal fluid secretions from the various sex glands; nourishes and activates sperm and prevent their clumping together 132
series circuit an electric circuit in which current passes through one component and then another 69
sex-linkage (X linkage) refers to characteristics which are determined by genes on the X chromosome 156
sex-linked genes genes on the X chromosome (see sex linkage) 158
sexual reproduction reproduction involving two individuals whose sex cells fuse to produce a new individual 124
short sight (myopia) eye defect where a person can see close objects but not far ones; the eyeball is too long 94
shutter opens to allow light into a camera; in more expensive cameras the length of time it is open can be controlled 102
simple machine device which makes something more convenient to do or reduces the effort required 106
single replacement reaction reaction where atoms of one element replace the atoms of another element within a compound 52
soluble a soluble substance will dissolve in a liquid; water-soluble substances dissolve in water 36
solution a substance dissolved in another substance 36
species single type of organism; individuals of the same species can interbreed to produce fertile offspring 124
spectrometer instrument for analysing the composition of light 98
spectrum seven colours seen when white light is passed through a prism; small part of the electromagnetic spectrum 96
sperm male gamete; organisms or parts of organisms which produce sperm are called male 126
spores small structures containing reproductive cells; a type of asexual reproduction 124
stamen male structure in flowering plants composed of filament and anther 126
state whether matter is solid, liquid or gas 24
sterile incapable of producing offspring 124
stigma tip of the style where the pollen sticks 127
strike-slip fault results when the blocks either side of a fault slide past each other horizontally 213
style part of the pistil through which the pollen tube grows to the ovary 127
subduction occurs when two plates collide and one overrides the other, which disappears into the mantle 216
sustainable development development which meets the needs of the present without compromising the ability of future generations to meet their own needs 200
symptoms any sign or outward appearance indicating the presence of disease 168
syncline downward bending fold 213
synthesis reaction (combination reaction) reaction where two substances react to form a third 52

temperature a measure of how hot something is; measured in degrees Celsius 62
terminals connection points on a component of an electric circuit 70
terrestrial living on land 186
testes male sex organs; produce gametes or sperm cells 132
testosterone male sex hormone 133
third-order consumer consumer that eats a second-order consumer 187
tiltmeter a device which measures the amount of tilt produced in rock layers by magma pushing up from below Earth's surface 218
total internal reflection when light hits a boundary between two materials at a large angle of incidence and is all reflected at the boundary with none transmitted; the smallest angle of incidence where this occurs is called the critical angle 88
toxin harmful chemical produced by pathogen 168
transgenic organism containing genes from another species 149

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transparent material through which light can travel 88
trend the direction and rate of change in data over a period of time 12
trial and error a process of experimentation; various methods or substances are tried and eliminated as unsuitable 6
tri-focal spectacles with lenses that are in three sections of different shapes and hence strengths 95
trophic level feeding level in an ecosystem 187
tsunami giant wave caused by underwater earthquake or volcanic eruption 206

universal indicator acid–base indicator which displays a range of colours which correspond to a number on the pH scale; can be used to determine the strength of an acid or base 38
uterus structure in which the embryo implants and develops 134

vaccination administration of a vaccine, usually swallowed or injected 178
vaccine substance usually containing a weakened form of a pathogen which is introduced into the body to stimulate the production of antibodies 178
vagina muscular tube leading from the outside to the uterus; receives the penis during copulation and is also the birth canal 134
valence electrons electrons in the outer or valence shell 31
valence shell outermost electron shell that contains electrons 31
valency number of electrons an atom gains or loses when it forms chemical bonds; charge an ion possesses 34
valency chart chart showing the valency of elements 34
variable any factor that may have an effect on the outcome of an experiment 2
vegetative propagation asexual reproduction in plants where the new individual is propagated or grown from part of the parent plant 125
vermicast end-product of vermiculture; has much less bulk and simpler molecules than the original organic waste 196
vermiculture process of worms breaking down organic waste to vermicast 196
virtual image image that can be seen but not projected onto a surface 92
virus very small non-cellular pathogen which only shows signs of life inside a host cell 166
voltage see potential difference 72
voltmeter device to measure potential difference or voltage 72
vulcanologist scientist who studies volcanoes 204

water cycle movement of water through the biosphere 198
wedge machine to help split wood; a double inclined plane 111, 118
wheel special type of lever with axle acting as the pivot 112
windrow aeration tray used in vermiculture 196

X chromosome one of two types of sex chromosome found in humans; females have two X chromosomes 152
X-ray crystallography technique which uses X-rays to infer the shape of molecules 144

Y chromosome one of two types of sex chromosome found in humans; males have one Y and one X chromosome 152

zooplankton tiny consumer organism found in the plankton of an aquatic ecosystem 189
zygote cell formed by the fusion of a sperm and egg; first cell of a multicellular organism 132

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